

Marine Water Quality in Hong Kong in 2016



Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Our Mission

To conduct a comprehensive and scientific monitoring programme that helps safeguard the health of Hong Kong's marine environment and achieve the Water Quality Objectives.



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1. Introduction

The Hong Kong Special Administrative Region (HKSAR) has a land area of 1,106 km² and marine waters of 1,649 km². It has long coastlines, including 468 km in the Kowloon Peninsula and New Territories and 732 km in Hong Kong Island, Lantau Island and other small islands. Excluding Hong Kong Island and Lantau Island, there are 261 islands in the territory, each with an area greater than 500m². Hong Kong's marine waters also support a variety of marine life as well as habitats.

To protect the marine environment of Hong Kong, the Environmental Protection Department (EPD) has been implementing a comprehensive marine water quality monitoring programme since 1986. The aims and objectives of the programme are to:

- evaluate the state of marine waters;
- monitor long-term changes in water quality;
- provide a scientific basis for planning water pollution control strategies; and
- assess the compliance with the key statutory Water Quality Objectives (WQOs).

Every month, the EPD monitors the marine water quality at 76 monitoring stations and collects and examines phytoplankton samples from 25 of these stations. We also monitor the water quality of 14 typhoon shelters, three sheltered anchorages and the Government Dockyard (GD) across Hong Kong bimonthly. In addition, sediment samples are collected and analyzed at 60 stations, covering open waters, typhoon shelters, sheltered anchorages and the GD, twice a year.



EPD's marine monitoring vessel "Dr. Catherine Lam"

The EPD has a dedicated marine monitoring vessel “*Dr. Catherine Lam*”. It is equipped with an advanced conductivity-temperature-depth (CTD) profiler linked to a computer-controlled rosette water sampler which takes field measurements and collects water samples simultaneously. Marine sediments are sampled and collected using a Van Veen sediment grab sampler. The water and sediment samples are analyzed by the EPD’s laboratory and the Government Laboratory for over 80 physical, chemical and biological parameters. More details are given in the Appendixes.

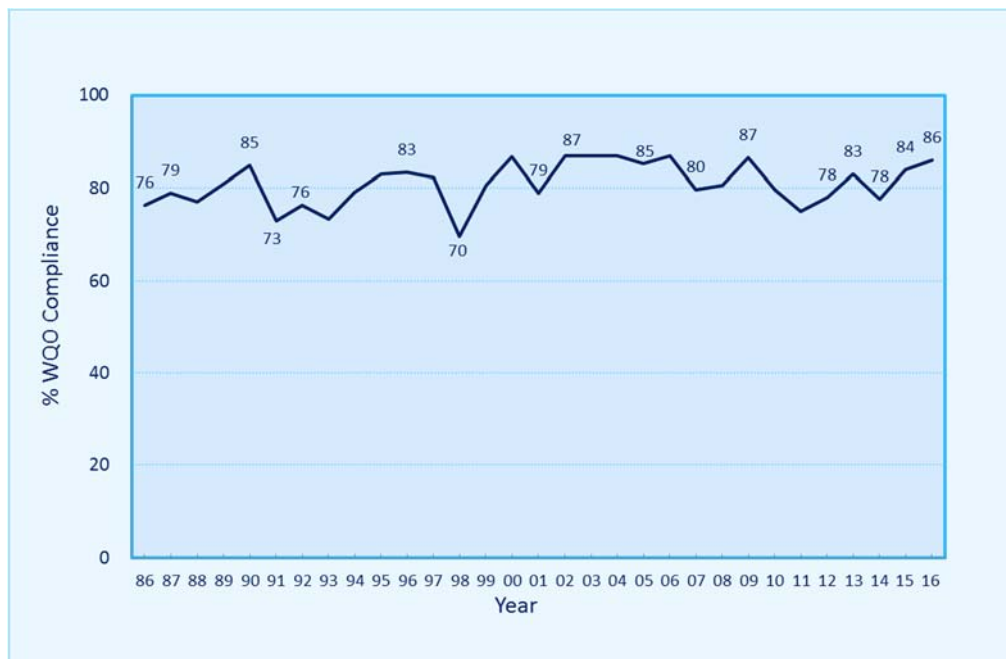


The water sampling process by the computer-controlled rosette water sampler.

2. The State of Hong Kong Marine Waters in 2016

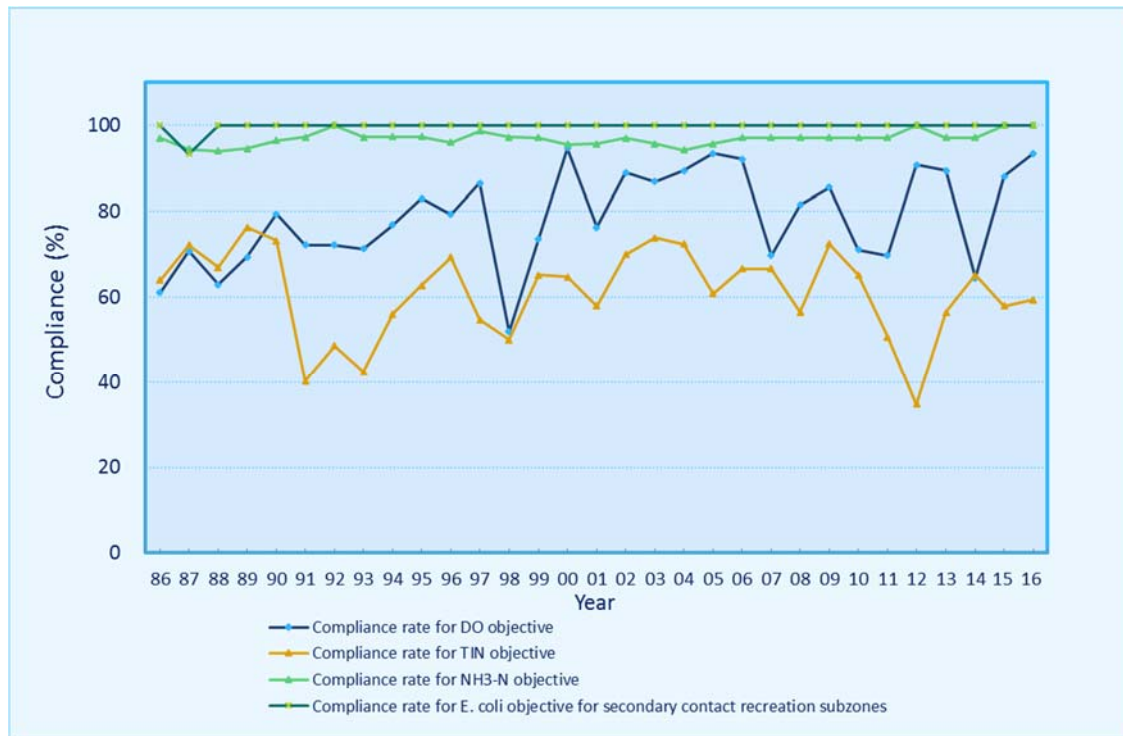
2.1 Compliance Rate of Water Quality Objectives (WQOs)

The calculation of the overall WQO compliance rate of the whole territory is based on the combined individual compliance rates of all stations with the marine WQOs for four key parameters namely Dissolved Oxygen (DO), Total Inorganic Nitrogen (TIN), Unionized Ammonia Nitrogen (NH₃-N) as well as *E. coli* bacteria. The overall marine water quality compliance with the WQOs in Hong Kong was 86% in 2016, compared with 84% in 2015.

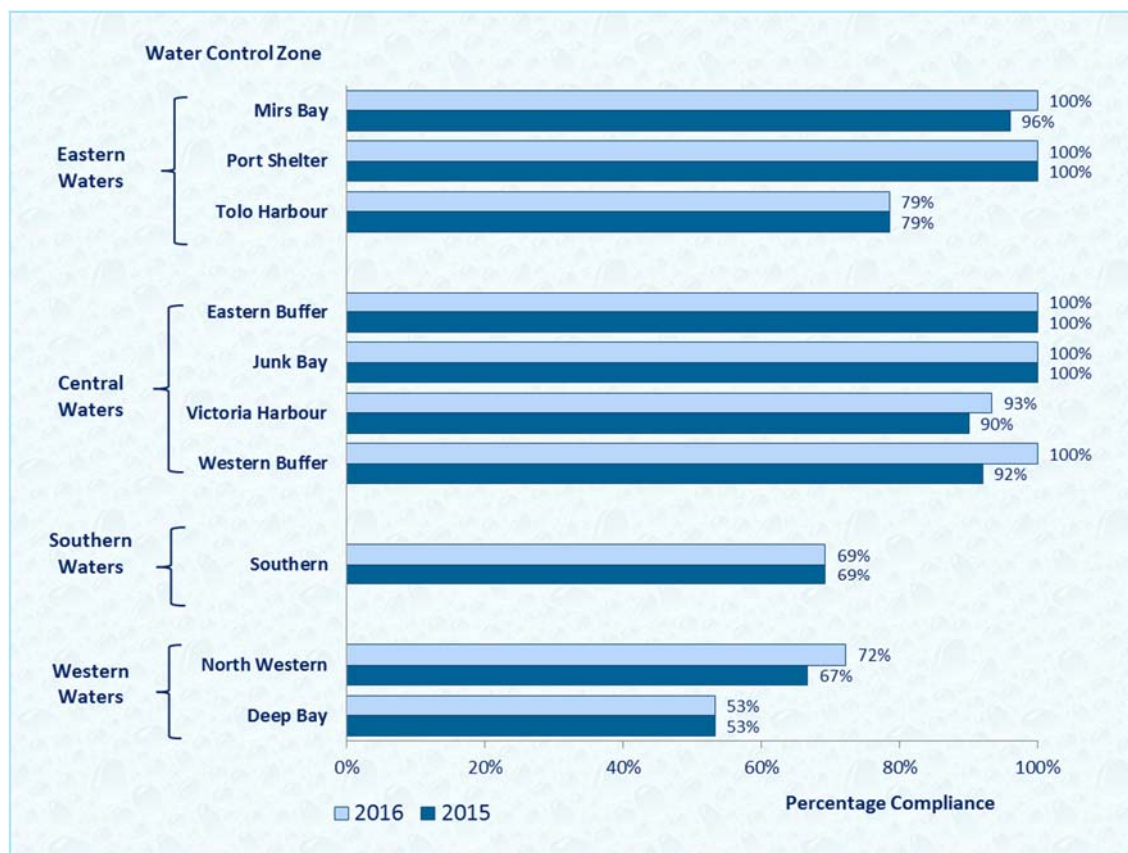


Overall compliance with the marine Water Quality Objectives (WQOs) in Hong Kong, 1986-2016

The compliance rates for the four parameters and the overall compliance rates for the ten water control zones (WCZ) are shown in the following figures. Notably, the compliance rates for *E. coli* and NH₃-N WQOs have achieved 100% in all applicable WCZs in 2016. There were also improvements of the compliance rates for the DO objective (from 88.2% in 2015 to 93.4% in 2016) and the TIN objective (from 58.0% in 2015 to 59.4% in 2016). For the DO WQO, the non-compliance was mainly observed in the Tolo Harbour and Channel and Deep Bay WCZs. For the TIN WQO, the non-compliance was observed in the Southern WCZ, North Western WCZ and to a less extent the Victoria Harbour WCZ. The cause of TIN non-compliance could be attributed to the seasonal discharges of the Pearl River as well as the stringent TIN WQO for these WCZs. It is believed that the TIN-non-compliances at these WCZs would not have any significant impact on their water quality as reflected by the low occurrence rates of red tides originated from these waters.



Compliance rates for the four key WQO objectives in Hong Kong, 1986-2016



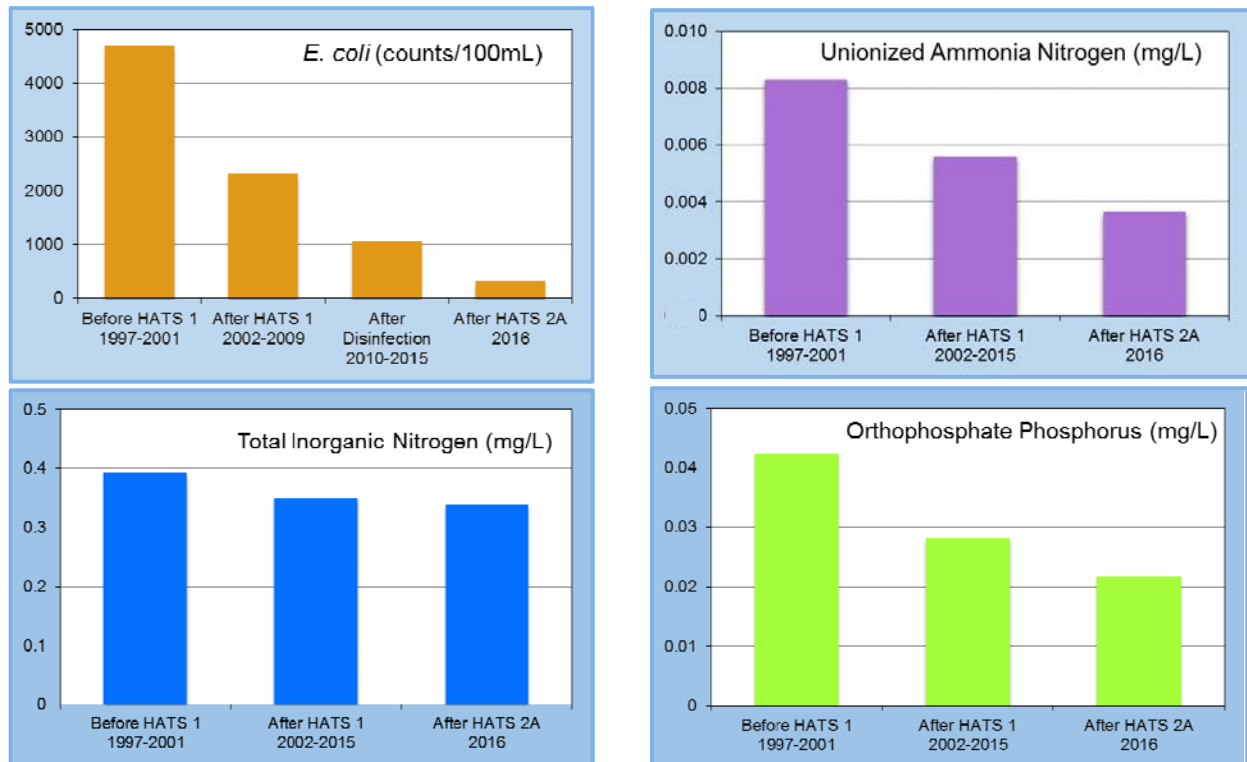
Overall level of compliance with the Water Quality Objectives in the ten Water Control Zones in 2015 and 2016

2.2 Highlight in 2016 - Harbour Area Treatment Scheme Stage 2A

Through the continuous efforts of the EPD, particularly through environmental law enforcement and implementing sewerage projects, the state of Hong Kong marine waters has shown a clear sign of gradual improvement.

For the Victoria Harbour, the water quality has been improving significantly with the progressive implementation of the Harbour Area Treatment Scheme (HATS), particularly after the commissioning of HATS Stage 2A in December 2015. In 2016, the WQO compliance rates for DO, NH₃-N, and TIN were 100%, 100% and 80% respectively. The overall WQO compliance rate of these three parameters in 2016 was 93%, as compared with 50-90% in 2011-2015. While there is no *E. coli* WQO in the Victoria Harbour WCZ, the overall annual geometric mean *E. coli* level of the WCZ has reduced to around 300 counts per 100mL, which was about 10 fold reduction as compared with that of the late 1990's. The level of *E. coli* in 2016 was also below the WQO of 610 count per 100 mL for the secondary contact recreation subzones.

Indeed, the EPD's monitoring data indicate that there were significant improvements in levels of key water quality parameters of the Victoria Harbour WCZ as compared with those before and during various stages of the HATS implementation as illustrated in the following figures.

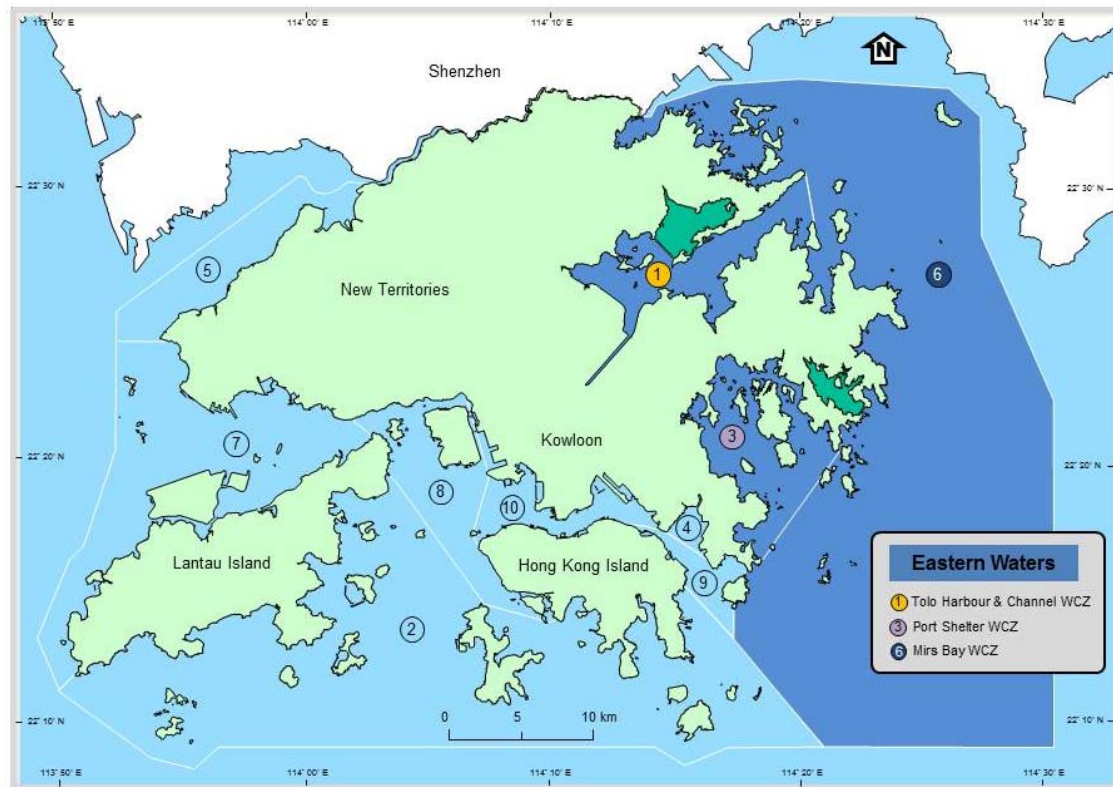


Reductions of *E. coli*, unionized ammonia nitrogen, total inorganic nitrogen and orthophosphate phosphorus levels in the Victoria Harbour WCZ as brought by the progressive implementation of the HATS

3. Water Quality of Ten Water Control Zones

Based on the hydrographical conditions of Hong Kong waters and the proximity to the Pearl River Estuary, the ten WCZs are broadly grouped into four main areas and their water quality in 2016 are reported in the following sections.

3.1 Eastern Waters



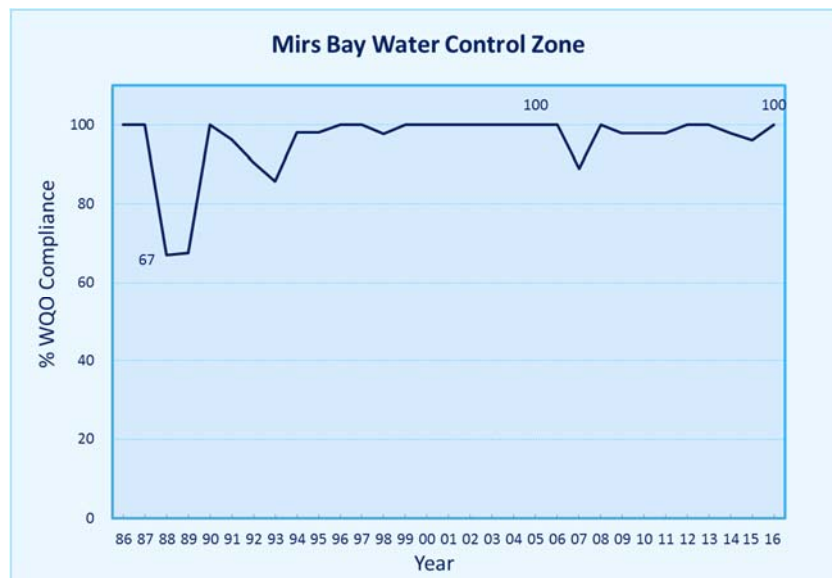
The eastern waters comprise the Mirs Bay, Port Shelter, and Tolo Harbour and Channel WCZs. These waterbodies have beautiful coastlines and good water quality which support a high diversity of marine life, fisheries and recreational activities, including six gazetted beaches, three marine parks and 21 fish culture zones.

Mirs Bay Water Control Zone

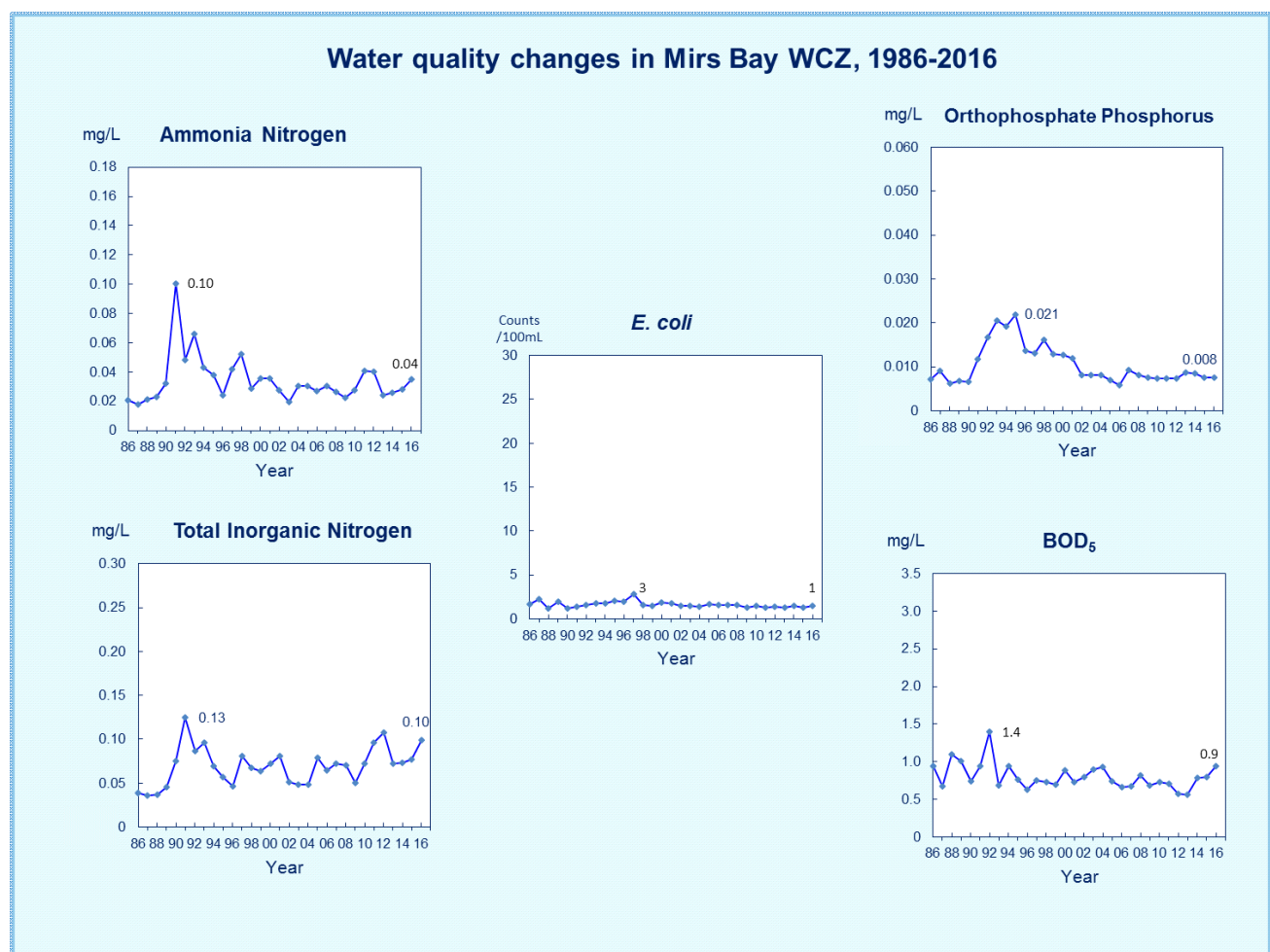
In 2016, the overall WQO compliance rate of the Mirs Bay WCZ was 100%. The water quality of the Mirs Bay WCZ was good in 2016 with high DO and low TIN levels. Furthermore, the Mirs Bay WCZ complied with the bacteriological WQO of ≤ 610 *E. coli* counts/100 mL (annual geometric mean) for secondary contact recreation.

The long term WQO compliance rates and trends of some key water quality parameters for the

Mirs Bay WCZ are illustrated in the following figures. In general, the water quality conditions in recent years were better than those in the late 80's.



Overall compliance with the marine WQOs in the Mirs Bay Water Control Zone, 1986-2016



Overall long term water quality changes in the Mirs Bay Water Control Zone, 1986-2016

Port Shelter Water Control Zone

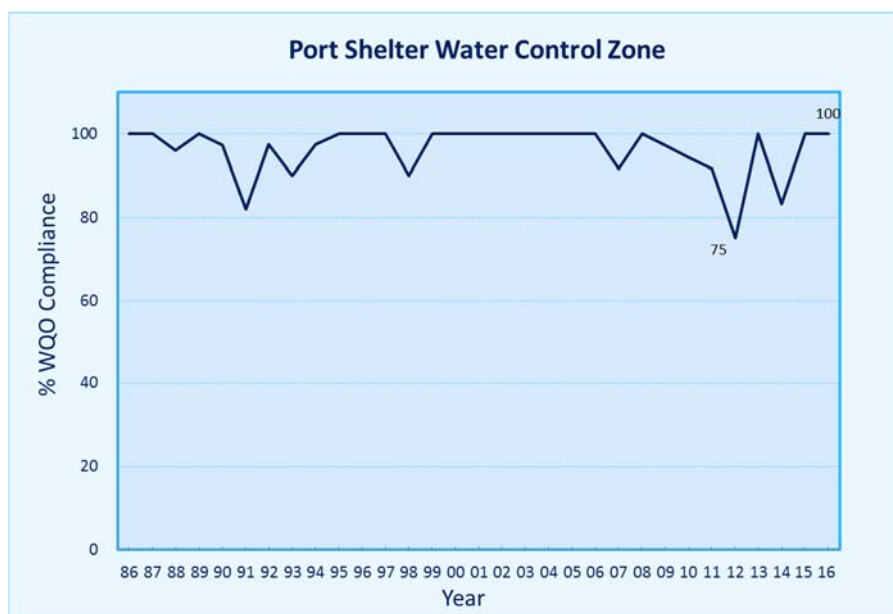
Port Shelter is a popular area for various water sports activities, including leisure fishing, swimming, yachting, water-skiing and diving. In 2016, the water quality of the Port Shelter WCZ was good and the overall WQO compliance rate was 100%, with full compliance with the WQOs for *E. coli*, DO, TIN and NH₃-N.

The six gazetted beaches which attracted 1.2 million visitors during the 2016 bathing season, complied with the bacteriological WQO for bathing in 2016.

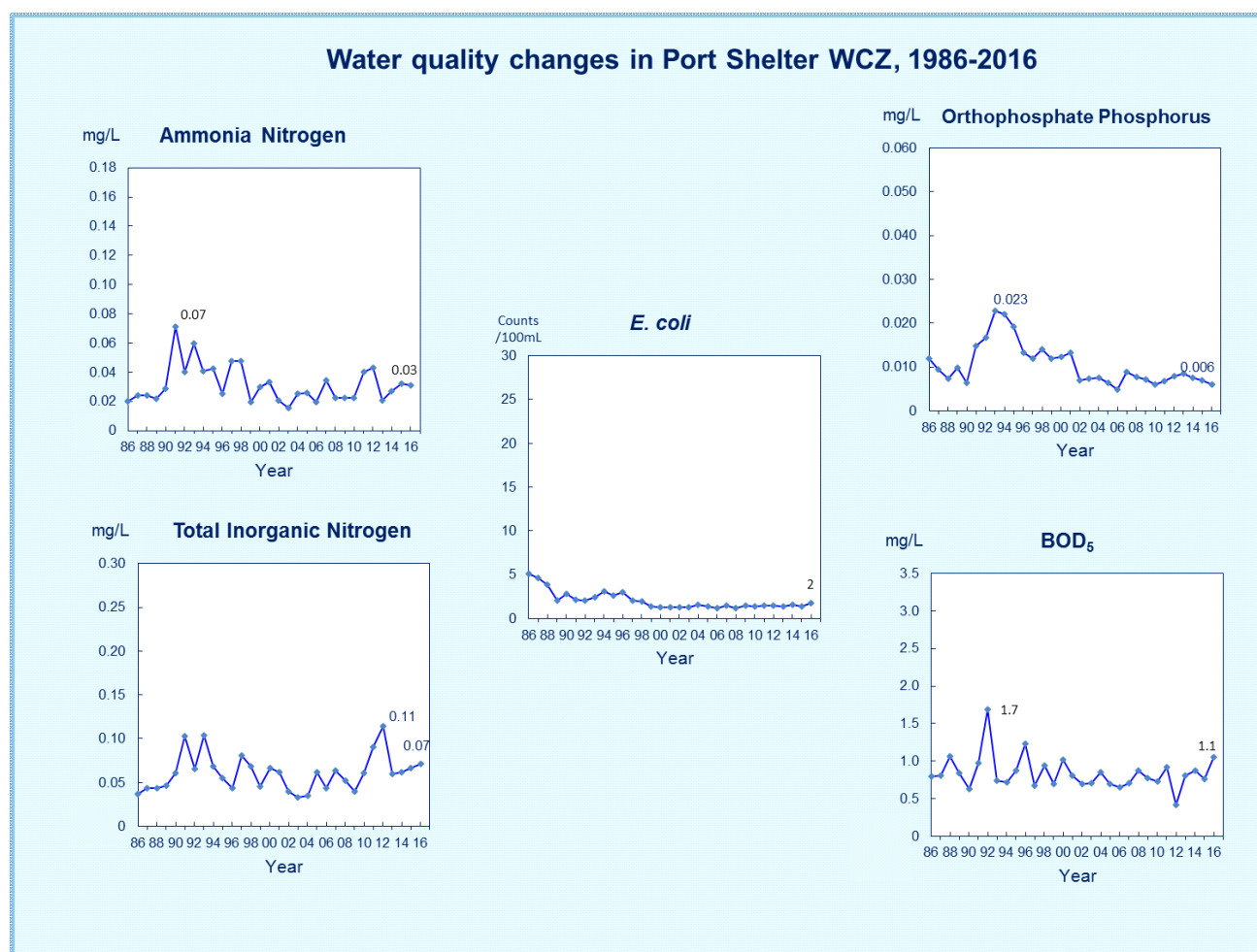
The long term WQO compliance rates and trends of some key water quality parameters for the Port Shelter WCZ are illustrated in the following figures. In general, the water quality in recent years was better than those in the late 80's. The levels of key water quality parameters, such as TIN and orthophosphate phosphorus (PO₄-P), remain steady and low in recent years.



The distinct marine cave features of the Hong Kong UNESCO Global Geopark in Port Shelter.



Overall compliance with the marine WQOs in the Port Shelter Water Control Zone, 1986-2016



Overall long term water quality changes in the Port Shelter Water Control Zone, 1986-2016

Tolo Harbour and Channel Water Control Zone

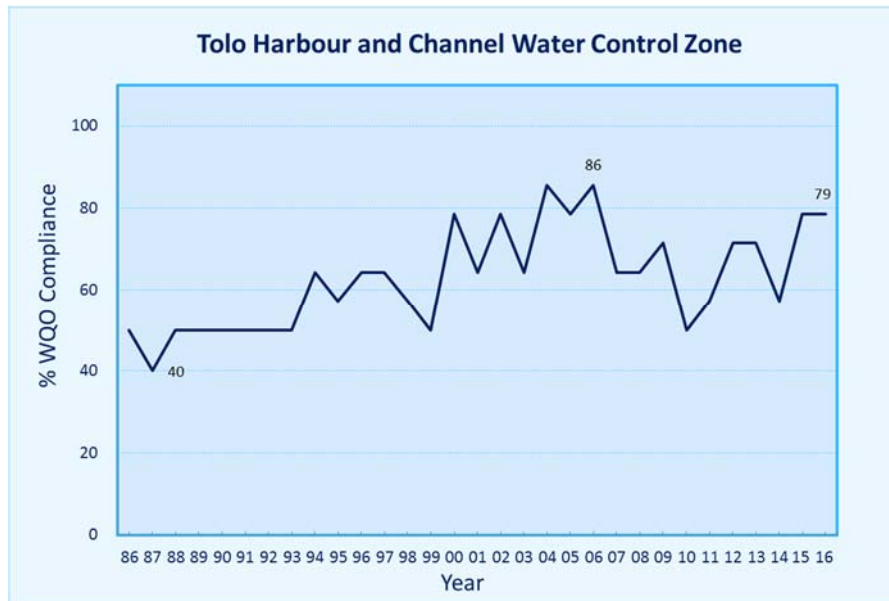
The monitoring results of key water quality parameters, including DO and *E. coli*, indicated that the overall WQO compliance rate of the Tolo Harbour and Channel WCZ in 2015 and 2016 was the same at 79%. The compliance rates for the DO objective of these two years were maintained at 57%. Furthermore, the Tolo Harbour and Channel WCZ also complied with the bacteriological WQO of ≤ 610 *E. coli* counts/100 mL (annual geometric mean) for the Secondary Contact Recreation Subzone.

Since Tolo Harbour is a shallow semi-enclosed water body with low water exchange rate with the Mirs Bay, the harbour's essentially landlocked situation often leads to the natural stratification of the water column and lower bottom DO levels, particularly during the hot summer months, hence resulting in non-compliance with the DO objective.

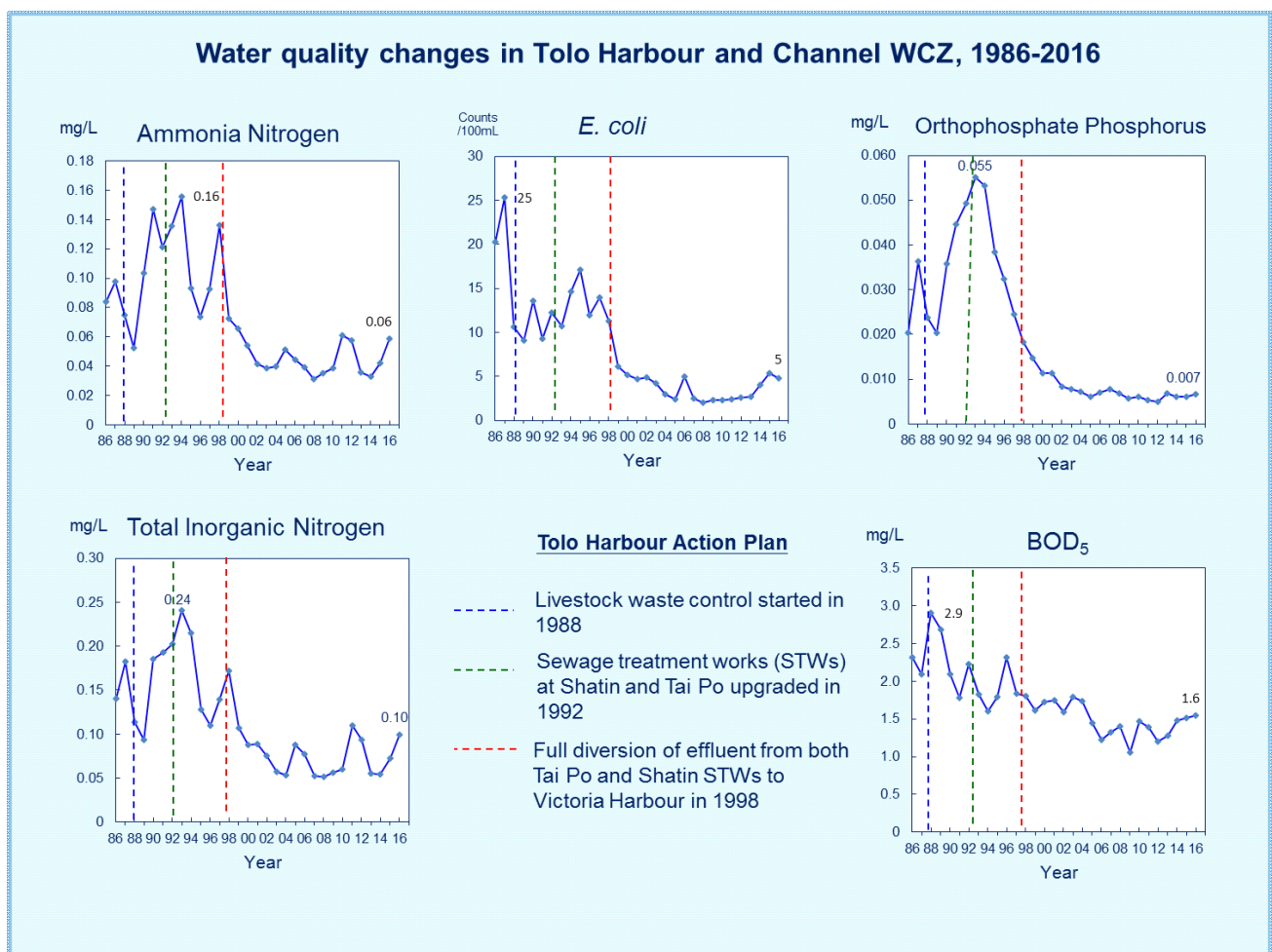
With the implementation of the Tolo Harbour Action Plan in the mid-80s, which includes the control of livestock waste, the provision and improvement of sewerage, the export of treated sewage effluent from Sha Tin and Tai Po Sewage Treatment Works outside the Tolo Harbour for discharging into the Victoria Harbour via the Kai Tak River and the extension of village sewerage in the catchment area, there has been a steady improvement in water quality in the Tolo Harbour in the past decade, including a decrease in the levels of 5-day biochemical oxygen demand (BOD₅), *E. coli*, TIN, NH₃-N and PO₄-P. The long term data as shown in the next figure indicate the decreasing trend of organic and nutrient levels in the Tolo Harbour and Channel WCZ.



A scenic view of the urbanized Ma On Shan in the Tolo Harbour and Channel Water Control Zone

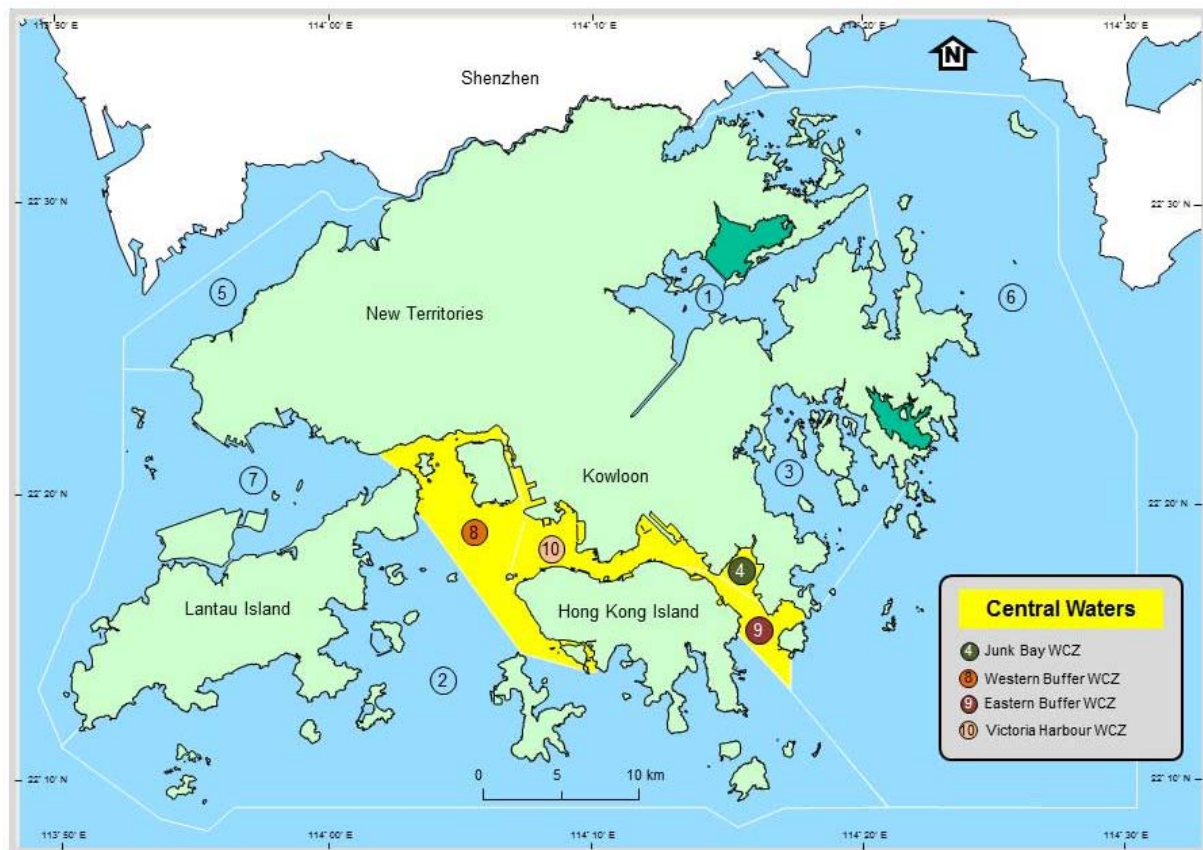


Overall compliance with the marine WQOs in the Tolo Harbour and Channel Water Control Zone, 1986-2016



Overall long term water quality changes in the Tolo Harbour and Channel Water Control Zone, 1986-2016

3.2 Central Waters



The central waters of Hong Kong, including the Victoria Harbour, Eastern Buffer, Western Buffer and Junk Bay WCZs, are important navigational channels and port areas.

After the commissioning of the Stonecutters Island Sewage Treatment Works (SCISTW) under the HATS Stage 1 in December 2001, about 75% of the sewage around Victoria Harbour receives chemically enhanced primary treatment, resulting in a 70% reduction of the pollution load (in terms of organic pollutants) into the harbour. This has improved the water quality in the harbour particularly in the Eastern Buffer and Junk Bay WCZs. With the Advance Disinfection Facilities (ADF) of the HATS Stage 2A commissioned in March 2010, the *E. coli* removal efficiency at the SCISTW has increased to over 99%. The *E. coli* level in the western Harbour area has been substantially reduced since then.

With the full commissioning of the HATS Stage 2A in December 2015, all the sewage generated from both sides of Victoria Harbour is transported to the SCISTW via deep tunnels for centralized chemically enhanced primary treatment and disinfection, and discharged into the western harbour area. Further improvement in water quality has been observed in terms of the reductions of *E. coli*, $\text{NH}_3\text{-N}$, TIN and $\text{PO}_4\text{-P}$ levels.

Victoria Harbour Water Control Zone

The overall WQO compliance rate of the Victoria Harbour WCZ in 2016 was 93%, as compared with 50-90% in 2011-2015. The WQO compliance rates for DO and NH₃-N in 2016 were both 100%. The WQO compliance rates for TIN of eight out of ten monitoring stations in 2016 were 100%. The two stations not meeting the TIN objective were VM12 and VM14. They are located at the enclosed navigation channel at Rambler Channel and Tsuen Wan Bay, respectively. As a result, the overall WQO compliance rate for TIN was 80%.

While there were clear improvement of NH₃-N, *E. coli*, PO₄-P and BOD₅ over the past 30 years, there were large fluctuations in the TIN levels in its long term trend. This is due to the year-to-year natural fluctuation of the discharge from surface run-off, as well as the higher background TIN level under the influence of Pearl River discharge, as reflected in the relatively higher TIN level in many stations in the Northwestern and Southern WCZs.

The *E. coli* level in the eastern side of Victoria Harbour as illustrated in the following figure has markedly improved since the HATS Stage 1 was commissioned in December 2001. The Cross Harbour Swim, suspended from 1979 because of poor water quality, has resumed as an annual event in October since 2011 in the eastern side of the Harbour. About 3000 participants took part in the 2016 Cross Harbour Swim. Since the full commissioning of the HATS Stage 2A in December 2015, the *E. coli* level of the central harbor area has dropped significantly. It was planned that the Cross Harbour Swim would be moved back to the central harbor area in 2017.



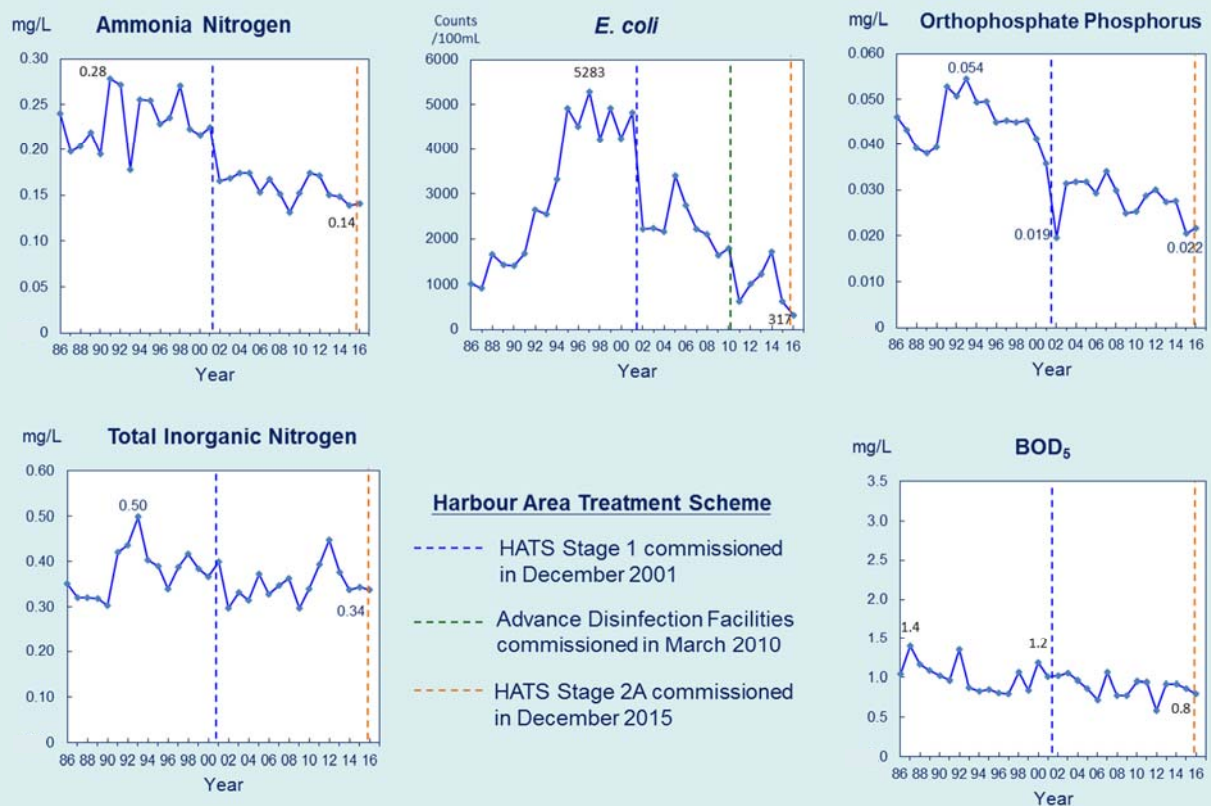
The *E. coli* level at the central (VM5) and eastern (VM1) parts of Victoria Harbour Water Control Zone, 1990-2016

The following figures illustrate the long term WQO compliance rates and trends of some important water quality parameters for Victoria Harbour since 1986 and after the implementation of the HATS Stage 1 in December 2001 and the HATS Stage 2A in December 2015.



Overall compliance with the marine WQOs in the Victoria Harbour Water Control Zone, 1986-2016

Water quality changes in Victoria Harbour WCZ, 1986-2016



Overall long term water quality changes in the Victoria Harbour Water Control Zone, 1986-2016

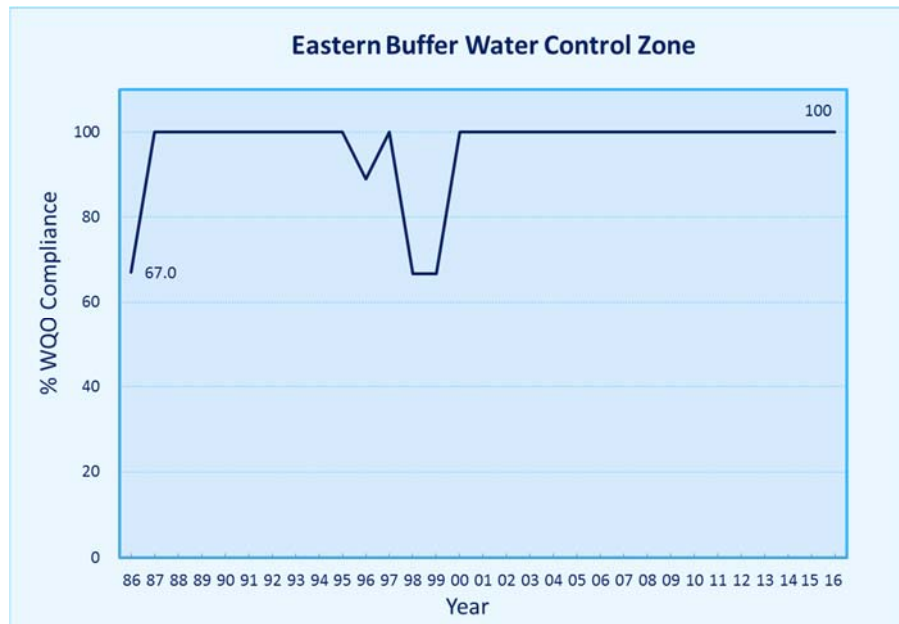
Eastern Buffer and Junk Bay Water Control Zones

Both the Eastern Buffer and Junk Bay WCZs have achieved full compliance (100%) with the WQOs in 2016 and for the last 15 years. Indeed, since the implementation of the HATS Stage 1 in December 2001 by which all sewage generated from Junk Bay (Tseung Kwan O), the Kowloon Peninsula and the eastern part of Hong Kong Island (Chai Wan) was diverted and treated at the SCISTW, the water quality of these two WCZs has improved markedly with increasing DO and decreasing nutrient and bacterial levels.



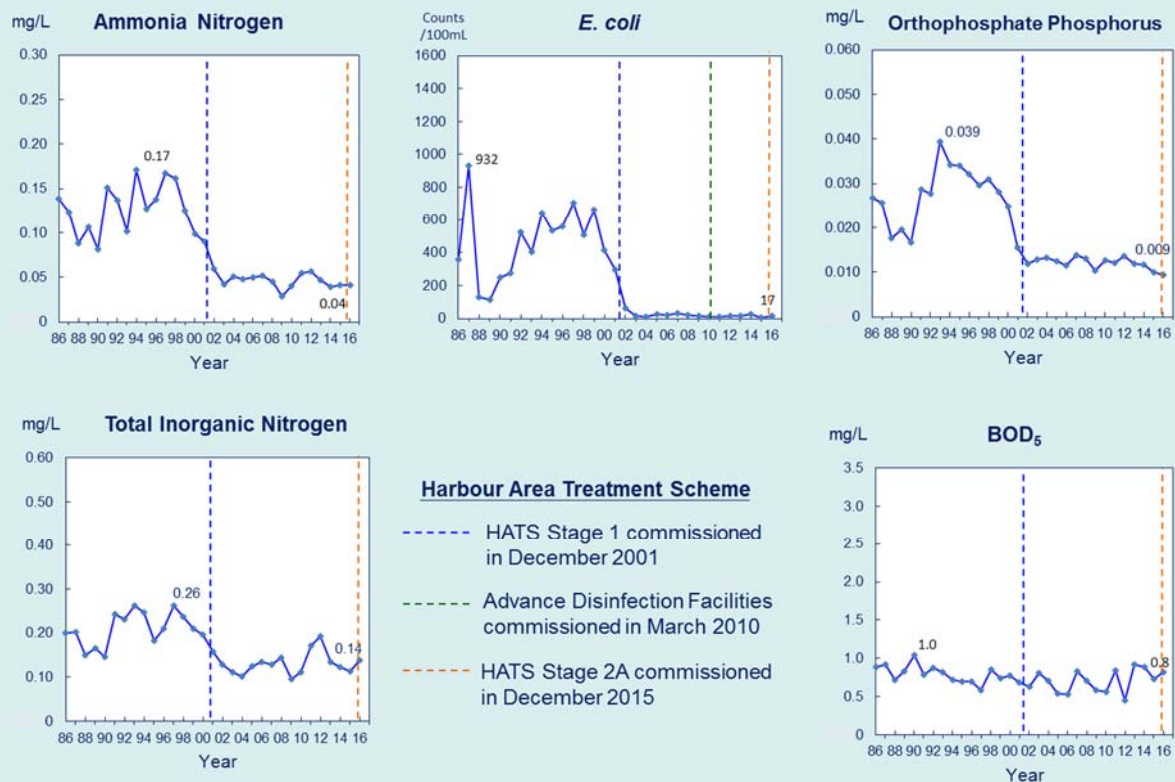
A scenic view of the Junk Bay Water Control Zone

The following figures illustrate the long term WQO compliance rates and improving trends of some important water quality parameters for Eastern Buffer and Junk Bay WCZs since 1986 and after the implementation of the HATS Stage 1 in December 2001 and the HATS Stage 2A in December 2015.

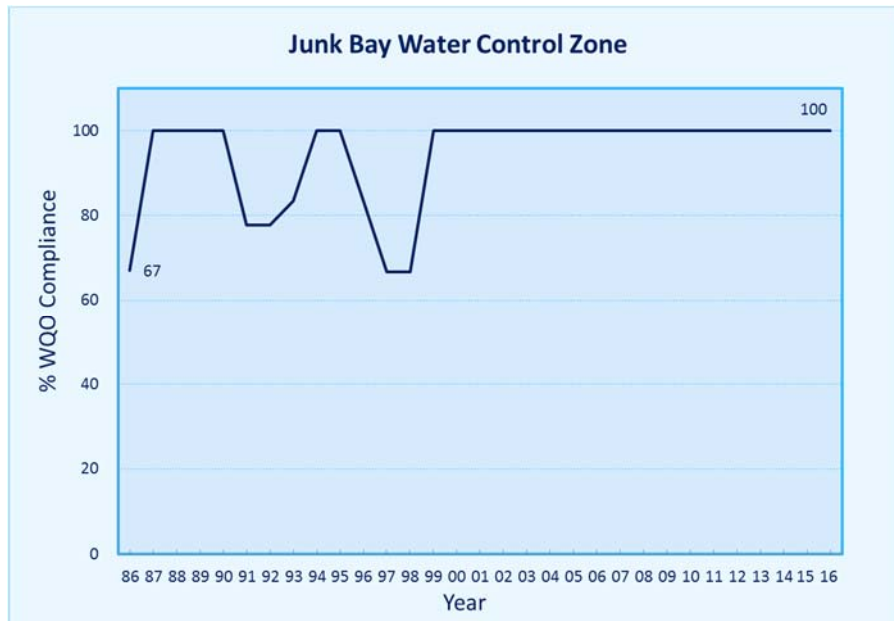


Overall compliance with the marine WQOs in the Eastern Buffer Water Control Zone, 1986-2016

Water quality changes in Eastern Buffer WCZ, 1986-2016

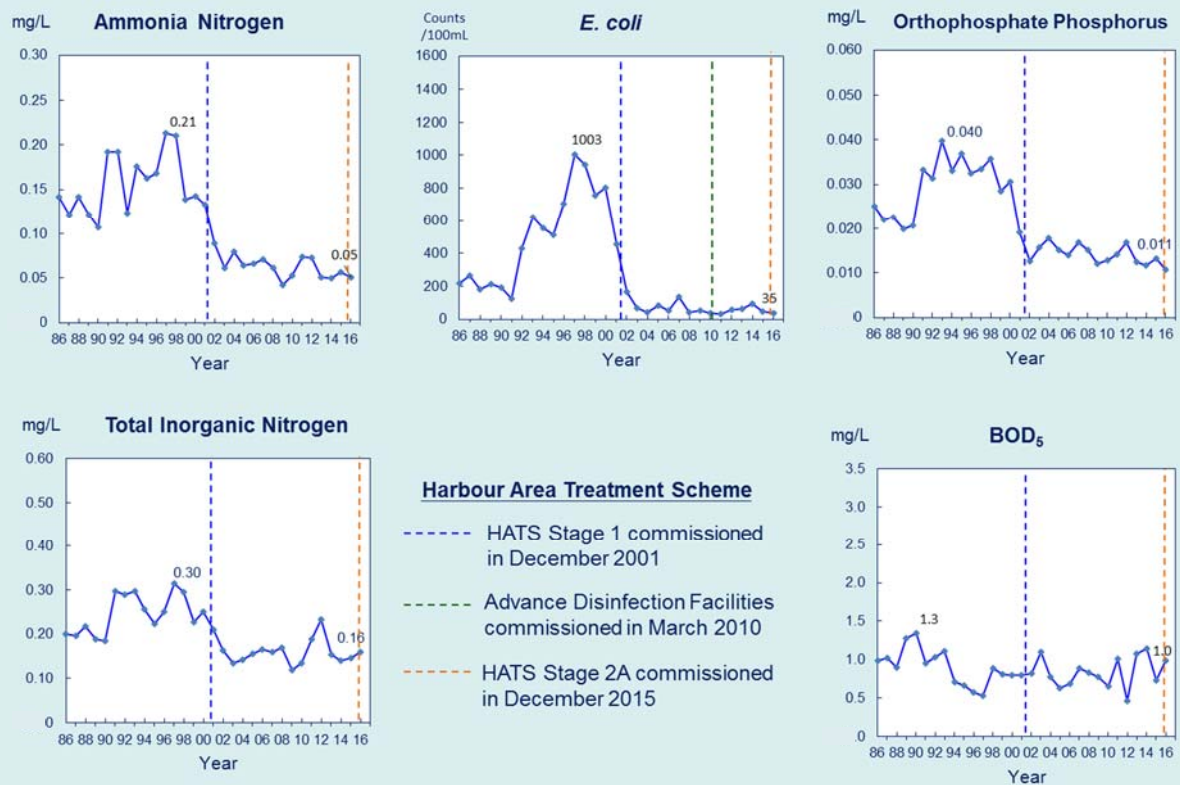


Overall long term water quality changes in the Eastern Buffer Water Control Zone, 1986-2016



Overall compliance with the marine WQOs in the Junk Bay Water Control Zone, 1986-2016

Water quality changes in Junk Bay WCZ, 1986-2016



Overall long term water quality changes in the Junk Bay Water Control Zone, 1986-2016

Western Buffer Water Control Zone

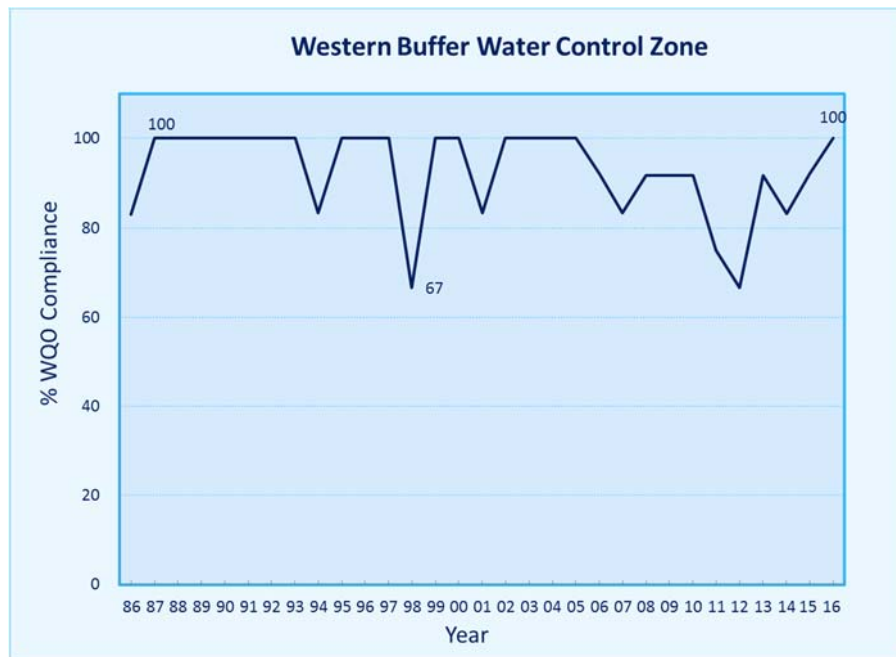
The Western Buffer WCZ achieved 100% overall compliance with the WQOs in 2016, with full compliance with the WQOs for DO, TIN and NH₃-N. The overall annual average level of TIN was 0.29 mg/L in 2016, as compared with 0.32 mg/L, 0.25 mg/L and 0.33 mg/L in 2015, 2014 and 2013 respectively. Similar to the Victoria Harbour WCZ, the TIN level in the Western Buffer WCZ was influenced by the discharges of Pearl River, the year to year fluctuation of surface run-off as well as the discharges from the SCISTW.

Since the commissioning of the Advance Disinfection Facilities at the SCISTW in March 2010, the *E. coli* level in the WCZ decreased by over 90% in 2011, as compared with the pre-commissioning period of 2009. In 2016, all the eight gazetted beaches located in the Tsuen Wan District of this WCZ complied with the bacteriological WQO for bathing, the seventh consecutive year since 2010 as reported in the *2016 Beach Water Quality in Hong Kong*.



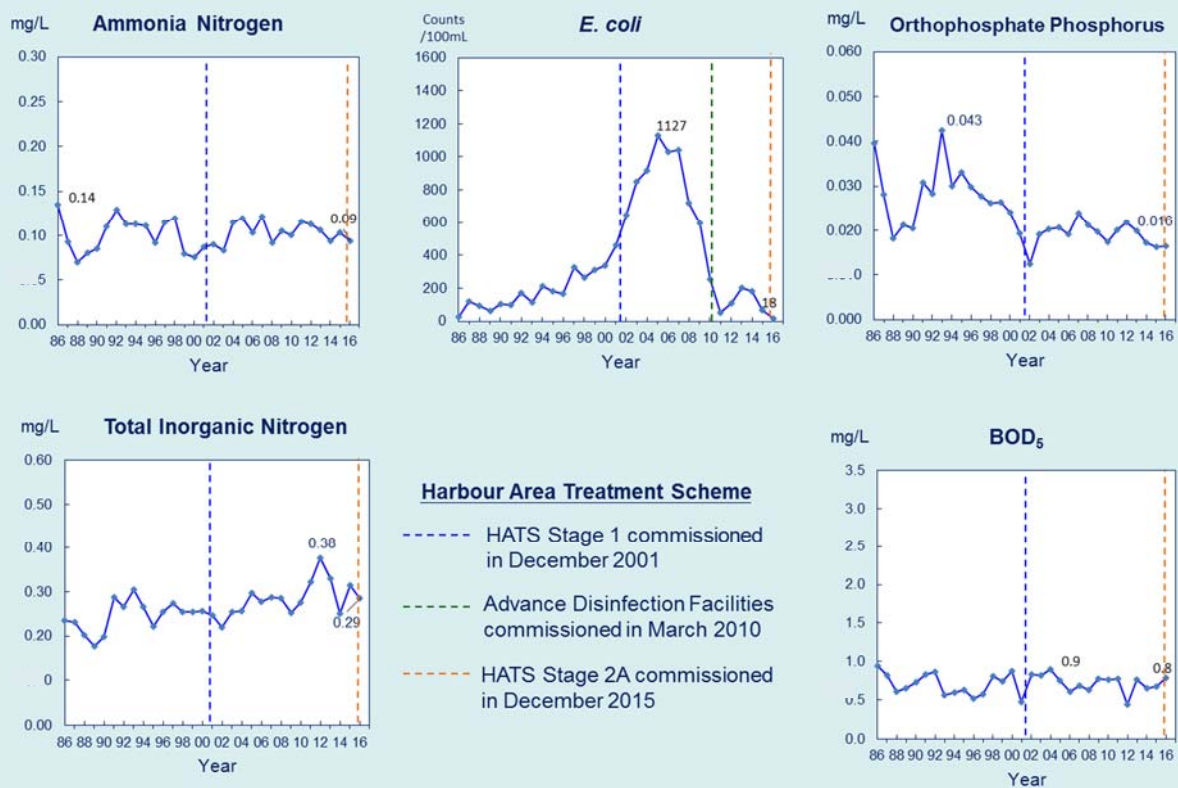
The busy shipping activities in the Western Buffer Water Control Zone

The following figures illustrate the long term WQO compliance rate and trends of some important water quality parameters for the Western Buffer WCZ since 1986 and after the implementation of the HATS Stage 1 in December 2001 and the HATS Stage 2A in December 2015. Similar to other WCZs in the central waters, significant improvements in *E. coli* and PO₄-P have been observed.



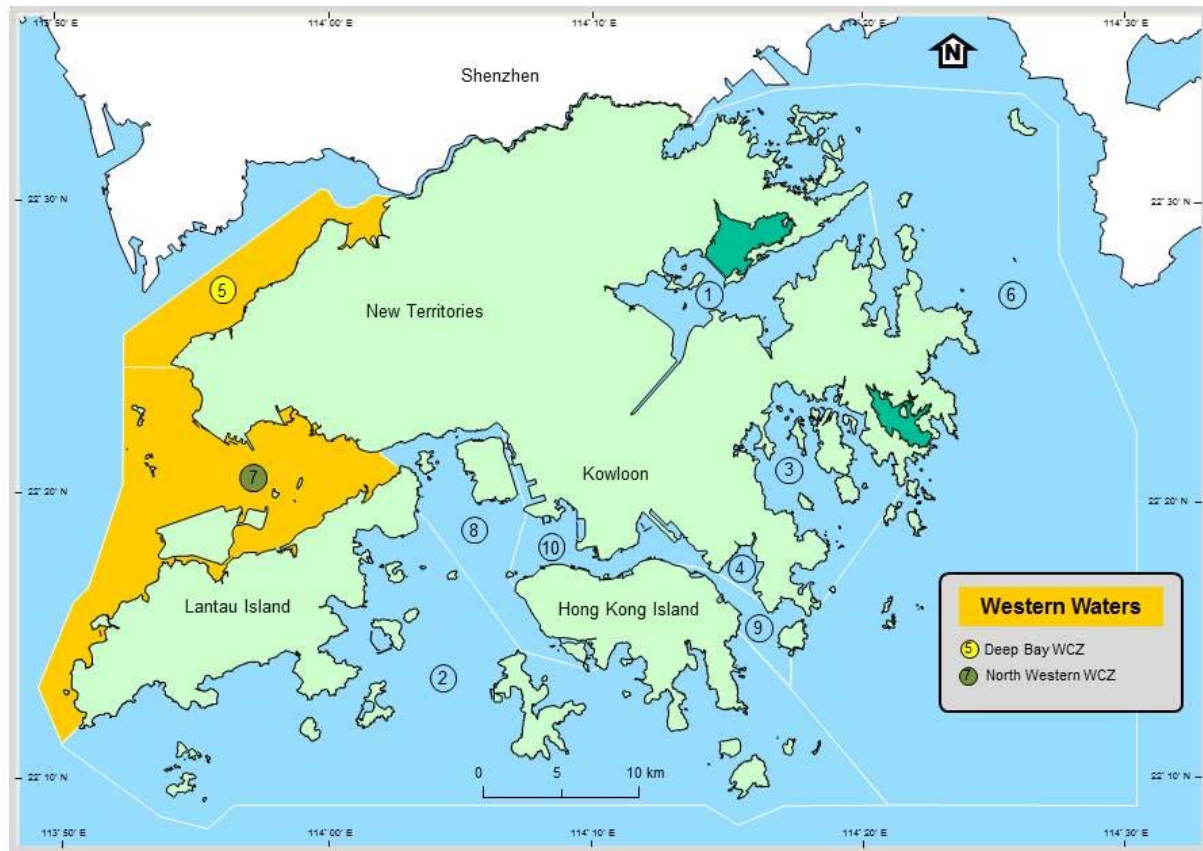
Overall compliance with the marine WQOs in the Western Buffer Water Control Zone, 1986-2016

Water quality changes in Western Buffer WCZ, 1986-2016



Overall long term water quality changes in the Western Buffer Water Control Zone, 1986-2016

3.3 Western Waters



The Deep Bay and North Western WCZs are located to the west of Hong Kong. Other than local discharges, both WCZs are influenced by the discharges from Pearl River, particularly during the wet season under the southwesterly monsoon wind effect. The Deep Bay WCZ also receives discharges from Shenzhen River all year round.

The Deep Bay WCZ includes the ecologically sensitive Mai Po and Inner Deep Bay Ramsar site, and areas of oyster culture. The Ramsar Site inside Deep Bay supports a high diversity of birds. It is an important feeding and resting ground for wintering and migratory birds, including a number of globally threatened species, such as Black-faced Spoonbill, Saunders's Gull and Imperial Eagle.

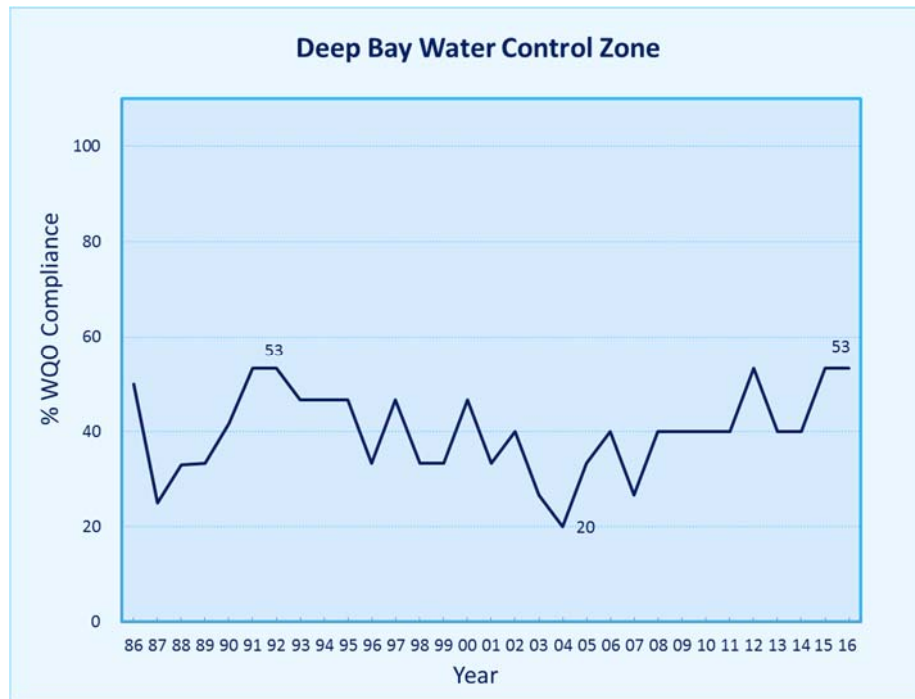
The North Western WCZ is also the water body where Chinese white dolphins are commonly spotted in areas around Lantau Island, Tuen Mun, Sha Chau and Lung Kwu Chau. The Sha Chau and Lung Kwu Chau Marine Park as well as the newly established The Brothers Marine Park are located in the western waters of Hong Kong.

Deep Bay Water Control Zone

Deep Bay is a large shallow bay on the eastern side of the Pearl River Estuary with an average depth of about 3m. Deep Bay has the largest and most important mangrove habitat in Hong Kong. The mudflats of inner Deep Bay also have high conservation value as an important feeding ground for a huge number of resident and migratory birds. Because of its shallowness, the presence of these muddy habitats as well as the strong riverine inputs from within and outside, the bay naturally has high suspended solids level.

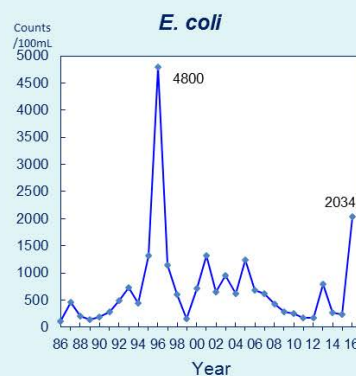
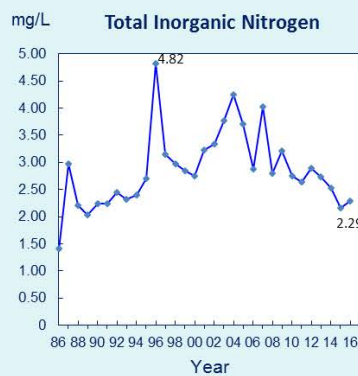
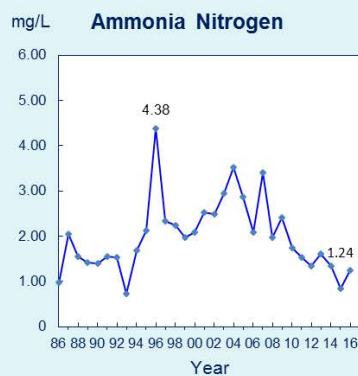
The overall WQO compliance rate of the Deep Bay WCZ was 53% in both 2015 and 2016 as compared to a ten-year average of 39% in 2005-2014. Similar to the previous years, the Deep Bay WCZ had relatively high nutrient levels in 2016. The Deep Bay WCZ is further divided into two subzones with different TIN objectives. The TIN levels in the Deep Bay Inner Marine Subzone (1.43 – 3.01 mg/L) and Deep Bay Outer Marine Subzone (0.89 – 1.12 mg/L) were higher than their respective TIN objectives of 0.7 mg/L and 0.5 mg/L. Although the TIN level are relatively high in this WCZ as compared with others in the region, there were, however, few reported cases of red tides from Deep Bay and no red tide incident was recorded in 2016. The high background suspended solids level has limited the photosynthesis and growth of phytoplankton in water. In 2016, non-compliances with the DO objective were mainly recorded in two out of the three stations in the Deep Bay Inner Marine Subzone (DM1 and DM2). The NH₃-N WQO was fully complied with in 2016.

While the TIN level in Deep Bay is relatively high, as a result of both Hong Kong and Shenzhen governments in reducing pollution loads, e.g., the voluntary farm surrender schemes for poultry and pig farms implemented in 2005-2008, there are notifiable signs of water quality improvement since the mid-2000s. A comparison of the monitoring results collected between 1997-2006 and 2007-2016 suggests that the water quality of Deep Bay has improved in terms of reduction in organic loading (i.e. BOD₅), nutrients (e.g. NH₃-N, TIN and PO₄-P) and *E. coli* levels. The following figures illustrate the long term WQO compliance rate and trends of some important water quality parameters for the Deep Bay WCZ since 1986.

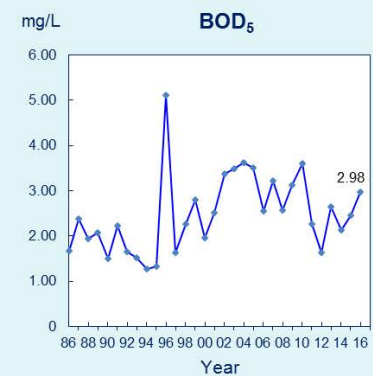
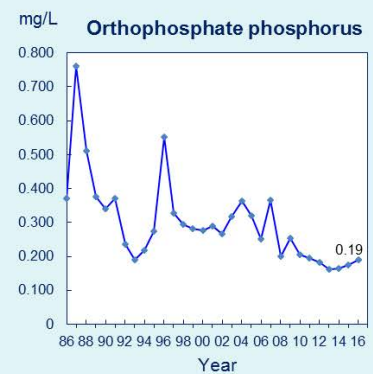


Overall compliance with the marine WQOs in the Deep Bay Water Control Zone, 1986-2016

Water quality changes in Inner Subzone (DM1-3) of the Deep Bay WCZ, 1986-2016

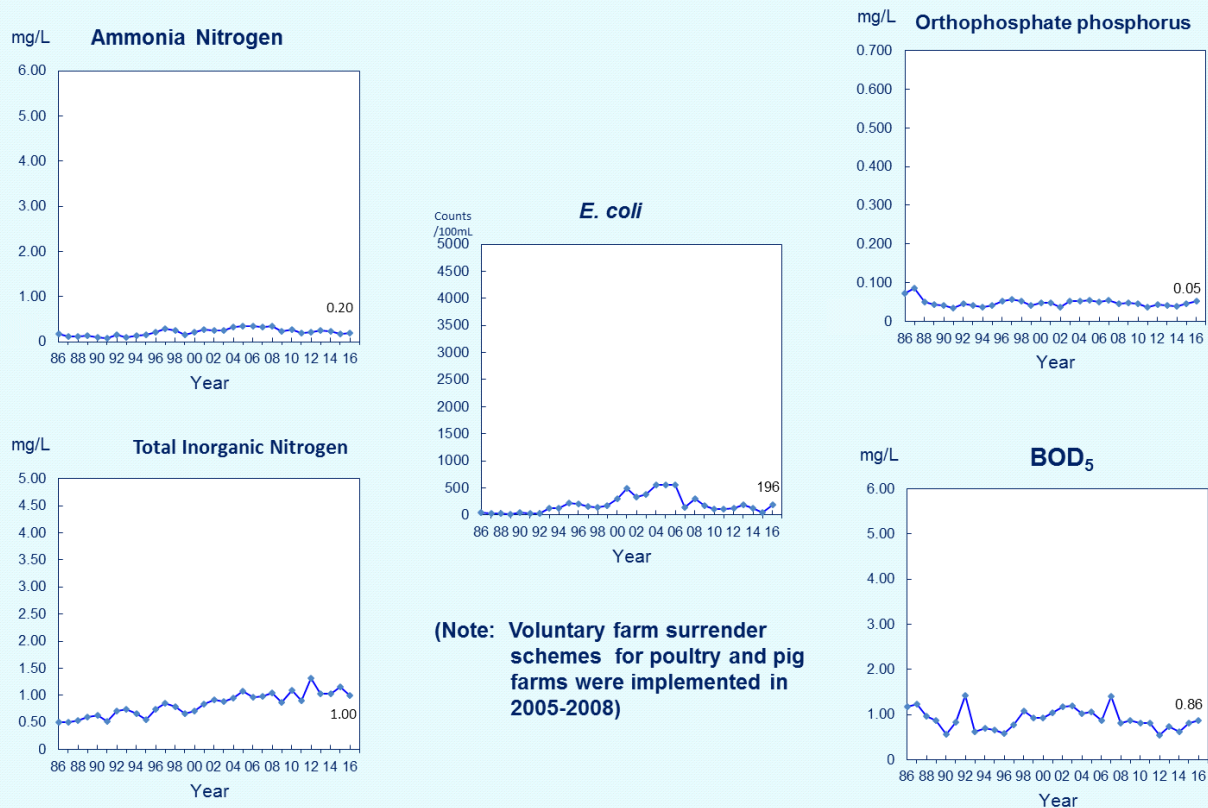


(Note: Voluntary farm surrender schemes for poultry and pig farms were implemented in 2005-2008)



Overall long term water quality changes in Inner Subzone the Deep Bay Water Control Zone, 1986-2016

Water quality changes in Outer Subzone (DM4-5) of the Deep Bay WCZ, 1986-2016



Overall long term water quality changes in the Outer Subzone of the Deep Bay Water Control Zone, 1986-2016



The oyster culture beds and rafts in the Deep Bay Water Control Zone

North Western Water Control Zone

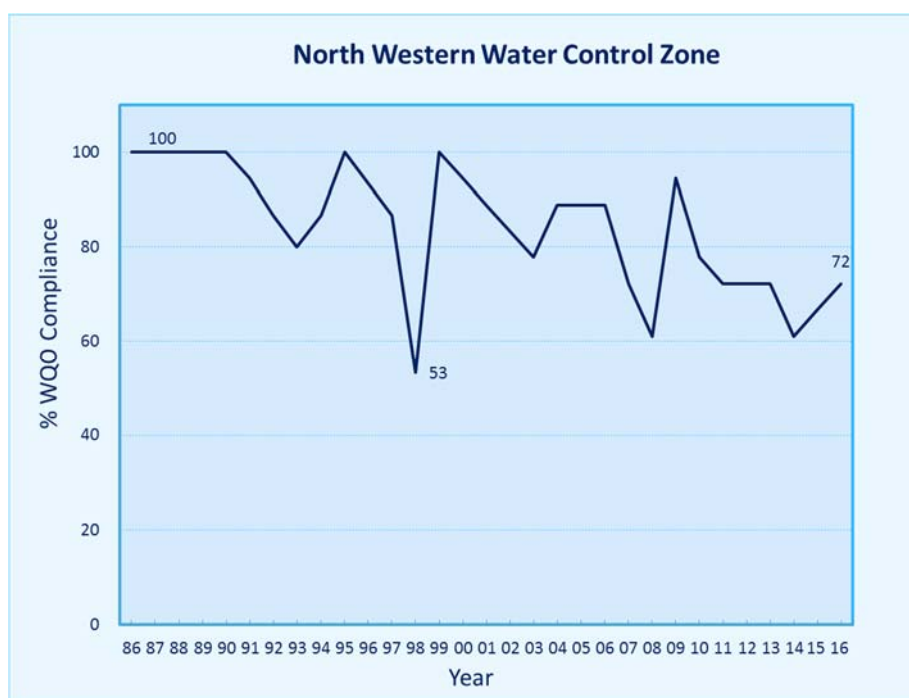
In 2016, the North Western WCZ attained an overall WQO compliance rate of 72%. Compliance with the NH₃-N and TIN objectives in this WCZ was 100% and 17% respectively. The relatively high levels of TIN (annual mean 0.44 - 0.78 mg/L) were likely attributed to the higher background level of Pearl River, and some local discharges and surface run-off from the Northwest New Territories as well as North Lantau.

The six gazetted beaches of the Tuen Mun District complied with the bacteriological WQO for bathing in 2016.



A scenic view of the Sha Chau and Lung Kwu Chau Marine Park in the North Western Water Control Zone

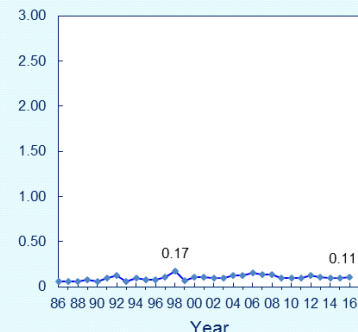
The long term WQO compliance rates and trends of some key water quality parameters for the North Western Water WCZ are illustrated in the following figures.



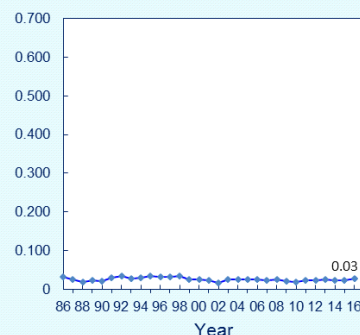
Overall compliance with the marine WQOs in the North Western Water Control Zone, 1986-2016

Water quality changes in North Western WCZ, 1986-2016

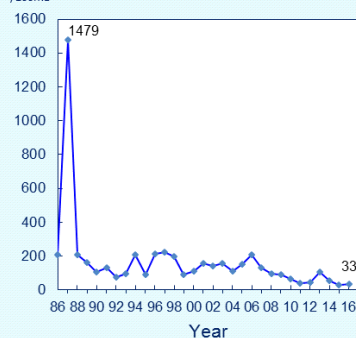
mg/L Ammonia Nitrogen



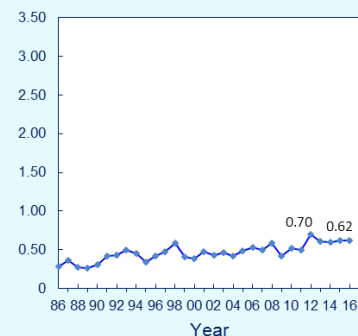
mg/L Orthophosphate phosphorus



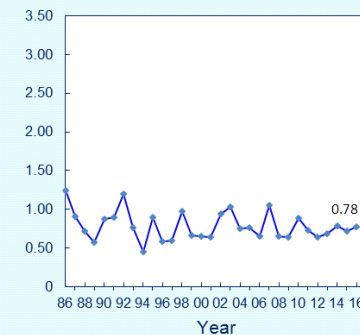
Counts /100mL *E. coli*



mg/L Total Inorganic Nitrogen



mg/L BOD₅



(Note: Voluntary farm surrender schemes for poultry and pig farms were implemented in 2005-2008)

Overall long term water quality changes in the North Western Water Control Zone, 1986-2016

3.4 Southern Waters

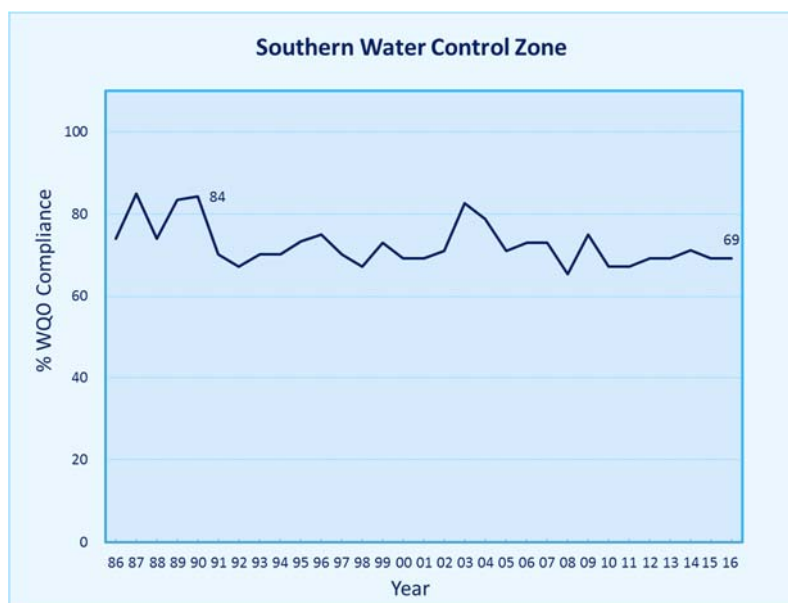


The southern waters consists of only one WCZ - the Southern WCZ. It stretches from south of Hong Kong Island to Lantau Island, facing the South China Sea. The western part of the WCZ is affected seasonally by the Pearl River and the effect diminishes gradually towards the east.

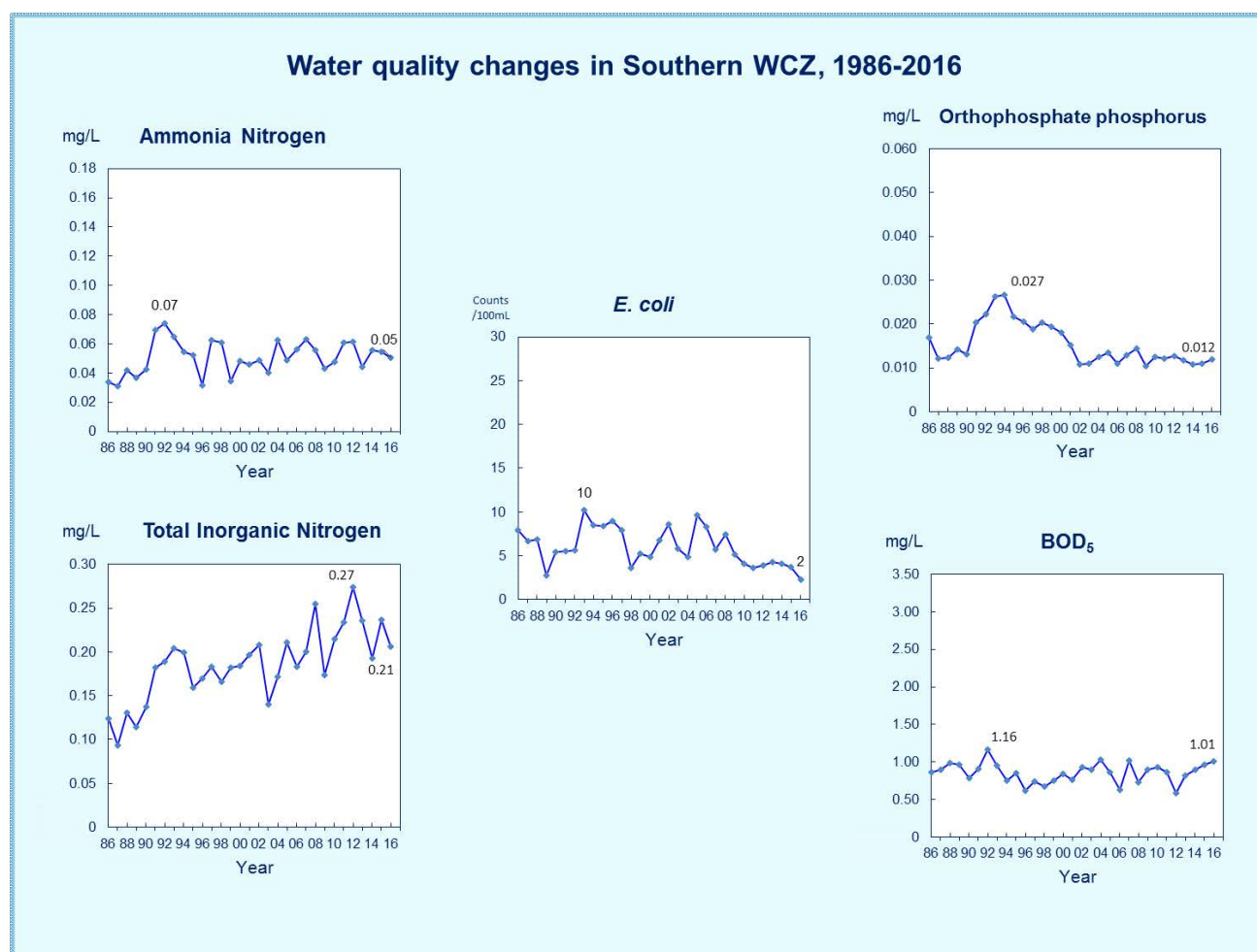
Southern Water Control Zone

The Southern WCZ attained an overall 69% compliance with the WQOs in 2016. All 16 stations in the WCZ fully complied with the DO and NH₃-N objectives. Due to the combined effect of the Pearl River and the South China Sea, stations nearby the south Lantau waters had TIN levels of 0.22 – 0.32 mg/L while those south of the Hong Kong Island had TIN levels of 0.12 – 0.16 mg/L. Both levels were higher than the TIN objective of the 0.1 mg/L, resulting in the non-compliance of the TIN WQO.

The Southern WCZ also covers a number of secondary contact recreation subzones which are mainly located along the coast in the Southern District (Hong Kong Island) and the outlying islands. In 2016, full compliance (100%) with the *E. coli* objective was attained in the secondary contact recreation subzones. All gazetted beaches in this WCZ, including nine of the Islands District and 12 of the Southern District, complied with the bacteriological WQO for bathing in 2016. The long term WQO compliance rates and trends of some key water quality parameters for the Southern WCZ are illustrated in the following figures.



Overall compliance with the marine WQOs in the Southern Water Control Zone, 1986-2016

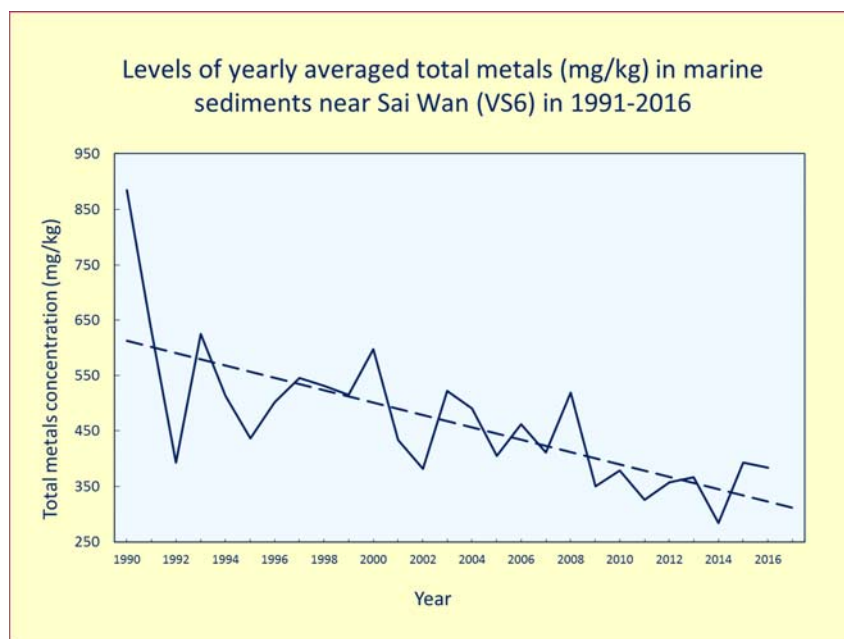


Overall long term water quality changes in the Southern Water Control Zone, 1986-2016

4. Sediment Quality

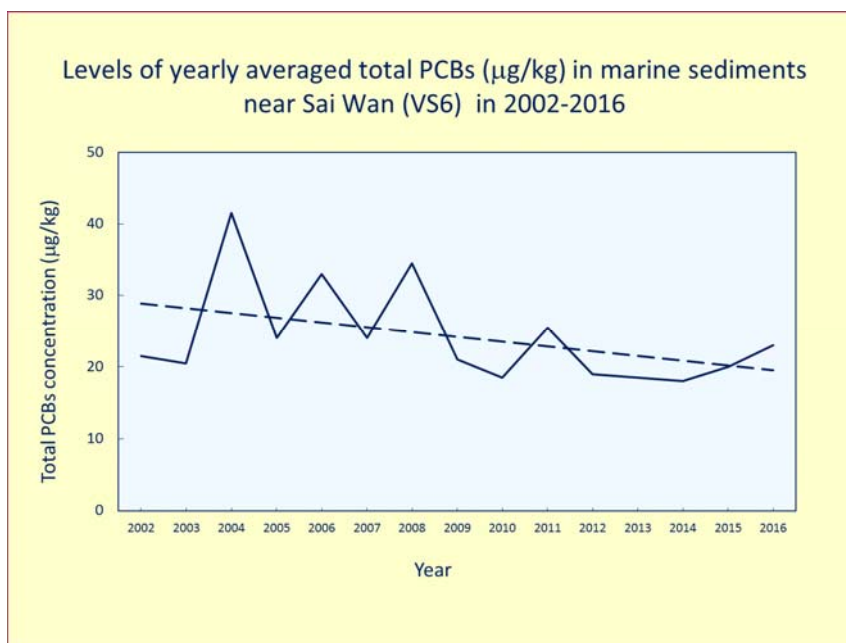
In addition to water quality analyses, the marine environment can be assessed through the quality of sediments since they are the ultimate sink of many pollutants. The quality of marine sediments can, therefore, reflect the historical and long term health of the marine environment.

The EPD samples and tests sediments in different parts of Hong Kong for over 60 physical, chemical and biological parameters. Since the implementation of the marine monitoring programme in 1986, elevated levels of selected heavy metals, in particular copper and silver, could often be detected in the sediments of Victoria Harbour and Tsuen Wan Bay. This was attributed to industrial pollution discharges in the 60s to 80s before pollution control was introduced. Nevertheless, through EPD's efforts in enforcing relevant pollution control ordinances, i.e. the Water Pollution Control Ordinance and Waste Disposal Ordinance, and the removal of various industrial pollution sources, we have observed a general decreasing trend of total metals in the marine sediments. The following figure gives an example for a station, VS6, in the western harbour area near Sai Wan with a decreasing trend of total metals since 1991.



Note: Total metals include cadmium, chromium, copper, mercury, nickel, lead, silver and zinc.

With regard to trace organic pollutants such as polychlorinated biphenyls (PCBs), they were generally present at low concentrations except for one station in western Victoria Harbour (VS6). The higher levels of PCB at VS6 could be related to the discharges from industries in the past. Similar to metals, a gradual decreasing trend of total PCBs in sediments could also be observed at VS6 since 2002 as illustrated in the following figure.



Note: Total PCBs were calculated based on the definition in the 2002 Technical Circular 'ETWB (W) No. 34/2002 Management of Dredged / Excavated Sediment'.

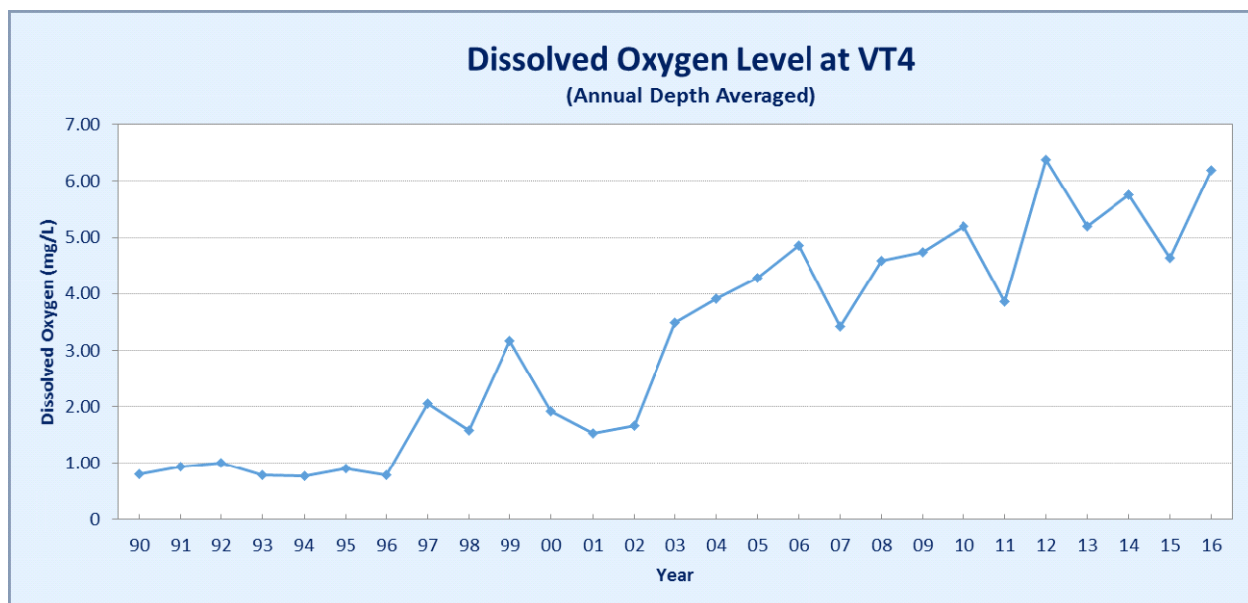


The sediment sampling process at a typhoon shelter.

5. Typhoon Shelters

Typhoon shelters in Hong Kong are used by small to medium vessels as protection against strong winds and rough sea conditions, particularly during the typhoon season. For this reason, typhoon shelters are often located in semi-enclosed water bodies. Since typhoon shelters are designed with breakwaters, the water circulation between the typhoon shelters and open waters is restricted, and hence more vulnerable to pollution impact from both land-based sources through the storm drain outlets, and discharge from vessels moored at the typhoon shelters.

In 2016, the EPD monitored the water quality of 14 typhoon shelters, as well as three sheltered anchorages and the Government Dockyard. Some of these typhoon shelters are located adjacent to populated residential and/or industrial areas, e.g. Causeway Bay Typhoon Shelter and Kwun Tong Typhoon Shelter. Others are located in outlying islands or far away from the urban areas, e.g. Cheung Chau Typhoon Shelter and Shuen Wan Typhoon Shelter. Since there is no bacteriological WQO laid down for typhoon shelters because their beneficial uses are mainly for mooring of vessels, the status of compliance with the WQOs for stations in typhoon shelters were checked mainly with the DO, the TIN and the $\text{NH}_3\text{-N}$. Regarding these WQO parameters in 2016, the $\text{NH}_3\text{-N}$ WQO could be fully complied with. Of these typhoon shelters, only one, i.e., New Yau Ma Tei Typhoon Shelter, has failed to meet the depth-averaged DO WQO in 2016. Because of their proximity to densely populated areas and/or high utilization by marine vessels, Tuen Mun, New Yau Ma Tei, Kwun Tong, Rambler Channel, Cheung Chau and Hei Ling Chau typhoon shelters have failed to comply with the TIN WQO. Nevertheless, the water quality of all typhoon shelters in Hong Kong as a whole was improving in general during the last decade. The following figure shows the long term trend in the depth-averaged DO level of the Kwun Tong Typhoon Shelter:



The long term improvement of the depth-averaged DO level at the Kwun Tong Typhoon Shelter (VT4)

6. Phytoplankton and Red Tides

Phytoplankton is a vast, highly diversified group of tiny plants that lives in the surface layer of aquatic environment. Like terrestrial plants, majority of them carry out photosynthesis and require sunlight, carbon dioxide, minerals and nutrients for their growth. Phytoplankton plays a very important role as a key primary producer to produce food for many aquatic lives ranging from tiny zooplankton to large fish. Under favourable conditions (including nutrients, water temperature, sunlight, water current and wind speed), phytoplankton can reproduce quickly and bloom, resulting in overly dense population which may “coloured” the waters with their photosynthetic pigment, a natural phenomenon commonly called “red tide”. Most red tides have no harmful impact to the marine community, while a few might have harmful impact due to the species involved (e.g., tiny spines on the cell wall, capability to produce toxin) and the scale of the bloom (e.g., excessive algal biomass can lower the oxygen level in the water column during its decomposition).

The EPD conducts monthly monitoring of phytoplankton at 25 stations to determine the changes and trends in their composition and densities in Hong Kong open marine waters. A total of 92 phytoplankton species was recorded in Hong Kong waters in 2016. Of these, 51 were diatoms (55%), 30 were dinoflagellates (33%) and 11 were from other seven minor algal groups¹ (12%). As in previous years, diatom was the dominant group in all WCZs both in terms of species richness (number of species) and population (cell density). The most abundant diatoms were *Thalassiosira* spp. which constituted 17-59% of the diatom population in these WCZs. The most abundant dinoflagellates were *Gymnodinium* spp. comprising 24-78% of the dinoflagellate populations in each WCZ. In 2016, the total phytoplankton densities were generally higher at the stations in the Tolo Harbour and Channel and Southern WCZs than those in the other WCZs. The densities of diatoms, dinoflagellates and other minor phytoplankton groups followed the similar pattern with the total phytoplankton density.

In Hong Kong, red tides occur throughout the year and generally peak during the spring months. They are also more common in semi-enclosed bays (e.g., Tolo Harbour) with low turbidity and flushing rate. In 2016, a total of 11 red tide incidents were reported in Hong Kong waters, as compared with an average of 15 incidents in the last five years (2011-2015). Of these, six incidents involved only one WCZ, three incidents involved 2-3 WCZs and two incidents involved

¹ Phytoplankton can be classified into 11 different groups based on their morphological characteristics, photosynthetic pigments and nutrition modes. In this report, phytoplankton other than the two major groups, i.e., diatoms and dinoflagellates, are collectively regarded as other minor phytoplankton groups to facilitate the data presentation.

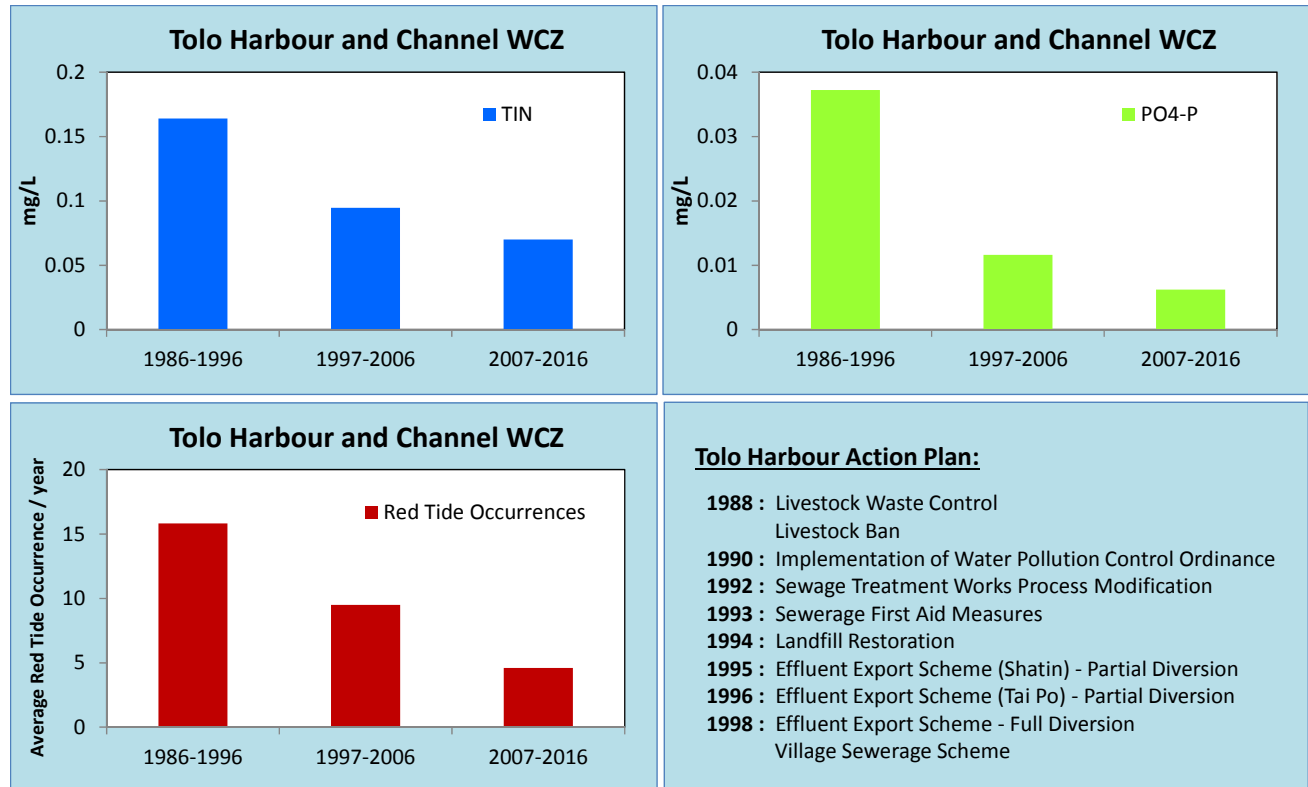
over five WCZs. As most incidents were reported in the eastern and southern waters which have relatively low nutrients levels, their occurrence in recent years was largely attributed to changes in climatic and/or hydrodynamic conditions (such as light intensity, water temperature, stratification, water currents, wind speed, etc). The 11 red tide incidents were caused by ten red tide species. Seven of them are non-toxic species while three of them, namely *Karenia mikimotoi*, *Karenia papilionacea* and *Phaeocystis globosa*, may cause fish kills. *Karenia mikimotoi*, for example, could produce toxins irritating the gill tissues of fish, thereby affecting their breathing and potentially resulting in suffocation to death. Its bloom in early 2016 caused an estimated loss of about 220 tonnes of cultured fish (mainly Sabah Grouper) at nine fish culture zones in Tolo Harbour and Long Harbour (i.e., Tai Tan Hoi). The following table shows the details of red tide incidents in 2016.

Incident No.	Sighting Period	Red Tide Species
1	1/1/2016 - 8/2/2016	<i>Karenia mikimotoi</i> <i>Karenia papilionacea</i>
2	21/1/2016 - 24/1/2016	<i>Noctiluca scintillans</i>
3	5/2/2016 - 28/4/2016	<i>Akashiwo sanguinea</i>
4	18/2/2016 - 13/4/2016	<i>Akashiwo sanguinea</i> <i>Noctiluca scintillans</i> <i>Dictyocha octonaria</i>
5	31/3/2016 - 12/5/2016	<i>Noctiluca scintillans</i>
6	1/4/2016 - 16/4/2016	<i>Akashiwo sanguinea</i> <i>Noctiluca scintillans</i>
7	4/8/2016 - 21/8/2016	<i>Gonyaulax verior</i> <i>Heterocapsa pygmaea</i>
8	6/9/2016 - 13/9/2016	<i>Tripes furca</i>
9	6/11/2016 - 14/11/2016	<i>Phaeocystis globosa</i>
10	21/12/2016 - 27/12/2016	<i>Mesodinium rubrum</i>
11	29/12/2016 - 19/1/2017	<i>Phaeocystis globosa</i>

Red tide incidents in 2016 (Source: Agriculture, Fisheries and Conservation Department)

Historically, Tolo Harbour had the highest number of red tide incidents. The number of red tide incidents in Tolo Harbour has, however, dropped significantly from the record high of 43 in 1988 to an annual average of about only 5 incidents for the past five years, as a result of the steady improvement in water quality in Tolo Harbour in the past decades due to the Tolo Harbour Action

Plan. The following figures illustrate the reductions of key nutrient parameters as well as the average annual occurrences of red tides in the Tolo Harbour as a result of the Government's effort in the implementation of the Tolo Harbour Action Plan.

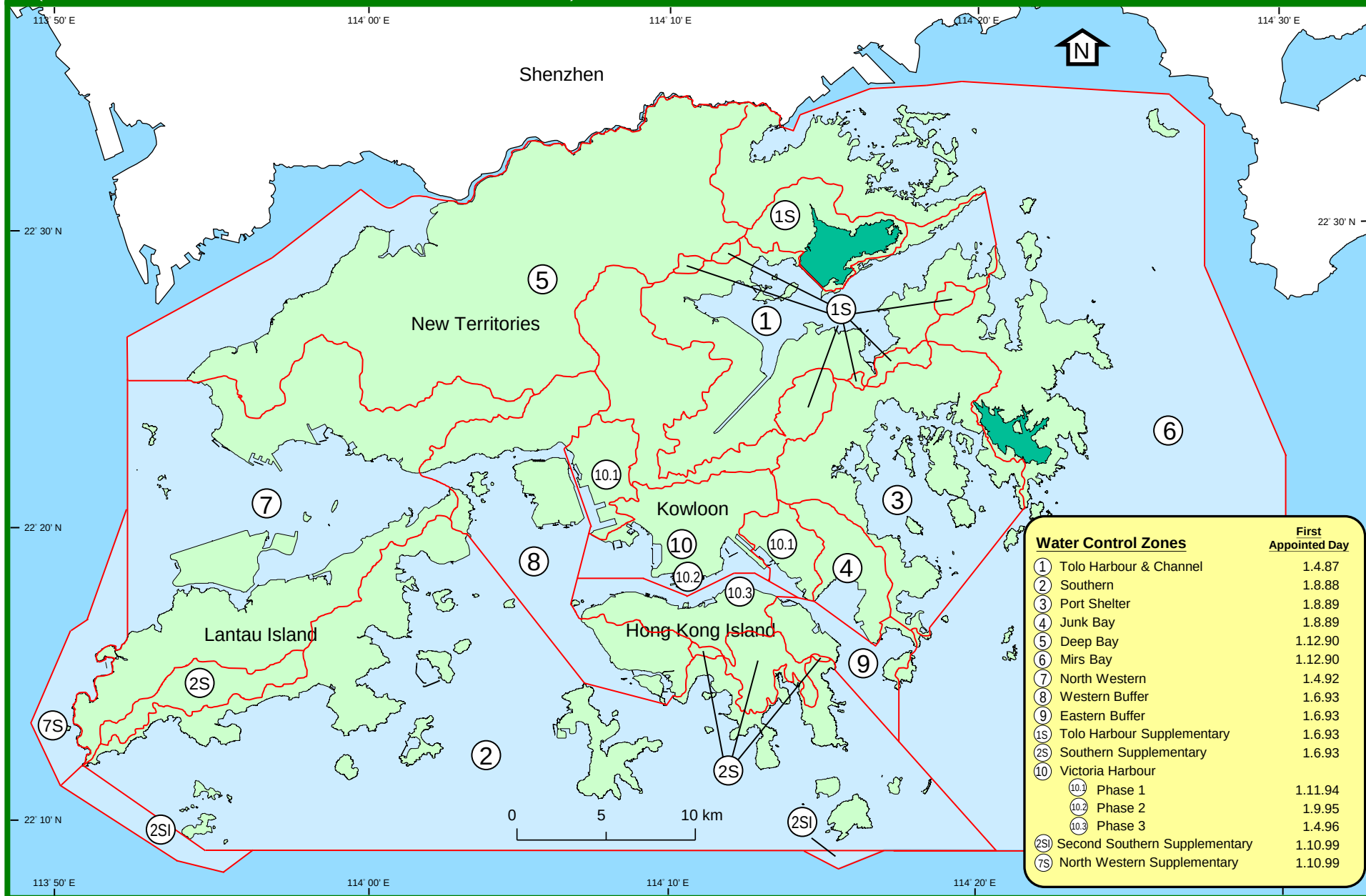


Reductions of total inorganic nitrogen (TIN) and orthophosphate phosphorus (PO₄-P) levels with the average annual red tide occurrences in the Tolo Harbour and Channel WCZ

Appendices

The Water Control Zones in Hong Kong

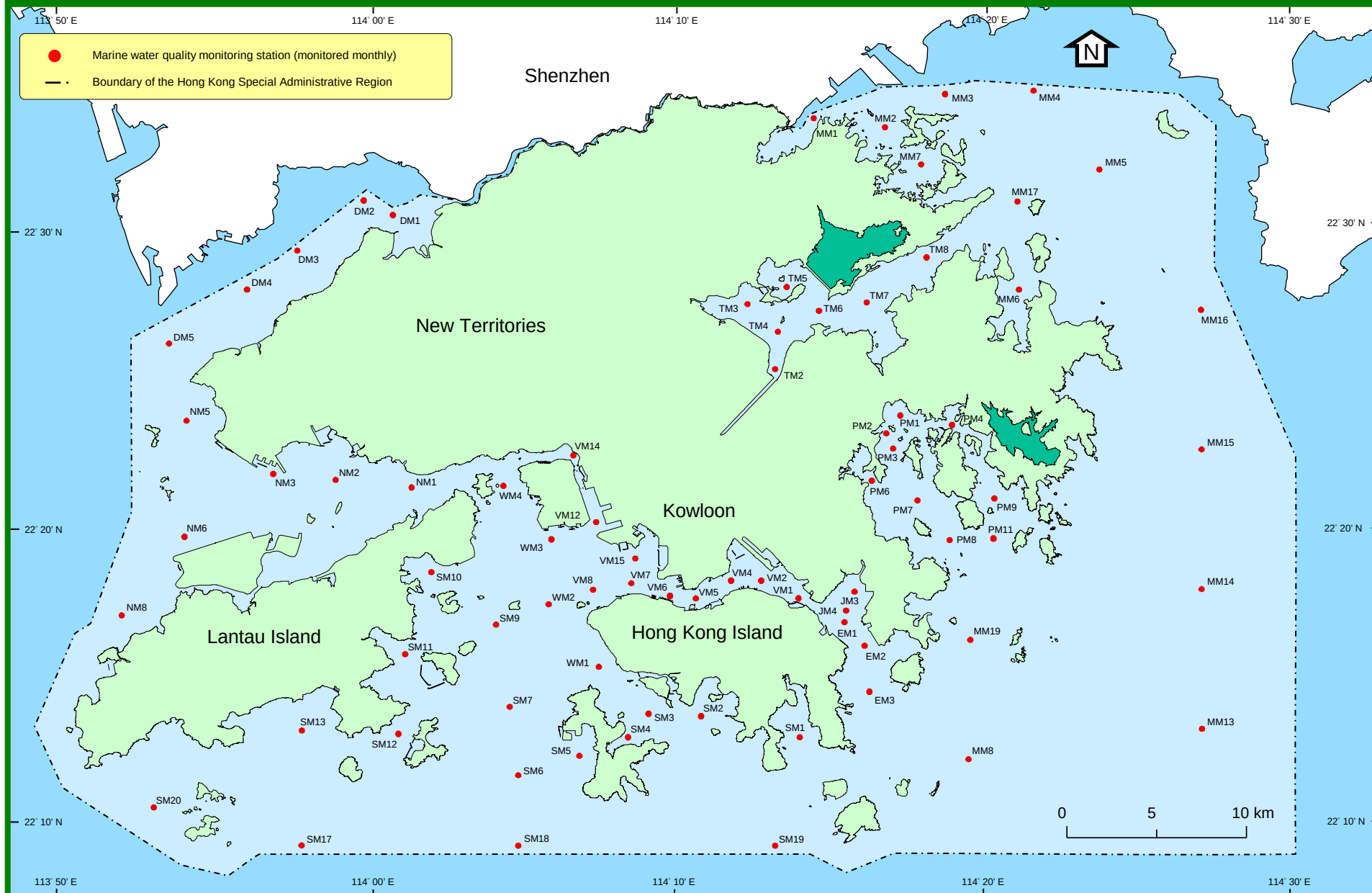
(Source: Environment Bureau— Plan No. WP/WP4/75, Oct 1999)



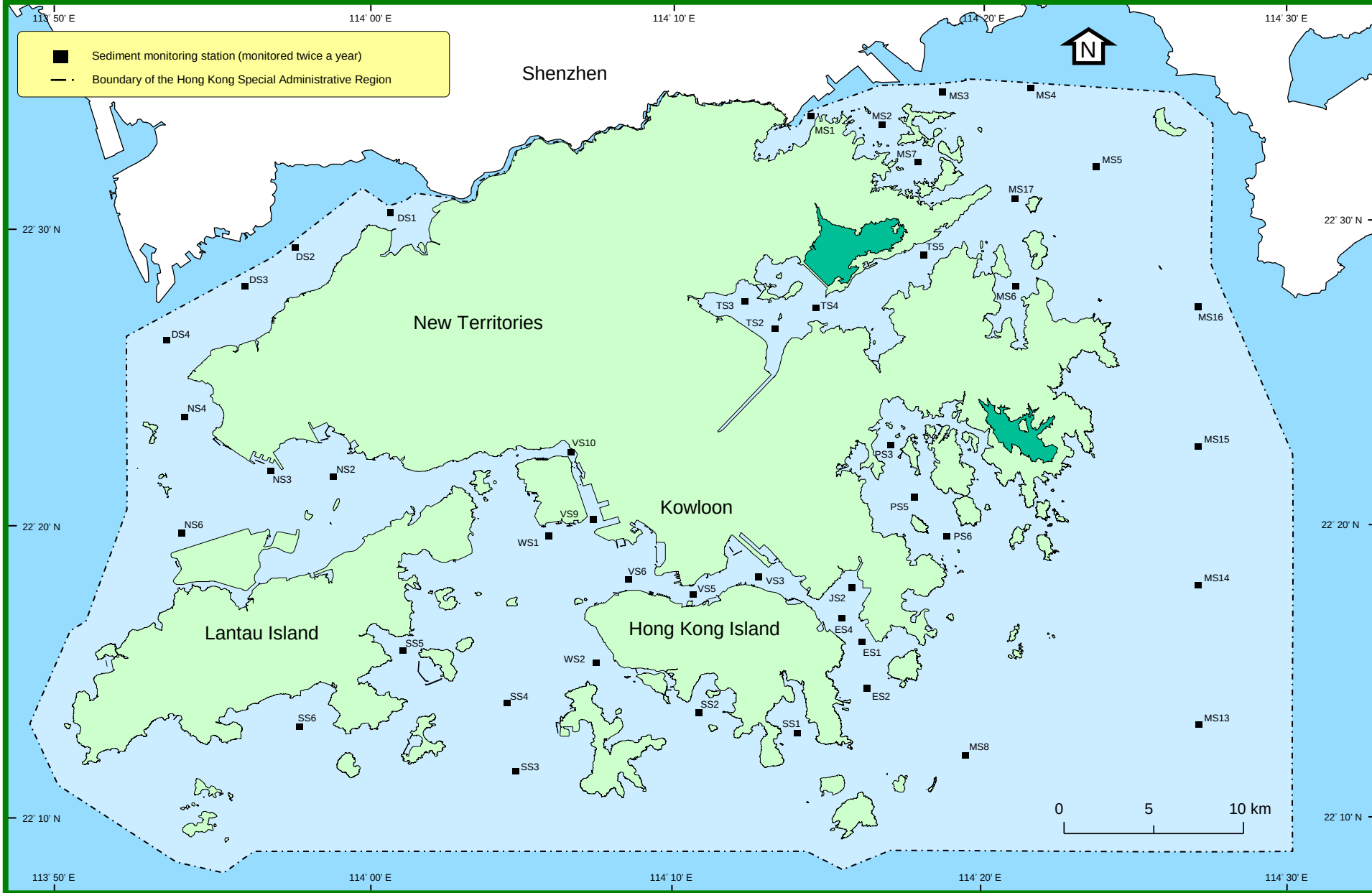
Water Control Zones

	First Appointed Day
① Tolo Harbour & Channel	1.4.87
② Southern	1.8.88
③ Port Shelter	1.8.89
④ Junk Bay	1.8.89
⑤ Deep Bay	1.12.90
⑥ Mirs Bay	1.12.90
⑦ North Western	1.4.92
⑧ Western Buffer	1.6.93
⑨ Eastern Buffer	1.6.93
1S Tolo Harbour Supplementary	1.6.93
2S Southern Supplementary	1.6.93
⑩ Victoria Harbour	
⑩.① Phase 1	1.11.94
⑩.② Phase 2	1.9.95
⑩.③ Phase 3	1.4.96
2SI Second Southern Supplementary	1.10.99
7S North Western Supplementary	1.10.99

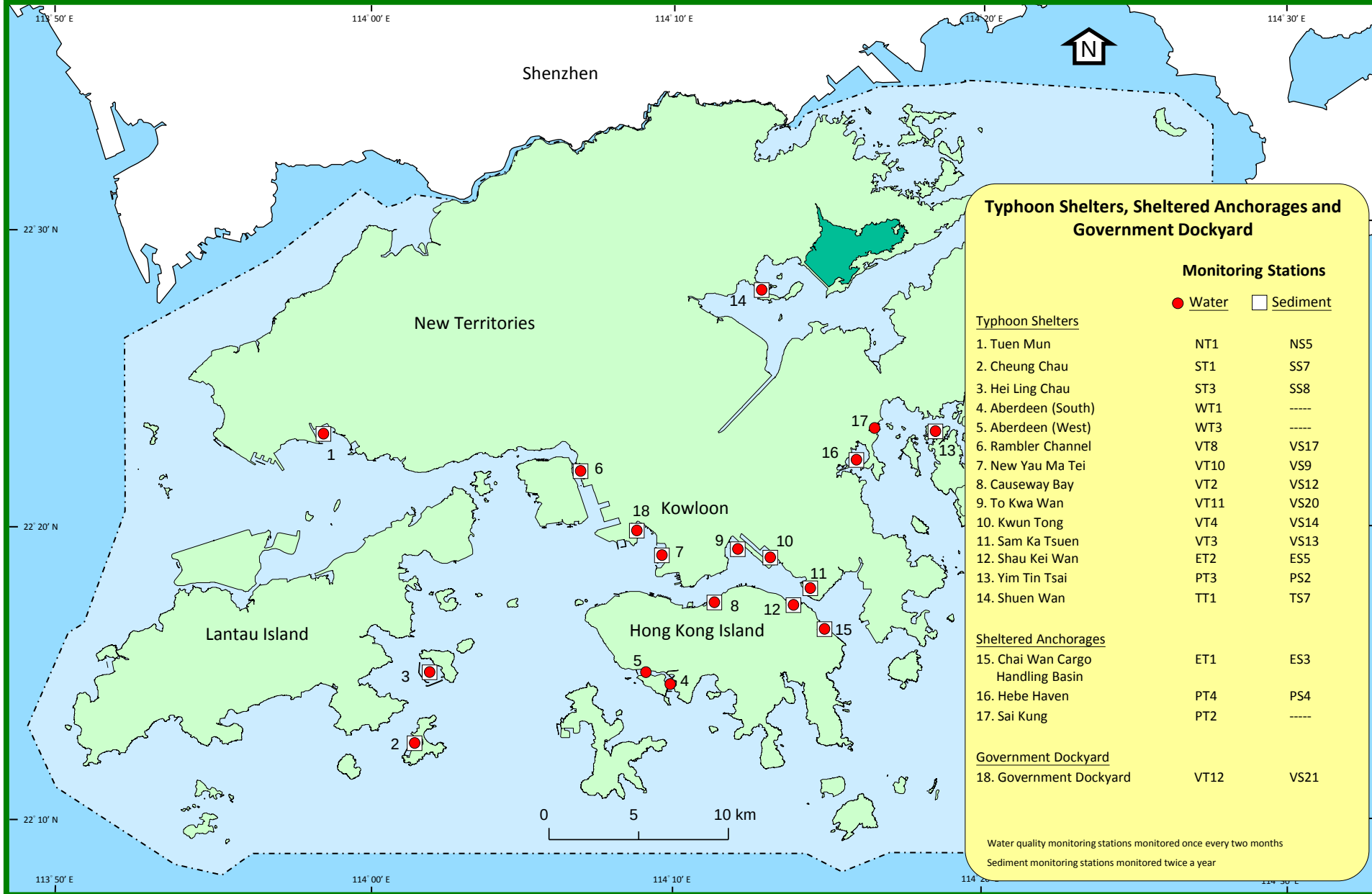
The 76 water quality monitoring stations in the open waters of Hong Kong in 2016



The 45 sediment monitoring stations in the open waters of Hong Kong in 2016



The 18 water quality monitoring stations and 15 sediment monitoring stations in the typhoon shelters, sheltered anchorages and Government Dockyard of Hong Kong in 2016

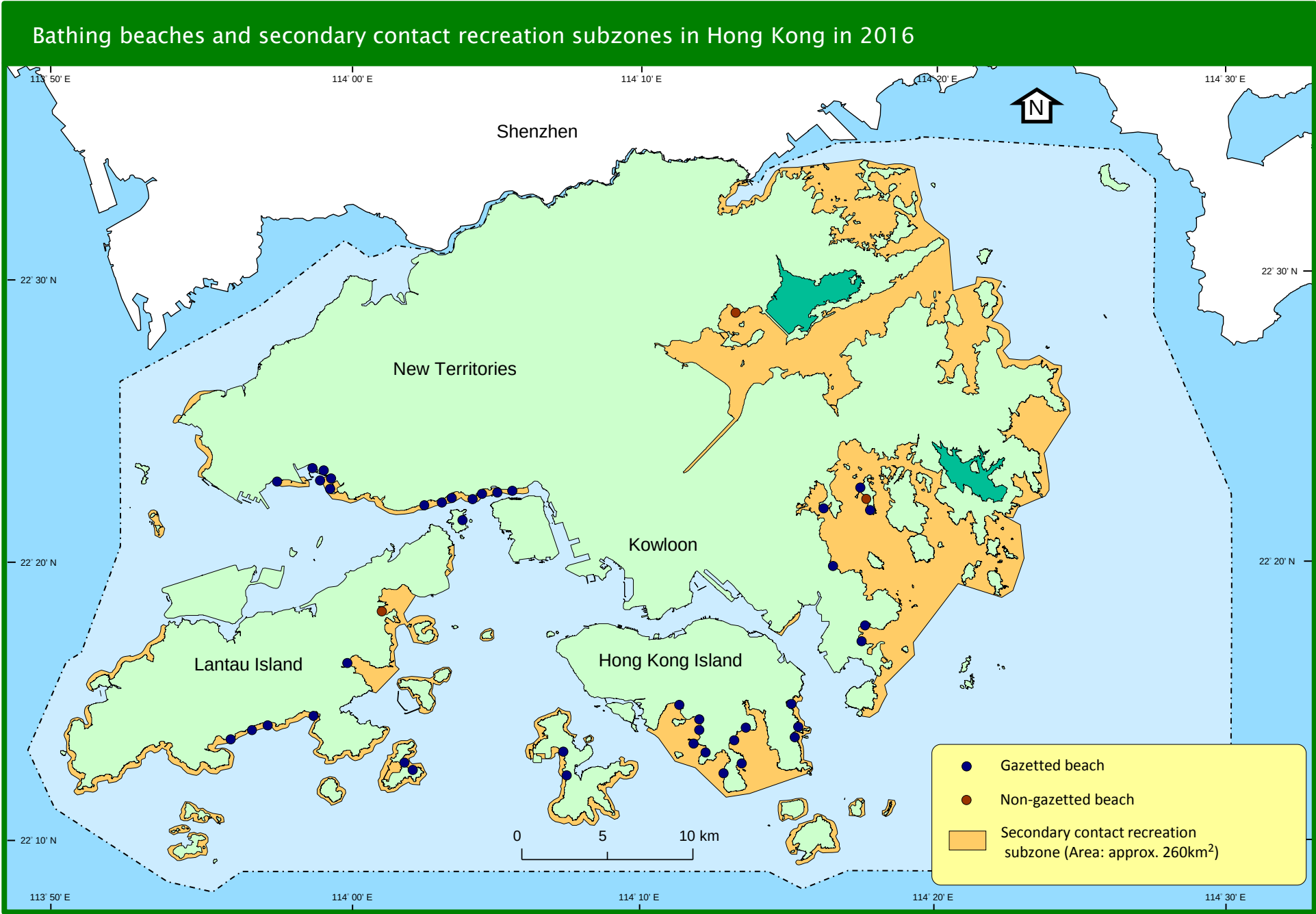


Location of marine water and sediment monitoring stations

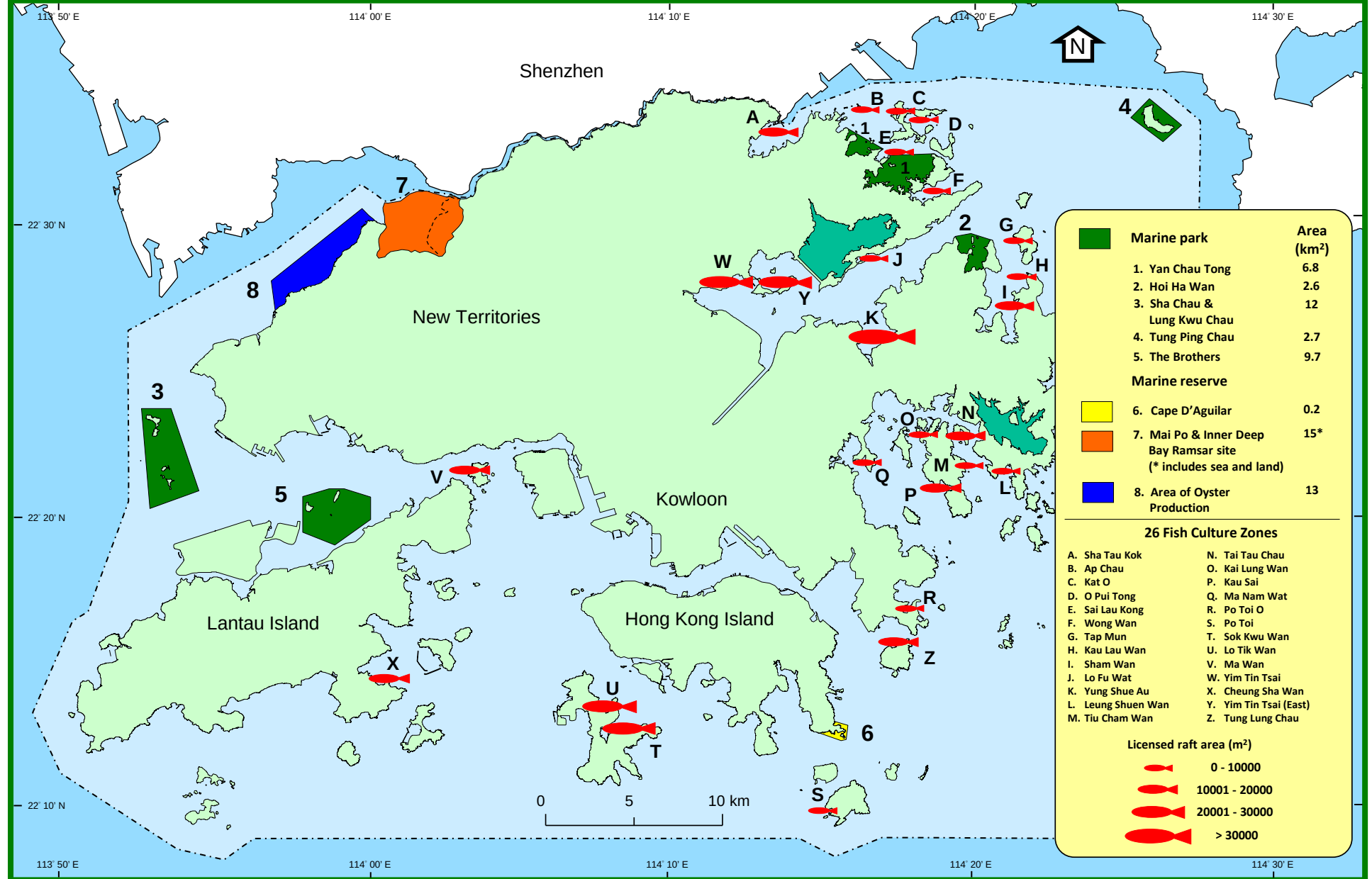
Water Control Zone	Station		Location		Depth (m) approx.
	Water	Sediment	Latitude	Longitude	
Tolo Harbour and Channel	TM2		22° 24.744' N	114° 13.085' E	4
	TM3	TS3	22° 26.857' N	114° 12.181' E	7
	TM4	TS2	22° 25.964' N	114° 13.176' E	8
	TM5		22° 27.426' N	114° 13.456' E	4
	TM6	TS4	22° 26.631' N	114° 14.506' E	12
	TM7		22° 26.907' N	114° 16.057' E	11
	TM8	TS5	22° 28.392' N	114° 18.003' E	22
	*TT1	*TS7	22° 27.270' N	114° 12.717' E	6
Southern Water	SM1	SS1	22° 12.738' N	114° 13.885' E	14
	SM2	SS2	22° 13.447' N	114° 10.691' E	14
	SM3		22° 13.527' N	114° 8.980' E	33
	SM4		22° 12.758' N	114° 8.315' E	11
	SM5		22° 12.141' N	114° 6.728' E	8
	SM6	SS3	22° 11.500' N	114° 4.743' E	14
	SM7	SS4	22° 13.740' N	114° 4.473' E	8
	SM9		22° 16.420' N	114° 4.024' E	8
	SM10		22° 18.125' N	114° 1.919' E	5
	SM11	SS5	22° 15.443' N	114° 1.078' E	8
	SM12		22° 12.861' N	114° 0.869' E	7
	SM13	SS6	22° 12.957' N	113° 57.724' E	6
	SM17		22° 9.211' N	113° 57.727' E	12
	SM18		22° 9.211' N	114° 4.746' E	21
	SM19		22° 9.211' N	114° 13.077' E	24
	SM20		22° 10.448' N	113° 52.932' E	7
	*ST1	*SS7	22° 12.517' N	114° 1.493' E	5
	*ST3	*SS8	22° 14.734' N	114° 1.928' E	6
Port Shelter	PM1		22° 23.242' N	114° 17.145' E	6
	PM2		22° 22.643' N	114° 16.687' E	8
	PM3	PS3	22° 22.156' N	114° 16.910' E	13
	PM4		22° 22.940' N	114° 18.819' E	6
	PM6		22° 21.102' N	114° 16.213' E	11
	PM7	PS5	22° 20.453' N	114° 17.703' E	17
	PM8	PS6	22° 19.168' N	114° 18.745' E	20
	PM9		22° 20.529' N	114° 20.196' E	15
	PM11		22° 19.240' N	114° 20.163' E	21
	*PT2		22° 22.798' N	114° 16.540' E	3
	*PT3	*PS2	22° 22.790' N	114° 18.400' E	6
	*PT4		22° 21.728' N	114° 15.879' E	5
Junk Bay	JM3	JS2	22° 17.490' N	114° 15.657' E	10
	JM4		22° 16.873' N	114° 15.378' E	16
Deep Bay	DM1	DS1	22° 29.769' N	114° 0.644' E	2
	DM2		22° 30.454' N	113° 59.549' E	2
	DM3	DS2	22° 28.600' N	113° 57.551' E	3
	DM4	DS3	22° 27.335' N	113° 55.937' E	4
	DM5	DS4	22° 25.561' N	113° 53.388' E	8
North Western	NM1		22° 20.877' N	114° 1.286' E	34
	NM2	NS2	22° 21.130' N	113° 58.815' E	11
	NM3	NS3	22° 21.324' N	113° 56.783' E	14
	NM5	NS4	22° 23.051' N	113° 53.972' E	20
	NM6	NS6	22° 19.281' N	113° 53.908' E	5
	NM8		22° 16.695' N	113° 51.886' E	8
	*NT1	*NS5	22° 22.475' N	113° 58.353' E	4
	MM1		22° 32.984' N	114° 14.271' E	6
Mirs Bay	MM2	MS2	22° 32.626' N	114° 16.648' E	11
	MM3	MS3	22° 33.714' N	114° 18.615' E	16
	MM4	MS4	22° 33.817' N	114° 21.483' E	18
	MM5	MS5	22° 31.233' N	114° 23.633' E	20
	MM6	MS6	22° 27.334' N	114° 20.997' E	12
	MM7	MS7	22° 31.409' N	114° 17.824' E	13
	MM8	MS8	22° 12.021' N	114° 19.345' E	31
	MM13	MS13	22° 13.000' N	114° 26.920' E	28
	MM14	MS14	22° 17.560' N	114° 26.920' E	25
	MM15	MS15	22° 22.120' N	114° 26.920' E	24
	MM16	MS16	22° 26.670' N	114° 26.920' E	22
	MM17	MS17	22° 30.192' N	114° 20.960' E	17
	MM19		22° 15.921' N	114° 19.411' E	28
Western Buffer	WM1	WS2	22° 15.044' N	114° 7.363' E	35
	WM2		22° 17.074' N	114° 5.730' E	13
	WM3	WS1	22° 19.203' N	114° 5.826' E	20
	WM4		22° 20.940' N	114° 4.256' E	26
	*WT1		22° 14.494' N	114° 9.737' E	7
	*WT3		22° 14.811' N	114° 8.918' E	10
	EM1	ES4	22° 16.506' N	114° 15.335' E	16
Eastern Buffer	EM2	ES1	22° 15.732' N	114° 15.971' E	21
	EM3	ES2	22° 14.237' N	114° 16.144' E	21
	*ET1	*ES3	22° 16.203' N	114° 14.624' E	6
	*ET2	*ES5	22° 17.078' N	114° 13.783' E	12
	VM1		22° 17.280' N	114° 13.839' E	38
Victoria Harbour	VM2		22° 17.862' N	114° 12.619' E	12
		VS3	22° 17.631' N	114° 12.526' E	8
	VM4		22° 17.860' N	114° 11.654' E	12
	VM5		22° 17.266' N	114° 10.510' E	11
		VS5	22° 17.077' N	114° 10.600' E	8
	VM6		22° 17.371' N	114° 9.665' E	14
	VM7	VS6	22° 17.771' N	114° 8.416' E	10
	VM8		22° 17.564' N	114° 7.175' E	11
	VM12	VS9	22° 19.757' N	114° 7.278' E	14
	VM14	VS10	22° 21.935' N	114° 6.527' E	11
	VM15		22° 18.579' N	114° 8.539' E	13
	*VT2	*VS12	22° 17.194' N	114° 11.304' E	5
	*VT3	*VS13	22° 17.448' N	114° 14.250' E	5
	*VT4	*VS14	22° 18.734' N	114° 12.814' E	6
	*VT8	*VS17	22° 21.360' N	114° 6.867' E	5
	*VT10	*VS19	22° 18.590' N	114° 9.430' E	5
	*VT11	*VS20	22° 18.981' N	114° 11.814' E	6
	*VT12	*VS21	22° 19.429' N	114° 8.587' E	5

Note : 1. All locations are based on WGS84 datum

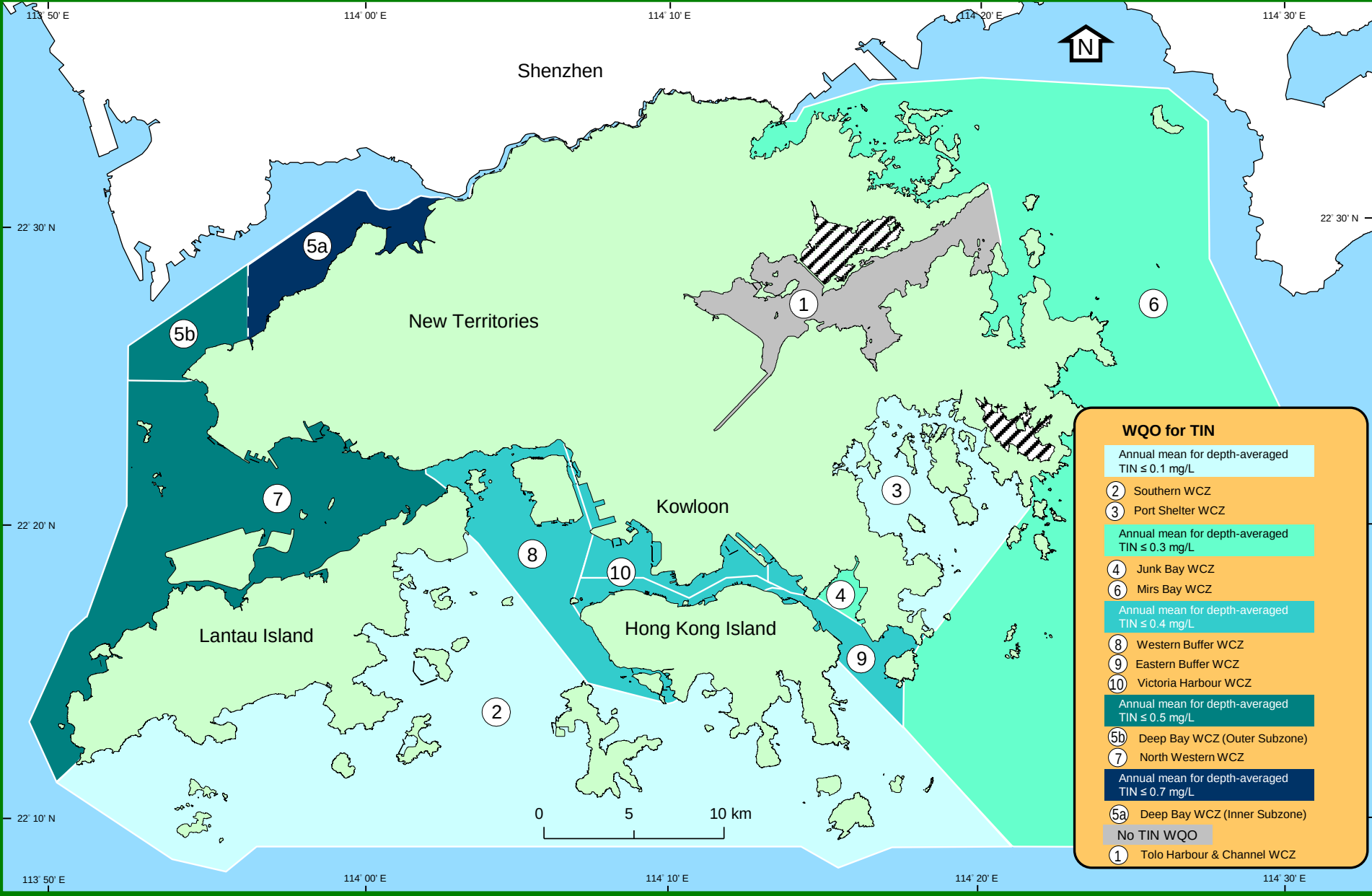
2. Water quality and sediment monitoring stations in typhoon shelters are marked with an asterisk *



Fish and shellfish culture zones and marine conservation sites in Hong Kong in 2016
(source: Agriculture, Fisheries and Conservation Department)



Water Quality Objective (WQO) for Total Inorganic Nitrogen (TIN) in the ten Water Control Zones



Summary of Water Quality Objectives (WQOs) for marine waters of Hong Kong		
Parameter	Water Quality Objective	Water Control Zone (WCZ) / Part(s) of zone / Subzone to which the WQO applies
Aesthetic Appearance	There should be no objectionable odours or discolouration of the water Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam. There should be no recognisable sewage-derived debris. Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent. The waters should not contain substances which settle to form objectionable deposits.	All WCZs (whole zone)
Dissolved Oxygen (bottom)	Not less than 2 mg/L for 90% of samples ;	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
Dissolved Oxygen (Depth-averaged)	Not less than 4 mg/L for 90% of samples ;	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
Dissolved Oxygen (bottom)	Not less than 2mg/L	Harbour Subzone in Tolo Harbour & Channel WCZ
	Not less than 3mg/L	Buffer Subzone in Tolo Harbour & Channel WCZ
	Not less than 4mg/L	Channel Subzone in Tolo Harbour & Channel WCZ
Dissolved Oxygen (surface to 2m above bottom)	Not less than 4mg/L	Harbour Subzone and Buffer Subzone in Tolo Harbour & Channel WCZ
Dissolved Oxygen (all depths)	Not less than 4mg/L	Channel Subzone in Tolo Harbour & Channel WCZ
Nutrients	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.1 mg/L	Marine waters of Southern WCZ and Port Shelter WCZ
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.3 mg/L	Marine waters of Mirs Bay WCZ, Junk Bay WCZ, North Western WCZ (Castle Peak Subzone)
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.4 mg/L	Marine waters of Eastern Buffer WCZ, Western Buffer WCZ, Victoria Harbour WCZ
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.5 mg/L	Marine waters of Deep Bay WCZ (Outer Subzone) and North Western WCZ (Whole zone except Castle Peak Subzone)
	Annual mean depth-averaged total inorganic nitrogen not to exceed 0.7 mg/L	Marine waters of Deep Bay WCZ (Inner Subzone)
Unionised ammonia	Annual mean not to exceed 0.021 mg/L	All WCZs (whole zone) except Tolo Harbour & Channel WCZ
<i>E. coli</i>	Annual geometric mean not to exceed 610 cfu/100mL	Secondary contact recreation subzones in Tolo Harbour & Channel WCZ, Southern WCZ, Port Shelter WCZ, Mirs Bay WCZ, Deep Bay WCZ, North Western WCZ, Western Buffer WCZ
	Annual geometric mean not to exceed 610 cfu/100mL	Fish culture subzones in Tolo Harbour & Channel WCZ, Southern WCZ, Port Shelter WCZ, Junk Bay WCZ, Mirs Bay WCZ, Deep Bay WCZ, Eastern Buffer WCZ, Western Buffer WCZ
pH	To be in the range 6.5 – 8.5, change due to waste discharge not to exceed 0.2	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 0.5	Harbour Subzone in Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 0.3	Buffer Subzone in Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 0.1	Channel Subzone in Tolo Harbour & Channel WCZ
Salinity	Change due to waste discharge not to exceed 10% of natural ambient level	All WCZs (Whole zone) except Tolo Harbour & Channel WCZ
	Change due to waste discharge not to be greater than ± 3 ppt	Tolo Harbour & Channel WCZ
Temperature	Change due to waste discharge not to exceed 2°C	All WCZs (Whole zone) except Tolo Harbour & Channel WCZ
	Change due to waste discharge not to exceed 1°C	Tolo Harbour & Channel WCZ
Suspended solids	Waste discharge not to raise the natural ambient level by 30% nor cause the accumulation of suspended solids which may adversely affect aquatic communities	Marine waters of all WCZs except Tolo Harbour & Channel WCZ
Toxicants	Not to be present at levels producing significant toxic effect	All WCZs (Whole zone)
Chlorophyll- <i>a</i>	Not to exceed 20mg/m ³ (µg/L) calculated as running arithmetic mean of 5 daily measurements for any location and depth	Harbour Subzone in Tolo Harbour & Channel WCZ
	Not to exceed 10mg/m ³ (µg/L) calculated as running arithmetic mean of 5 daily measurements for any location and depth	Buffer Subzone in Tolo Harbour & Channel WCZ
	Not to exceed 6mg/m ³ (µg/L) calculated as running arithmetic mean of 5 daily measurements for any location and depth	Channel Subzone in Tolo Harbour & Channel WCZ

Sediment Quality Criteria for the Classification of Sediments¹

Contaminants	Lower Chemical Exceedance Level (LCEL)	Upper Chemical Exceedance Level (UCEL)
Metals (<i>mg/kg dry weight</i>)		
Cadmium (Cd)	1.5	4
Chromium (Cr)	80	160
Copper (Cu)	65	110
Mercury (Hg)	0.5	1
Nickel (Ni) ²	40	40
Lead (Pb)	75	110
Silver (Ag)	1	2
Zinc (Zn)	200	270
Metalloid (<i>mg/kg dry weight</i>)		
Arsenic (As)	12	42
Organic-PAHs (<i>µg/kg dry weight</i>)		
Low Molecular Weight PAHs ³	550	3160
High Molecular Weight PAHs ⁴	1700	9600
Organic-non-PAHs (<i>µg/kg dry weight</i>)		
Total PCBs	23	180
Organometallics (<i>mg TBT/L in Interstitial water</i>)		
Tributyltin ²	0.15	0.15
Note: 1 The table is extracted from Appendix A of WBTC (W) No. 34/2002 Management of Dredged / Excavated Sediment (http://www.devb-wb.gov.hk). 2 When the LCEL and UCEL for a contaminant are the same, the contaminant level is considered to have exceeded UCEL if it is greater than the value shown. 3 Low molecular weight PAHs include acenaphthene, acenaphthylene, anthracene, fluorene, naphthalene, and phenanthrene. 4 High molecular weight PAHs include benzo[a]anthracene, benzo[a]pyrene, chrysene, dibenzo[a,h]anthracene, fluoranthene, pyrene, benzo[b]fluoranthene, benzo[k]fluoranthene, indeno[1,2,3-c,d]pyrene and benzo[g,h,i]perylene. 5 Total PCBs include 18 congeners: PCB 8, 18, 28, 44, 52, 66, 77, 101, 105, 118, 126, 128, 138, 153, 169, 170, 180, 187.		

Summary of marine water quality parameters

	Parameter	Unit	Reporting Limit	Sampling Depth	Standard Method / Techniques used ²⁰	Analysed by
Physical and Aggregate Properties	Temperature ¹	°C	0.1	Depth Profiling ¹⁰	Instrumental (thermistor), SEACAT19+ CTD and Water Quality Profiler	MMT/EPD ¹⁵
	Salinity ^{1,8}	-	0.1	Depth Profiling	Instrumental (electrical conductivity), SEACAT19+ CTD and Water Quality Profiler	MMT/EPD
	Dissolved Oxygen ¹	mg/L % saturation ⁹	0.1 1	Depth Profiling	Instrumental (membrane electrode), SBE23Y dissolved oxygen sensor linked to SEACAT19+ CTD and Water Quality Profiler	MMT/EPD
	Turbidity ²	NTU	0.1	Depth Profiling	Instrumental (nephelometric / infrared backscattering), OBS-3 turbidity sensor linked to SEACAT 19+ CTD and Water Quality Profiler	MMT/EPD
	pH ¹	-	0.1	Depth Profiling	Instrumental (electrodeometric), SBE18 pH sensor linked to SEACAT19+ CTD and Water Quality Profiler	MMT/EPD
	Secchi Disc Depth ²	m	0.1	---	Manual	MMT/EPD
	Suspended Solids ²	mg/L	0.5	S,M,B ¹¹	In-house method GL-PH-23 based on APHA 22nd 2540D (weighing)	GL ¹⁸
	Volatile Suspended Solids ³	mg/L	0.5	S,M,B	In-house method GL-PH-23 based on APHA 22nd 2540E (weighing)	GL
Aggregate Organic Constituents	5-day Biochemical Oxygen Demand (BOD ₅) ⁴	mg/L	0.1	S,M,B	In-house method based on APHA 20ed 5210B	EML/EPD ¹⁶
Nutrients and Inorganic Constituents	Ammonia Nitrogen ⁵	mg/L	0.005	S,M,B	In-house method GL-IN-15 based on ASTM D 3590-11 B (FIA)	GL
	Unionised Ammonia ⁵	mg/L	0.001	S,M,B	By calculation ¹²	MMT/EPD
	Nitrite Nitrogen ⁵	mg/L	0.002	S,M,B	In-house method GL-IN-18 based on APHA 22ed 4500-NO ₂ B (FIA)	GL
	Nitrate Nitrogen ⁵	mg/L	0.002	S,M,B	In-house method GL-IN-18 based on APHA 22ed 4500-NO ₃ I (FIA)	GL
	Total Inorganic Nitrogen ⁵	mg/L	0.01	S,M,B	By calculation ¹³	MMT/EPD
	Total Kjeldahl Nitrogen ⁵ (soluble; soluble & particulate)	mg/L	0.05	S,M,B	In-house methods GL-IN-14 and GL-IN-15 respectively based on ASTM D 3590-11 B (FIA)	GL
	Total Nitrogen ⁵	mg/L	0.05	S,M,B	By calculation ¹³	MMT/EPD
	Orthophosphate Phosphorus ⁵	mg/L	0.002	S,M,B	In-house method GL-IN-16 based on APHA 22ed 4500-P G (FIA)	GL
	Total Phosphorus ⁵ (soluble; soluble & particulate)	mg/L	0.02	S,M,B	In-house methods GL-IN-14 and GL-IN-16 respectively based on ASTM D 515-88 B (FIA)	GL
Biological and Microbiological Examination	Silica (as SiO ₂) (soluble) ⁵	mg/L	0.05	S,M,B	In-house method GL-IN-17 based on APHA 22ed 4500-SiO ₂ F (FIA)	GL
	Chlorophyll-a ⁶	µg/L	0.2	S,M,B	In-house method GL-OR-34 based on APHA 20ed 10200H 2 (spectrophotometric)	GL
	<i>Escherichia coli</i> (<i>E. coli</i>) ⁷	cfu/100mL	1	S,M,B	In-house method, membrane filtration with CHROMagar Liquid <i>E. coli</i> -coliform culture ¹⁴	EML/EPD
	Faecal Coliforms ⁷	cfu/100mL	1	S,M,B	In-house method, membrane filtration with CHROMagar Liquid <i>E. coli</i> -coliform culture ¹⁴	EML/EPD
	Phytoplankton	cell/mL	1	S	In-house method, 10 ml settled sub-sample using plankton chamber and inverted microscope ¹⁹	WSL/EPD ¹⁷

- Note:
- 1 Indicate general oceanographic conditions of marine water
 - 2 Low transparency and light penetration would affect aesthetic value and photosynthesis in marine water
 - 3 Indicate the amount of particulate organic matters in marine water
 - 4 Indicate the amount of organic pollutants in marine water
 - 5 Major nutrients (nitrogen, phosphorus, silica) promoting algal growth in marine water
 - 6 Indicate the amount of algal biomass in marine water
 - 7 Sewage bacteria indicate the extent of faecal pollution in marine water
 - 8 Measuring and reporting of Salinity (S) are based on the Practical Salinity Scale and International Equation of State of Seawater (UNESCO Technical Papers in Marine Science No. 30 (1981) ; No. 36 (1981) and No. 45 (1985))
 - 9 Percent saturation of dissolved oxygen is calculated from dissolved oxygen in mg/L based on Weiss R.F. (1970); The solubility of nitrogen, oxygen and argon in water and seawater. Deep Sea Res. Vol. 17, pp.721-735
 - 10 Depth profiling – continuous measurements at downcast are processed and presented at 1m intervals from 1m below the surface to 1m above the seabed
 - 11 If water depth is 6m or above, sampling is taken at three depths during upcast: S – 1m below water surface; M – mid-depth of water column; B – 1m above seabed. If water depth is 4 to 5 m, "M" is skipped; If water depth is 3m or less, "M" and "B" are skipped.
 - 12 i) Bower C.E. and Bidwell J.P. (1978), Ionization of ammonia in seawater: Effect of temperature, pH and salinity. J. Fish. Res. Board Can. Vol.35, pp.1012-1016;
ii) K., Russo R.C. & et. al. (1975), Aqueous ammonia equilibrium calculations: effect of pH and temperature. J. Fish. Res. Board Can. Vol.32, pp.2379-2383
 - 13 Total Inorganic Nitrogen = Ammonia Nitrogen + Nitrite Nitrogen + Nitrate Nitrogen ; Total Nitrogen = Total Kjeldahl Nitrogen (soluble & particulate) + Nitrite Nitrogen
 - 14 i) DoE, DHSS & PHLS (1983); The Bacteriological Examination of Drinking Water Supplies 1982, Sec.7.8 & 7.9;
ii) B.S.W. Ho and T.Y. Tam (1997), Enumeration of *E. coli* in environmental waters and wastewater using a chromogenic medium. Wat. Sci. Tech.Vol.35, No.11-12, pp.409-413; method adopted in 1997.
 - 15 MMT/EPD – Marine Monitoring Team, Water Policy and Science Group, Environmental Protection Department.
 - 16 EML/EPD – Environmental Microbiology Laboratory, Water Policy and Science Group, Environmental Protection Department.
 - 17 WSL/EPD – Water Sciences Laboratory, Water Policy and Science Group, Environmental Protection Department
 - 18 GL – Government Laboratory.
 - 19 i) Lund, J.H., Kipling, C. and Le Cren, E.D. 1958. The inverted microscope method of estimating algal numbers, and the statistical basis of estimations by counting. Hydrobiologia Vol. 11, pp. 143-170.
ii) Utermohl, H. 1958. Zur Vervollkommnung der Quantitativen Phytoplankton-Methodik. Mitt. Inter. Verein. Lim. Vol. 9, pp. 1-38.
 - 20 Mention of brand names and commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.

Summary of marine sediment¹ parameters

	Parameter	Unit ²	Reporting Limit	Standard Method / Techniques used ⁸	Analysed by
Physical and Aggregate Properties	Particle Size Fractionation	% w/w	1	In-house method, sieving and weighing ; 8 fractions: >4000 µm, <4000µm, <2000µm, <1000µm, <500µm, <250µm, <125µm and <63µm	MMT/EPD ⁶
	Electrochemical Potential ⁴	mV	1	Instrumental, Orion Model 250A pH/Redox Meter (electrodeometric)	MMT/EPD
	Total Solids (TS) ³	% w/w	0.1	In-house method GL-PH-22 based on APHA 20ed 2540G (weighing)	GL ⁷
	Total Volatile Solids (TVS) ³	% TS	0.1	In-house method GL-PH-22 based on APHA 20ed 2540G (weighing)	GL
Aggregate Organic Constituents ³	Dry Wet Ratio	-	0.01	In-house method GL-PH-22 based on APHA 20ed 2540G (weighing)	GL
	Chemical Oxygen Demand (COD)	mg/kg	2	In-house method GL-OR-47 based on ASTM D1252-00 A (open reflux)	GL
	Total Carbon (TC)	% w/w	0.1	In-house method GL-OR-33 based on APHA 20ed 5310 B (combustion - IR)	GL
Nutrients and Inorganic Constituents ³	Ammonia Nitrogen (NH ₄ -N)	mg/kg	0.05	In-house method GL-IN-19 based on ASTM D3590-11 B (FIA)	GL
	Total Kjeldahl Nitrogen (TKN)	mg/kg	0.5	In-house methods GL-IN-14 and GL-IN-15 respectively based on ASTM D3590-11 B (FIA)	GL
	Total Phosphorus	mg/kg	0.2	In-house methods GL-IN-14 and GL-IN-16 respectively based on ASTM D515-88 B (FIA)	GL
	Total Sulphide	mg/kg	0.2	In-house method GL-IN-45 based on APHA 20ed 4500-S ² -A&D (spectrophotometric)	GL
Metals & Metalloids ⁵	Total Cyanide	mg/kg	0.1	In-house method GL-IN-44 based on APHA 20ed 4500-CN ⁻ A&E (distillation and amperometric)	GL
	Aluminium (Al)	mg/kg	1	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Arsenic (As)	mg/kg	0.1	In-house methods GL-TE-64 and GL-TE-66 based on USEPA method 6020 (ICP-MS)	GL
	Barium (Ba)	mg/kg	0.2	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Boron (B)	mg/kg	5	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Cadmium (Cd)	mg/kg	0.1	In-house method GL-TE-64 based on USEPA method 6020 (ICP-MS)	GL
	Chromium (Cr)	mg/kg	0.2	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Copper (Cu)	mg/kg	0.2	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Iron (Fe)	mg/kg	5	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Lead (Pb)	mg/kg	0.2	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Manganese (Mn)	mg/kg	1	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Mercury (Hg)	mg/kg	0.05	In-house methods GL-TE-64 and GL-TE-66 based on USEPA method 6020 (ICP-MS)	GL
	Nickel (Ni)	mg/kg	0.2	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Silver (Ag)	mg/kg	0.2	In-house method GL-TE-64 based on USEPA method 6020 (ICP-MS)	GL
	Vanadium (V)	mg/kg	0.1	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
	Zinc (Zn)	mg/kg	0.2	In-house methods GL-TE-60 and GL-TE-64 respectively based on USEPA methods 6010B (ICP-AES) and 6020 (ICP-MS)	GL
Trace Organic Compounds	Polychlorinated Biphenyls (PCBs)				
	18 PCB congeners : PCB 8, 18, 28, 44, 52, 66, 77, 101, 105, 118, 126, 128, 138, 153, 169, 170, 180, 187	µg/kg	2	In-house method GL-OR-25 based on Reference Method for the Analysis of Polychlorinated Biphenyls, Environmental Protection Series: Report EPS 1/RM/31, March 1997, Environment Canada (GC-MS)	GL
	Polyaromatic Hydrocarbons (PAHs)				
	- Acenaphthene	µg/kg	50	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Acenaphthylene	µg/kg	50	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Naphthalene	µg/kg	60	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Fluorene	µg/kg	10	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Phenanthrene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Anthracene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Fluoranthene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Pyrene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(a)anthracene	µg/kg	3	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Chrysene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(b)fluoranthene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(k)fluoranthene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(a)pyrene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Dibenzo(a,h)anthracene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Benzo(ghi)perylene	µg/kg	1	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL
	- Indeno(1,2,3-cd)pyrene	µg/kg	5	In-house method GL-OR-15 based on USEPA method 610, 1984 (UV-FLUO)	GL

Note: 1 Birge-Ekman (0.023sq.m) grab / Van Veen (0.1sq.m) grab / Smith-McIntyre (0.1sq.m) grab is employed to collect sediment samples from the top 10cm of seabed.

2 All parameters are reported on a dry weight basis unless otherwise stated.

3 Determinants are reported on a wet weight basis.

4 Electrochemical potential (Eh) is measured "on-site" at 3cm below the surface of freshly collected sediment samples (Reference : Handbook of Techniques for Aquatic Sediment Sampling. By A. Mudrock & S.D. MacKnight, 1994, CRC Press).

5 Digestion procedure for metals and metalloids in sediment follows Government Laboratory's in-house method GL-TE-51.

6 MMT/EPD – Marine Monitoring Team, Water Policy and Science Group, Environmental Protection Department.

7 GL – Government Laboratory.

8 Mention of brand names and commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.

Summary of water quality statistics for the Mirs Bay WCZ in 2016

Parameter	Starling Inlet	Crooked Island		Port Island	Mirs Bay North		
	MM1	MM2	MM7	MM17	MM3	MM4	MM5
Number of samples	12	12	12	12	12	12	12
Temperature (°C)	24.0 (15.5 - 29.7)	23.6 (15.5 - 29.0)	23.5 (15.1 - 28.6)	23.1 (14.9 - 28.2)	23.1 (15.0 - 28.4)	23.1 (15.3 - 28.1)	23.1 (15.4 - 27.6)
Salinity	30.4 (27.8 - 31.5)	31.1 (30.2 - 32.5)	31.3 (29.6 - 32.9)	31.6 (30.8 - 33.6)	31.6 (30.7 - 33.3)	31.7 (31.1 - 33.6)	31.8 (31.1 - 33.4)
Dissolved Oxygen (mg/L)	6.9 (5.4 - 9.4)	6.7 (5.1 - 11.0)	6.2 (4.9 - 8.7)	6.1 (4.3 - 8.6)	6.2 (4.5 - 8.9)	6.2 (4.1 - 8.3)	6.0 (4.7 - 8.1)
	Bottom						
	6.7 (4.8 - 9.1)	6.0 (2.1 - 9.5)	5.2 (2.1 - 8.6)	5.3 (1.3 - 8.6)	5.2 (1.6 - 8.6)	5.4 (2.2 - 7.9)	5.0 (1.6 - 7.6)
Dissolved Oxygen (% saturation)	97 (72 - 126)	94 (78 - 134)	87 (76 - 107)	84 (65 - 103)	85 (68 - 107)	86 (61 - 101)	84 (70 - 100)
	Bottom						
	94 (65 - 140)	83 (30 - 115)	72 (31 - 104)	71 (19 - 103)	70 (23 - 103)	73 (31 - 101)	68 (24 - 99)
pH	8.0 (7.6 - 8.3)	8.0 (7.5 - 8.3)	7.9 (7.5 - 8.2)	7.9 (7.5 - 8.2)	8.0 (7.5 - 8.2)	7.9 (7.5 - 8.3)	7.9 (7.5 - 8.2)
Secchi Disc Depth (m)	2.7 (1.5 - 3.0)	3.4 (2.0 - 6.0)	3.4 (2.5 - 4.5)	4.0 (2.5 - 7.0)	3.5 (2.0 - 5.0)	3.8 (2.5 - 6.0)	3.6 (2.0 - 5.0)
Turbidity (NTU)	4.0 (0.8 - 23.8)	2.3 (0.6 - 6.6)	1.6 (0.4 - 4.4)	1.8 (0.3 - 5.5)	2.9 (0.4 - 10.3)	2.7 (0.5 - 9.4)	2.6 (0.5 - 8.9)
Suspended Solids (mg/L)	8.0 (1.3 - 17.7)	8.1 (1.0 - 19.7)	7.0 (1.1 - 17.7)	5.7 (0.6 - 16.0)	6.6 (0.6 - 15.5)	6.8 (0.7 - 18.0)	5.8 (0.7 - 14.8)
5-day Biochemical Oxygen Demand (mg/L)	1.5 (0.5 - 3.7)	1.2 (0.3 - 1.8)	1.1 (0.3 - 1.7)	1.0 (0.3 - 2.0)	0.9 (0.2 - 1.7)	0.9 (<0.1 - 1.9)	1.1 (0.2 - 3.0)
Ammonia Nitrogen (mg/L)	0.061 (0.016 - 0.159)	0.043 (0.015 - 0.096)	0.046 (0.020 - 0.121)	0.037 (0.015 - 0.078)	0.038 (0.017 - 0.077)	0.034 (0.011 - 0.074)	0.041 (0.010 - 0.117)
Unionised Ammonia (mg/L)	0.002 (<0.001 - 0.006)	0.002 (<0.001 - 0.003)	0.002 (<0.001 - 0.004)	0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)	0.014 (<0.002 - 0.053)	0.014 (0.002 - 0.038)	0.015 (0.002 - 0.040)	0.011 (0.002 - 0.023)	0.014 (<0.002 - 0.044)	0.013 (<0.002 - 0.031)	0.011 (<0.002 - 0.025)
Nitrate Nitrogen (mg/L)	0.045 (<0.002 - 0.120)	0.022 (0.004 - 0.062)	0.026 (0.002 - 0.063)	0.031 (0.003 - 0.083)	0.033 (<0.002 - 0.086)	0.034 (0.003 - 0.094)	0.036 (0.004 - 0.095)
Total Inorganic Nitrogen (mg/L)	0.12 (0.02 - 0.29)	0.08 (0.03 - 0.13)	0.09 (0.04 - 0.16)	0.08 (0.05 - 0.15)	0.08 (0.04 - 0.12)	0.08 (0.04 - 0.14)	0.09 (0.05 - 0.14)
Total Kjeldahl Nitrogen (mg/L)	0.48 (0.28 - 0.82)	0.45 (0.24 - 0.64)	0.44 (0.26 - 0.60)	0.47 (0.25 - 0.69)	0.41 (0.20 - 0.66)	0.48 (0.33 - 0.91)	0.42 (0.29 - 0.64)
Total Nitrogen (mg/L)	0.54 (0.32 - 0.95)	0.48 (0.28 - 0.72)	0.48 (0.29 - 0.64)	0.51 (0.26 - 0.72)	0.46 (0.21 - 0.69)	0.52 (0.36 - 0.94)	0.47 (0.31 - 0.68)
Orthophosphate Phosphorus (mg/L)	0.007 (<0.002 - 0.024)	0.006 (0.002 - 0.009)	0.007 (0.003 - 0.013)	0.007 (0.002 - 0.014)	0.007 (0.002 - 0.018)	0.008 (0.003 - 0.017)	0.008 (0.003 - 0.014)
Total Phosphorus (mg/L)	0.05 (0.02 - 0.11)	0.03 (0.02 - 0.05)	0.03 (0.02 - 0.07)	0.03 (0.02 - 0.06)	0.03 (0.02 - 0.04)	0.04 (0.02 - 0.06)	0.04 (<0.02 - 0.06)
Silica (as SiO ₂) (mg/L)	0.75 (0.16 - 1.37)	0.74 (0.33 - 1.10)	0.85 (0.26 - 1.25)	0.81 (0.37 - 1.10)	0.83 (0.35 - 1.20)	0.85 (0.46 - 1.34)	0.86 (0.47 - 1.28)
Chlorophyll-a (µg/L)	7.5 (0.6 - 34.3)	3.9 (0.4 - 7.1)	4.5 (1.1 - 11.7)	2.4 (1.1 - 4.7)	2.8 (1.1 - 6.2)	2.3 (1.0 - 6.6)	4.2 (1.1 - 22.8)
<i>E.coli</i> (cfu/100mL)	15 (1 - 1100)	2 (<1 - 21)	1 (<1 - 4)	1 (<1 - 20)	1 (<1 - 5)	1 (<1 - 10)	1 (<1 - 6)
Faecal Coliforms (cfu/100mL)	30 (2 - 3200)	4 (<1 - 58)	2 (<1 - 9)	2 (<1 - 47)	2 (<1 - 13)	2 (<1 - 19)	2 (<1 - 12)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Mirs Bay WCZ in 2016 (continued)

Parameter	Ninepin Group MM19	Waglan Isalnd MM8	Mirs Bay (South) MM13	Mirs Bay (Central) MM14 MM15 MM16			Long Harbour MM6
Number of samples	12	12	12	12	12	12	12
Temperature (°C)	22.1 (15.2 - 26.7)	22.4 (15.4 - 26.6)	22.3 (15.4 - 26.4)	22.1 (15.3 - 26.4)	22.3 (15.2 - 26.7)	22.4 (15.2 - 27.0)	23.4 (14.9 - 28.5)
Salinity	32.1 (30.9 - 33.8)	31.7 (29.8 - 33.2)	31.9 (30.5 - 33.6)	32.1 (30.5 - 33.9)	32.2 (30.6 - 34.1)	31.9 (30.8 - 33.8)	31.3 (29.2 - 33.1)
Dissolved Oxygen (mg/L)	6.4 (3.8 - 9.9)	6.3 (4.7 - 8.5)	6.7 (4.6 - 9.2)	6.6 (4.3 - 9.4)	6.3 (4.2 - 9.2)	6.3 (4.2 - 9.1)	6.1 (5.1 - 8.5)
Bottom	5.8 (1.4 - 8.5)	5.6 (2.0 - 8.2)	6.1 (2.0 - 8.6)	6.0 (1.4 - 8.7)	5.8 (0.8 - 8.6)	5.5 (0.9 - 8.5)	5.7 (3.4 - 8.3)
Dissolved Oxygen (% saturation)	88 (56 - 123)	86 (69 - 106)	92 (69 - 133)	91 (65 - 117)	86 (63 - 114)	86 (65 - 112)	85 (75 - 110)
Bottom	78 (20 - 103)	76 (29 - 100)	83 (29 - 112)	81 (20 - 106)	78 (12 - 104)	74 (13 - 103)	79 (52 - 104)
pH	7.8 (7.1 - 8.1)	7.8 (7.1 - 8.1)	7.8 (7.0 - 8.1)	7.8 (6.9 - 8.1)	7.8 (6.8 - 8.1)	7.7 (6.6 - 8.0)	7.9 (7.6 - 8.2)
Secchi Disc Depth (m)	3.7 (2.0 - 7.0)	4.0 (2.5 - 7.0)	3.8 (2.5 - 7.0)	3.9 (3.0 - 7.0)	4.2 (2.0 - 9.0)	3.9 (2.5 - 6.0)	3.8 (2.5 - 5.0)
Turbidity (NTU)	3.2 (1.0 - 5.0)	4.6 (1.0 - 15.6)	4.9 (1.3 - 10.1)	4.0 (0.6 - 8.5)	4.7 (0.8 - 12.5)	4.4 (0.3 - 18.4)	2.0 (0.3 - 9.5)
Suspended Solids (mg/L)	5.0 (0.7 - 11.7)	7.1 (0.6 - 24.7)	6.6 (0.7 - 20.7)	6.7 (0.8 - 20.3)	5.0 (0.7 - 16.7)	5.1 (1.2 - 18.3)	6.1 (1.2 - 15.0)
5-day Biochemical Oxygen Demand (mg/L)	0.9 (0.1 - 2.1)	0.6 (0.2 - 1.3)	0.6 (0.2 - 1.8)	0.8 (0.2 - 3.1)	0.9 (0.1 - 3.9)	0.7 (<0.1 - 2.0)	1.0 (0.2 - 1.7)
Ammonia Nitrogen (mg/L)	0.025 (<0.005 - 0.061)	0.023 (0.008 - 0.044)	0.022 (0.006 - 0.045)	0.024 (0.008 - 0.052)	0.027 (0.007 - 0.072)	0.029 (0.008 - 0.057)	0.040 (0.011 - 0.079)
Unionised Ammonia (mg/L)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.001)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.002)
Nitrite Nitrogen (mg/L)	0.013 (0.002 - 0.032)	0.014 (0.002 - 0.035)	0.011 (<0.002 - 0.030)	0.011 (<0.002 - 0.022)	0.011 (<0.002 - 0.023)	0.011 (<0.002 - 0.024)	0.011 (<0.002 - 0.030)
Nitrate Nitrogen (mg/L)	0.078 (0.007 - 0.183)	0.096 (0.007 - 0.173)	0.091 (0.003 - 0.183)	0.076 (0.006 - 0.183)	0.066 (0.007 - 0.170)	0.064 (0.013 - 0.147)	0.027 (0.004 - 0.073)
Total Inorganic Nitrogen (mg/L)	0.12 (0.05 - 0.21)	0.13 (0.04 - 0.23)	0.12 (0.02 - 0.19)	0.11 (0.03 - 0.20)	0.10 (0.04 - 0.19)	0.10 (0.06 - 0.18)	0.08 (0.05 - 0.14)
Total Kjeldahl Nitrogen (mg/L)	0.41 (0.10 - 0.79)	0.43 (0.20 - 0.85)	0.47 (0.18 - 1.05)	0.53 (0.20 - 0.96)	0.43 (0.20 - 0.69)	0.45 (0.23 - 0.77)	0.39 (0.16 - 0.65)
Total Nitrogen (mg/L)	0.50 (0.30 - 0.87)	0.54 (0.35 - 1.01)	0.58 (0.33 - 1.17)	0.62 (0.27 - 1.03)	0.51 (0.35 - 0.73)	0.52 (0.28 - 0.86)	0.43 (0.20 - 0.66)
Orthophosphate Phosphorus (mg/L)	0.009 (0.004 - 0.024)	0.008 (0.003 - 0.021)	0.009 (<0.002 - 0.022)	0.008 (<0.002 - 0.023)	0.008 (0.002 - 0.022)	0.009 (0.004 - 0.021)	0.006 (0.002 - 0.011)
Total Phosphorus (mg/L)	0.04 (0.02 - 0.09)	0.04 (0.02 - 0.07)	0.04 (0.02 - 0.08)	0.05 (0.02 - 0.17)	0.03 (0.02 - 0.05)	0.04 (0.02 - 0.08)	0.03 (<0.02 - 0.05)
Silica (as SiO ₂) (mg/L)	0.94 (0.56 - 1.50)	0.93 (0.41 - 1.37)	0.83 (0.34 - 1.40)	0.87 (0.41 - 1.47)	0.90 (0.54 - 1.53)	0.95 (0.62 - 1.43)	0.87 (0.43 - 1.32)
Chlorophyll-a (µg/L)	3.6 (0.7 - 12.4)	3.2 (0.7 - 13.0)	3.1 (0.8 - 14.4)	5.2 (0.7 - 32.3)	3.7 (1.1 - 25.0)	2.4 (0.8 - 8.7)	2.2 (1.0 - 4.3)
<i>E.coli</i> (cfu/100mL)	1 (<1 - 1)	1 (<1 - 1)	1 (<1 - 1)	1 (<1 - 1)	1 (<1 - 1)	1 (<1 - 2)	1 (<1 - 9)
Faecal Coliforms (cfu/100mL)	1 (<1 - 4)	1 (<1 - 3)	1 (<1 - 2)	1 (<1 - 1)	1 (<1 - 1)	1 (<1 - 3)	2 (<1 - 16)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Port Shelter WCZ in 2016

Parameter	Inner Port Shelter				Hebe Haven
	PM1	PM2	PM3	PM4	PM6
Number of samples	12	12	12	12	12
Temperature (°C)	23.6 (15.8 - 28.8)	23.3 (15.7 - 28.7)	23.0 (15.7 - 28.3)	23.4 (15.6 - 29.1)	23.3 (15.8 - 28.9)
Salinity	31.5 (29.8 - 33.6)	31.7 (30.4 - 33.7)	31.9 (30.9 - 34.3)	31.5 (29.6 - 34.1)	31.5 (29.7 - 33.7)
Dissolved Oxygen (mg/L)	6.7 (5.1 - 11.4)	6.4 (5.1 - 10.0)	6.2 (4.7 - 9.6)	6.6 (5.1 - 10.9)	6.2 (5.3 - 8.0)
Bottom	6.5 (4.6 - 9.2)	6.1 (4.7 - 8.4)	5.4 (2.5 - 8.0)	6.4 (4.7 - 8.9)	5.6 (4.2 - 7.7)
Dissolved Oxygen (% saturation)	93 (75 - 148)	90 (76 - 129)	86 (72 - 123)	91 (73 - 141)	86 (77 - 98)
Bottom	89 (71 - 119)	84 (71 - 114)	74 (37 - 110)	88 (71 - 113)	77 (53 - 101)
pH	7.9 (7.6 - 8.2)	7.9 (7.5 - 8.1)	7.9 (7.5 - 8.1)	7.9 (7.6 - 8.2)	7.9 (7.5 - 8.2)
Secchi Disc Depth (m)	3.2 (2.0 - 5.0)	3.1 (1.5 - 4.0)	3.6 (2.5 - 5.0)	3.2 (2.5 - 5.0)	3.4 (2.5 - 4.5)
Turbidity (NTU)	2.5 (1.0 - 3.8)	2.5 (0.8 - 3.9)	2.6 (0.9 - 5.1)	2.8 (1.0 - 5.0)	2.5 (0.7 - 3.6)
Suspended Solids (mg/L)	3.2 (1.2 - 8.3)	3.8 (1.2 - 10.1)	3.7 (0.8 - 9.4)	4.5 (1.5 - 11.4)	3.6 (0.6 - 10.1)
5-day Biochemical Oxygen Demand (mg/L)	1.5 (0.4 - 4.9)	1.2 (0.3 - 4.0)	1.2 (0.3 - 5.0)	1.4 (0.4 - 4.2)	1.0 (0.3 - 3.6)
Ammonia Nitrogen (mg/L)	0.033 (0.005 - 0.070)	0.033 (0.008 - 0.063)	0.034 (0.014 - 0.056)	0.032 (0.006 - 0.077)	0.036 (0.010 - 0.080)
Unionised Ammonia (mg/L)	0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)
Nitrite Nitrogen (mg/L)	0.006 (<0.002 - 0.017)	0.007 (<0.002 - 0.024)	0.008 (<0.002 - 0.016)	0.005 (<0.002 - 0.014)	0.009 (<0.002 - 0.041)
Nitrate Nitrogen (mg/L)	0.021 (<0.002 - 0.085)	0.028 (<0.002 - 0.097)	0.032 (<0.002 - 0.103)	0.015 (<0.002 - 0.081)	0.031 (<0.002 - 0.117)
Total Inorganic Nitrogen (mg/L)	0.06 (<0.01 - 0.14)	0.07 (0.01 - 0.15)	0.07 (0.03 - 0.16)	0.05 (0.01 - 0.14)	0.08 (0.02 - 0.18)
Total Kjeldahl Nitrogen (mg/L)	0.57 (0.21 - 1.04)	0.53 (0.18 - 0.92)	0.57 (0.19 - 1.01)	0.58 (0.24 - 0.93)	0.49 (0.18 - 0.83)
Total Nitrogen (mg/L)	0.59 (0.30 - 1.05)	0.57 (0.27 - 0.95)	0.61 (0.22 - 1.03)	0.60 (0.27 - 0.94)	0.53 (0.19 - 0.85)
Orthophosphate Phosphorus (mg/L)	0.004 (<0.002 - 0.011)	0.005 (<0.002 - 0.013)	0.006 (0.002 - 0.013)	0.004 (<0.002 - 0.010)	0.006 (0.003 - 0.014)
Total Phosphorus (mg/L)	0.03 (<0.02 - 0.05)	0.04 (0.02 - 0.10)	0.03 (0.02 - 0.05)	0.03 (0.02 - 0.06)	0.03 (<0.02 - 0.07)
Silica (as SiO ₂) (mg/L)	0.67 (0.06 - 1.23)	0.68 (0.07 - 1.05)	0.73 (0.13 - 1.13)	0.72 (0.05 - 1.30)	0.80 (0.43 - 1.11)
Chlorophyll-a (µg/L)	6.5 (1.0 - 22.4)	6.7 (1.0 - 33.5)	5.3 (0.9 - 19.8)	8.3 (1.4 - 34.7)	5.3 (1.0 - 25.2)
<i>E.coli</i> (cfu/100mL)	1 (<1 - 5)	15 (1 - 150)	2 (<1 - 16)	2 (<1 - 14)	2 (<1 - 14)
Faecal Coliforms (cfu/100mL)	3 (<1 - 18)	82 (4 - 610)	6 (<1 - 68)	4 (<1 - 54)	5 (<1 - 63)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics of the Port Shelter WCZ in 2016 (continued)

Parameter	Outer Port Shelter		Rocky Harbour	Bluff Island
	PM7	PM8	PM9	PM11
Number of samples	12	12	12	12
Temperature (°C)	22.8 (16.0 - 27.9)	22.7 (16.0 - 27.5)	22.8 (15.8 - 28.1)	22.5 (16.0 - 27.3)
Salinity	32.1 (31.1 - 34.2)	32.2 (31.0 - 34.3)	32.1 (31.0 - 34.3)	32.3 (31.0 - 34.3)
Dissolved Oxygen (mg/L)	6.2 (4.4 - 8.2)	6.4 (3.7 - 8.5)	6.4 (5.0 - 8.6)	6.3 (4.3 - 8.6)
Bottom	5.3 (0.6 - 7.4)	5.5 (1.8 - 7.6)	5.8 (2.9 - 8.0)	5.7 (1.9 - 7.7)
Dissolved Oxygen (% saturation)	86 (68 - 105)	88 (56 - 109)	88 (75 - 111)	87 (63 - 109)
Bottom	72 (8 - 101)	75 (26 - 97)	79 (42 - 99)	77 (27 - 98)
pH	7.9 (7.5 - 8.2)	7.9 (7.5 - 8.3)	7.9 (7.5 - 8.2)	7.9 (7.5 - 8.3)
Secchi Disc Depth (m)	4.1 (2.5 - 5.5)	3.9 (3.0 - 5.0)	3.9 (3.0 - 5.0)	3.9 (3.0 - 5.0)
Turbidity (NTU)	2.5 (0.8 - 6.0)	2.5 (0.9 - 4.0)	2.4 (0.7 - 4.1)	2.8 (1.2 - 4.9)
Suspended Solids (mg/L)	3.4 (1.0 - 10.6)	2.8 (1.2 - 8.7)	2.9 (1.2 - 7.2)	3.1 (1.1 - 9.7)
5-day Biochemical Oxygen Demand (mg/L)	0.7 (<0.1 - 1.7)	0.7 (0.2 - 2.2)	0.9 (0.1 - 2.3)	0.7 (0.1 - 1.7)
Ammonia Nitrogen (mg/L)	0.027 (0.010 - 0.047)	0.030 (0.012 - 0.058)	0.028 (0.013 - 0.049)	0.028 (0.009 - 0.058)
Unionised Ammonia (mg/L)	<0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.002)	<0.001 (<0.001 - 0.003)
Nitrite Nitrogen (mg/L)	0.011 (<0.002 - 0.023)	0.011 (0.003 - 0.027)	0.009 (<0.002 - 0.024)	0.012 (<0.002 - 0.028)
Nitrate Nitrogen (mg/L)	0.036 (<0.002 - 0.110)	0.041 (0.008 - 0.110)	0.039 (<0.002 - 0.120)	0.043 (0.005 - 0.110)
Total Inorganic Nitrogen (mg/L)	0.07 (0.03 - 0.16)	0.08 (0.04 - 0.16)	0.08 (0.03 - 0.16)	0.08 (0.03 - 0.16)
Total Kjeldahl Nitrogen (mg/L)	0.53 (0.23 - 0.98)	0.46 (0.16 - 0.90)	0.47 (0.12 - 0.93)	0.50 (0.24 - 0.88)
Total Nitrogen (mg/L)	0.58 (0.25 - 1.01)	0.51 (0.18 - 0.95)	0.51 (0.14 - 0.95)	0.55 (0.29 - 0.91)
Orthophosphate Phosphorus (mg/L)	0.007 (<0.002 - 0.014)	0.007 (0.004 - 0.015)	0.007 (0.003 - 0.013)	0.007 (<0.002 - 0.013)
Total Phosphorus (mg/L)	0.03 (<0.02 - 0.04)	0.03 (<0.02 - 0.06)	0.03 (<0.02 - 0.09)	0.03 (<0.02 - 0.09)
Silica (as SiO ₂) (mg/L)	0.72 (0.48 - 1.20)	0.75 (0.38 - 1.44)	0.75 (0.27 - 1.20)	0.74 (0.34 - 1.29)
Chlorophyll-a (µg/L)	3.0 (0.8 - 10.7)	3.4 (0.8 - 13.3)	4.3 (0.6 - 15.3)	3.4 (0.9 - 8.0)
<i>E.coli</i> (cfu/100mL)	1 (<1 - 2)	1 (<1 - 2)	1 (<1 - 1)	1 (<1 - 1)
Faecal Coliforms (cfu/100mL)	1 (<1 - 5)	1 (<1 - 8)	1 (<1 - 3)	1 (<1 - 2)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Tolo Harbour and Channel WCZ in 2016

Parameter	Harbour Subzone			Buffer Subzone		Channel Subzone	
	TM2	TM3	TM4	TM5	TM6	TM7	TM8
Number of samples	12	12	12	12	12	12	12
Temperature (°C)	23.5 (15.9 - 28.4)	23.7 (15.8 - 28.3)	23.4 (16.1 - 28.3)	23.9 (15.6 - 29.2)	23.4 (15.8 - 28.5)	23.4 (15.7 - 28.6)	23.0 (15.4 - 28.4)
Salinity	28.5 (23.5 - 32.4)	29.6 (26.6 - 32.1)	30.2 (27.8 - 33.3)	29.8 (26.8 - 31.8)	30.8 (29.1 - 32.5)	30.9 (30.0 - 32.1)	31.4 (28.6 - 33.2)
Dissolved Oxygen (mg/L)	6.2 (4.4 - 9.8)	6.4 (4.3 - 9.1)	6.2 (4.3 - 8.8)	6.3 (4.1 - 8.6)	5.8 (4.1 - 8.3)	5.9 (4.5 - 8.8)	5.5 (2.9 - 7.8)
Bottom	6.2 (4.0 - 11.5)	5.3 (3.2 - 7.2)	5.6 (3.5 - 7.1)	6.1 (4.0 - 9.6)	4.6 (1.5 - 7.9)	4.9 (2.0 - 8.3)	4.8 (1.1 - 8.3)
Dissolved Oxygen (% saturation)	85 (63 - 126)	88 (66 - 118)	87 (65 - 114)	87 (62 - 113)	80 (63 - 110)	82 (68 - 115)	76 (44 - 102)
Bottom	85 (61 - 149)	74 (49 - 109)	78 (54 - 107)	85 (61 - 118)	64 (21 - 104)	67 (28 - 108)	66 (16 - 105)
pH	7.9 (7.4 - 8.1)	7.9 (7.6 - 8.2)	7.8 (7.5 - 8.1)	7.9 (7.6 - 8.2)	7.9 (7.6 - 8.2)	7.9 (7.4 - 8.2)	7.8 (7.6 - 8.2)
Secchi Disc Depth (m)	2.7 (1.5 - 4.0)	2.6 (1.5 - 5.0)	2.7 (1.5 - 5.0)	2.6 (2.0 - 3.5)	3.0 (2.0 - 5.0)	3.2 (2.3 - 6.0)	3.5 (2.0 - 6.0)
Turbidity (NTU)	2.4 (0.7 - 5.3)	3.1 (0.8 - 14.9)	1.8 (0.7 - 3.6)	2.4 (0.4 - 4.0)	1.7 (0.4 - 3.0)	1.9 (0.4 - 3.2)	2.6 (0.6 - 6.2)
Suspended Solids (mg/L)	2.9 (1.1 - 7.9)	3.0 (1.1 - 6.1)	2.2 (0.9 - 3.6)	5.3 (1.1 - 25.0)	1.6 (0.6 - 2.5)	2.3 (0.8 - 8.3)	3.0 (1.0 - 6.9)
5-day Biochemical Oxygen Demand (mg/L)	2.5 (0.7 - 6.3)	1.9 (0.6 - 4.7)	1.6 (0.6 - 3.2)	1.6 (0.6 - 3.5)	1.2 (0.5 - 2.9)	1.2 (0.5 - 2.6)	0.8 (0.2 - 1.8)
Ammonia Nitrogen (mg/L)	0.079 (0.024 - 0.150)	0.074 (0.024 - 0.149)	0.072 (0.013 - 0.154)	0.049 (0.011 - 0.098)	0.051 (0.014 - 0.097)	0.043 (0.012 - 0.078)	0.046 (0.011 - 0.083)
Unionised Ammonia (mg/L)	0.002 (<0.001 - 0.006)	0.003 (<0.001 - 0.005)	0.002 (<0.001 - 0.005)	0.002 (<0.001 - 0.004)	0.002 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)
Nitrite Nitrogen (mg/L)	0.008 (<0.002 - 0.028)	0.010 (<0.002 - 0.040)	0.009 (<0.002 - 0.037)	0.004 (<0.002 - 0.011)	0.011 (<0.002 - 0.034)	0.009 (<0.002 - 0.030)	0.013 (0.002 - 0.047)
Nitrate Nitrogen (mg/L)	0.057 (0.007 - 0.270)	0.032 (<0.002 - 0.130)	0.030 (0.003 - 0.149)	0.018 (<0.002 - 0.062)	0.025 (0.003 - 0.093)	0.021 (<0.002 - 0.061)	0.031 (0.002 - 0.121)
Total Inorganic Nitrogen (mg/L)	0.14 (0.05 - 0.35)	0.12 (0.05 - 0.29)	0.11 (0.02 - 0.31)	0.07 (0.02 - 0.16)	0.09 (0.05 - 0.12)	0.07 (0.04 - 0.12)	0.09 (0.05 - 0.13)
Total Kjeldahl Nitrogen (mg/L)	0.55 (0.24 - 1.10)	0.49 (0.20 - 0.86)	0.46 (0.17 - 0.92)	0.45 (0.14 - 1.10)	0.48 (0.18 - 0.97)	0.46 (0.17 - 0.95)	0.45 (0.13 - 0.97)
Total Nitrogen (mg/L)	0.61 (0.31 - 1.12)	0.54 (0.22 - 0.88)	0.50 (0.19 - 0.93)	0.48 (0.15 - 1.10)	0.52 (0.19 - 1.07)	0.49 (0.18 - 1.01)	0.49 (0.15 - 1.10)
Orthophosphate Phosphorus (mg/L)	0.007 (<0.002 - 0.029)	0.006 (<0.002 - 0.016)	0.006 (<0.002 - 0.015)	0.005 (<0.002 - 0.012)	0.007 (0.002 - 0.020)	0.007 (<0.002 - 0.022)	0.008 (0.002 - 0.019)
Total Phosphorus (mg/L)	0.05 (0.03 - 0.09)	0.05 (<0.02 - 0.12)	0.05 (0.02 - 0.09)	0.05 (<0.02 - 0.18)	0.03 (<0.02 - 0.06)	0.04 (<0.02 - 0.08)	0.03 (<0.02 - 0.06)
Silica (as SiO ₂) (mg/L)	1.67 (0.86 - 3.10)	1.24 (0.72 - 2.07)	1.30 (0.70 - 1.93)	1.12 (0.59 - 1.75)	1.16 (0.56 - 1.80)	0.98 (0.51 - 1.34)	1.01 (0.40 - 1.75)
Chlorophyll-a (µg/L)	10.8 (1.0 - 35.5)	7.8 (1.6 - 23.3)	6.2 (0.5 - 17.6)	5.4 (0.9 - 16.0)	5.2 (1.6 - 18.8)	4.7 (0.3 - 12.8)	2.4 (0.5 - 5.3)
<i>E.coli</i> (cfu/100mL)	36 (3 - 3900)	12 (2 - 380)	7 (<1 - 210)	4 (<1 - 510)	3 (<1 - 72)	1 (<1 - 4)	1 (<1 - 3)
Faecal Coliforms (cfu/100mL)	150 (12 - 10000)	48 (5 - 1200)	25 (2 - 530)	16 (2 - 1100)	9 (1 - 690)	2 (<1 - 18)	1 (<1 - 6)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Southern WCZ in 2016

Parameter	Hong Kong Island (South)			East Lamma Channel	
	SM1	SM2	SM19	SM3	SM4
Number of samples	12	12	12	12	12
Temperature (°C)	23.3 (16.2 - 27.9)	23.2 (16.2 - 28.2)	23.1 (16.3 - 27.8)	22.7 (15.7 - 28.5)	23.3 (16.4 - 28.1)
Salinity	31.1 (26.3 - 32.7)	30.8 (26.9 - 32.7)	31.2 (26.6 - 32.8)	31.7 (30.2 - 33.2)	30.4 (26.2 - 32.4)
Dissolved Oxygen (mg/L)	6.8 (4.8 - 9.1)	6.7 (4.8 - 9.4)	6.8 (5.0 - 9.8)	6.2 (4.2 - 8.2)	6.5 (4.3 - 8.7)
Bottom	6.3 (2.1 - 8.5)	6.3 (3.6 - 9.3)	5.9 (2.8 - 8.4)	5.9 (2.3 - 8.6)	6.4 (4.7 - 8.7)
Dissolved Oxygen (% saturation)	95 (72 - 115)	93 (73 - 119)	94 (76 - 121)	86 (60 - 105)	90 (65 - 108)
Bottom	87 (30 - 106)	87 (52 - 116)	81 (39 - 103)	81 (32 - 109)	88 (70 - 108)
pH	7.9 (7.6 - 8.3)	7.9 (7.6 - 8.4)	8.0 (7.6 - 8.3)	7.9 (7.4 - 8.2)	7.9 (7.6 - 8.3)
Secchi Disc Depth (m)	3.4 (2.5 - 5.0)	3.3 (2.0 - 5.5)	3.3 (2.4 - 5.0)	3.3 (2.0 - 6.0)	3.3 (2.0 - 4.5)
Turbidity (NTU)	2.8 (0.7 - 7.9)	3.1 (0.8 - 8.6)	4.1 (2.0 - 9.2)	5.5 (1.4 - 19.4)	3.0 (0.9 - 8.3)
Suspended Solids (mg/L)	7.0 (0.6 - 25.0)	7.5 (0.7 - 26.3)	7.6 (1.8 - 26.7)	3.8 (1.2 - 6.9)	6.6 (0.9 - 17.0)
5-day Biochemical Oxygen Demand (mg/L)	1.0 (0.1 - 2.8)	1.0 (0.4 - 3.1)	0.8 (0.1 - 2.0)	0.8 (0.3 - 1.7)	1.0 (0.4 - 3.0)
Ammonia Nitrogen (mg/L)	0.032 (0.011 - 0.052)	0.037 (0.006 - 0.061)	0.027 (0.006 - 0.042)	0.037 (0.013 - 0.057)	0.050 (0.023 - 0.079)
Unionised Ammonia (mg/L)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.005)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.002)	0.002 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)	0.014 (0.002 - 0.034)	0.022 (0.002 - 0.072)	0.015 (0.003 - 0.044)	0.021 (0.003 - 0.056)	0.030 (0.002 - 0.097)
Nitrate Nitrogen (mg/L)	0.075 (0.005 - 0.160)	0.099 (0.009 - 0.250)	0.089 (0.007 - 0.173)	0.097 (0.019 - 0.260)	0.131 (0.016 - 0.297)
Total Inorganic Nitrogen (mg/L)	0.12 (0.04 - 0.22)	0.16 (0.05 - 0.31)	0.13 (0.04 - 0.21)	0.15 (0.08 - 0.30)	0.21 (0.05 - 0.36)
Total Kjeldahl Nitrogen (mg/L)	0.39 (0.13 - 0.68)	0.41 (0.12 - 0.69)	0.43 (0.18 - 0.98)	0.46 (0.21 - 0.72)	0.38 (0.07 - 0.67)
Total Nitrogen (mg/L)	0.48 (0.30 - 0.70)	0.54 (0.38 - 0.78)	0.54 (0.36 - 1.03)	0.58 (0.40 - 0.77)	0.55 (0.39 - 0.79)
Orthophosphate Phosphorus (mg/L)	0.008 (0.002 - 0.020)	0.009 (0.003 - 0.016)	0.009 (0.002 - 0.020)	0.012 (0.004 - 0.020)	0.011 (<0.002 - 0.023)
Total Phosphorus (mg/L)	0.04 (<0.02 - 0.08)	0.05 (0.02 - 0.11)	0.05 (<0.02 - 0.07)	0.03 (0.02 - 0.05)	0.05 (0.03 - 0.12)
Silica (as SiO ₂) (mg/L)	0.78 (0.33 - 1.40)	0.87 (0.30 - 1.70)	0.86 (0.33 - 1.40)	1.02 (0.40 - 2.23)	0.94 (0.21 - 1.93)
Chlorophyll-a (µg/L)	4.6 (0.9 - 12.6)	5.5 (0.7 - 15.8)	5.2 (0.9 - 14.3)	2.1 (0.8 - 5.5)	5.9 (0.6 - 20.0)
<i>E.coli</i> (cfu/100mL)	2 (<1 - 9)	2 (1 - 13)	1 (<1 - 3)	1 (<1 - 4)	2 (<1 - 37)
Faecal Coliforms (cfu/100mL)	3 (<1 - 37)	5 (1 - 49)	1 (<1 - 5)	3 (1 - 17)	7 (1 - 89)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Southern WCZ in 2016 (continued)

Parameter	West Lamma Channel				
	SM5	SM6	SM7	SM9	SM18
Number of samples	12	12	12	12	12
Temperature (°C)	23.7 (16.7 - 29.4)	23.3 (16.2 - 28.1)	23.6 (16.4 - 28.0)	23.3 (16.1 - 28.4)	23.4 (16.6 - 28.0)
Salinity	29.8 (23.5 - 32.4)	30.5 (24.5 - 32.4)	29.7 (24.3 - 32.3)	29.4 (23.8 - 31.3)	30.5 (24.9 - 32.7)
Dissolved Oxygen (mg/L)	6.9 (4.5 - 11.7)	6.8 (4.6 - 10.4)	6.8 (4.4 - 9.3)	6.0 (4.2 - 8.4)	6.9 (4.6 - 9.9)
Bottom	6.5 (3.7 - 10.2)	6.2 (2.1 - 9.6)	6.5 (4.4 - 8.4)	5.8 (3.0 - 8.8)	6.1 (2.1 - 8.1)
Dissolved Oxygen (% saturation)	97 (67 - 171)	95 (69 - 132)	94 (66 - 136)	82 (60 - 104)	96 (69 - 126)
Bottom	90 (54 - 148)	85 (30 - 120)	90 (66 - 110)	79 (43 - 110)	84 (29 - 100)
pH	8.0 (7.6 - 8.3)	7.9 (7.6 - 8.4)	7.9 (7.7 - 8.3)	7.8 (7.4 - 8.3)	7.9 (7.6 - 8.3)
Secchi Disc Depth (m)	2.9 (1.5 - 5.0)	3.1 (2.4 - 4.0)	2.9 (2.0 - 5.0)	2.9 (1.8 - 5.0)	3.3 (1.7 - 5.0)
Turbidity (NTU)	3.4 (1.2 - 6.6)	4.0 (1.0 - 11.4)	4.2 (1.1 - 11.4)	6.0 (1.5 - 29.4)	4.2 (1.1 - 10.4)
Suspended Solids (mg/L)	7.4 (1.5 - 19.3)	6.8 (0.9 - 16.3)	8.7 (0.9 - 22.7)	5.0 (2.4 - 9.7)	6.7 (1.2 - 17.7)
5-day Biochemical Oxygen Demand (mg/L)	1.0 (0.3 - 2.5)	0.9 (0.2 - 2.7)	1.0 (0.3 - 2.0)	1.0 (0.2 - 3.0)	0.9 (0.2 - 3.5)
Ammonia Nitrogen (mg/L)	0.036 (0.008 - 0.063)	0.037 (0.005 - 0.077)	0.054 (0.026 - 0.082)	0.096 (0.042 - 0.160)	0.030 (<0.005 - 0.065)
Unionised Ammonia (mg/L)	0.002 (<0.001 - 0.006)	0.001 (<0.001 - 0.005)	0.002 (<0.001 - 0.004)	0.003 (<0.001 - 0.005)	0.001 (<0.001 - 0.003)
Nitrite Nitrogen (mg/L)	0.026 (<0.002 - 0.071)	0.024 (0.002 - 0.071)	0.037 (0.009 - 0.110)	0.036 (0.008 - 0.093)	0.022 (0.003 - 0.058)
Nitrate Nitrogen (mg/L)	0.126 (0.012 - 0.293)	0.116 (0.011 - 0.280)	0.170 (0.062 - 0.340)	0.212 (0.022 - 0.503)	0.095 (0.006 - 0.223)
Total Inorganic Nitrogen (mg/L)	0.19 (0.03 - 0.37)	0.18 (0.05 - 0.38)	0.26 (0.12 - 0.44)	0.34 (0.10 - 0.60)	0.15 (0.05 - 0.27)
Total Kjeldahl Nitrogen (mg/L)	0.35 (0.10 - 0.66)	0.43 (0.13 - 0.99)	0.35 (0.10 - 0.65)	0.43 (0.13 - 0.98)	0.38 (0.09 - 0.87)
Total Nitrogen (mg/L)	0.50 (0.39 - 0.73)	0.57 (0.38 - 1.01)	0.55 (0.31 - 0.91)	0.68 (0.46 - 1.11)	0.49 (0.31 - 0.90)
Orthophosphate Phosphorus (mg/L)	0.009 (0.003 - 0.017)	0.010 (<0.002 - 0.018)	0.011 (0.004 - 0.026)	0.019 (0.003 - 0.034)	0.009 (<0.002 - 0.017)
Total Phosphorus (mg/L)	0.05 (0.02 - 0.11)	0.04 (0.02 - 0.10)	0.06 (0.03 - 0.12)	0.07 (0.03 - 0.19)	0.04 (<0.02 - 0.06)
Silica (as SiO ₂) (mg/L)	0.91 (0.44 - 1.63)	1.00 (0.18 - 1.57)	1.05 (0.19 - 2.00)	1.52 (0.72 - 3.37)	0.96 (0.34 - 1.63)
Chlorophyll-a (µg/L)	6.7 (0.8 - 32.0)	5.5 (0.8 - 18.3)	8.9 (1.1 - 29.7)	4.5 (0.4 - 18.7)	5.6 (0.8 - 17.0)
<i>E.coli</i> (cfu/100mL)	1 (<1 - 2)	1 (<1 - 1)	2 (1 - 44)	5 (1 - 23)	1 (<1 - 1)
Faecal Coliforms (cfu/100mL)	2 (<1 - 4)	1 (<1 - 5)	5 (1 - 93)	11 (2 - 62)	1 (<1 - 2)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Southern WCZ in 2016 (continued)

Parameter	Lantau Island (East)		Lantau Island (South)			Soko Islands
	SM10	SM11	SM12	SM13	SM17	SM20
Number of samples	12	12	12	12	12	12
Temperature (°C)	23.8 (15.5 - 28.7)	23.6 (15.5 - 28.6)	23.5 (15.5 - 28.2)	23.6 (15.4 - 28.2)	23.5 (15.5 - 28.3)	23.6 (15.6 - 28.0)
Salinity	29.3 (20.5 - 32.2)	29.6 (20.5 - 32.4)	29.9 (21.8 - 32.4)	29.9 (22.0 - 32.5)	30.8 (23.9 - 32.7)	29.9 (21.4 - 32.6)
Dissolved Oxygen (mg/L)	6.9 (4.6 - 9.8)	6.9 (4.7 - 10.3)	6.8 (4.8 - 9.0)	6.8 (4.7 - 9.1)	6.8 (5.2 - 8.4)	6.5 (5.0 - 8.6)
Bottom	6.9 (4.4 - 10.2)	6.9 (4.3 - 10.1)	6.9 (4.7 - 10.9)	6.8 (4.8 - 10.0)	6.2 (1.1 - 8.5)	6.5 (4.4 - 8.5)
Dissolved Oxygen (% saturation)	96 (67 - 141)	96 (68 - 149)	94 (72 - 132)	94 (72 - 134)	95 (79 - 108)	90 (71 - 113)
Bottom	96 (64 - 149)	95 (62 - 145)	95 (69 - 157)	94 (72 - 145)	86 (16 - 103)	91 (64 - 119)
pH	7.9 (7.5 - 8.3)	7.9 (7.5 - 8.3)	7.9 (7.5 - 8.2)	8.0 (7.6 - 8.2)	7.9 (7.6 - 8.1)	7.9 (7.6 - 8.2)
Secchi Disc Depth (m)	2.3 (1.5 - 3.0)	2.6 (1.8 - 4.0)	2.6 (1.8 - 4.0)	2.7 (2.0 - 4.0)	3.0 (2.4 - 4.0)	2.7 (2.0 - 4.5)
Turbidity (NTU)	4.3 (1.0 - 10.0)	4.4 (2.1 - 9.0)	5.5 (2.3 - 10.7)	5.6 (2.6 - 14.0)	3.6 (1.3 - 7.6)	5.2 (1.6 - 14.0)
Suspended Solids (mg/L)	6.2 (1.2 - 18.5)	7.5 (1.6 - 18.0)	7.9 (2.9 - 18.7)	7.9 (2.4 - 24.7)	5.5 (1.2 - 15.1)	8.4 (2.5 - 21.3)
5-day Biochemical Oxygen Demand (mg/L)	1.2 (0.4 - 3.1)	1.4 (0.5 - 4.4)	1.1 (0.3 - 3.5)	1.1 (0.2 - 2.5)	0.8 (0.2 - 1.4)	1.2 (0.2 - 4.5)
Ammonia Nitrogen (mg/L)	0.088 (0.022 - 0.175)	0.087 (0.030 - 0.153)	0.061 (0.006 - 0.118)	0.053 (0.009 - 0.108)	0.037 (0.008 - 0.090)	0.048 (0.006 - 0.104)
Unionised Ammonia (mg/L)	0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.009)	0.003 (<0.001 - 0.005)	0.002 (<0.001 - 0.006)	0.002 (<0.001 - 0.005)	0.002 (<0.001 - 0.005)
Nitrite Nitrogen (mg/L)	0.033 (0.006 - 0.074)	0.030 (0.004 - 0.065)	0.026 (<0.002 - 0.071)	0.027 (<0.002 - 0.072)	0.015 (<0.002 - 0.044)	0.023 (0.007 - 0.051)
Nitrate Nitrogen (mg/L)	0.198 (0.055 - 0.525)	0.181 (0.030 - 0.433)	0.140 (0.039 - 0.393)	0.142 (0.031 - 0.500)	0.089 (0.011 - 0.350)	0.129 (0.019 - 0.410)
Total Inorganic Nitrogen (mg/L)	0.32 (0.12 - 0.66)	0.30 (0.07 - 0.59)	0.23 (0.12 - 0.49)	0.22 (0.10 - 0.61)	0.14 (0.05 - 0.40)	0.20 (0.08 - 0.47)
Total Kjeldahl Nitrogen (mg/L)	0.50 (0.21 - 0.89)	0.48 (0.22 - 0.93)	0.49 (0.30 - 1.01)	0.48 (0.25 - 0.95)	0.50 (0.23 - 0.96)	0.53 (0.23 - 0.87)
Total Nitrogen (mg/L)	0.73 (0.42 - 1.08)	0.69 (0.43 - 1.19)	0.65 (0.37 - 1.20)	0.65 (0.36 - 1.17)	0.60 (0.29 - 1.10)	0.69 (0.41 - 0.99)
Orthophosphate Phosphorus (mg/L)	0.019 (0.002 - 0.056)	0.018 (0.003 - 0.026)	0.014 (0.006 - 0.021)	0.011 (0.005 - 0.019)	0.009 (0.002 - 0.019)	0.012 (0.006 - 0.027)
Total Phosphorus (mg/L)	0.08 (0.03 - 0.18)	0.07 (0.03 - 0.19)	0.07 (0.02 - 0.19)	0.07 (0.03 - 0.25)	0.04 (0.02 - 0.08)	0.07 (0.02 - 0.15)
Silica (as SiO ₂) (mg/L)	1.41 (0.32 - 2.75)	1.35 (0.10 - 2.77)	1.11 (0.29 - 2.27)	1.18 (0.21 - 2.77)	0.91 (0.13 - 1.93)	1.14 (0.11 - 2.97)
Chlorophyll-a (µg/L)	9.1 (0.6 - 37.0)	10.4 (0.4 - 28.5)	6.0 (0.6 - 23.5)	5.8 (1.4 - 26.3)	3.6 (0.9 - 14.7)	5.1 (1.2 - 20.0)
<i>E.coli</i> (cfu/100mL)	2 (<1 - 14)	4 (<1 - 27)	49 (3 - 440)	5 (<1 - 67)	1 (<1 - 2)	1 (<1 - 3)
Faecal Coliforms (cfu/100mL)	5 (1 - 33)	7 (<1 - 58)	87 (6 - 900)	9 (1 - 110)	2 (<1 - 7)	1 (<1 - 10)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics of the Victoria Harbour WCZ in 2016

Parameter	Victoria Harbour (East)		Victoria Harbour (Central)		
	VM1	VM2	VM4	VM5	VM6
Number of samples	12	12	12	12	12
Temperature (°C)	22.7 (15.6 - 28.2)	22.8 (15.6 - 28.2)	22.9 (15.7 - 28.2)	23.1 (15.8 - 28.3)	23.1 (15.9 - 28.2)
Salinity	31.7 (29.5 - 33.4)	31.3 (27.2 - 32.7)	31.1 (27.4 - 32.7)	30.5 (25.4 - 31.6)	30.4 (26.4 - 31.6)
Dissolved Oxygen (mg/L)	5.9 (4.0 - 8.1)	5.8 (4.2 - 8.0)	5.6 (4.0 - 7.9)	5.5 (4.1 - 7.5)	5.6 (4.4 - 7.4)
Bottom	5.7 (2.7 - 8.3)	5.8 (3.9 - 8.3)	5.5 (3.0 - 8.2)	5.5 (3.2 - 8.1)	5.5 (3.2 - 8.0)
Dissolved Oxygen (% saturation)	81 (57 - 100)	79 (60 - 97)	76 (57 - 96)	76 (59 - 95)	77 (65 - 92)
Bottom	78 (38 - 105)	80 (56 - 104)	76 (43 - 104)	75 (48 - 102)	75 (48 - 101)
pH	7.8 (7.1 - 8.2)	7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)
Secchi Disc Depth (m)	3.5 (2.4 - 5.2)	2.9 (2.0 - 5.0)	3.1 (2.0 - 5.0)	3.0 (2.0 - 5.0)	3.1 (2.3 - 5.0)
Turbidity (NTU)	3.4 (1.0 - 6.1)	3.2 (1.0 - 6.1)	3.8 (0.9 - 12.7)	3.5 (1.0 - 6.7)	3.6 (1.5 - 9.1)
Suspended Solids (mg/L)	4.4 (1.2 - 9.6)	4.5 (0.6 - 10.1)	4.8 (1.4 - 9.6)	4.5 (1.2 - 9.9)	5.0 (1.5 - 9.4)
5-day Biochemical Oxygen Demand (mg/L)	0.8 (0.1 - 1.4)	0.8 (<0.1 - 1.8)	0.7 (<0.1 - 1.6)	0.7 (0.1 - 1.4)	0.9 (0.1 - 1.7)
Ammonia Nitrogen (mg/L)	0.072 (0.016 - 0.133)	0.088 (0.018 - 0.173)	0.112 (0.020 - 0.197)	0.152 (0.049 - 0.287)	0.172 (0.066 - 0.315)
Unionised Ammonia (mg/L)	0.002 (<0.001 - 0.004)	0.002 (<0.001 - 0.006)	0.003 (<0.001 - 0.007)	0.004 (0.001 - 0.009)	0.004 (0.001 - 0.009)
Nitrite Nitrogen (mg/L)	0.022 (0.007 - 0.051)	0.023 (0.007 - 0.052)	0.025 (0.007 - 0.053)	0.028 (0.009 - 0.054)	0.030 (0.011 - 0.061)
Nitrate Nitrogen (mg/L)	0.106 (0.036 - 0.273)	0.118 (0.042 - 0.313)	0.137 (0.061 - 0.347)	0.164 (0.088 - 0.387)	0.167 (0.089 - 0.353)
Total Inorganic Nitrogen (mg/L)	0.20 (0.12 - 0.38)	0.23 (0.13 - 0.45)	0.27 (0.14 - 0.49)	0.34 (0.18 - 0.55)	0.37 (0.20 - 0.52)
Total Kjeldahl Nitrogen (mg/L)	0.41 (0.18 - 0.82)	0.39 (0.17 - 0.65)	0.44 (0.17 - 0.78)	0.44 (0.20 - 0.83)	0.42 (0.25 - 0.66)
Total Nitrogen (mg/L)	0.54 (0.35 - 0.91)	0.53 (0.35 - 0.74)	0.60 (0.30 - 0.88)	0.64 (0.41 - 0.97)	0.62 (0.42 - 0.86)
Orthophosphate Phosphorus (mg/L)	0.015 (0.003 - 0.028)	0.016 (0.005 - 0.031)	0.020 (0.005 - 0.045)	0.024 (0.006 - 0.045)	0.024 (0.005 - 0.040)
Total Phosphorus (mg/L)	0.04 (0.02 - 0.07)	0.04 (0.03 - 0.09)	0.05 (0.03 - 0.09)	0.06 (0.03 - 0.09)	0.06 (0.03 - 0.10)
Silica (as SiO ₂) (mg/L)	0.97 (0.37 - 1.53)	1.02 (0.36 - 1.67)	1.12 (0.37 - 2.07)	1.23 (0.48 - 2.10)	1.29 (0.55 - 2.10)
Chlorophyll-a (µg/L)	2.1 (0.5 - 5.4)	2.6 (0.5 - 8.9)	2.2 (0.5 - 6.1)	2.4 (0.4 - 8.6)	2.4 (0.7 - 6.9)
<i>E.coli</i> (cfu/100mL)	200 (95 - 720)	220 (100 - 540)	440 (110 - 3700)	460 (110 - 1700)	410 (52 - 2100)
Faecal Coliforms (cfu/100mL)	410 (150 - 1900)	510 (140 - 1900)	1200 (230 - 12000)	1100 (160 - 4000)	1100 (220 - 5700)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics of the Victoria Harbour WCZ in 2016 (continued)

Parameter	Victoria Harbour (West)		Stonecutters Island	Rambler Channel	
	VM7	VM8	VM15	VM12	VM14
Number of samples	12	12	12	12	12
Temperature (°C)	23.2 (15.9 - 28.2)	23.2 (15.8 - 28.2)	23.1 (16.2 - 28.4)	23.2 (16.2 - 28.3)	23.4 (16.1 - 28.4)
Salinity	30.2 (25.4 - 31.5)	30.0 (24.0 - 31.7)	30.5 (27.1 - 31.6)	29.9 (26.6 - 31.2)	28.6 (23.2 - 31.1)
Dissolved Oxygen (mg/L)	5.6 (4.2 - 7.9)	5.9 (4.2 - 8.6)	5.4 (4.0 - 7.3)	5.4 (4.0 - 7.5)	5.4 (4.0 - 8.0)
Bottom	5.5 (3.5 - 8.0)	5.9 (3.6 - 8.7)	5.3 (3.5 - 7.6)	5.3 (3.3 - 7.9)	5.3 (3.1 - 8.3)
Dissolved Oxygen (% saturation)	77 (60 - 100)	81 (59 - 108)	74 (57 - 91)	74 (54 - 94)	74 (57 - 99)
Bottom	76 (52 - 101)	81 (54 - 110)	73 (50 - 95)	72 (46 - 100)	73 (44 - 104)
pH	7.8 (7.4 - 8.2)	7.8 (7.5 - 8.2)	7.8 (7.4 - 8.2)	7.8 (7.2 - 8.2)	7.8 (7.3 - 8.3)
Secchi Disc Depth (m)	2.9 (2.0 - 4.0)	2.7 (2.0 - 4.0)	3.0 (2.0 - 5.0)	2.5 (1.8 - 3.1)	2.8 (1.9 - 4.0)
Turbidity (NTU)	5.1 (1.1 - 13.3)	4.2 (1.1 - 8.9)	16.7 (1.6 - 153.1)	18.1 (2.1 - 164.0)	4.1 (1.4 - 7.8)
Suspended Solids (mg/L)	4.7 (1.0 - 9.9)	5.2 (2.4 - 8.7)	5.9 (1.2 - 11.5)	5.8 (2.3 - 12.2)	5.2 (1.4 - 9.9)
5-day Biochemical Oxygen Demand (mg/L)	0.9 (0.1 - 2.2)	0.9 (<0.1 - 2.4)	0.8 (0.3 - 1.9)	0.8 (0.3 - 1.7)	0.7 (0.3 - 1.3)
Ammonia Nitrogen (mg/L)	0.184 (0.072 - 0.333)	0.136 (0.077 - 0.270)	0.177 (0.080 - 0.263)	0.176 (0.123 - 0.290)	0.136 (0.064 - 0.220)
Unionised Ammonia (mg/L)	0.005 (0.002 - 0.011)	0.004 (0.001 - 0.007)	0.005 (0.001 - 0.008)	0.005 (0.001 - 0.007)	0.004 (0.001 - 0.006)
Nitrite Nitrogen (mg/L)	0.029 (0.010 - 0.051)	0.030 (0.009 - 0.059)	0.031 (0.011 - 0.056)	0.036 (0.011 - 0.079)	0.045 (0.010 - 0.127)
Nitrate Nitrogen (mg/L)	0.177 (0.088 - 0.420)	0.174 (0.063 - 0.410)	0.173 (0.107 - 0.340)	0.202 (0.120 - 0.360)	0.255 (0.123 - 0.453)
Total Inorganic Nitrogen (mg/L)	0.39 (0.25 - 0.58)	0.34 (0.22 - 0.54)	0.38 (0.26 - 0.51)	0.41 (0.30 - 0.54)	0.44 (0.27 - 0.58)
Total Kjeldahl Nitrogen (mg/L)	0.45 (0.20 - 0.67)	0.41 (0.16 - 0.55)	0.51 (0.35 - 0.71)	0.44 (0.30 - 0.62)	0.33 (0.13 - 0.49)
Total Nitrogen (mg/L)	0.66 (0.47 - 0.86)	0.61 (0.43 - 0.75)	0.71 (0.52 - 0.89)	0.68 (0.54 - 0.88)	0.63 (0.48 - 0.87)
Orthophosphate Phosphorus (mg/L)	0.024 (0.005 - 0.042)	0.020 (0.004 - 0.041)	0.025 (0.005 - 0.041)	0.025 (0.005 - 0.038)	0.024 (0.004 - 0.034)
Total Phosphorus (mg/L)	0.07 (0.04 - 0.14)	0.06 (0.03 - 0.10)	0.06 (0.03 - 0.09)	0.06 (0.03 - 0.10)	0.08 (0.03 - 0.23)
Silica (as SiO ₂) (mg/L)	1.28 (0.53 - 2.10)	1.31 (0.64 - 2.10)	1.26 (0.69 - 2.43)	1.44 (0.79 - 2.67)	1.70 (0.85 - 3.23)
Chlorophyll-a (µg/L)	2.7 (0.5 - 9.1)	3.0 (0.7 - 9.6)	1.8 (0.4 - 5.3)	1.6 (0.3 - 5.3)	1.6 (0.4 - 5.8)
<i>E.coli</i> (cfu/100mL)	390 (96 - 2800)	130 (27 - 710)	490 (52 - 2200)	320 (58 - 1300)	370 (25 - 4700)
Faecal Coliforms (cfu/100mL)	980 (140 - 11000)	330 (66 - 1100)	1200 (130 - 7700)	730 (110 - 3600)	960 (48 - 15000)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Eastern Buffer WCZ in 2016

Parameter	Chai Wan	Tathong Channel	
	EM1	EM2	EM3
Number of samples	12	12	12
Temperature (°C)	22.7 (15.4 - 27.6)	22.7 (15.3 - 27.7)	22.7 (15.4 - 27.7)
Salinity	31.6 (29.5 - 32.9)	31.8 (29.1 - 33.3)	31.9 (29.5 - 33.2)
Dissolved Oxygen (mg/L)	6.2 (4.1 - 8.9)	6.4 (4.3 - 9.1)	6.5 (4.1 - 9.2)
Bottom	6.0 (3.5 - 8.6)	5.9 (3.4 - 8.4)	6.1 (3.7 - 8.3)
Dissolved Oxygen (% saturation)	86 (60 - 110)	88 (63 - 113)	90 (60 - 114)
Bottom	82 (50 - 107)	81 (48 - 104)	84 (53 - 103)
pH	7.9 (7.6 - 8.2)	7.9 (7.6 - 8.2)	7.9 (7.6 - 8.2)
Secchi Disc Depth (m)	3.3 (2.0 - 5.0)	3.6 (2.4 - 6.0)	3.5 (2.2 - 6.0)
Turbidity (NTU)	2.5 (0.5 - 4.6)	2.6 (0.4 - 5.2)	2.2 (0.5 - 3.9)
Suspended Solids (mg/L)	6.7 (0.7 - 21.3)	6.8 (0.6 - 25.3)	5.6 (0.6 - 13.4)
5-day Biochemical Oxygen Demand (mg/L)	0.8 (<0.1 - 1.6)	0.8 (<0.1 - 1.3)	0.9 (<0.1 - 2.9)
Ammonia Nitrogen (mg/L)	0.049 (0.017 - 0.084)	0.042 (0.016 - 0.071)	0.034 (0.009 - 0.064)
Unionised Ammonia (mg/L)	0.001 (<0.001 - 0.003)	0.001 (<0.001 - 0.003)	<0.001 (<0.001 - 0.002)
Nitrite Nitrogen (mg/L)	0.023 (<0.002 - 0.089)	0.020 (0.003 - 0.089)	0.016 (0.003 - 0.048)
Nitrate Nitrogen (mg/L)	0.090 (0.022 - 0.200)	0.078 (0.016 - 0.183)	0.065 (0.008 - 0.143)
Total Inorganic Nitrogen (mg/L)	0.16 (0.06 - 0.24)	0.14 (0.05 - 0.22)	0.11 (0.03 - 0.19)
Total Kjeldahl Nitrogen (mg/L)	0.44 (0.18 - 0.93)	0.47 (0.12 - 1.09)	0.46 (0.14 - 0.98)
Total Nitrogen (mg/L)	0.55 (0.38 - 1.07)	0.56 (0.31 - 1.21)	0.55 (0.29 - 1.10)
Orthophosphate Phosphorus (mg/L)	0.011 (<0.002 - 0.021)	0.009 (<0.002 - 0.018)	0.008 (0.002 - 0.018)
Total Phosphorus (mg/L)	0.04 (0.02 - 0.09)	0.03 (0.02 - 0.06)	0.04 (<0.02 - 0.12)
Silica (as SiO ₂) (mg/L)	0.89 (0.36 - 1.57)	0.83 (0.36 - 1.53)	0.77 (0.34 - 1.16)
Chlorophyll-a (µg/L)	2.7 (0.6 - 6.2)	2.5 (0.3 - 6.4)	3.2 (0.5 - 10.8)
<i>E.coli</i> (cfu/100mL)	65 (6 - 270)	17 (2 - 150)	4 (1 - 140)
Faecal Coliforms (cfu/100mL)	140 (11 - 650)	42 (3 - 380)	9 (2 - 240)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Western Buffer WCZ in 2016

Parameter	Hong Kong Island (West)		Tsing Yi (South)	Tsing Yi (West)
	WM1	WM2	WM3	WM4
Number of samples	12	12	12	12
Temperature (°C)	22.8 (15.8 - 28.3)	23.1 (16.0 - 28.3)	23.0 (16.0 - 28.4)	22.9 (16.0 - 28.4)
Salinity	31.5 (29.9 - 32.9)	30.1 (26.5 - 31.6)	30.5 (28.2 - 31.6)	30.2 (27.9 - 31.5)
Dissolved Oxygen (mg/L)	6.1 (4.2 - 8.6)	5.9 (4.3 - 8.5)	5.7 (4.0 - 8.3)	5.5 (4.0 - 7.6)
Bottom	5.8 (2.3 - 8.6)	5.9 (3.4 - 8.6)	5.5 (2.3 - 8.6)	5.3 (2.6 - 7.7)
Dissolved Oxygen (% saturation)	84 (60 - 108)	81 (62 - 107)	78 (57 - 104)	75 (51 - 98)
Bottom	79 (32 - 108)	81 (49 - 109)	76 (32 - 108)	72 (31 - 99)
pH	7.9 (7.4 - 8.3)	7.8 (7.3 - 8.3)	7.8 (7.3 - 8.3)	7.8 (7.3 - 8.3)
Secchi Disc Depth (m)	3.3 (2.0 - 6.0)	3.0 (2.0 - 6.0)	2.8 (1.9 - 5.0)	3.0 (2.0 - 5.0)
Turbidity (NTU)	4.7 (1.8 - 11.1)	15.5 (1.2 - 130.1)	7.9 (2.2 - 36.7)	5.5 (2.0 - 12.2)
Suspended Solids (mg/L)	5.5 (1.7 - 13.1)	4.3 (1.2 - 6.9)	6.1 (1.8 - 10.0)	6.9 (1.7 - 11.5)
5-day Biochemical Oxygen Demand (mg/L)	0.7 (<0.1 - 1.6)	0.8 (<0.1 - 1.7)	0.8 (<0.1 - 1.7)	0.7 (0.2 - 1.4)
Ammonia Nitrogen (mg/L)	0.051 (0.009 - 0.099)	0.097 (0.048 - 0.167)	0.117 (0.073 - 0.197)	0.111 (0.057 - 0.160)
Unionised Ammonia (mg/L)	0.002 (<0.001 - 0.005)	0.003 (<0.001 - 0.005)	0.003 (<0.001 - 0.006)	0.003 (<0.001 - 0.006)
Nitrite Nitrogen (mg/L)	0.024 (0.004 - 0.062)	0.035 (0.004 - 0.087)	0.034 (0.006 - 0.081)	0.036 (0.007 - 0.073)
Nitrate Nitrogen (mg/L)	0.110 (0.028 - 0.230)	0.180 (0.043 - 0.420)	0.168 (0.075 - 0.330)	0.178 (0.097 - 0.297)
Total Inorganic Nitrogen (mg/L)	0.19 (0.08 - 0.27)	0.31 (0.12 - 0.52)	0.32 (0.18 - 0.42)	0.33 (0.23 - 0.41)
Total Kjeldahl Nitrogen (mg/L)	0.37 (0.14 - 0.65)	0.38 (0.10 - 0.64)	0.46 (0.27 - 0.69)	0.39 (0.25 - 0.71)
Total Nitrogen (mg/L)	0.50 (0.30 - 0.88)	0.60 (0.42 - 0.76)	0.66 (0.39 - 0.96)	0.60 (0.49 - 1.01)
Orthophosphate Phosphorus (mg/L)	0.012 (0.004 - 0.025)	0.018 (0.003 - 0.030)	0.017 (0.004 - 0.030)	0.019 (0.004 - 0.033)
Total Phosphorus (mg/L)	0.04 (0.02 - 0.08)	0.05 (0.03 - 0.11)	0.06 (0.03 - 0.15)	0.04 (0.03 - 0.07)
Silica (as SiO ₂) (mg/L)	1.12 (0.37 - 2.03)	1.40 (0.55 - 2.47)	1.37 (0.73 - 2.33)	1.42 (0.72 - 2.23)
Chlorophyll-a (µg/L)	2.0 (0.9 - 6.0)	2.4 (0.7 - 7.6)	2.1 (0.4 - 8.2)	1.9 (0.6 - 6.4)
<i>E.coli</i> (cfu/100mL)	4 (1 - 42)	13 (1 - 220)	49 (4 - 540)	38 (6 - 340)
Faecal Coliforms (cfu/100mL)	10 (1 - 54)	32 (1 - 400)	110 (5 - 760)	83 (9 - 1100)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

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3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Junk Bay WCZ in 2016

Parameter	Junk Bay	
	JM3	JM4
Number of samples	12	12
Temperature (°C)	22.9 (15.4 - 28.2)	22.7 (15.4 - 27.6)
Salinity	31.6 (29.7 - 33.0)	31.7 (29.7 - 33.3)
Dissolved Oxygen (mg/L)	6.5 (4.4 - 9.1)	6.2 (4.0 - 9.2)
Bottom	6.4 (3.7 - 8.9)	6.0 (3.6 - 8.9)
Dissolved Oxygen (% saturation)	89 (64 - 113)	85 (58 - 114)
Bottom	89 (54 - 111)	83 (51 - 110)
pH	7.9 (7.6 - 8.3)	7.9 (7.6 - 8.2)
Secchi Disc Depth (m)	2.9 (2.0 - 6.0)	3.1 (2.3 - 5.0)
Turbidity (NTU)	2.2 (0.5 - 4.6)	2.6 (0.5 - 5.3)
Suspended Solids (mg/L)	4.5 (0.6 - 16.7)	4.3 (0.8 - 13.7)
5-day Biochemical Oxygen Demand (mg/L)	1.1 (0.2 - 2.2)	0.8 (0.2 - 1.4)
Ammonia Nitrogen (mg/L)	0.052 (0.008 - 0.105)	0.051 (0.010 - 0.121)
Unionised Ammonia (mg/L)	0.001 (<0.001 - 0.003)	0.002 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)	0.020 (0.002 - 0.060)	0.023 (0.002 - 0.085)
Nitrate Nitrogen (mg/L)	0.082 (0.015 - 0.163)	0.091 (0.022 - 0.187)
Total Inorganic Nitrogen (mg/L)	0.15 (0.05 - 0.25)	0.16 (0.06 - 0.24)
Total Kjeldahl Nitrogen (mg/L)	0.45 (0.19 - 0.71)	0.44 (0.15 - 1.01)
Total Nitrogen (mg/L)	0.55 (0.22 - 0.89)	0.56 (0.19 - 1.19)
Orthophosphate Phosphorus (mg/L)	0.010 (0.003 - 0.019)	0.012 (0.005 - 0.019)
Total Phosphorus (mg/L)	0.04 (<0.02 - 0.10)	0.04 (<0.02 - 0.06)
Silica (as SiO ₂) (mg/L)	0.71 (0.22 - 1.13)	0.86 (0.36 - 1.43)
Chlorophyll-a (µg/L)	4.6 (0.7 - 9.8)	2.8 (0.3 - 7.3)
<i>E.coli</i> (cfu/100mL)	22 (3 - 140)	55 (7 - 200)
Faecal Coliforms (cfu/100mL)	56 (12 - 300)	130 (15 - 480)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for the Deep Bay WCZ in 2016

Parameter	Inner Deep Bay			Outer Deep Bay	
	DM1	DM2	DM3	DM4	DM5
Number of samples	12	12	12	12	12
Temperature (°C)	24.3 (15.3 - 31.4)	24.4 (14.8 - 31.5)	24.3 (13.8 - 31.1)	24.3 (14.9 - 30.1)	24.1 (15.6 - 29.4)
Salinity	10.8 (0.2 - 20.6)	12.1 (0.3 - 21.6)	17.7 (8.0 - 25.5)	19.9 (9.0 - 29.0)	23.0 (11.1 - 30.3)
Dissolved Oxygen (mg/L)	4.5 (1.9 - 7.0)	4.9 (2.8 - 7.4)	5.8 (4.4 - 6.9)	5.7 (4.9 - 7.2)	5.8 (4.6 - 7.3)
Bottom	NA	NA	NA	5.6 (3.8 - 7.2)	5.8 (4.4 - 7.3)
Dissolved Oxygen (% saturation)	58 (24 - 94)	63 (34 - 100)	76 (53 - 96)	76 (65 - 86)	79 (65 - 93)
Bottom	NA	NA	NA	75 (55 - 86)	79 (63 - 93)
pH	7.1 (6.6 - 7.8)	7.2 (6.7 - 8.0)	7.5 (7.0 - 8.1)	7.6 (7.1 - 8.2)	7.7 (7.3 - 8.2)
Secchi Disc Depth (m)	1.3 (1.0 - 1.7)	1.2 (0.9 - 1.5)	1.4 (1.0 - 2.0)	1.8 (1.5 - 2.5)	1.9 (1.5 - 2.5)
Turbidity (NTU)	36.2 (11.9 - 108.2)	33.5 (9.1 - 84.7)	8.7 (2.6 - 21.4)	7.2 (1.8 - 21.3)	4.3 (1.1 - 7.5)
Suspended Solids (mg/L)	49.2 (15.0 - 150.0)	47.6 (12.0 - 110.0)	12.2 (3.0 - 39.0)	9.2 (3.4 - 34.0)	5.7 (2.2 - 10.4)
5-day Biochemical Oxygen Demand (mg/L)	4.3 (1.3 - 11.0)	3.5 (1.3 - 11.0)	1.2 (0.6 - 2.6)	0.8 (0.4 - 1.4)	0.9 (0.4 - 1.6)
Ammonia Nitrogen (mg/L)	1.961 (0.730 - 3.600)	1.324 (0.510 - 2.600)	0.431 (0.110 - 0.830)	0.241 (0.105 - 0.565)	0.162 (0.074 - 0.433)
Unionised Ammonia (mg/L)	0.013 (0.006 - 0.030)	0.012 (0.001 - 0.026)	0.006 (0.002 - 0.013)	0.004 (0.002 - 0.014)	0.004 (0.001 - 0.014)
Nitrite Nitrogen (mg/L)	0.388 (0.100 - 0.760)	0.300 (<0.002 - 1.000)	0.168 (0.070 - 0.300)	0.117 (0.039 - 0.200)	0.085 (0.017 - 0.133)
Nitrate Nitrogen (mg/L)	0.750 (0.045 - 1.400)	0.711 (0.011 - 1.300)	0.831 (0.590 - 1.200)	0.757 (0.330 - 1.400)	0.645 (0.207 - 1.300)
Total Inorganic Nitrogen (mg/L)	3.10 (1.55 - 4.48)	2.34 (1.40 - 3.02)	1.43 (1.18 - 1.95)	1.11 (0.60 - 1.61)	0.89 (0.36 - 1.46)
Total Kjeldahl Nitrogen (mg/L)	3.57 (1.40 - 6.20)	2.51 (1.10 - 7.10)	0.75 (0.35 - 1.50)	0.44 (0.19 - 0.95)	0.38 (0.19 - 0.86)
Total Nitrogen (mg/L)	4.70 (2.22 - 6.97)	3.52 (1.99 - 7.11)	1.75 (1.43 - 2.71)	1.32 (0.68 - 1.89)	1.11 (0.61 - 1.67)
Orthophosphate Phosphorus (mg/L)	0.244 (0.180 - 0.410)	0.212 (0.110 - 0.330)	0.115 (0.060 - 0.240)	0.062 (0.035 - 0.125)	0.040 (0.008 - 0.062)
Total Phosphorus (mg/L)	0.40 (0.26 - 0.79)	0.36 (0.20 - 0.86)	0.18 (0.09 - 0.31)	0.12 (0.06 - 0.22)	0.09 (0.03 - 0.15)
Silica (as SiO ₂) (mg/L)	10.36 (3.00 - 21.00)	8.81 (1.90 - 15.00)	5.55 (1.30 - 9.50)	4.68 (2.35 - 8.60)	4.09 (1.33 - 8.23)
Chlorophyll-a (µg/L)	6.2 (1.1 - 20.0)	6.7 (1.3 - 21.0)	3.0 (1.0 - 7.6)	2.0 (0.3 - 4.5)	2.0 (0.4 - 6.4)
<i>E.coli</i> (cfu/100mL)	14000 (530 - 550000)	4500 (600 - 690000)	140 (5 - 5100)	130 (7 - 880)	290 (80 - 1200)
Faecal Coliforms (cfu/100mL)	23000 (1100 - 1100000)	9100 (1300 - 1300000)	300 (21 - 9300)	270 (15 - 3300)	570 (160 - 2300)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

4. NA (Not Applicable) indicates the measurement was not made due to shallow water

Summary of water quality statistics for the North Western WCZ in 2016

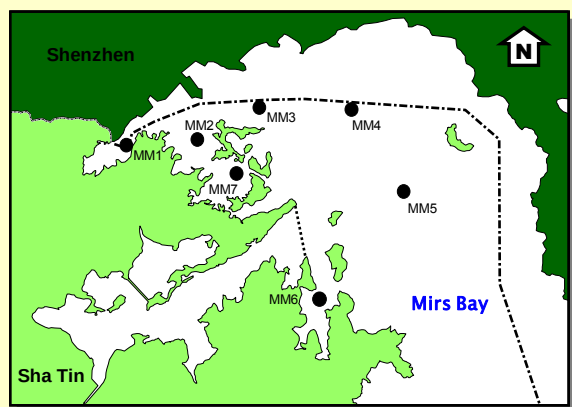
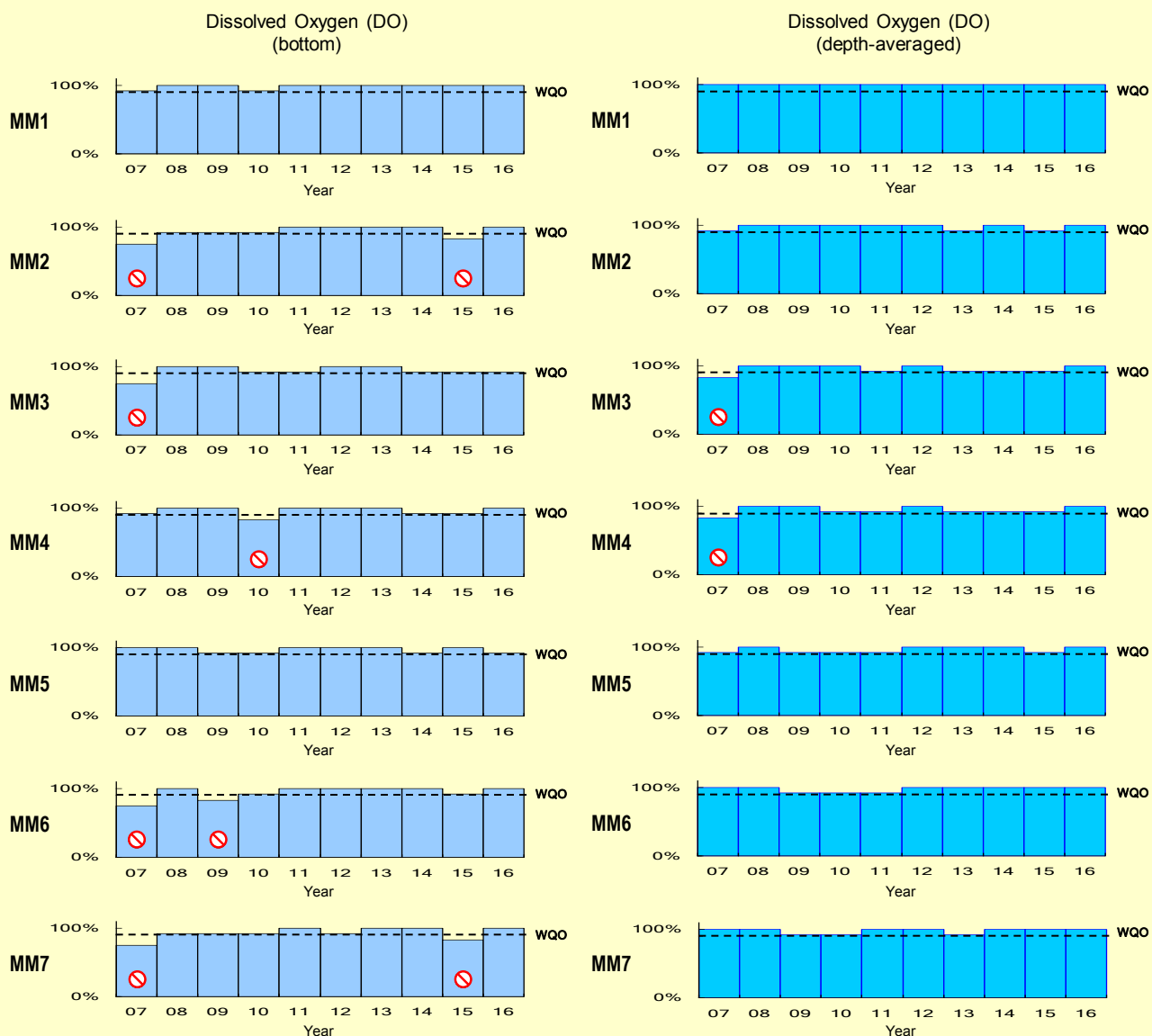
	Lantau Island (North)	Pearl Island	Pillar Point	Urmston Road	Chek Lap Kok (North) (West)	
Parameter	NM1	NM2	NM3	NM5	NM6	NM8
Number of samples	12	12	12	12	12	12
Temperature (°C)	23.3 (16.2 - 28.5)	23.6 (15.6 - 28.7)	23.7 (15.9 - 28.6)	23.7 (15.8 - 28.8)	24.0 (15.5 - 28.8)	24.0 (15.8 - 28.7)
Salinity	28.6 (22.4 - 30.7)	26.0 (15.9 - 30.2)	26.2 (16.0 - 30.7)	24.9 (17.2 - 29.7)	22.5 (13.5 - 29.7)	24.2 (9.4 - 30.8)
Dissolved Oxygen (mg/L)	5.6 (4.0 - 8.0)	5.8 (4.0 - 8.2)	5.6 (4.2 - 8.1)	5.7 (4.1 - 7.9)	6.0 (4.8 - 8.5)	6.1 (4.6 - 7.8)
Bottom	5.2 (2.2 - 8.1)	5.5 (3.7 - 8.3)	5.4 (3.0 - 8.1)	5.3 (2.6 - 8.0)	5.8 (4.1 - 8.2)	5.8 (4.1 - 7.5)
Dissolved Oxygen (% saturation)	76 (57 - 98)	78 (60 - 100)	76 (62 - 99)	76 (56 - 96)	81 (64 - 103)	83 (65 - 99)
Bottom	71 (31 - 99)	75 (53 - 102)	73 (42 - 99)	72 (37 - 98)	77 (57 - 99)	79 (56 - 99)
pH	7.8 (7.4 - 8.3)	7.9 (7.4 - 8.3)	7.8 (7.4 - 8.3)	7.8 (7.4 - 8.1)	7.8 (7.4 - 8.3)	7.9 (7.4 - 8.3)
Secchi Disc Depth (m)	2.9 (2.5 - 4.5)	2.6 (2.0 - 4.0)	2.5 (2.0 - 4.0)	2.4 (1.6 - 4.0)	2.2 (1.5 - 3.0)	2.3 (1.5 - 3.0)
Turbidity (NTU)	7.0 (0.8 - 20.6)	66.2 (1.1 - 744.7)	15.4 (1.8 - 116.8)	9.2 (2.2 - 32.7)	14.7 (4.0 - 108.0)	48.5 (2.2 - 430.7)
Suspended Solids (mg/L)	7.5 (1.6 - 24.3)	6.7 (1.5 - 18.7)	9.3 (1.9 - 25.0)	10.2 (2.4 - 20.3)	8.6 (2.5 - 23.1)	13.9 (1.9 - 31.5)
5-day Biochemical Oxygen Demand (mg/L)	0.8 (0.3 - 1.5)	0.7 (0.3 - 1.5)	0.7 (0.3 - 1.1)	0.7 (0.4 - 1.6)	1.1 (0.3 - 5.0)	0.7 (0.1 - 1.7)
Ammonia Nitrogen (mg/L)	0.103 (0.033 - 0.197)	0.114 (0.053 - 0.273)	0.122 (0.054 - 0.283)	0.138 (0.054 - 0.303)	0.112 (0.030 - 0.347)	0.071 (0.024 - 0.227)
Unionised Ammonia (mg/L)	0.003 (0.001 - 0.008)	0.003 (0.001 - 0.010)	0.004 (0.002 - 0.011)	0.003 (0.002 - 0.006)	0.003 (0.001 - 0.007)	0.002 (<0.001 - 0.004)
Nitrite Nitrogen (mg/L)	0.045 (0.016 - 0.086)	0.058 (0.019 - 0.096)	0.056 (0.019 - 0.107)	0.072 (0.024 - 0.115)	0.077 (0.027 - 0.143)	0.056 (0.019 - 0.094)
Nitrate Nitrogen (mg/L)	0.296 (0.167 - 0.547)	0.434 (0.183 - 0.893)	0.397 (0.190 - 0.643)	0.516 (0.217 - 0.837)	0.591 (0.280 - 1.133)	0.486 (0.103 - 1.367)
Total Inorganic Nitrogen (mg/L)	0.44 (0.31 - 0.63)	0.61 (0.34 - 1.04)	0.58 (0.34 - 0.80)	0.73 (0.39 - 1.15)	0.78 (0.41 - 1.31)	0.61 (0.20 - 1.54)
Total Kjeldahl Nitrogen (mg/L)	0.50 (0.18 - 0.75)	0.36 (0.14 - 0.54)	0.49 (0.20 - 0.80)	0.44 (0.25 - 0.69)	0.38 (0.13 - 0.72)	0.40 (0.20 - 0.69)
Total Nitrogen (mg/L)	0.84 (0.56 - 1.10)	0.85 (0.42 - 1.22)	0.95 (0.54 - 1.35)	1.03 (0.57 - 1.57)	1.05 (0.60 - 1.47)	0.94 (0.45 - 1.72)
Orthophosphate Phosphorus (mg/L)	0.025 (0.004 - 0.053)	0.029 (0.004 - 0.054)	0.030 (0.008 - 0.056)	0.033 (0.005 - 0.060)	0.031 (0.002 - 0.062)	0.021 (0.003 - 0.048)
Total Phosphorus (mg/L)	0.07 (0.04 - 0.12)	0.10 (0.04 - 0.14)	0.10 (0.05 - 0.16)	0.11 (0.06 - 0.17)	0.11 (0.05 - 0.16)	0.09 (0.03 - 0.13)
Silica (as SiO ₂) (mg/L)	2.06 (0.93 - 3.63)	2.72 (1.10 - 5.73)	2.62 (1.20 - 4.60)	3.22 (1.67 - 5.47)	3.63 (1.47 - 7.00)	3.02 (0.77 - 8.20)
Chlorophyll-a (µg/L)	2.7 (0.7 - 12.4)	3.0 (0.5 - 13.3)	2.1 (0.7 - 7.1)	2.2 (0.5 - 6.7)	5.2 (0.7 - 24.0)	4.1 (1.0 - 18.0)
<i>E.coli</i> (cfu/100mL)	27 (3 - 78)	35 (5 - 70)	54 (7 - 420)	210 (46 - 2000)	26 (2 - 120)	5 (<1 - 79)
Faecal Coliforms (cfu/100mL)	66 (5 - 160)	79 (7 - 190)	130 (9 - 1500)	480 (100 - 4900)	63 (5 - 310)	12 (2 - 270)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Levels of compliance with key Water Quality Objectives in the Mirs Bay WCZ

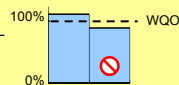


Dissolved Oxygen (DO)

1. Bottom

WQO: 90% sample with bottom DO ≥ 2 mg/L

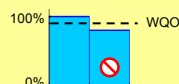
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

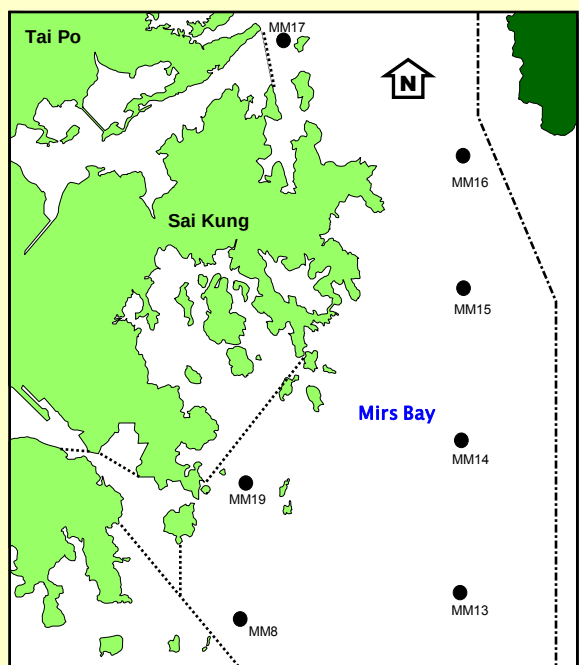
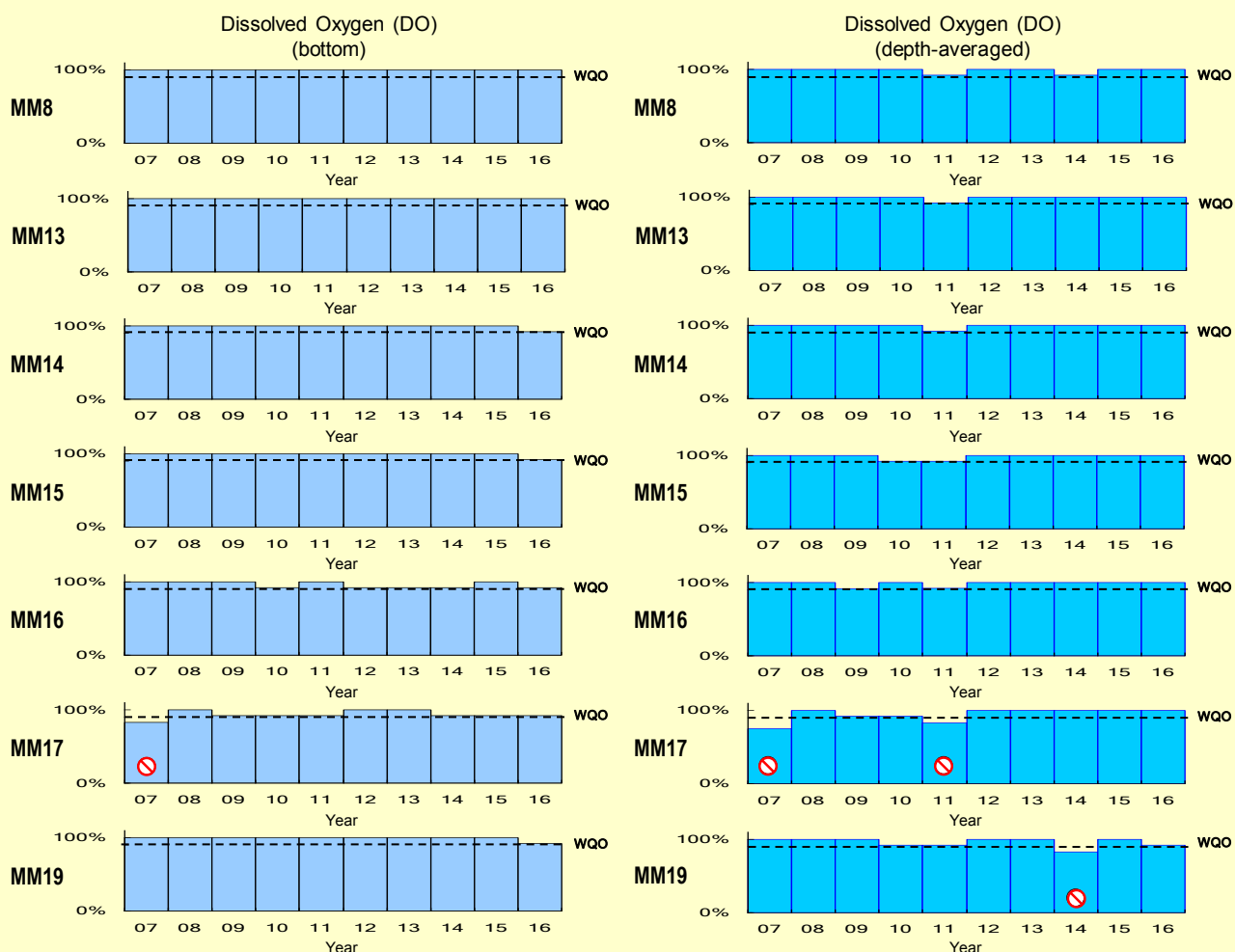
WQO: 90% sample with depth-averaged DO ≥ 4 mg/L

% sample with depth-averaged DO ≥ 4 mg/L



⊘ Non-compliance

Levels of compliance with key Water Quality Objectives in the Mirs Bay WCZ (continued)

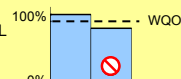


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

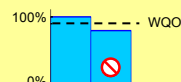
Blue bar : % sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

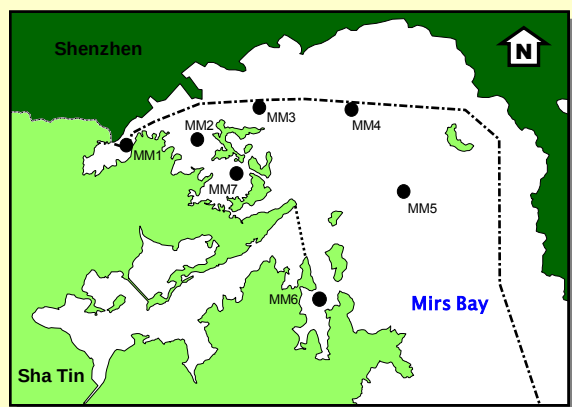
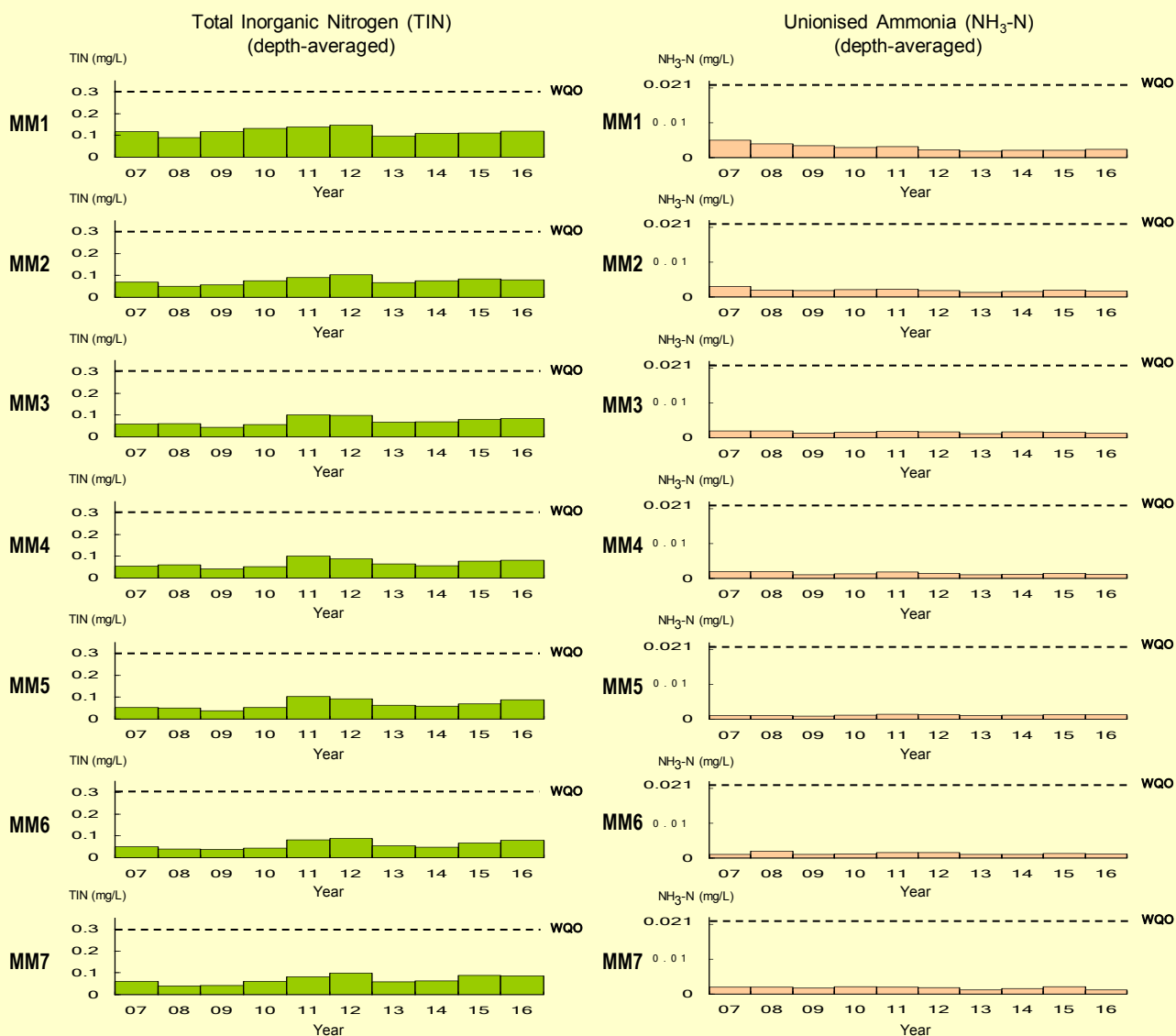
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

Blue bar : % sample with depth-averaged DO ≥ 4 mg/L



Red circle with slash : Non-compliance

Levels of compliance with key Water Quality Objectives in the Mirs Bay WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.3 mg/L

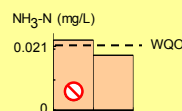
■ annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

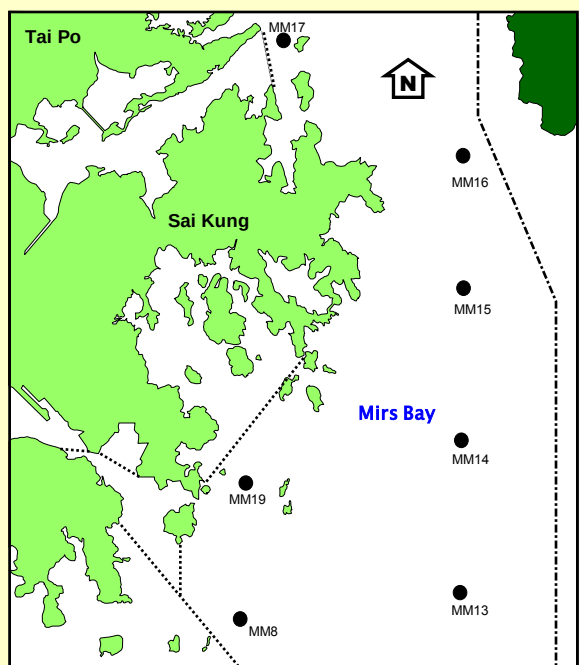
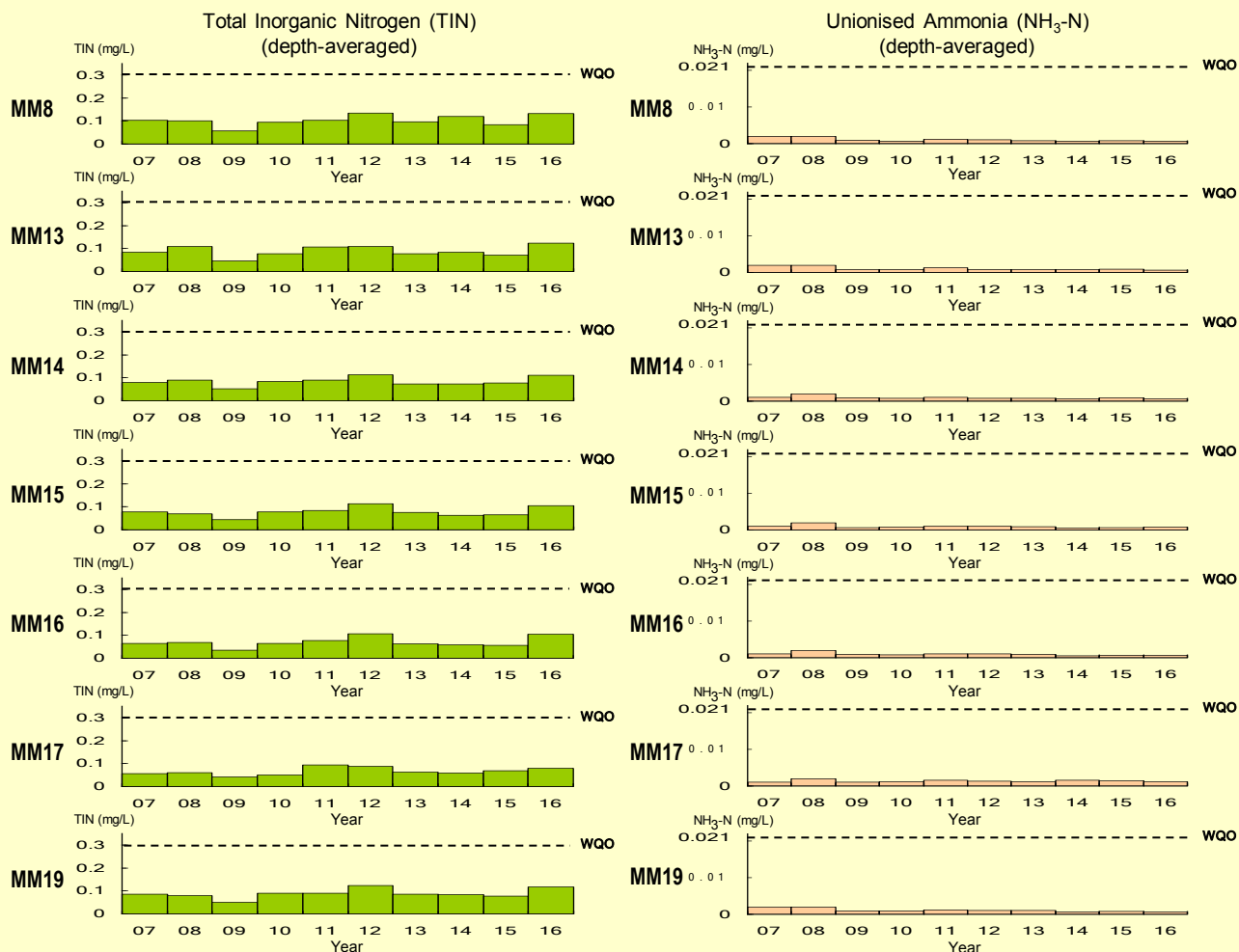
WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

■ annual mean for depth-averaged NH₃-N



⊘ Non-compliance

Levels of compliance with key Water Quality Objectives in the Mirs Bay WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.3 mg/L

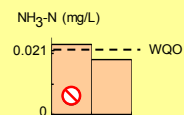
■ annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

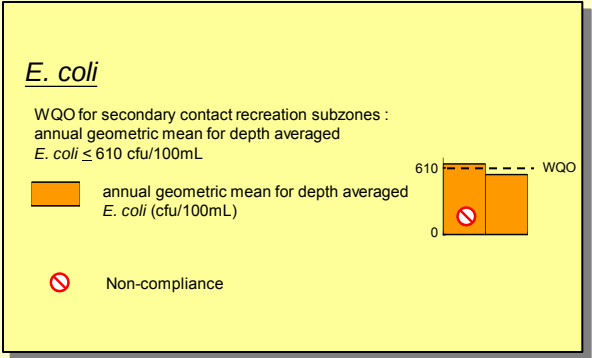
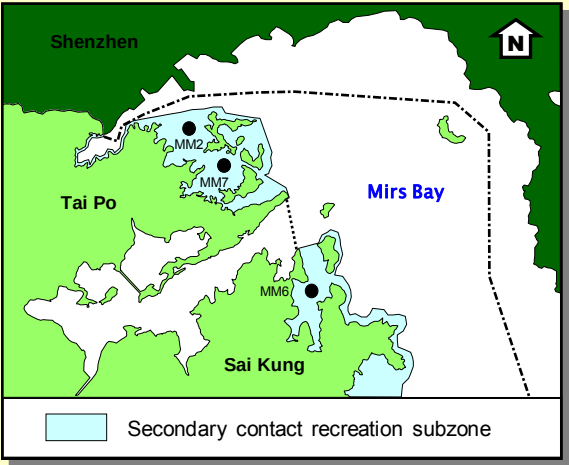
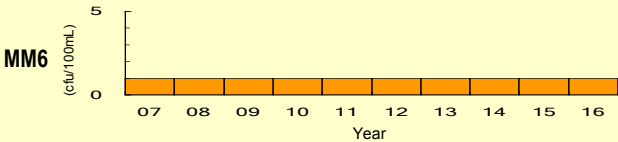
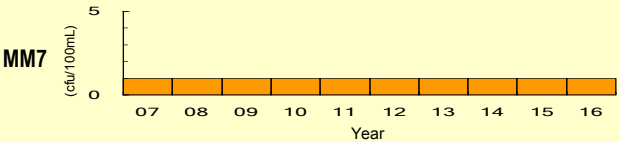
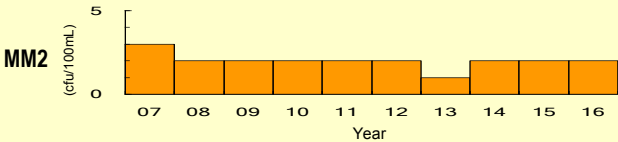
■ annual mean for depth-averaged NH₃-N



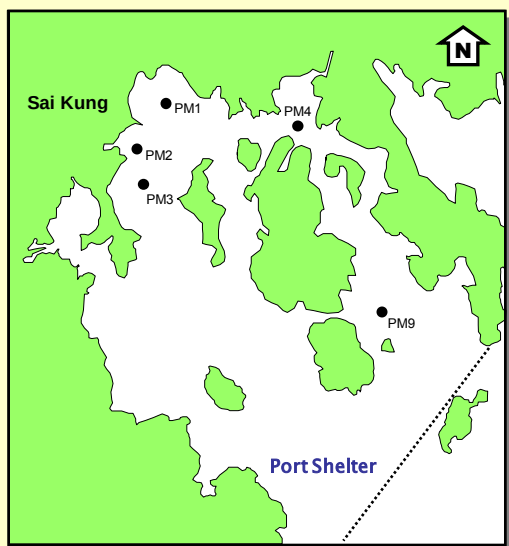
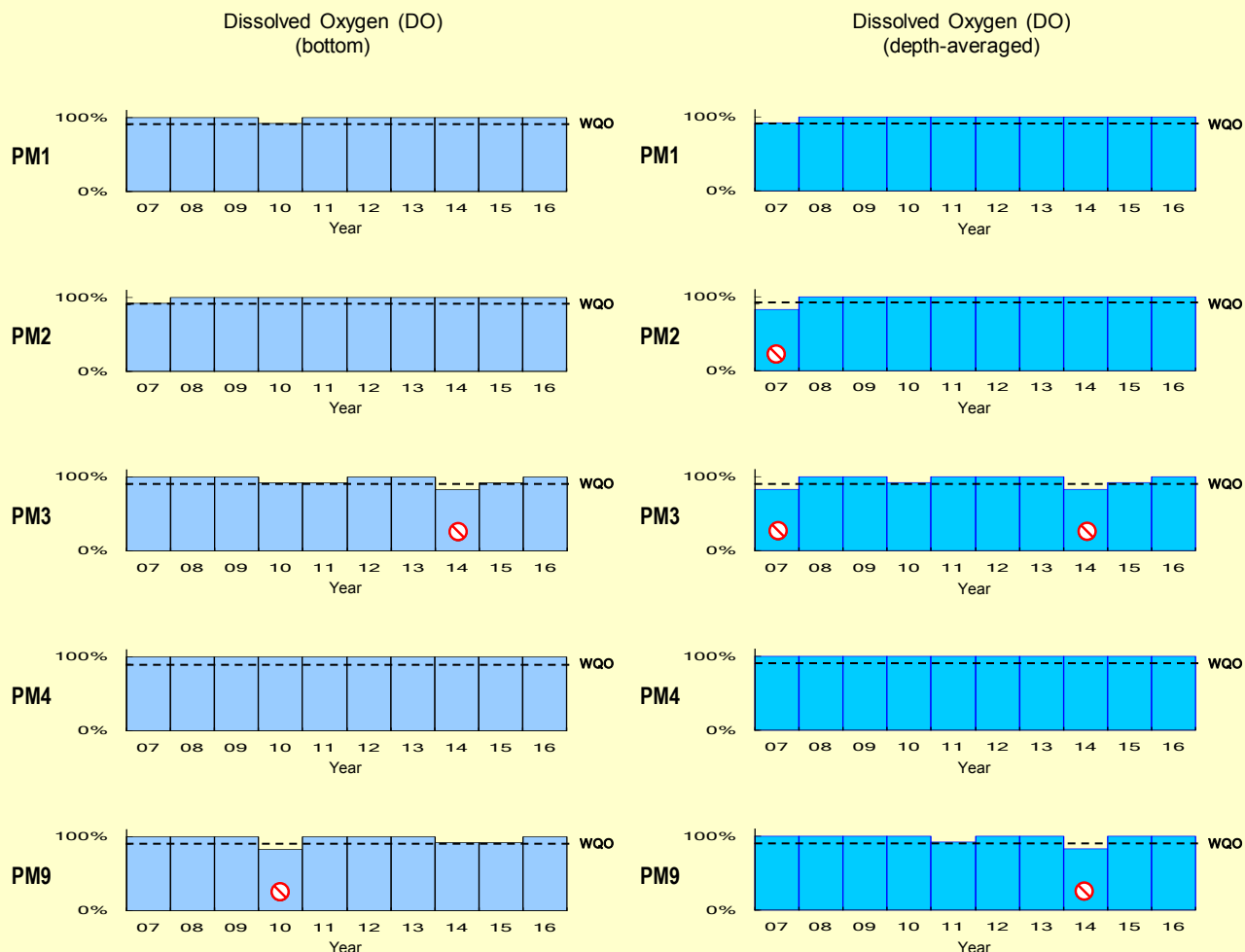
⊘ Non-compliance

Levels of compliance with key Water Quality Objectives in the Mirs Bay WCZ (continued)

E. coli
(annual geometric mean)



Levels of compliance with key Water Quality Objectives in the Port Shelter WCZ

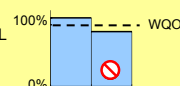


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO $\geq$ 2 mg/L

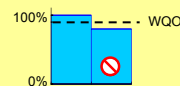
% sample with bottom DO $\geq$ 2 mg/L



2. Depth-averaged

WQO : 90% sample with depth-averaged DO $\geq$ 4 mg/L

% sample with depth-averaged DO $\geq$ 4 mg/L

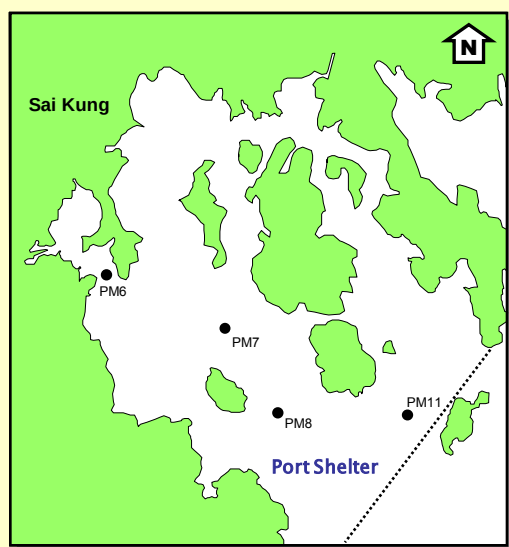
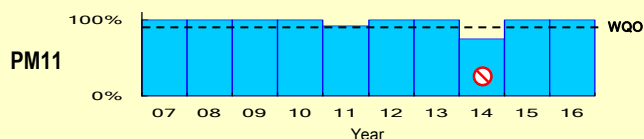
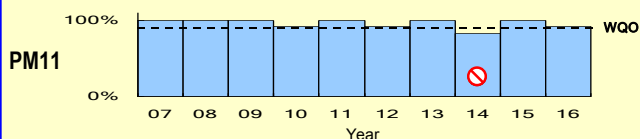
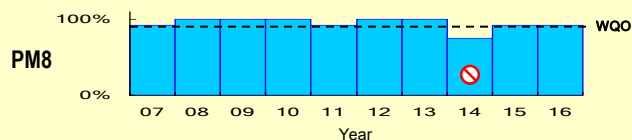
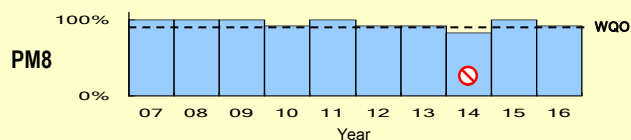
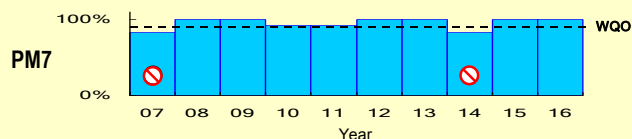
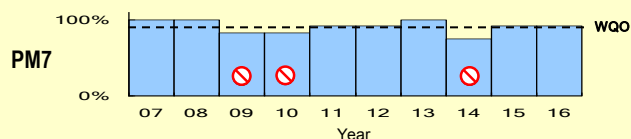
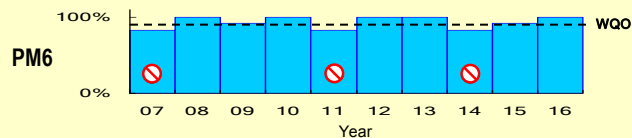
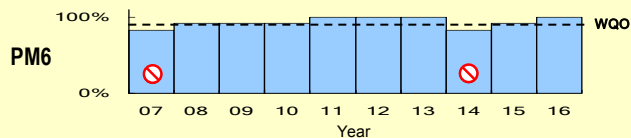


Non-compliance

Levels of compliance with key Water Quality Objectives in the Port Shelter WCZ (continued)

Dissolved Oxygen (DO)
(bottom)

Dissolved Oxygen (DO)
(depth-averaged)

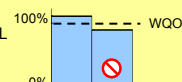


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO $\geq$ 2 mg/L

% sample with bottom DO $\geq$ 2 mg/L



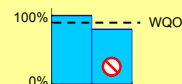
2. Depth-averaged

WQO : 90% sample with depth-averaged DO $\geq$ 4 mg/L

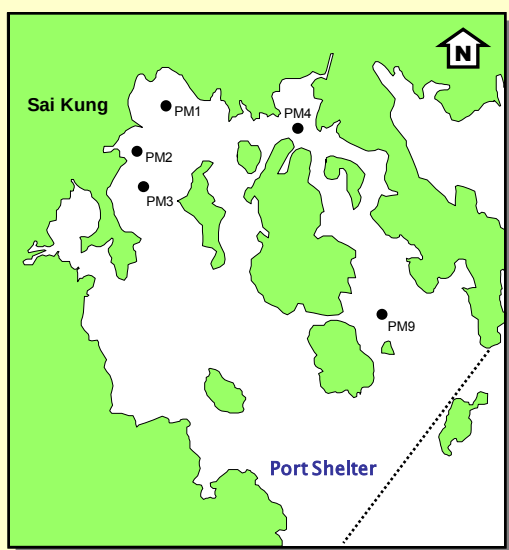
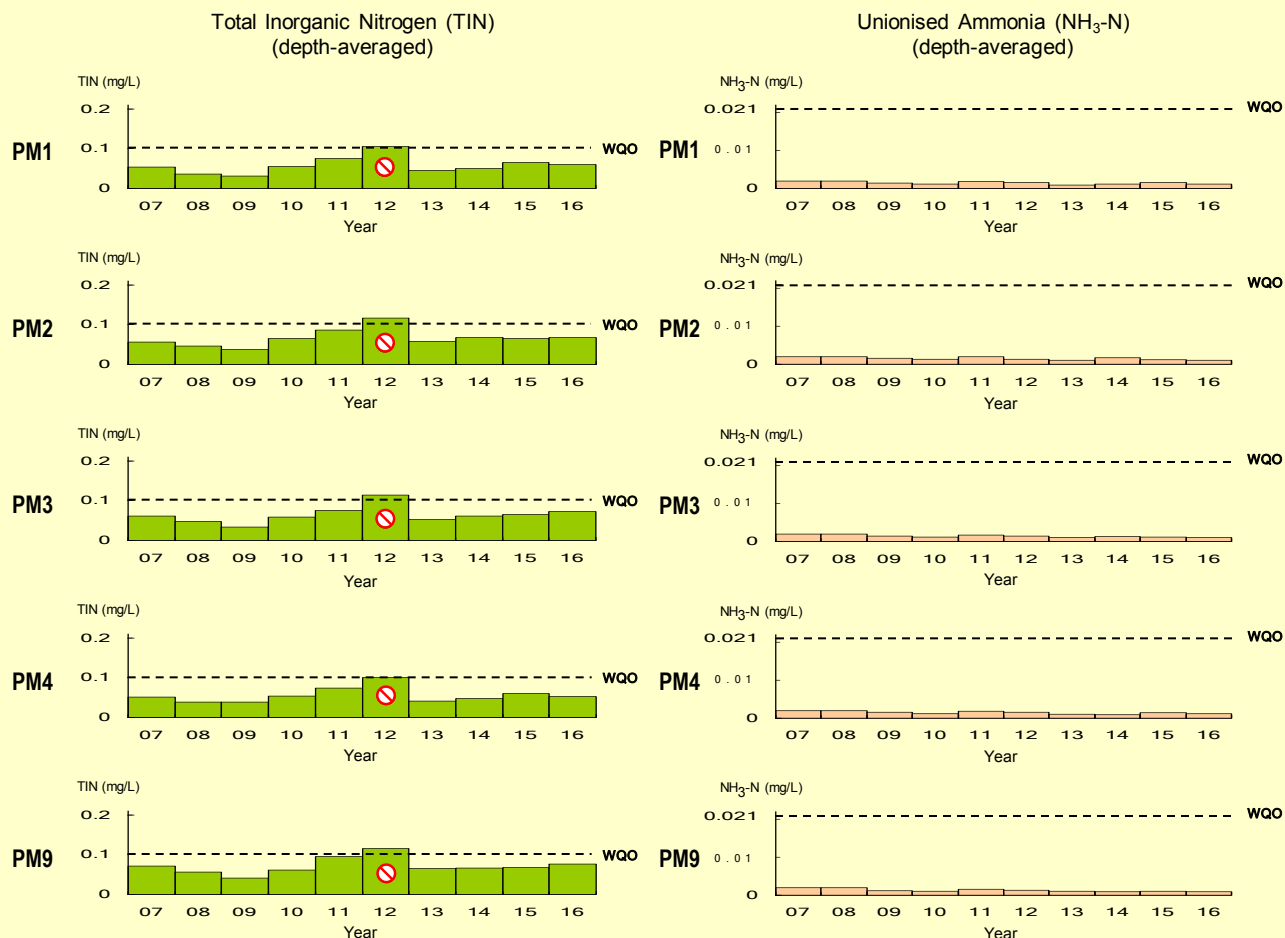
% sample with depth-averaged DO $\geq$ 4 mg/L



Non-compliance



Levels of compliance with key Water Quality Objectives in the Port Shelter WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.1 mg/L

■ annual mean for depth-averaged TIN



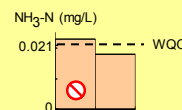
Unionised Ammonia (NH₃-N)

WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

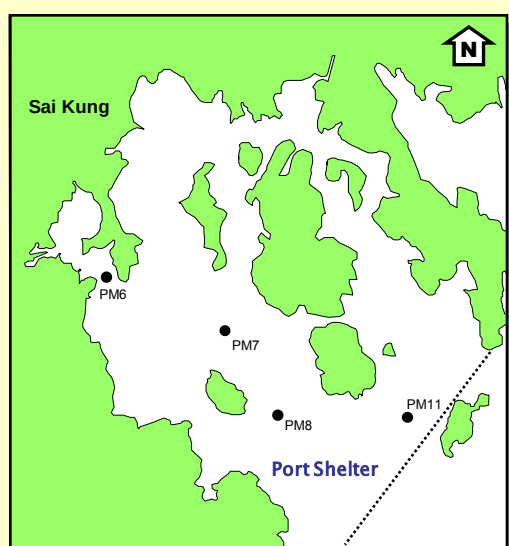
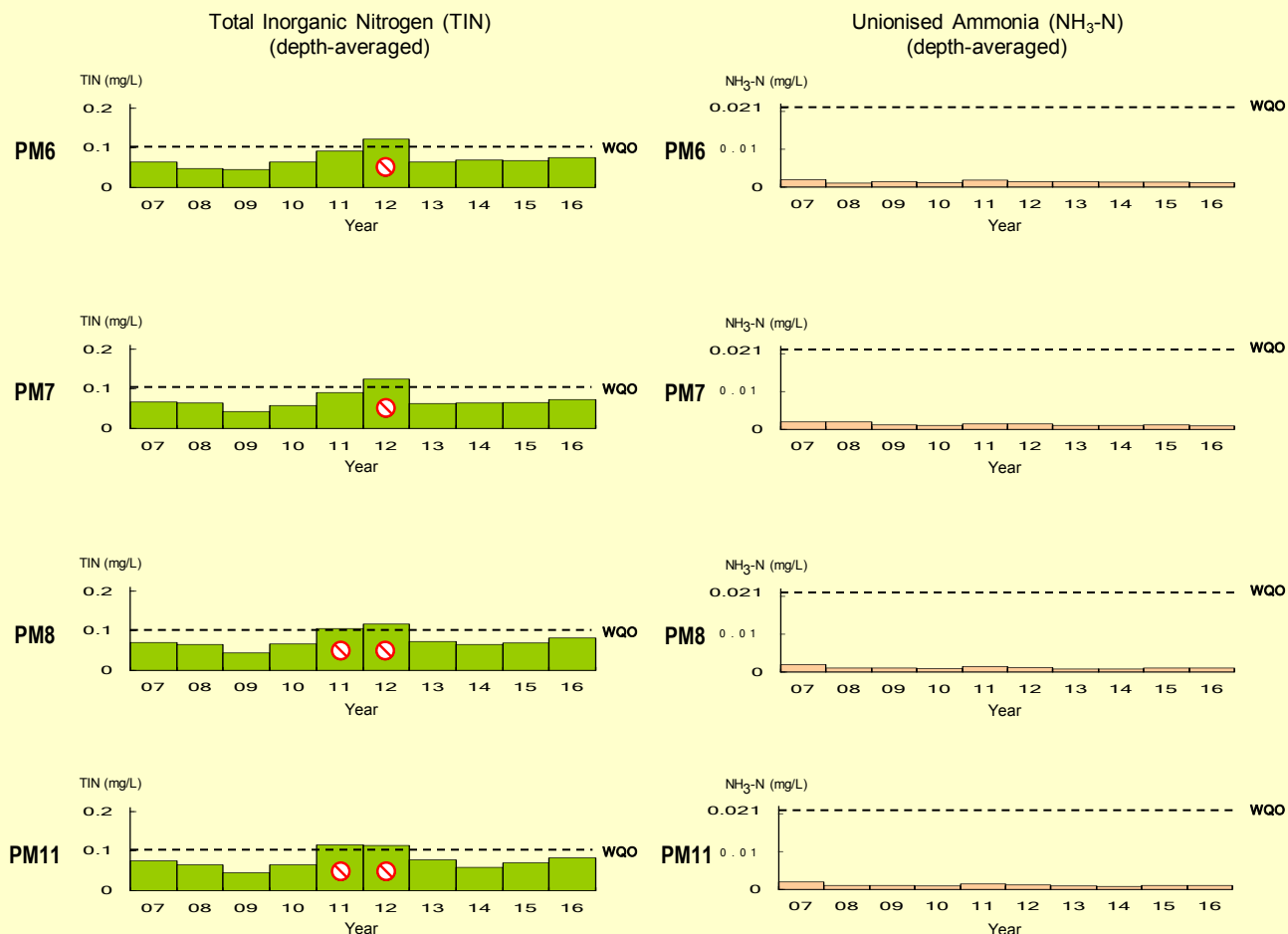
■ annual mean for depth-averaged NH₃-N



Non-compliance



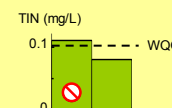
Levels of compliance with key Water Quality Objectives in the Port Shelter WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.1 mg/L

■ annual mean for depth-averaged TIN

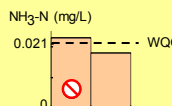


Unionised Ammonia (NH₃-N)

WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

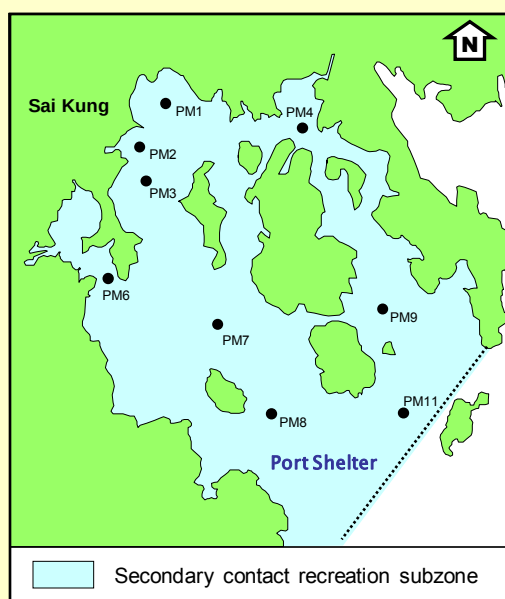
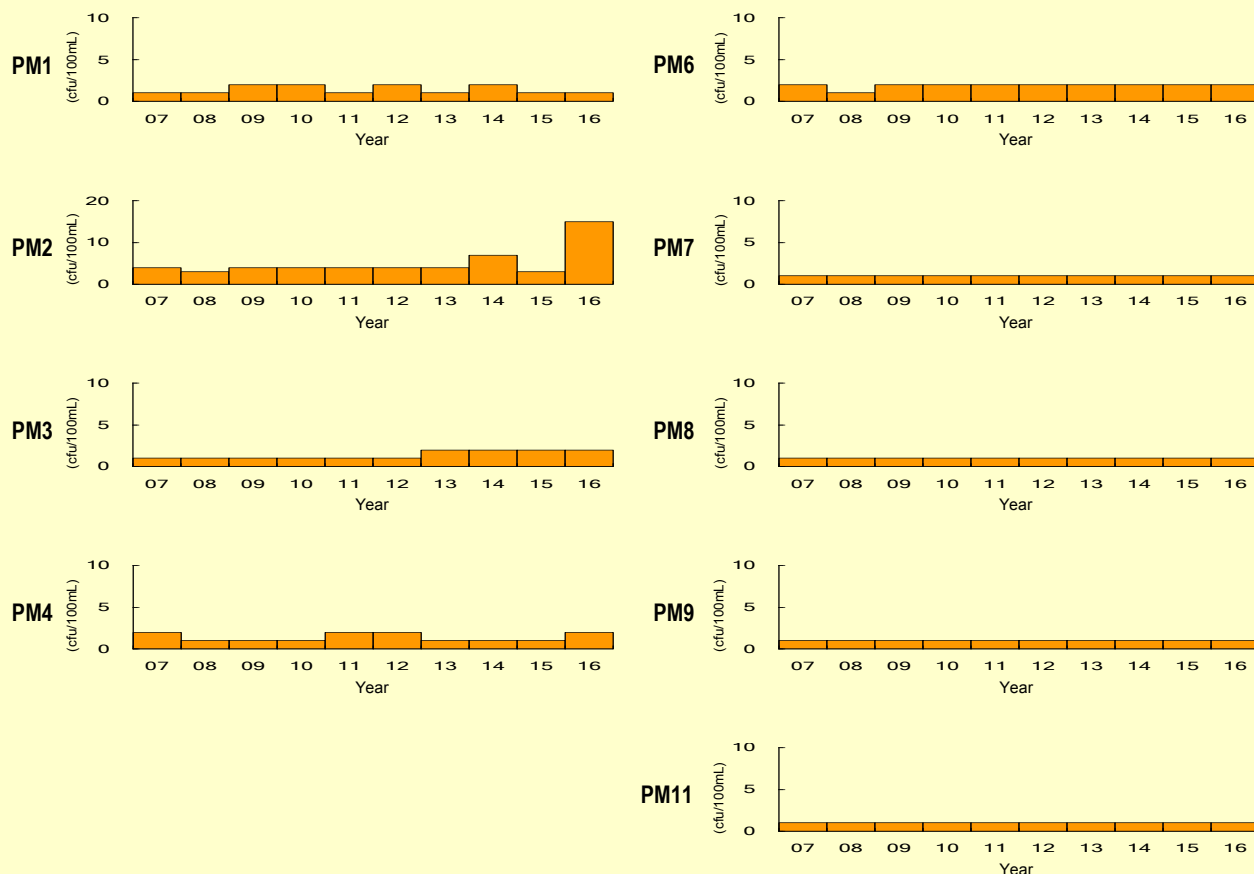
■ annual mean for depth-averaged NH₃-N

○ Non-compliance



Levels of compliance with key Water Quality Objectives in the Port Shelter WCZ (continued)

E. coli
(annual geometric mean)



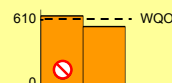
E. coli

WQO for secondary contact recreation subzones :
annual geometric mean for depth averaged
E. coli ≤ 610 cfu/100mL

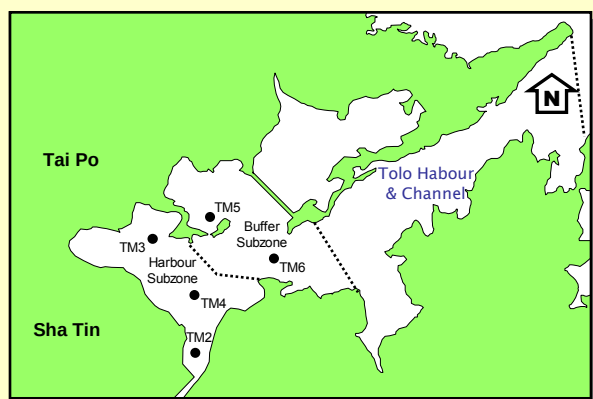
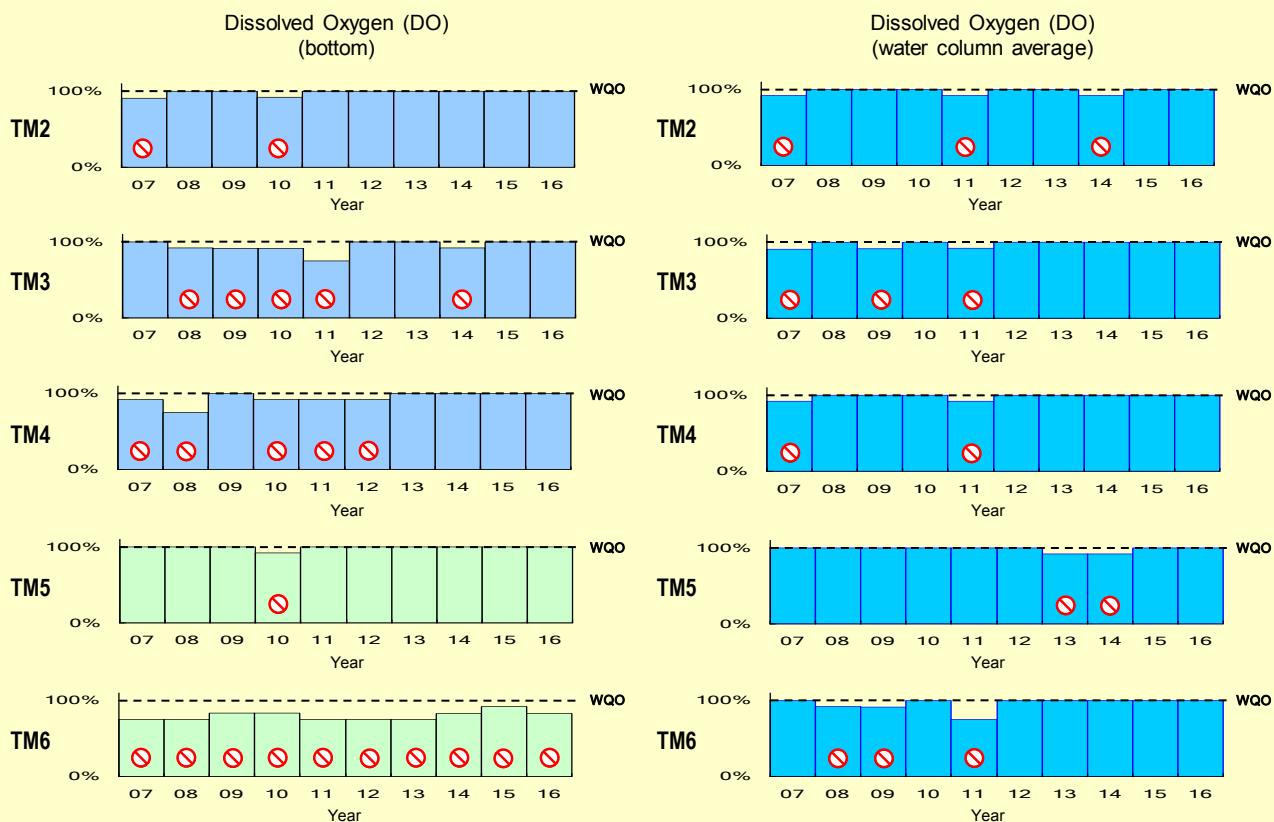
annual geometric mean for depth averaged
E. coli (cfu/100mL)



Non-compliance



Levels of compliance with key Water Quality Objectives in the Tolo Harbour and Channel WCZ



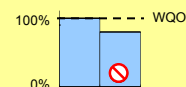
Dissolved Oxygen (DO)

Harbour Subzone (TM2 - TM4)

1. Bottom

WQO: 100% sample with bottom DO ≥ 2 mg/L

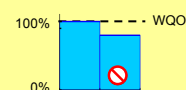
% sample with bottom DO ≥ 2 mg/L



2. Water column average (surface to 2m above bottom)

WQO: 100% sample with water column average DO ≥ 4 mg/L

% sample with water column average DO ≥ 4 mg/L

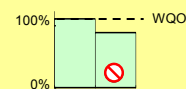


Buffer Subzone (TM5 - TM6)

1. Bottom

WQO: 100% sample with bottom DO ≥ 3 mg/L

% sample with bottom DO ≥ 3 mg/L



2. Water column average (surface to 2m above bottom)

WQO: 100% sample with water column average DO ≥ 4 mg/L

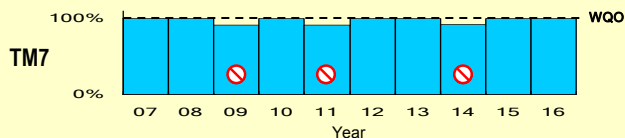
% sample with water column average DO ≥ 4 mg/L



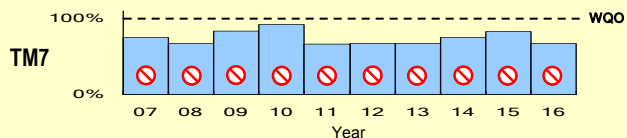
Non-compliance

Levels of compliance with key Water Quality Objectives in the Tolo Harbour and Channel WCZ (continued)

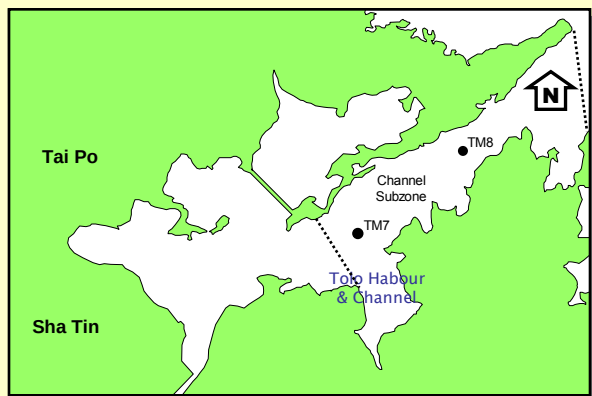
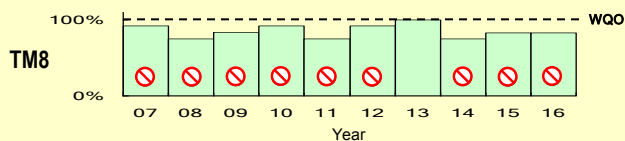
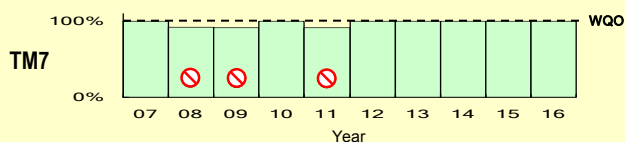
Dissolved Oxygen (DO)
(surface)



Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(middle)



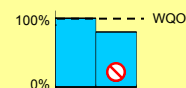
Dissolved Oxygen (DO)

Channel Subzone (TM7 - TM8)

1. Surface

WQO: 100% sample with surface DO $\geq$ 4 mg/L

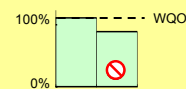
% sample with surface DO $\geq$ 4 mg/L



2. Middle

WQO: 100% sample with middle DO $\geq$ 4 mg/L

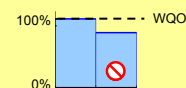
% sample with middle DO $\geq$ 4 mg/L



3. Bottom

WQO: 100% sample with bottom DO $\geq$ 4 mg/L

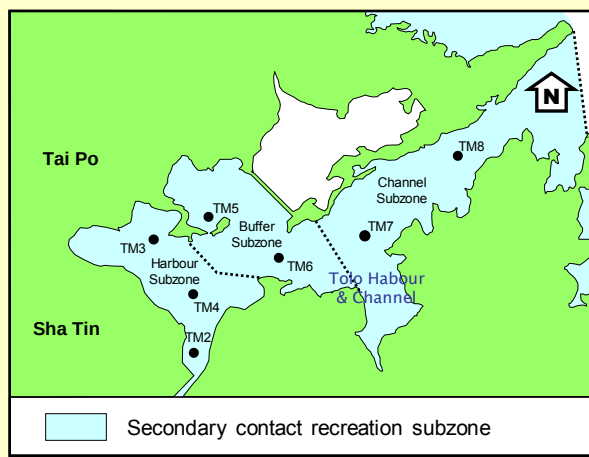
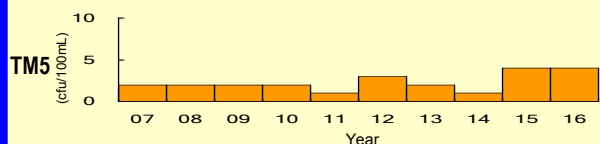
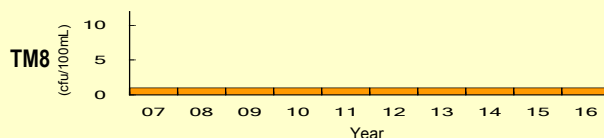
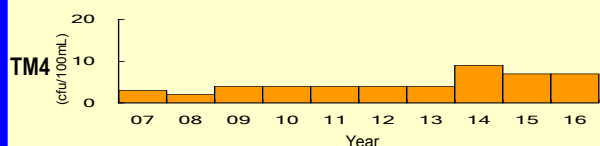
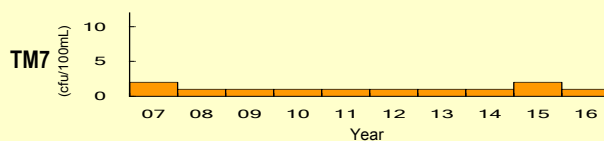
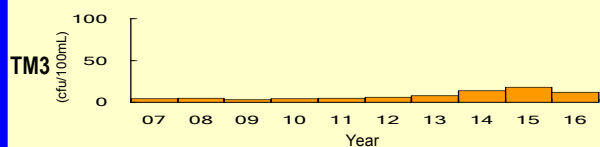
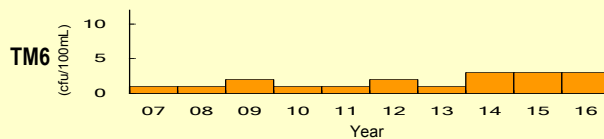
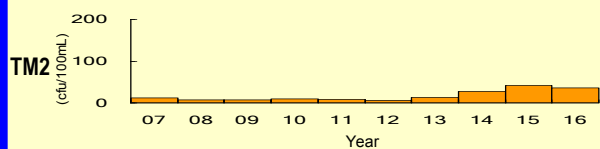
% sample with bottom DO $\geq$ 4 mg/L



Non-compliance

Levels of compliance with key Water Quality Objectives in the Tolo Harbour and Channel WCZ (continued)

E. coli
(annual geometric mean)



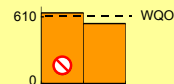
E. coli

WQO for secondary contact recreation subzones :
annual geometric mean for depth averaged
E. coli ≤ 610 cfu/100mL

annual geometric mean for depth averaged
E. coli (cfu/100mL)



Non-compliance

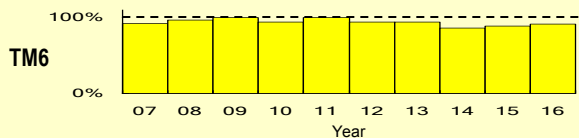
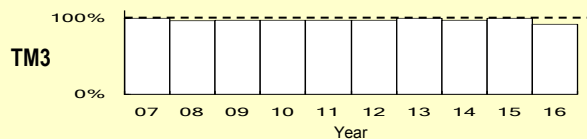
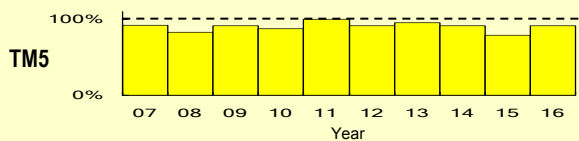
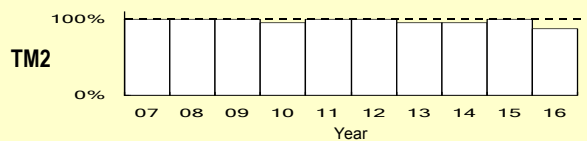


Levels of chlorophyll-*a* in the Tolo Harbour and Channel WCZ

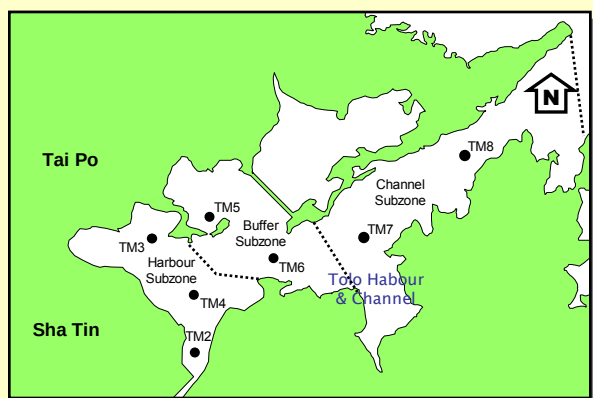
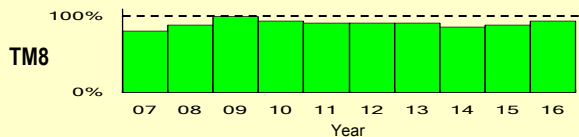
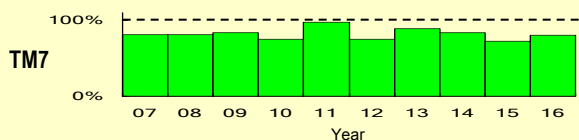
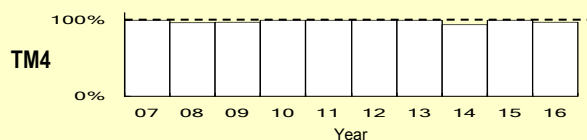
Chlorophyll-*a*

1. Harbour Subzone

2. Buffer Subzone



3. Channel Subzone



Chlorophyll-*a*

1. Harbour Subzone

% sample (S, M, B) with Chlorophyll-*a* ≤ 20 $\mu\text{g/L}$

WQO: Chlorophyll-*a* ≤ 20 $\mu\text{g/L}$

2. Buffer Subzone

% sample (S, M, B) with Chlorophyll-*a* ≤ 10 $\mu\text{g/L}$

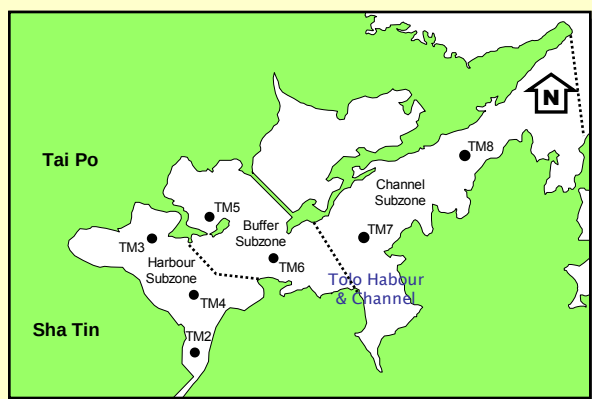
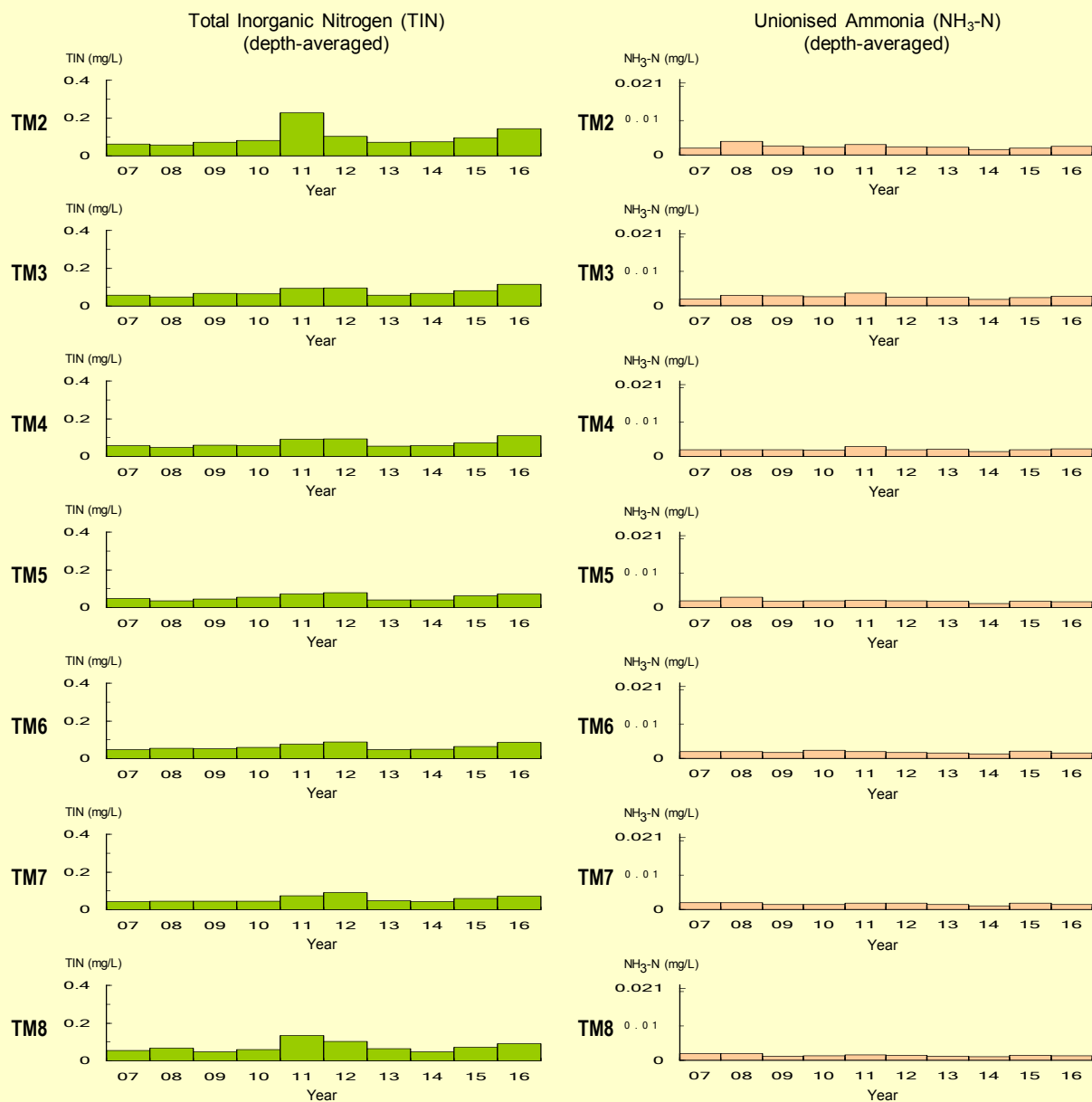
WQO: Chlorophyll-*a* ≤ 10 $\mu\text{g/L}$

3. Channel Subzone

% sample (S, M, B) with Chlorophyll-*a* ≤ 6 $\mu\text{g/L}$

WQO: Chlorophyll-*a* ≤ 6 $\mu\text{g/L}$

Levels of total inorganic nitrogen and unionised ammonia in the Tolo Harbour and Channel WCZ



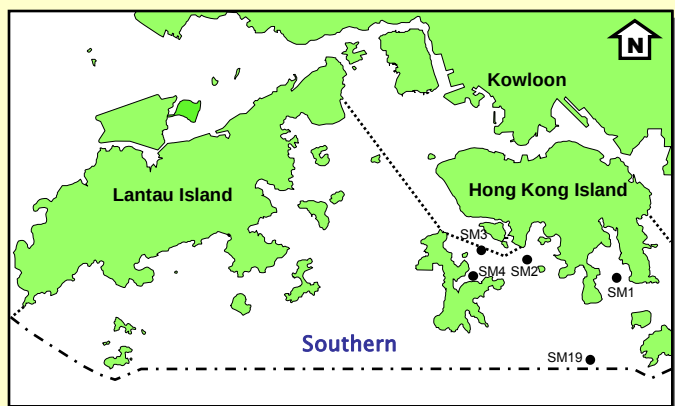
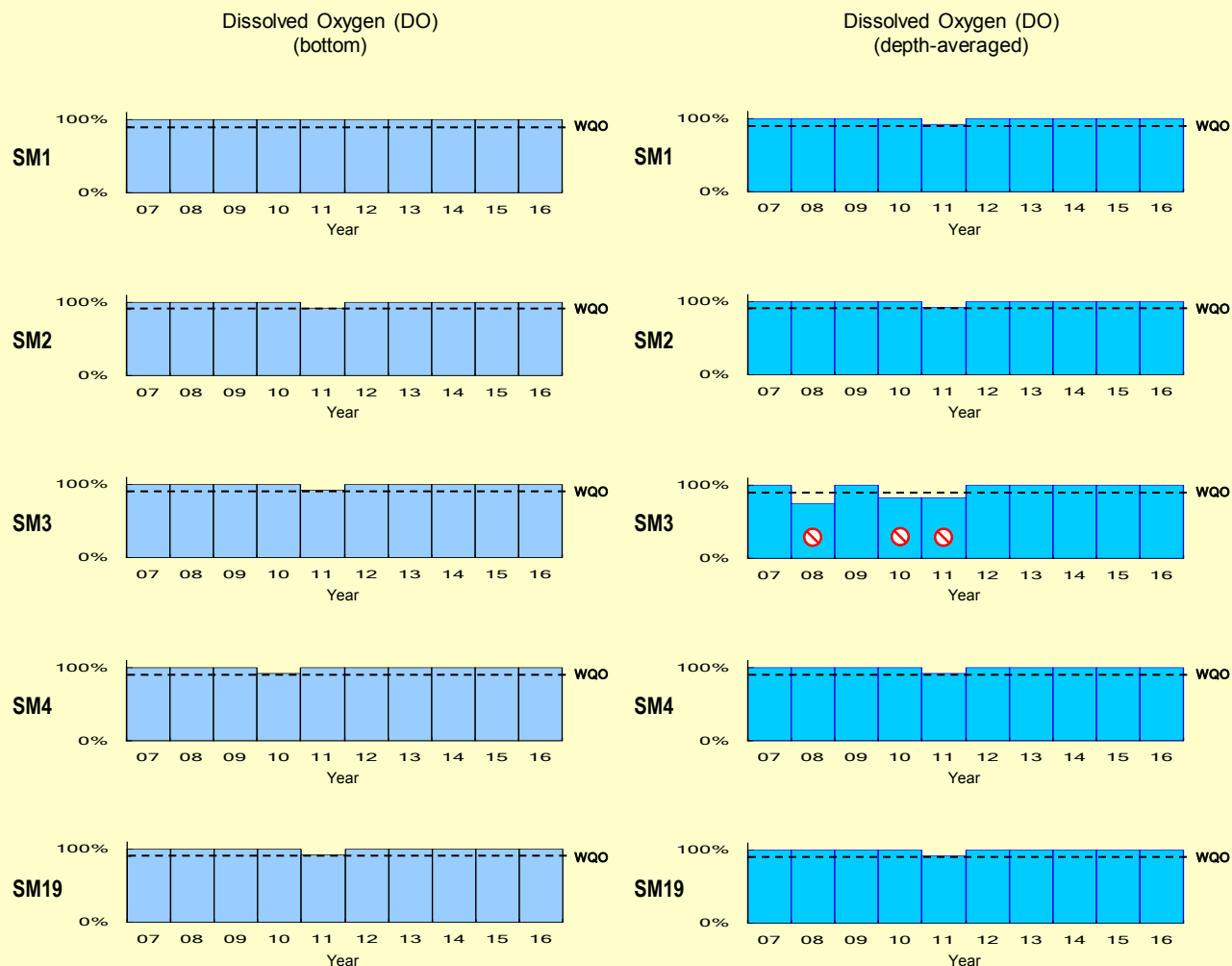
Total Inorganic Nitrogen (TIN)

■ annual mean for depth-averaged TIN (mg/L)

Unionised Ammonia (NH₃-N)

■ annual mean for depth-averaged NH₃-N (mg/L)

Levels of compliance with key Water Quality Objectives in the Southern WCZ

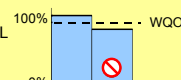


Dissolved Oxygen (DO)

1. Bottom

WQO: 90% sample with bottom DO ≥ 2 mg/L

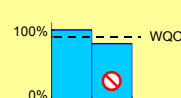
Blue bar: % sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

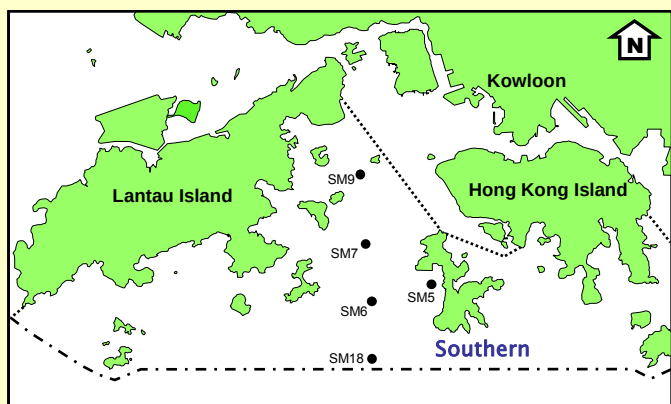
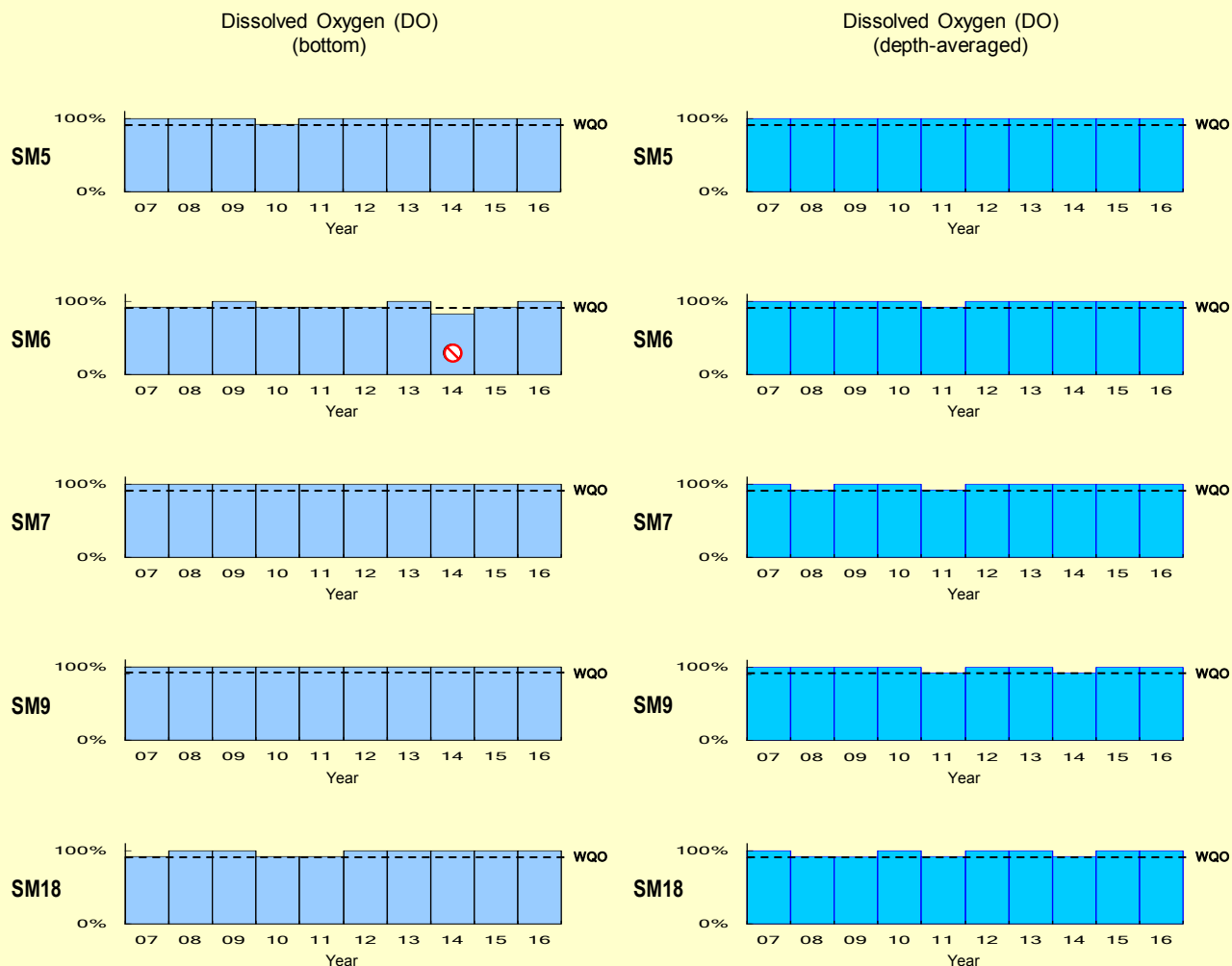
WQO: 90% sample with depth-averaged DO ≥ 4 mg/L

Blue bar: % sample with depth-averaged DO ≥ 4 mg/L



Red circle with slash: Non-compliance

Levels of compliance with key Water Quality Objectives in the Southern WCZ (continued)

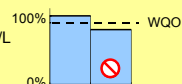


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO $\geq$ 2 mg/L

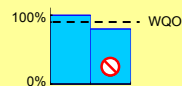
Blue bar : % sample with bottom DO $\geq$ 2 mg/L



2. Depth-averaged

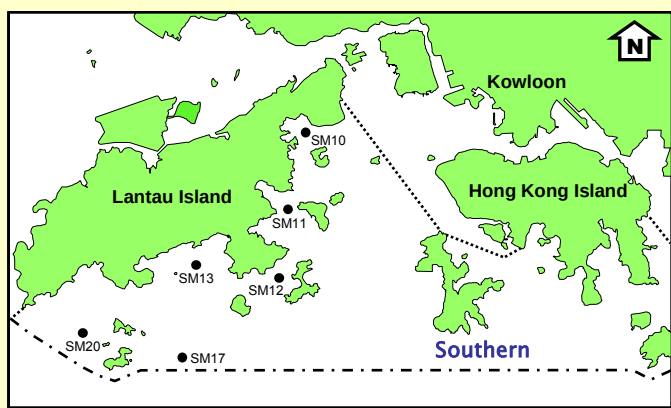
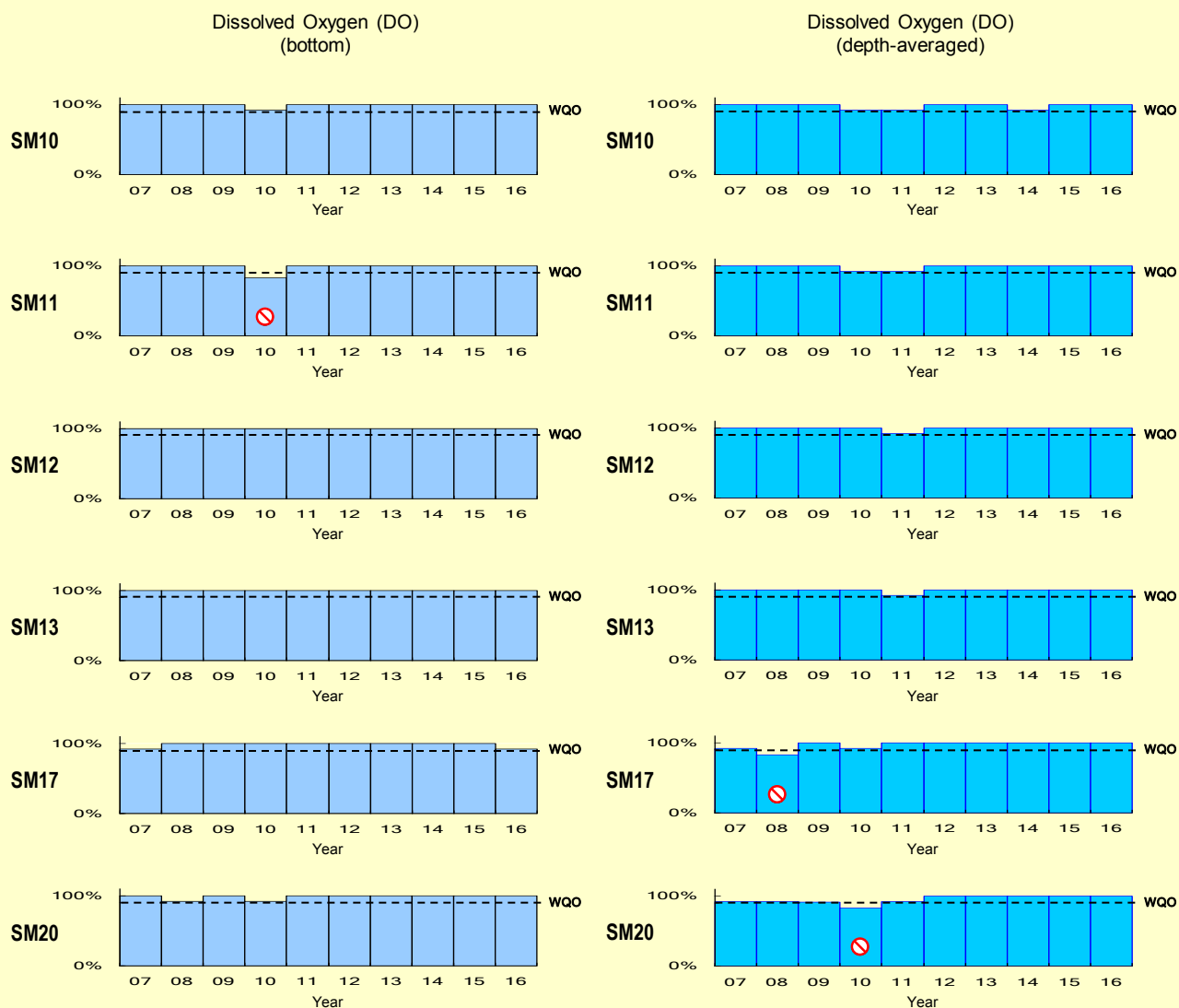
WQO : 90% sample with depth-averaged DO $\geq$ 4 mg/L

Blue bar : % sample with depth-averaged DO $\geq$ 4 mg/L



Red circle with slash : Non-compliance

Levels of compliance with key Water Quality Objectives in the Southern WCZ (continued)

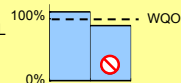


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

■ % sample with bottom DO ≥ 2 mg/L



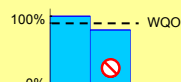
2. Depth-averaged

WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

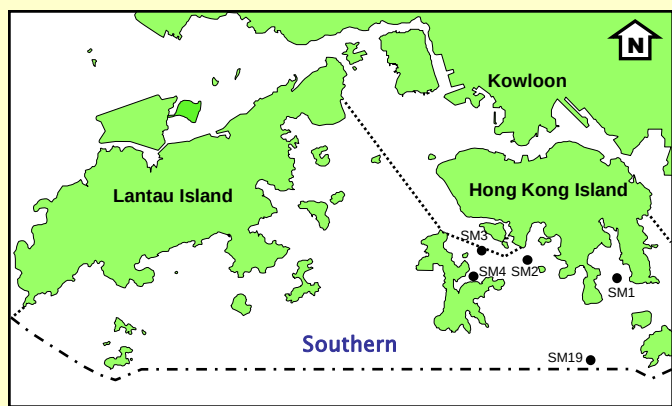
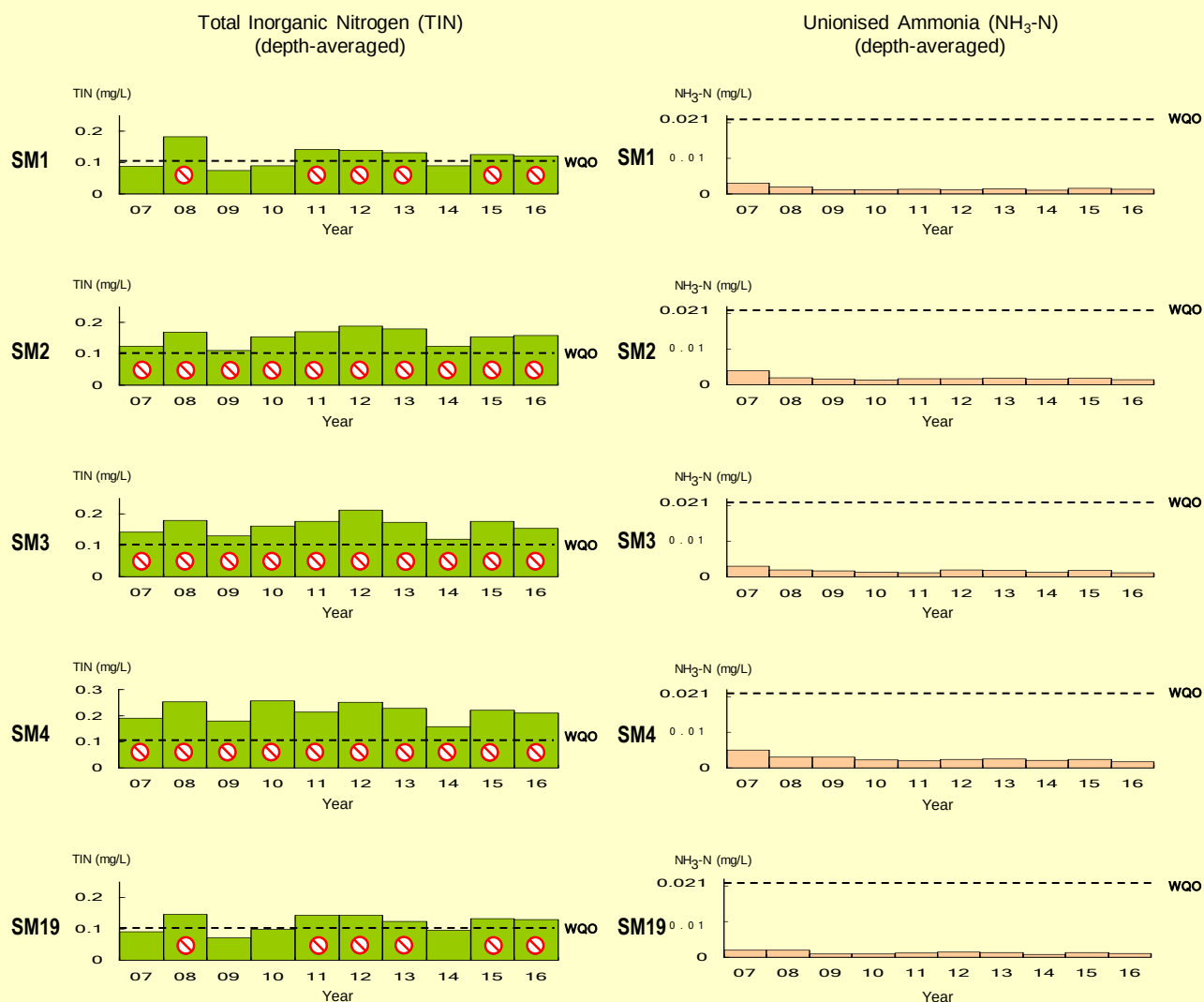
■ % sample with depth-averaged DO ≥ 4 mg/L



Non-compliance



Levels of compliance with key Water Quality Objectives in the Southern WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged
TIN ≤ 0.1 mg/L

■ annual mean for depth-averaged
TIN

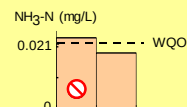


Unionised Ammonia (NH₃-N)

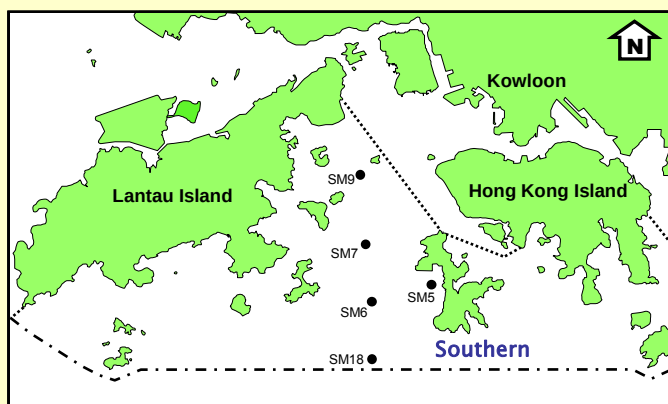
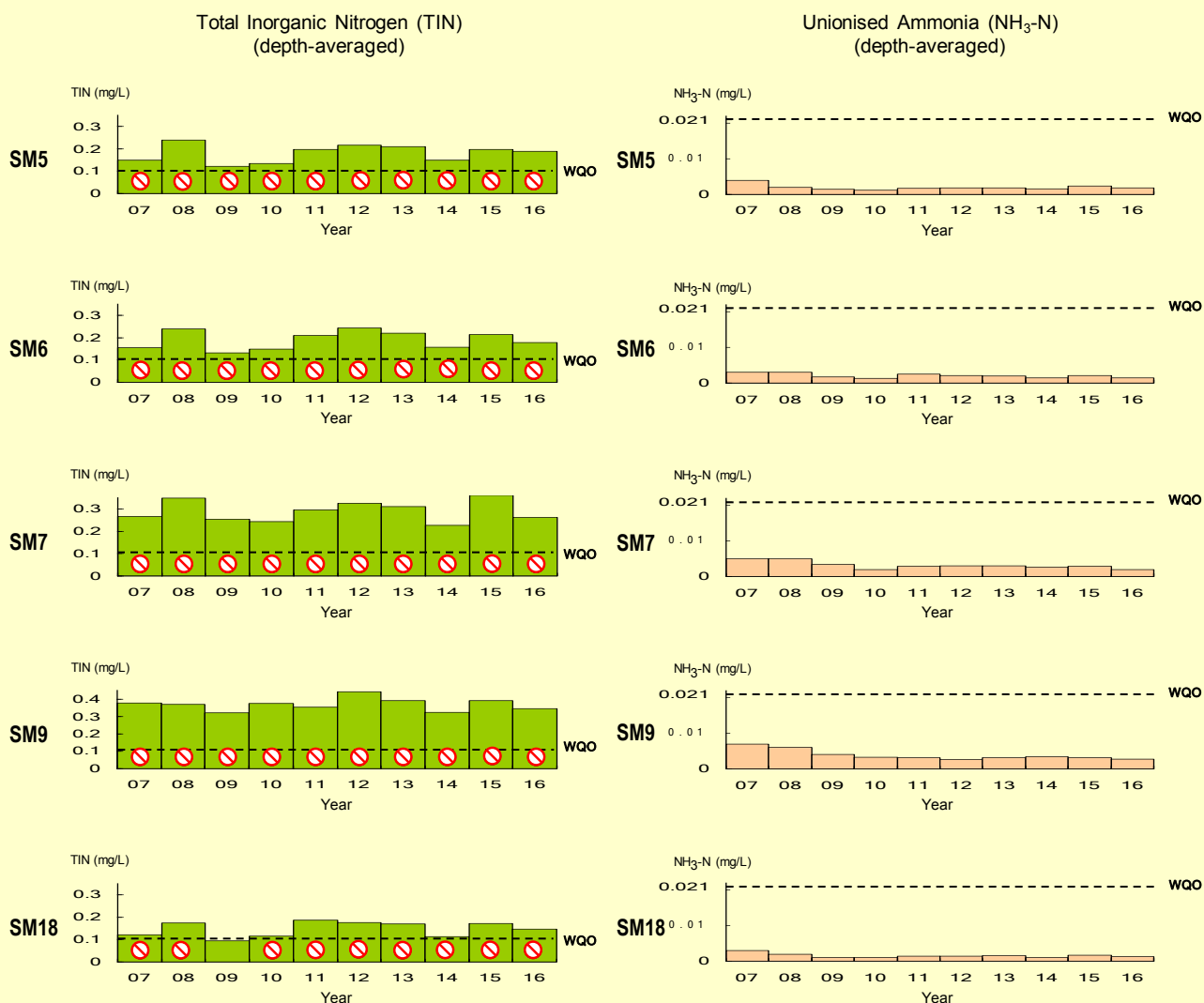
WQO: annual mean for depth-averaged
NH₃-N ≤ 0.021 mg/L

■ annual mean for depth-averaged
NH₃-N

⊘ Non-compliance



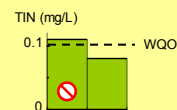
Levels of compliance with key Water Quality Objectives in the Southern WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO : annual mean for depth-averaged
TIN ≤ 0.1 mg/L

■ annual mean for depth-averaged
TIN

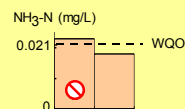


Unionised Ammonia (NH₃-N)

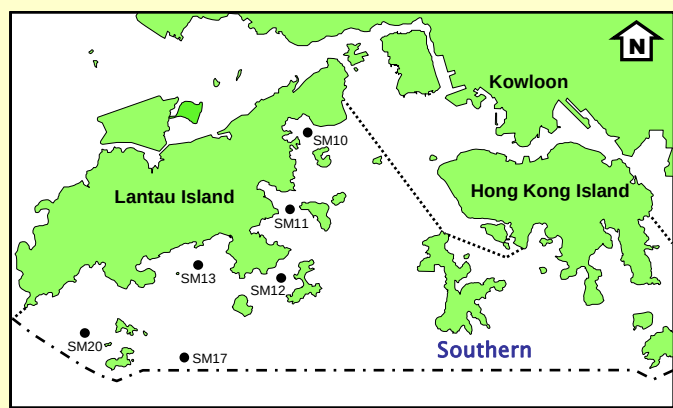
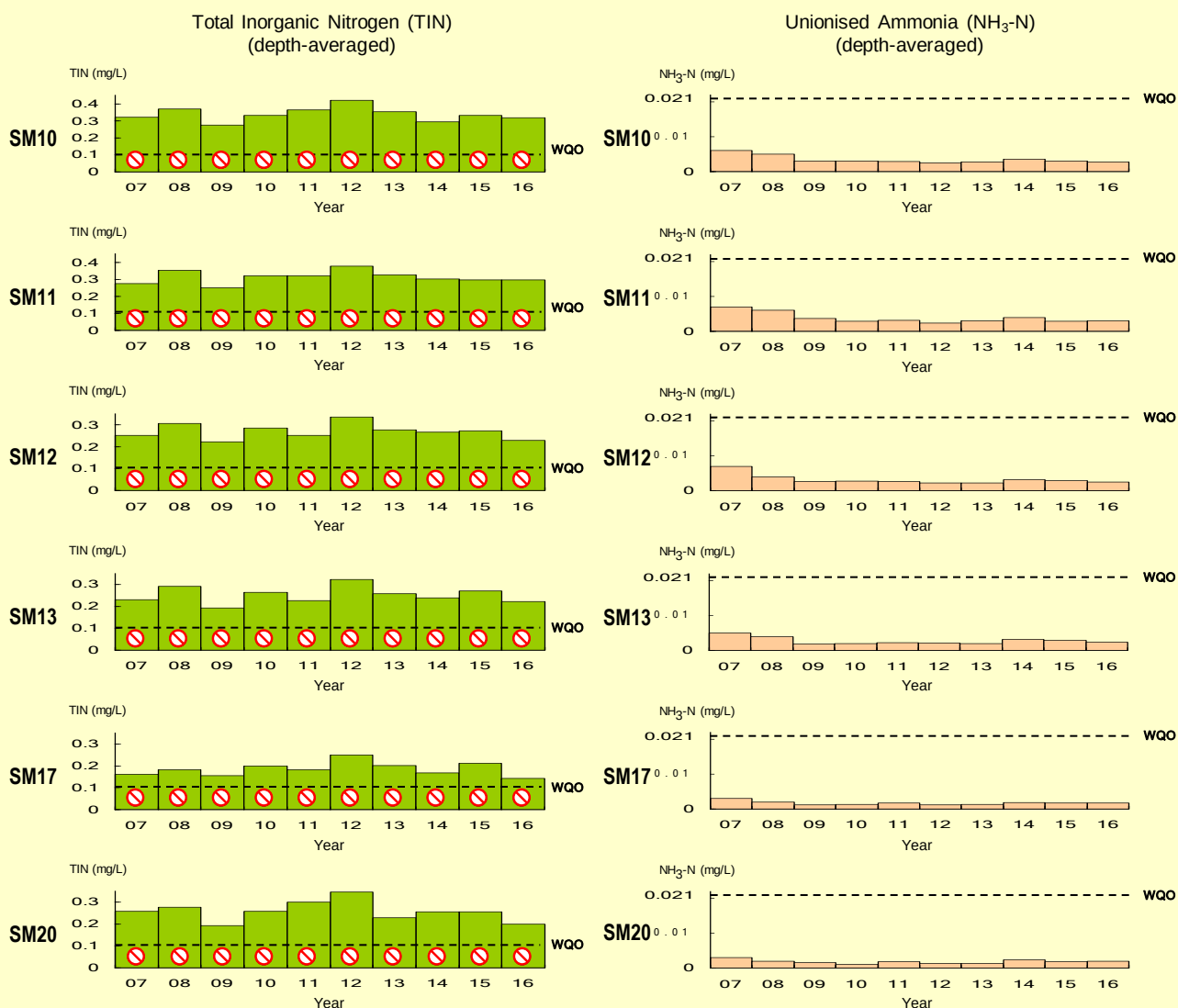
WQO : annual mean for depth-averaged
NH₃-N ≤ 0.021 mg/L

■ annual mean for depth-averaged
NH₃-N

⊘ Non-compliance



Levels of compliance with key Water Quality Objectives in the Southern WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.1 mg/L

■ annual mean for depth-averaged TIN

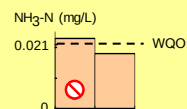


Unionised Ammonia (NH₃-N)

WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

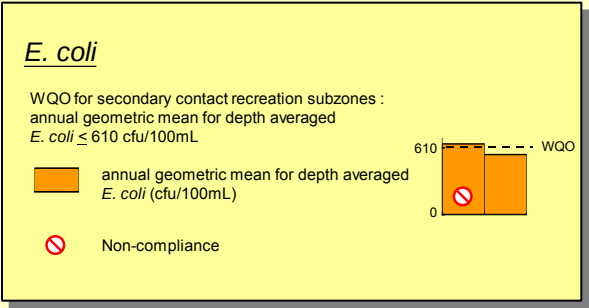
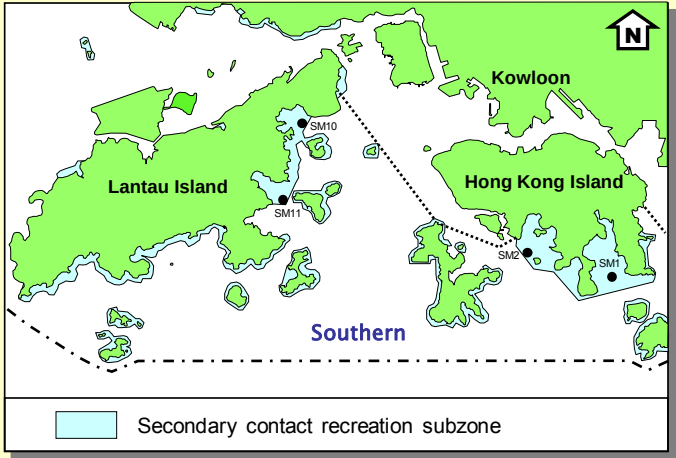
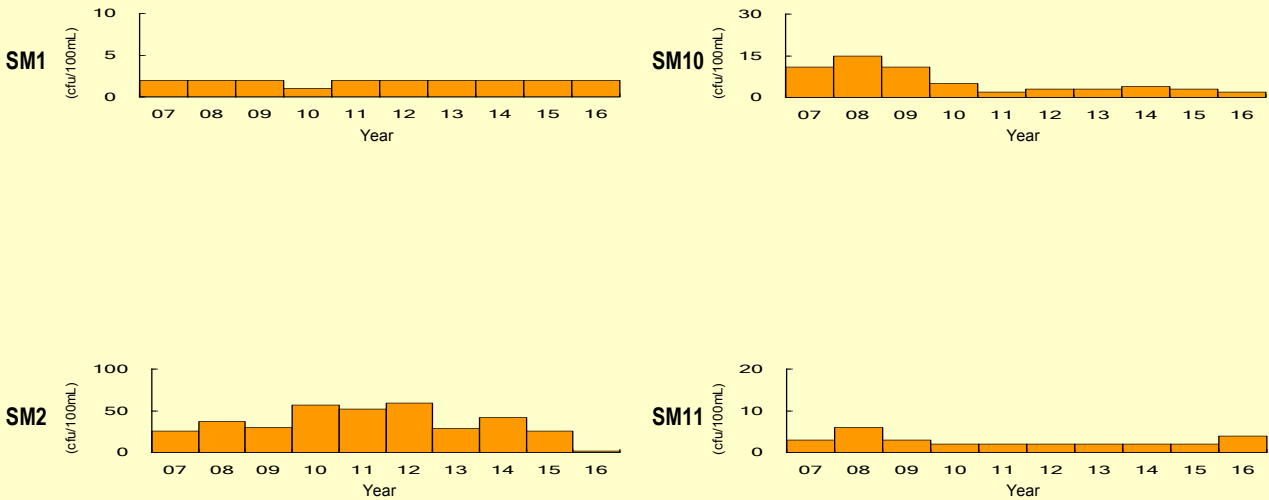
■ annual mean for depth-averaged NH₃-N

⊘ Non-compliance



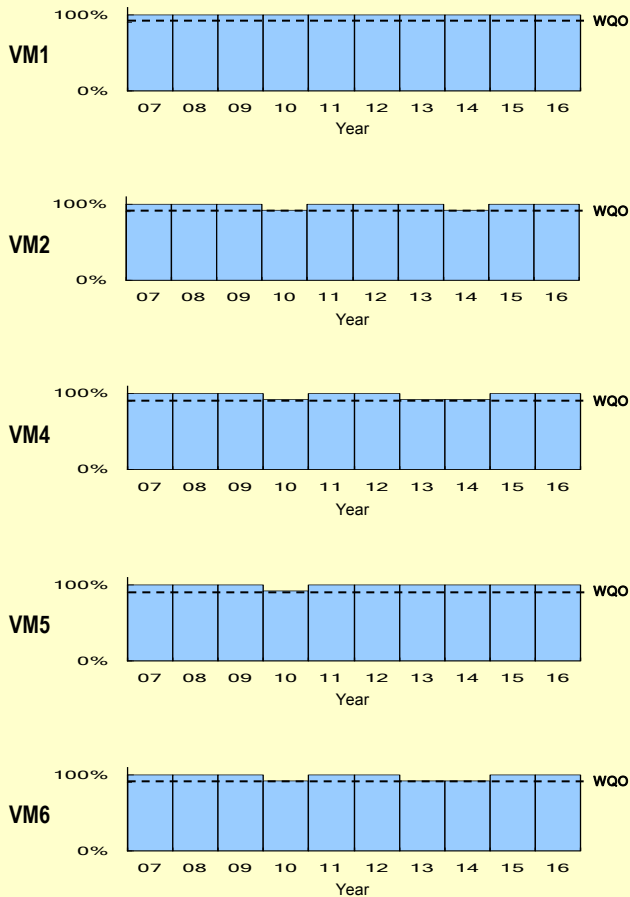
Levels of compliance with key Water Quality Objectives in the Southern WCZ (continued)

E. coli
(annual geometric mean)

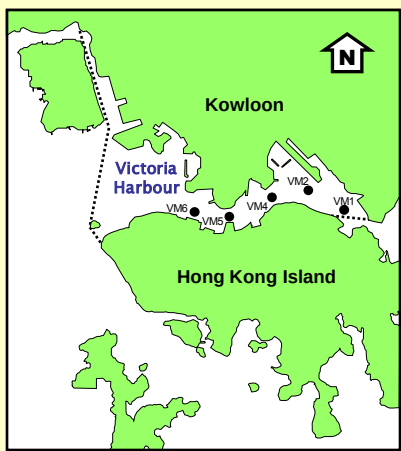
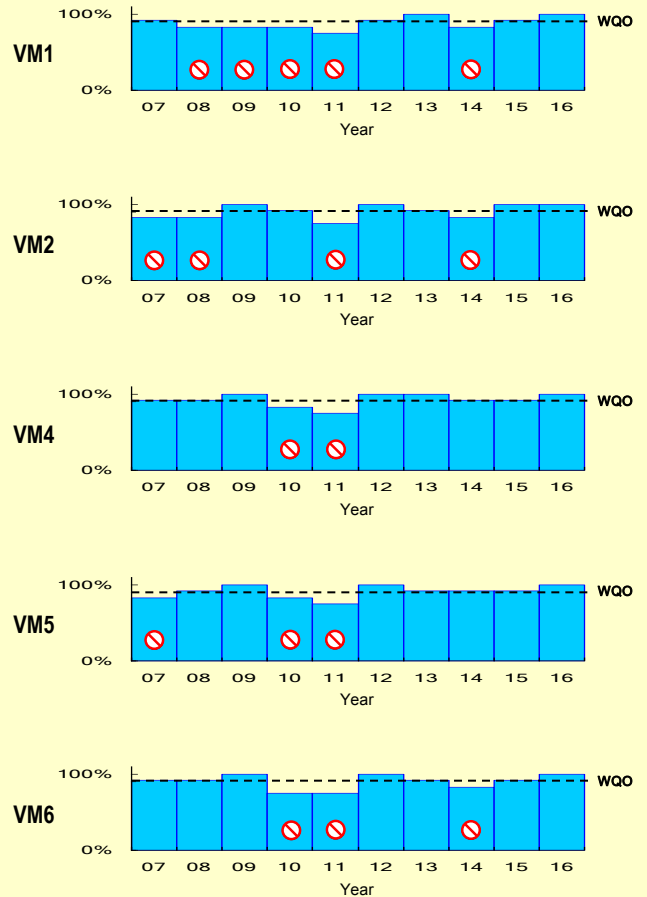


Levels of compliance with key Water Quality Objectives in the Victoria Harbour WCZ

Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(depth-averaged)

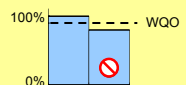


Dissolved Oxygen (DO)

1. Bottom

WQO: 90% sample with bottom DO ≥ 2 mg/L

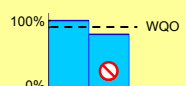
■ % sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

WQO: 90% sample with depth-averaged DO ≥ 4 mg/L

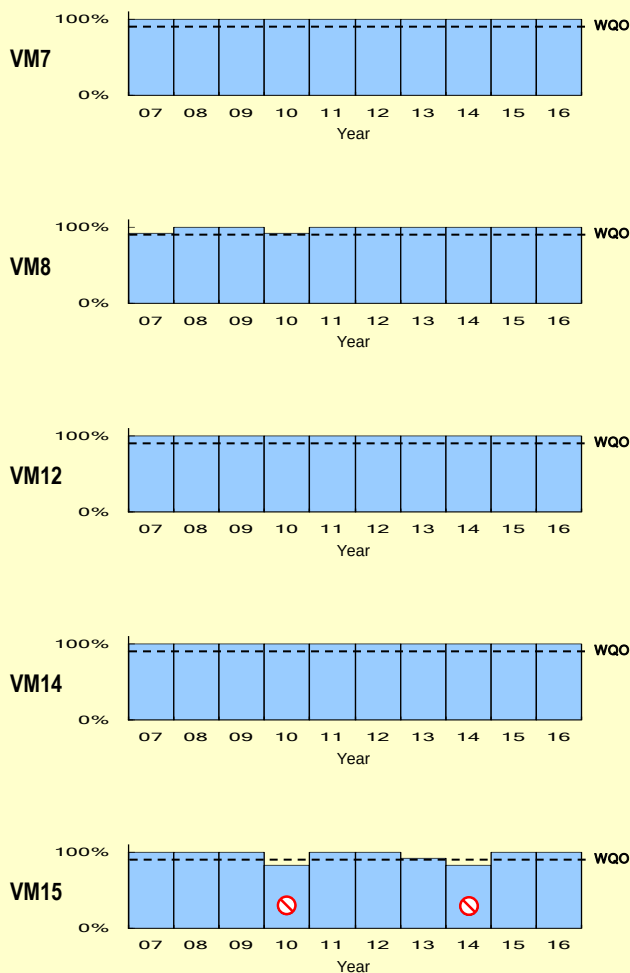
■ % sample with depth-averaged DO ≥ 4 mg/L



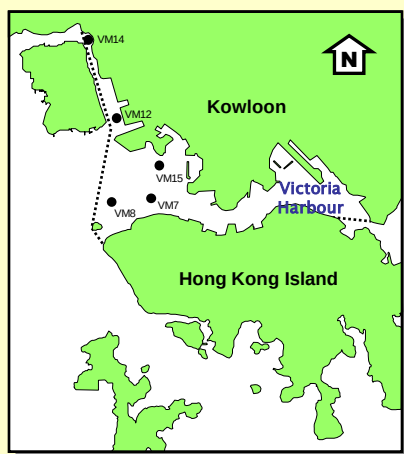
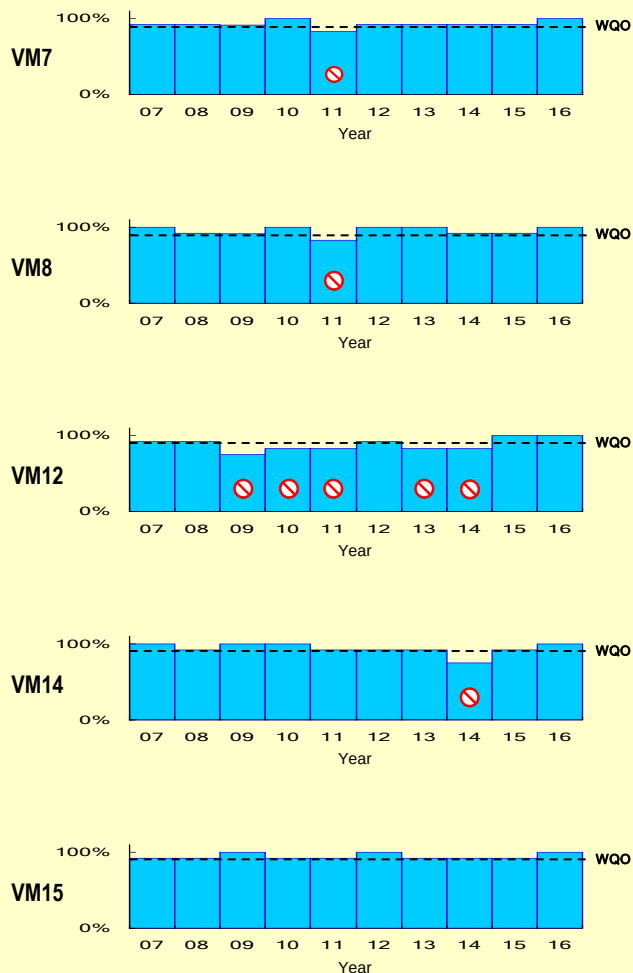
○ Non-compliance

Levels of compliance with key Water Quality Objectives in the Victoria Harbour WCZ (continued)

Dissolved Oxygen (DO)
(bottom)



Dissolved Oxygen (DO)
(depth-averaged)

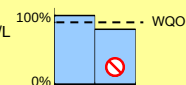


Dissolved Oxygen (DO)

1. Bottom

WQO: 90% sample with bottom DO $\geq$ 2 mg/L

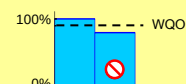
Blue bar: % sample with bottom DO $\geq$ 2 mg/L



2. Depth-averaged

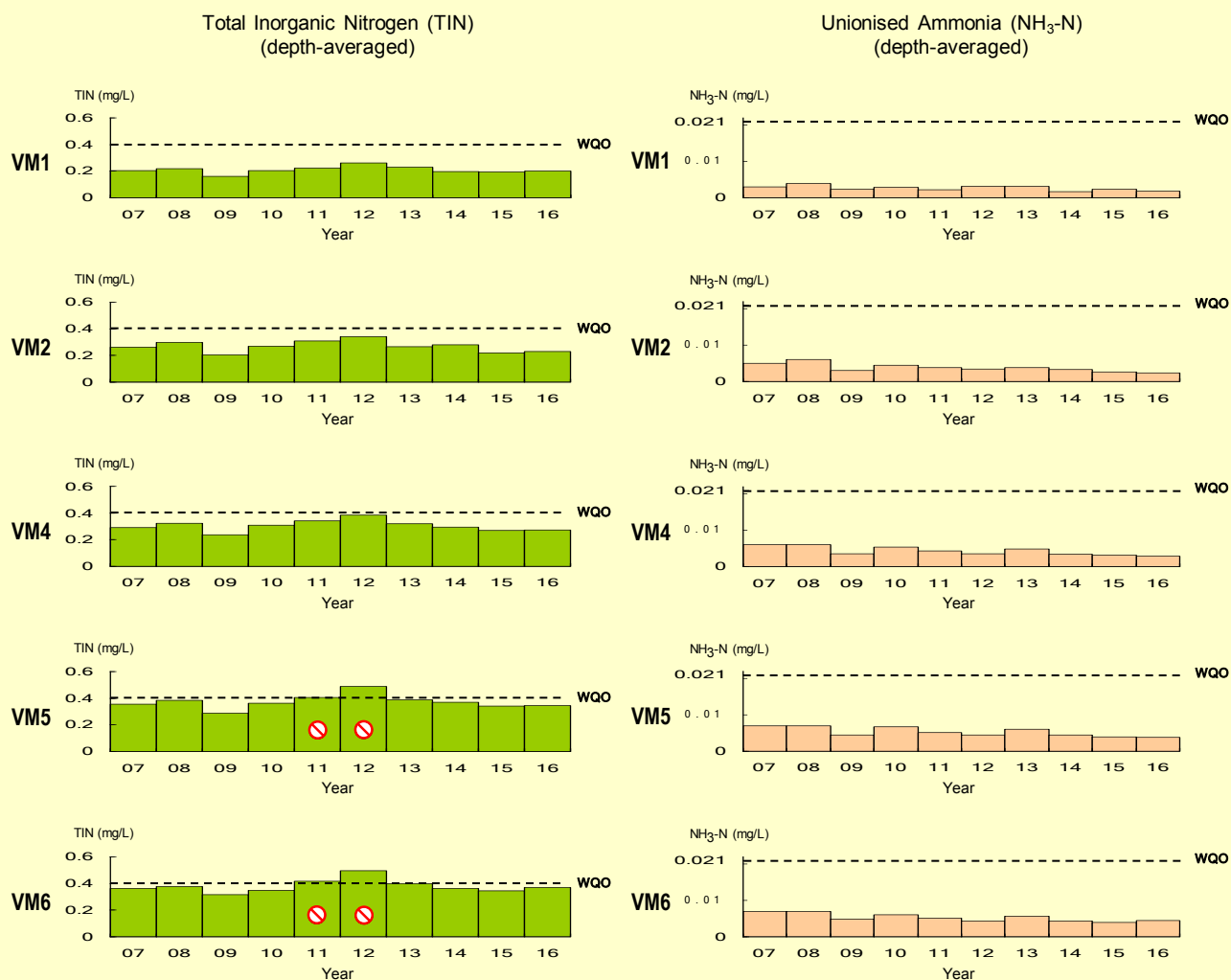
WQO: 90% sample with depth-averaged DO $\geq$ 4 mg/L

Blue bar: % sample with depth-averaged DO $\geq$ 4 mg/L



Red circle with slash: Non-compliance

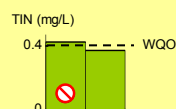
Levels of compliance with key Water Quality Objectives in the Victoria Harbour WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.4 mg/L

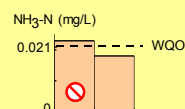
■ annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

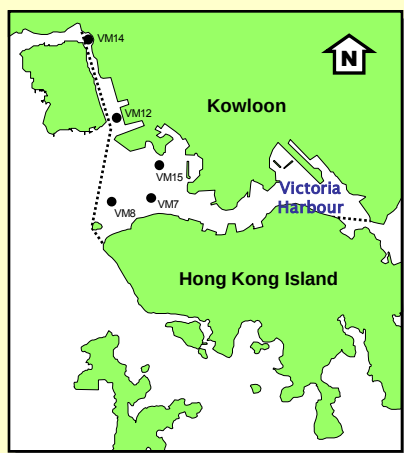
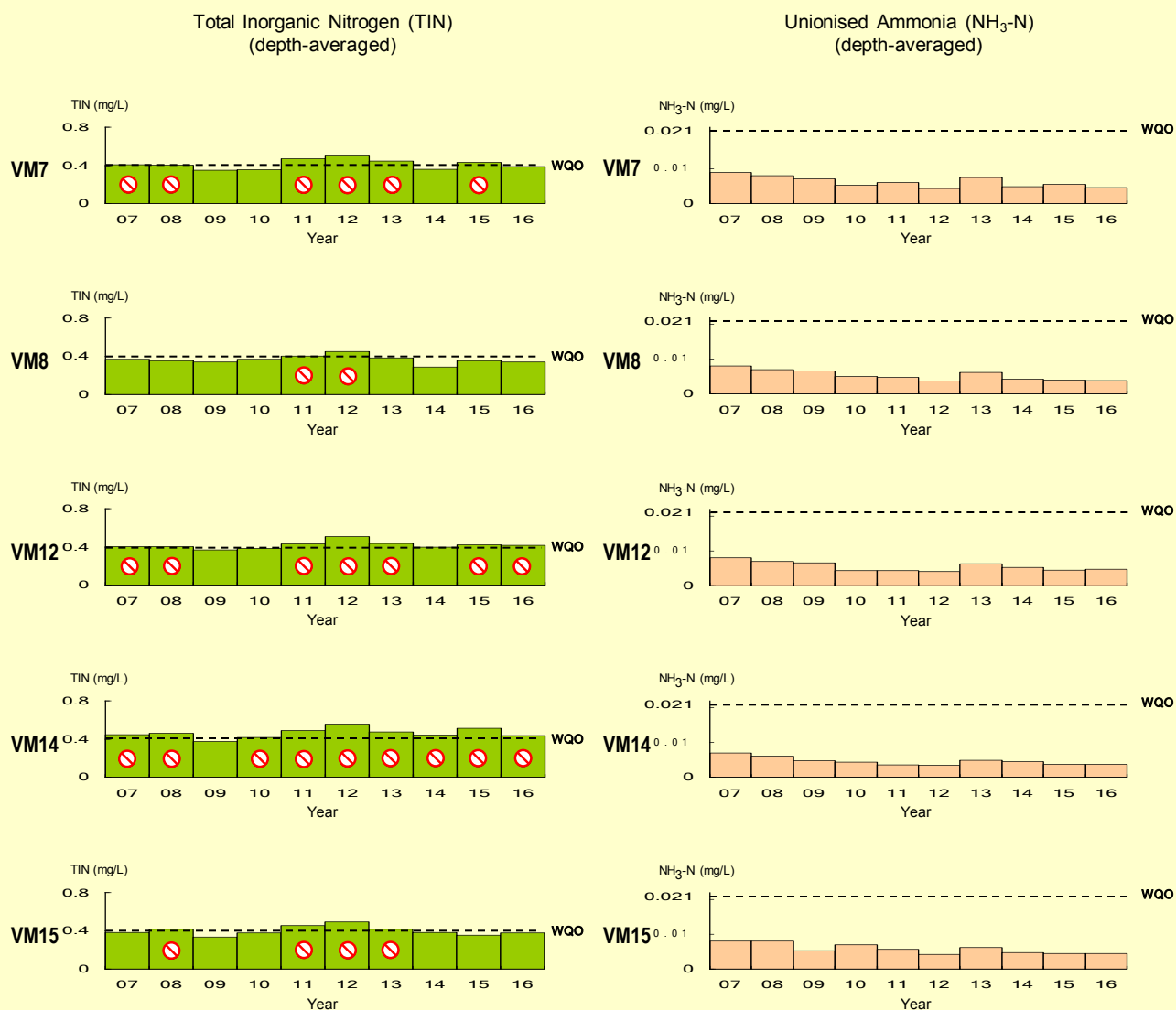
WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

■ annual mean for depth-averaged NH₃-N



⊘ Non-compliance

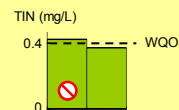
Levels of compliance with key Water Quality Objectives in the Victoria Harbour WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.4 mg/L

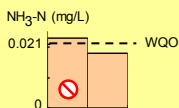
annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

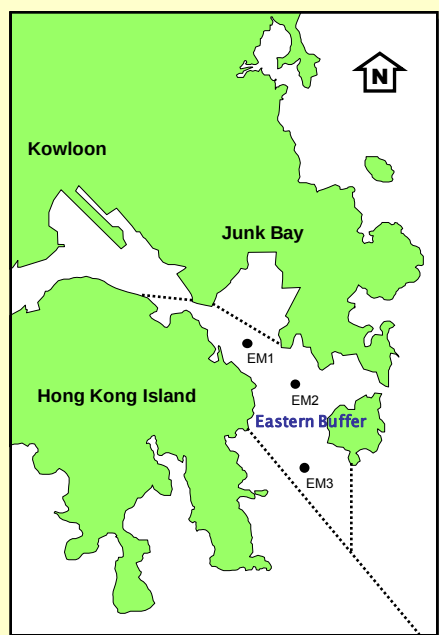
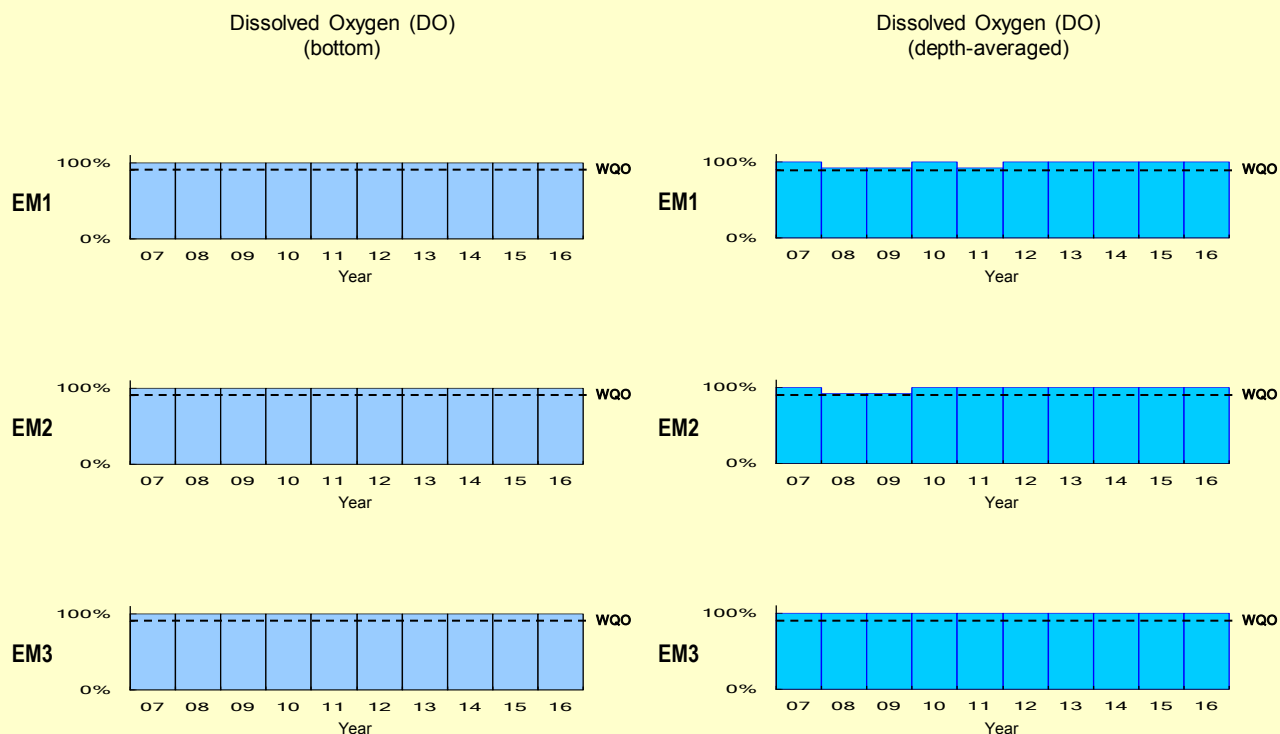
WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

annual mean for depth-averaged NH₃-N



Non-compliance

Levels of compliance with key Water Quality Objectives in the Eastern Buffer WCZ

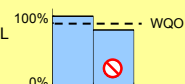


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

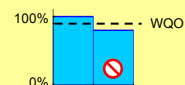
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

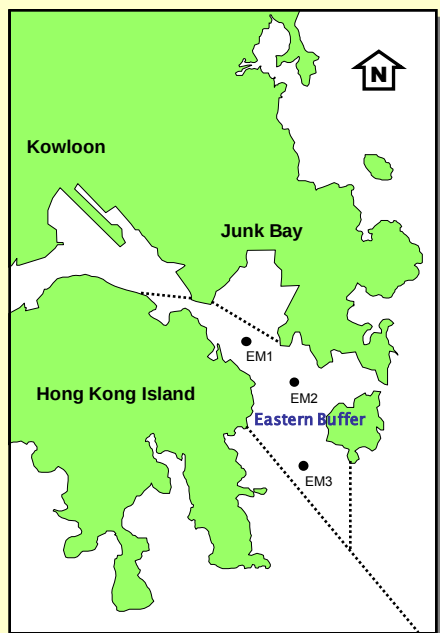
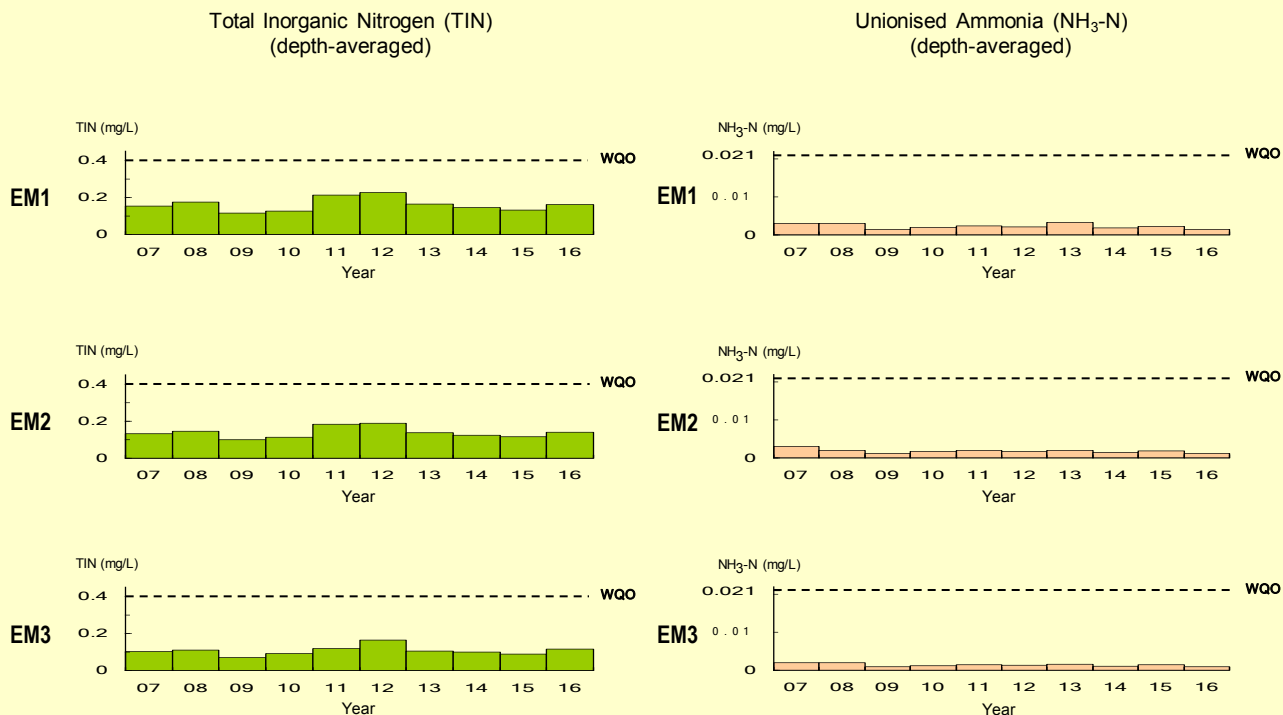
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

% sample with depth-averaged DO ≥ 4 mg/L



Non-compliance

Levels of compliance with key Water Quality Objectives in the Eastern Buffer WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.4 mg/L

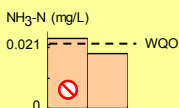
annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

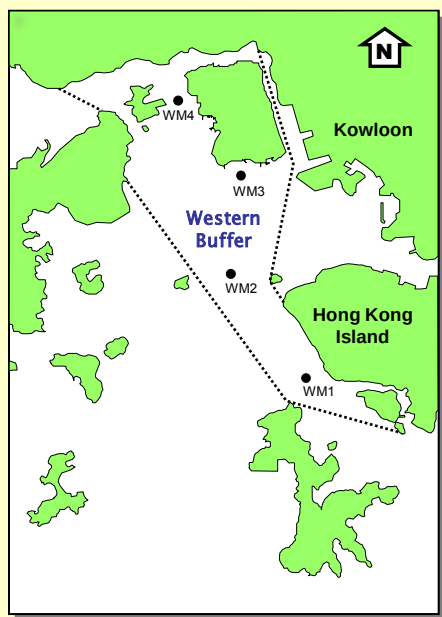
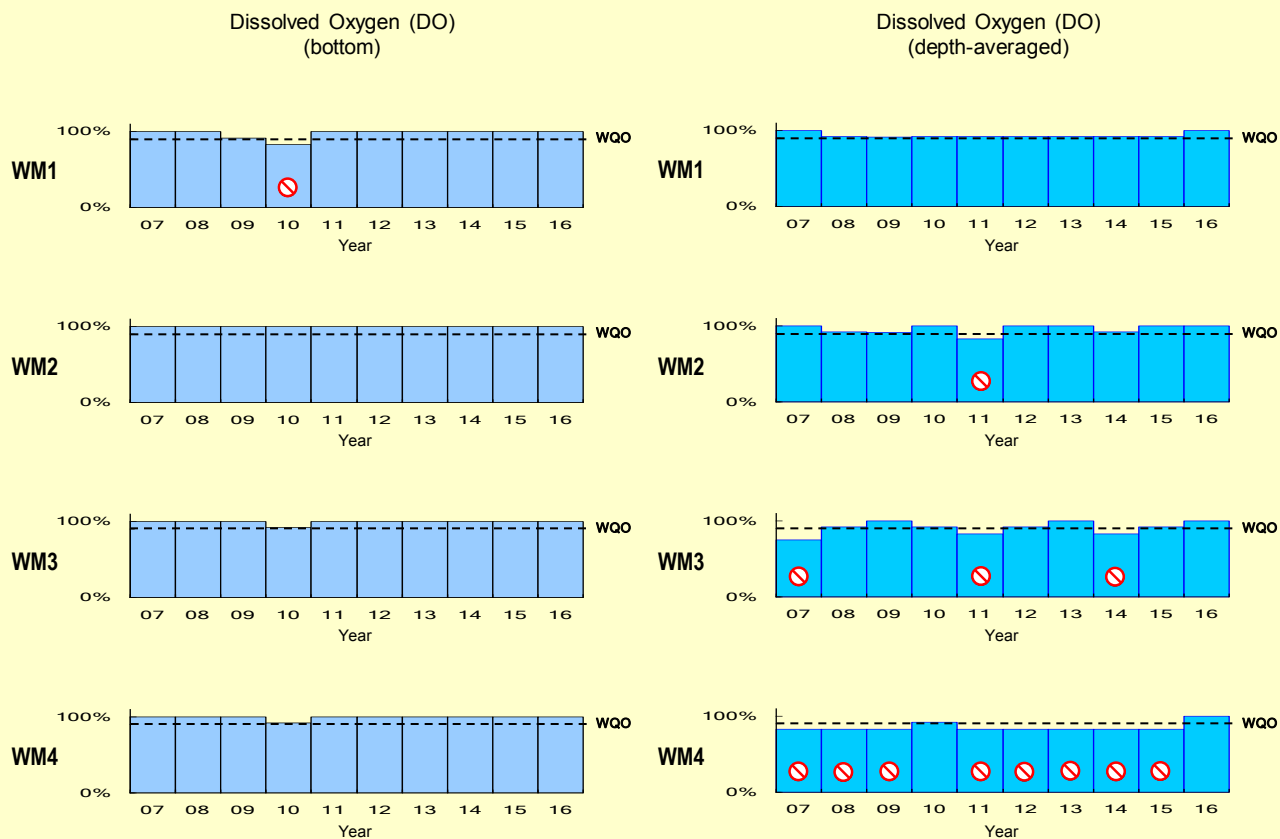
WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

annual mean for depth-averaged NH₃-N



Non-compliance

Levels of compliance with key Water Quality Objectives in the Western Buffer WCZ

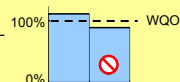


Dissolved Oxygen (DO)

1. Bottom

WQO: 90% sample with bottom DO ≥ 2 mg/L

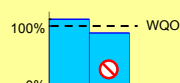
% sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

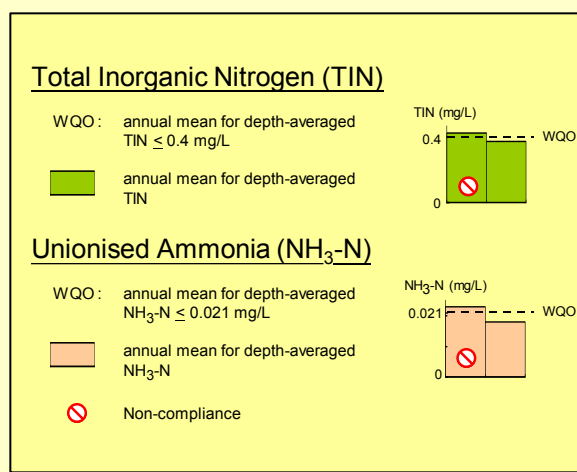
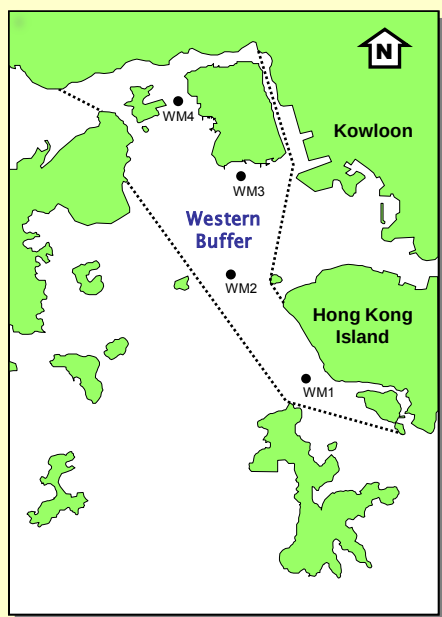
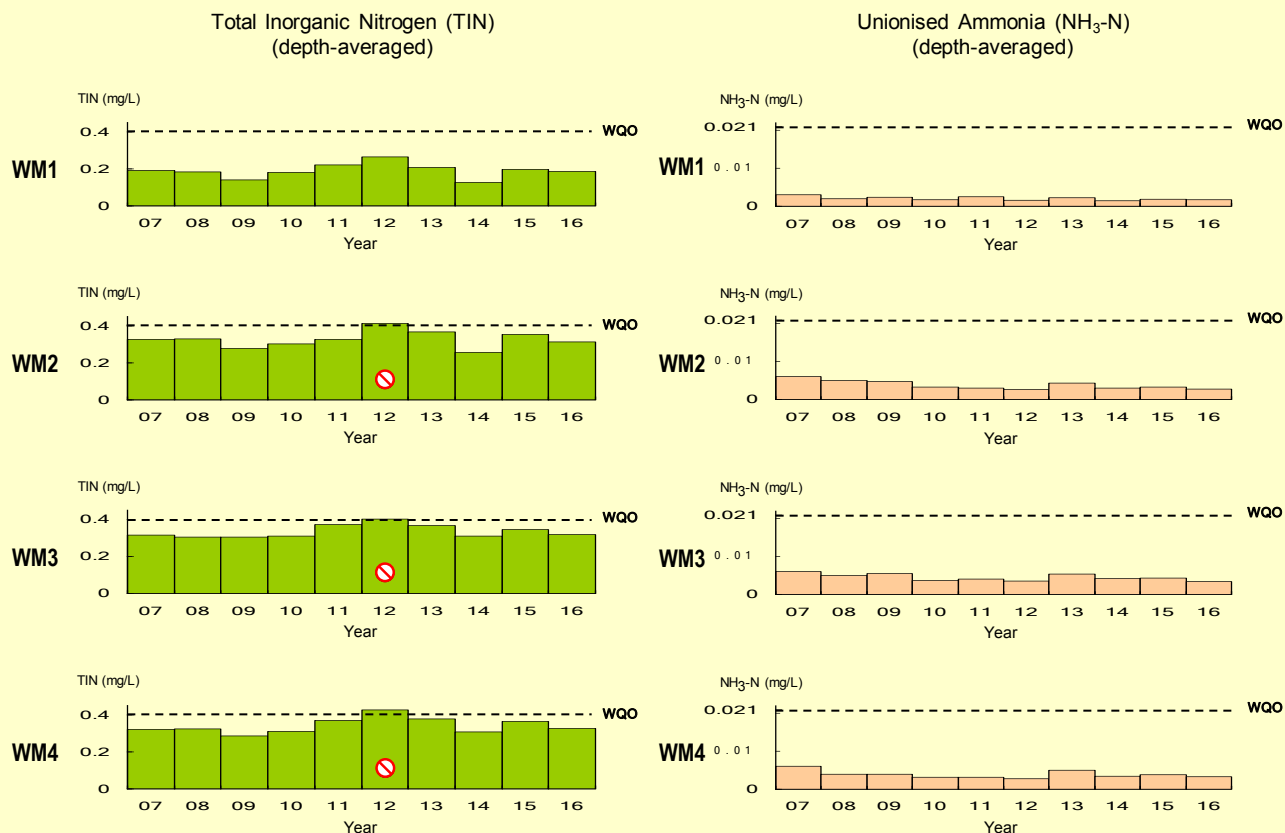
WQO: 90% sample with depth-averaged DO ≥ 4 mg/L

% sample with depth-averaged DO ≥ 4 mg/L

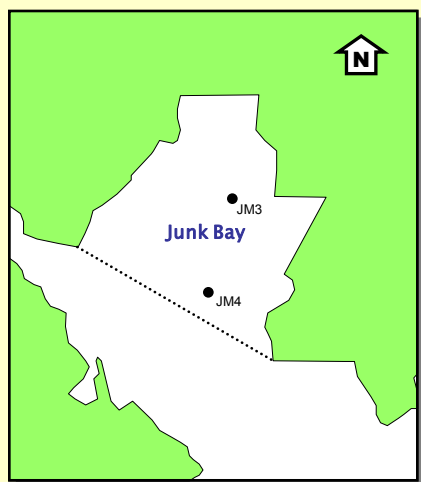
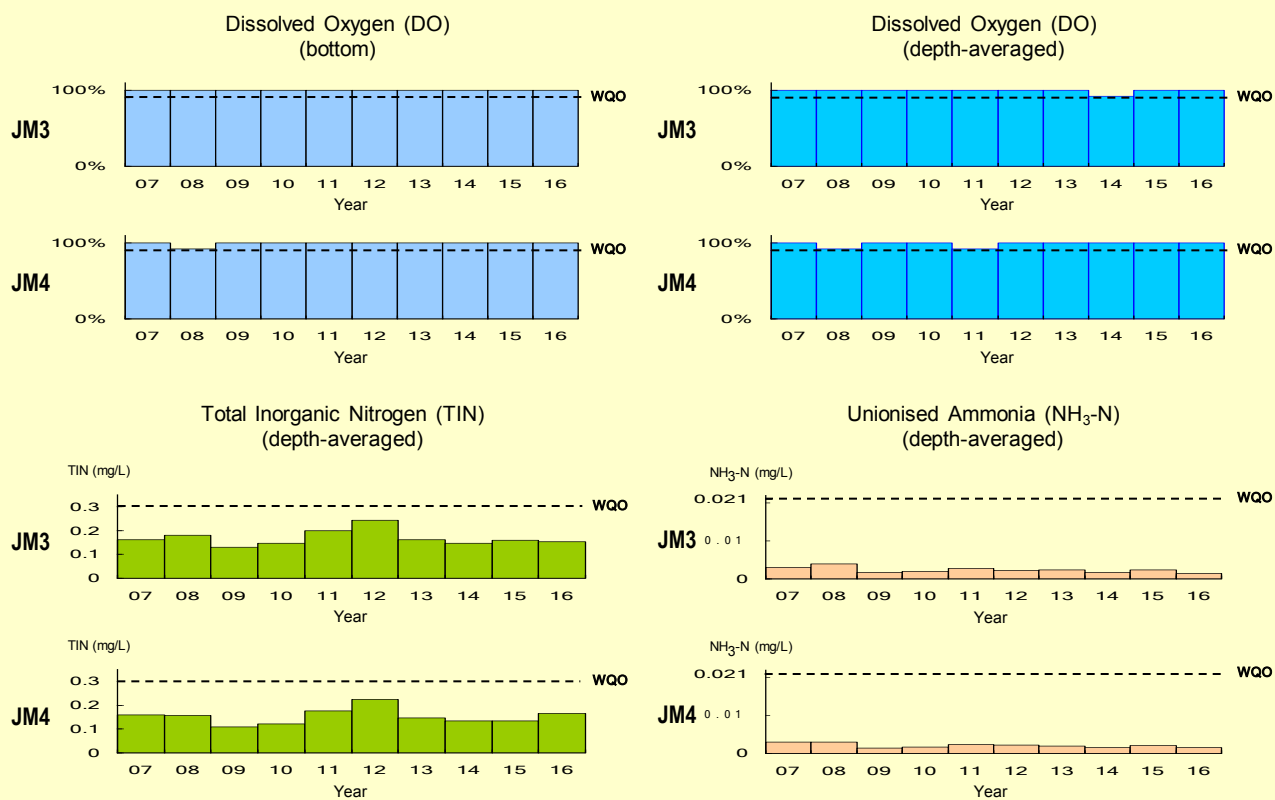


Non-compliance

Levels of compliance with key Water Quality Objectives in the Western Buffer WCZ (continued)



Levels of compliance with key Water Quality Objectives in the Junk Bay WCZ

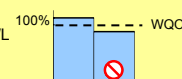


Dissolved Oxygen (DO)

1. Bottom

WQO: 90% sample with bottom DO $\geq$ 2 mg/L

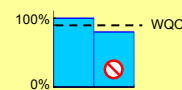
% sample with bottom DO $\geq$ 2 mg/L



2. Depth-averaged

WQO: 90% sample with depth-averaged DO $\geq$ 4 mg/L

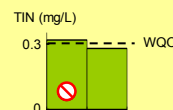
% sample with depth-averaged DO $\geq$ 4 mg/L



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN $\leq$ 0.3 mg/L

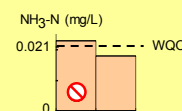
annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

WQO: annual mean for depth-averaged NH₃-N $\leq$ 0.021 mg/L

annual mean for depth-averaged NH₃-N



Non-compliance

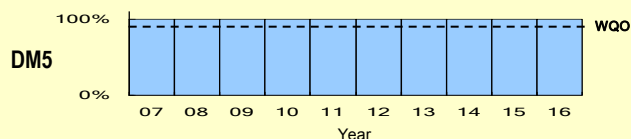
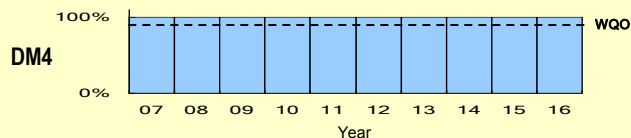
Levels of compliance with key Water Quality Objectives in the Deep Bay WCZ

Dissolved Oxygen (DO) (bottom)

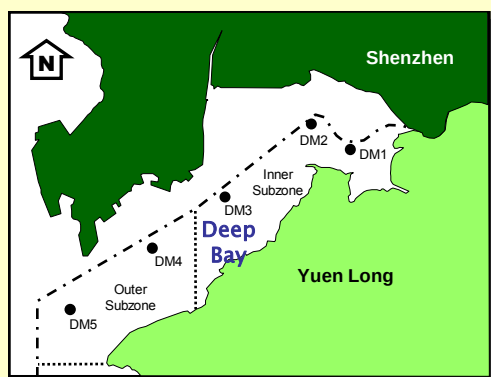
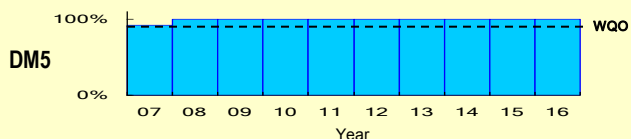
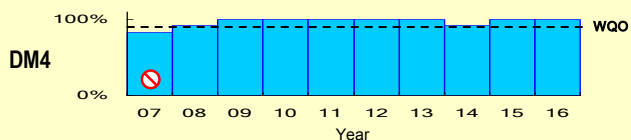
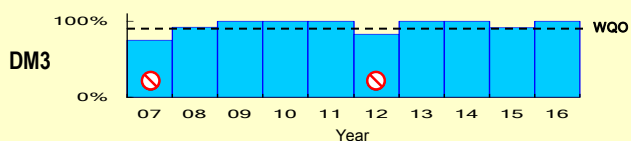
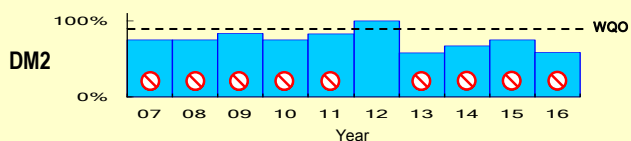
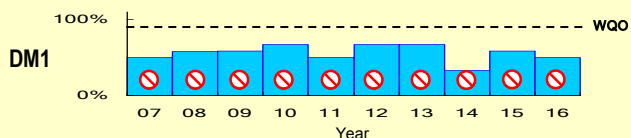
DM1 DO (bottom) not measured, water depth $\leq 3\text{m}$

DM2 DO (bottom) not measured, water depth $\leq 3\text{m}$

DM3 DO (bottom) not measured, water depth $\leq 3\text{m}$



Dissolved Oxygen (DO) (depth-averaged)

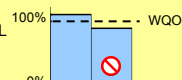


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO $\geq 2\text{ mg/L}$

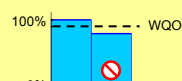
% sample with bottom DO $\geq 2\text{ mg/L}$



2. Depth-averaged

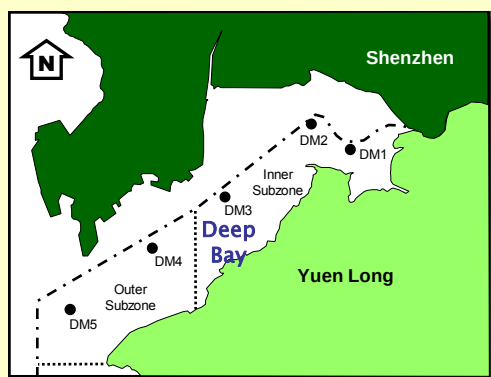
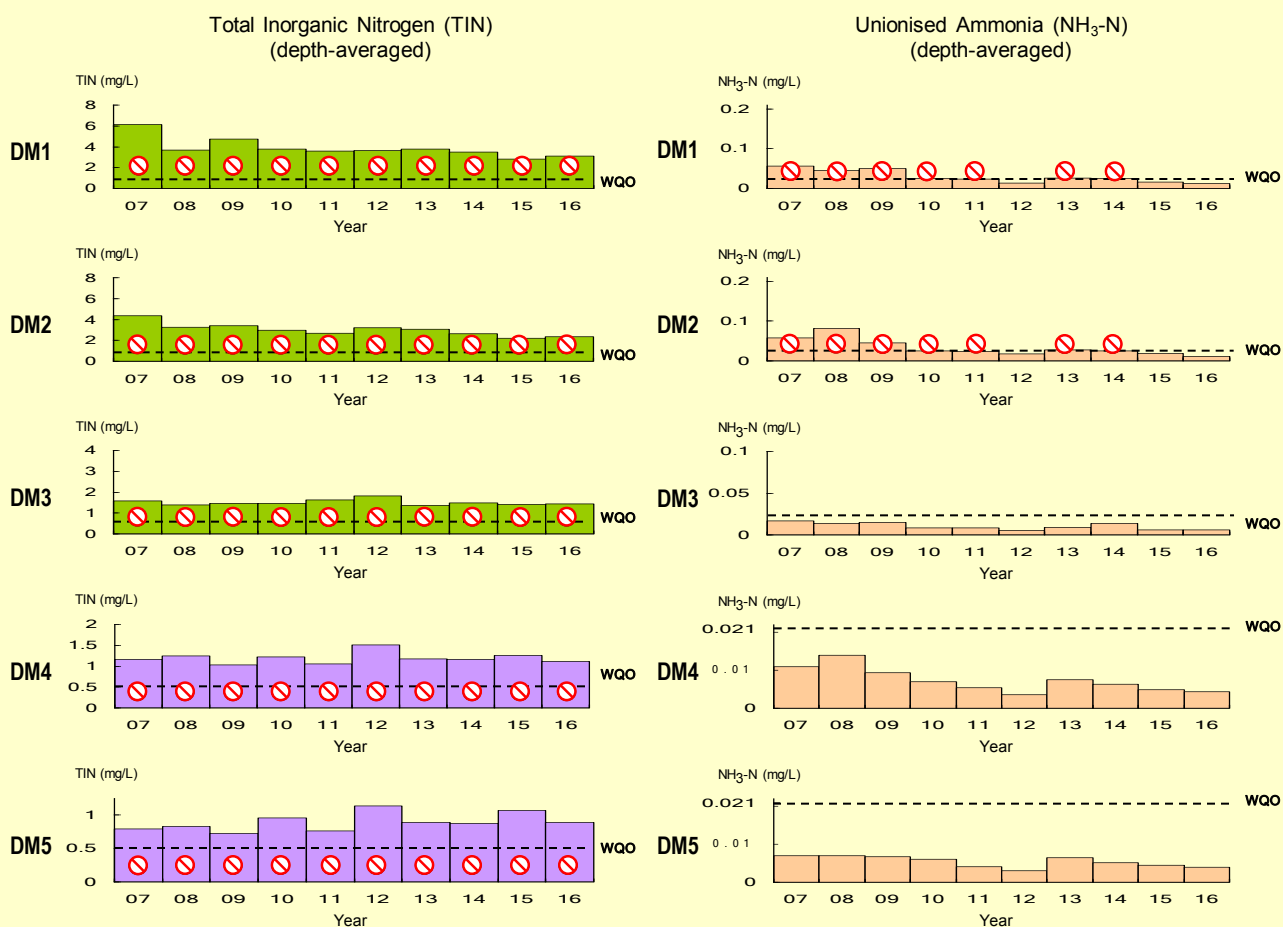
WQO : 90% sample with depth-averaged DO $\geq 4\text{ mg/L}$

% sample with depth-averaged DO $\geq 4\text{ mg/L}$



Non-compliance

Levels of compliance with key Water Quality Objectives in the Deep Bay WCZ (continued)



Total Inorganic Nitrogen (TIN)

Inner Subzone (DM1 - DM3)

WQO: annual mean for depth-averaged TIN ≤ 0.7 mg/L

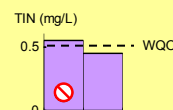
■ annual mean for depth-averaged TIN



Outer Subzone (DM4 - DM5)

WQO: annual mean for depth-averaged TIN ≤ 0.5 mg/L

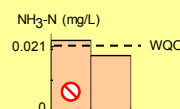
■ annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

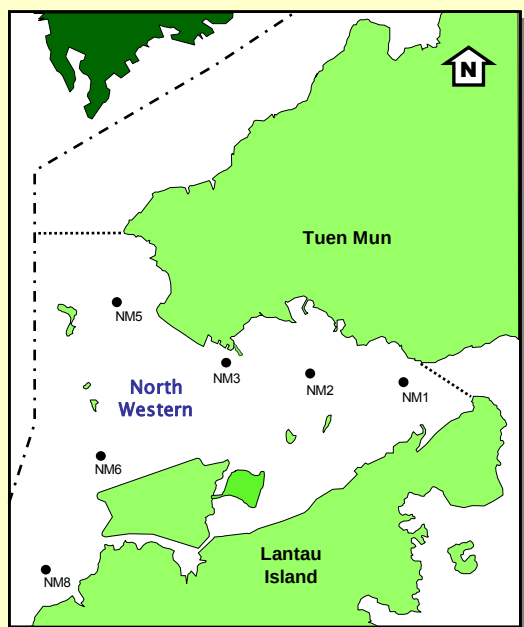
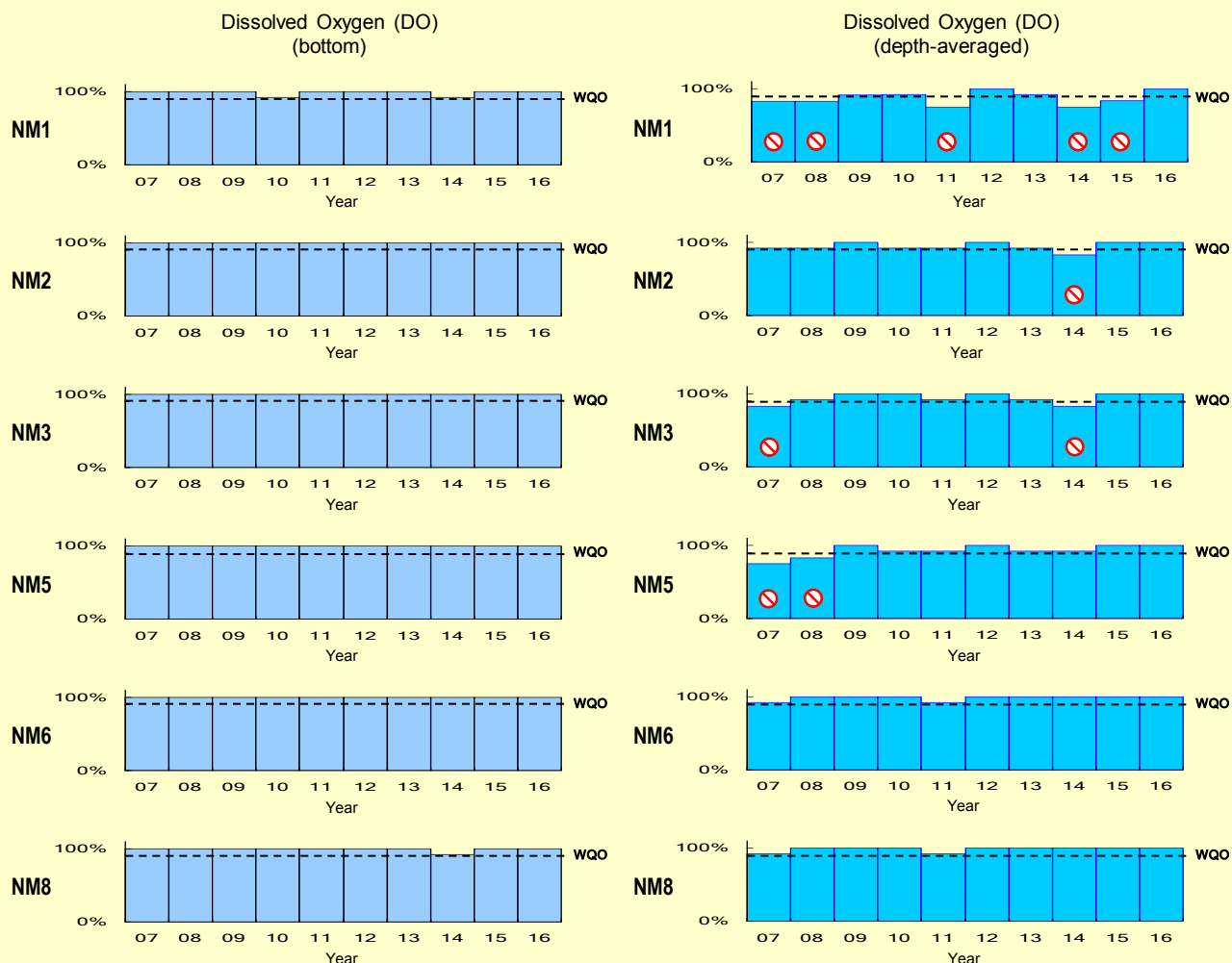
WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

■ annual mean for depth-averaged NH₃-N



○ Non-compliance

Levels of compliance with key Water Quality Objectives in the North Western WCZ

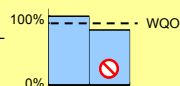


Dissolved Oxygen (DO)

1. Bottom

WQO : 90% sample with bottom DO ≥ 2 mg/L

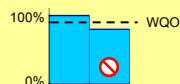
■ % sample with bottom DO ≥ 2 mg/L



2. Depth-averaged

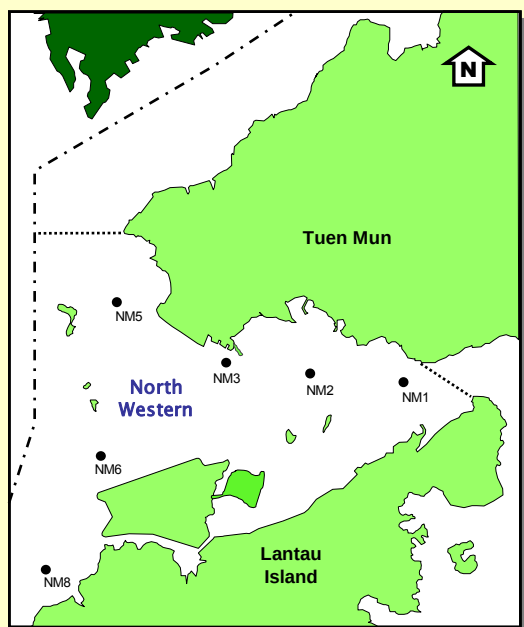
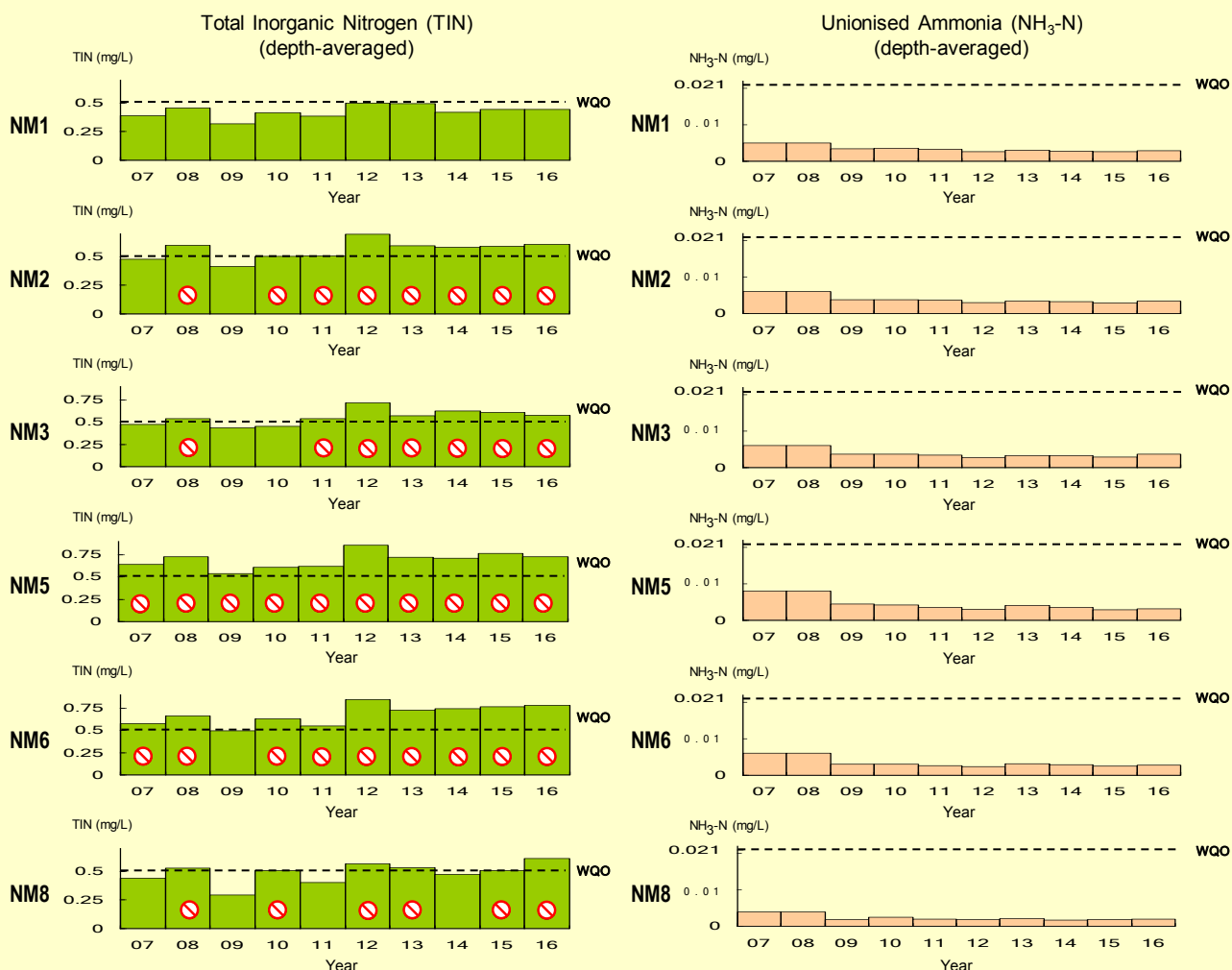
WQO : 90% sample with depth-averaged DO ≥ 4 mg/L

■ % sample with depth-averaged DO ≥ 4 mg/L



Non-compliance

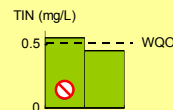
Levels of compliance with key Water Quality Objectives in the North Western WCZ (continued)



Total Inorganic Nitrogen (TIN)

WQO: annual mean for depth-averaged TIN ≤ 0.5 mg/L

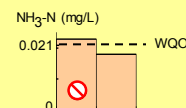
Green bar: annual mean for depth-averaged TIN



Unionised Ammonia (NH₃-N)

WQO: annual mean for depth-averaged NH₃-N ≤ 0.021 mg/L

Orange bar: annual mean for depth-averaged NH₃-N



Red circle with slash: Non-compliance

Results of the Seasonal Kendall Test for water quality trends in the Mirs Bay WCZ, 1991–2016

Monitoring Station		MM1	MM2	MM3	MM4	MM5	MM6	MM7
Monitoring Period		1991 I 2016	1991 I 2016	1991 I 2016	1991 I 2016	1991 I 2016	1991 I 2016	1991 I 2016
Parameter	Water Depth							
Temperature (°C)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	↗	-	-	-	-	↗
	Average	-	-	-	-	-	-	-
Salinity	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (% saturation)	Surface	↘	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↘	↘	↘	↘	-	↘
Turbidity (NTU)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Suspended Solids (mg/L)	Surface	-	↗	↗	↗	↗	-	↗
	Middle	-	↗	↗	↗	↗	-	↗
	Bottom	↘	↗	↗	↗	↗	-	↗
	Average	-	↗	↗	↗	↗	-	↗
Total volatile solids (mg/L)	Surface	-	↗	↗	-	-	-	↗
	Middle	-	↗	↗	-	-	-	↗
	Bottom	-	↗	↗	-	-	-	↗
	Average	-	↗	↗	-	↗	-	↗
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	-	-	↘	-
	Middle	↘	↘	↘	-	-	↘	-
	Bottom	↘	↘	↘	-	-	↘	-
	Average	↘	↘	↘	-	-	↘	-
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	↗	-	-	-	-	-	-
	Average	↗	↗	↗	↗	-	-	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	-	↗
	Middle	↗	↗	↗	↗	↗	-	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	↗	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Total nitrogen (mg/L)	Surface	↘	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	-	-	-	↘	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	↘	↘	↘	-	-	-
	Average	↘	↘	↘	↘	↘	-	-
Silica (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	↗	↗	↗	↗	-	↗
	Middle	-	↗	↗	↗	↗	-	↗
	Bottom	-	↗	↗	↗	↗	-	↗
	Average	-	↗	↗	↗	↗	↗	↗
<i>E. coli</i> (cfu/100mL)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	-	-	-	-	-	-
	Middle	↘	-	-	-	-	-	-
	Bottom	↘	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Mirs Bay WCZ, 1986 – 2016

Monitoring Station		MM8	MM13	MM14	MM15	MM16	MM17	MM19
Monitoring Period		1991 2016	1991 2016	1994 2016	1994 2016	1994 2016	1986 2016	2001 2016
Parameter	Water Depth							
Temperature (°C)	Surface	↗	-	-	-	-	↗↗↗	-
	Middle	-	-	-	-	-	↗↗↗	-
	Bottom	-	-	-	-	-	↗↗↗	-
	Average	-	-	-	-	-	↗↗↗	-
Salinity	Surface	-	-	-	-	-	-	↘
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	-	-	-	↘	-
	Middle	↘↘↘	-	-	-	↘	-	-
	Bottom	↘↘↘	-	-	-	↘	-	-
	Average	↘	-	-	-	-	-	-
Dissolved Oxygen (% saturation)	Surface	↗	-	-	-	-	↘	-
	Middle	-	-	-	-	↘	-	-
	Bottom	-	-	-	-	↘	-	-
	Average	-	-	-	-	-	-	-
pH	Surface	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘
	Middle	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘
	Bottom	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘
	Average	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘
Secchi disc depth (m)		-	↘	↘	-	↘	↘	-
Turbidity (NTU)	Surface	-	-	-	-	-	-	↘↘↘
	Middle	-	-	-	-	-	-	↘↘↘
	Bottom	-	-	↘	↘	↘	-	↘↘↘
	Average	-	-	-	↘	↘	-	↘↘↘
Suspended Solids (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	↗	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Total volatile solids (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	↗	↗	-	-	-
	Average	-	-	↗	↗	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘↘↘	↘↘↘	↘	↘	-	-	↘
	Middle	↘↘↘	↘↘↘	↘	↘	-	-	↘
	Bottom	↘↘↘	↘↘↘	↘	↘	-	↘	↘
	Average	↘↘↘	↘↘↘	-	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘↘↘	↘↘↘	↘↘↘	-	↘	-	↘
	Middle	↘↘↘	↘↘↘	↘↘↘	-	↘	-	↘
	Bottom	↘↘↘	↘↘↘	↘↘↘	↘	↘↘	-	↘
	Average	↘↘↘	↘↘↘	↘↘↘	-	↘↘	-	↘
Nitrite nitrogen (mg/L)	Surface	↗↗↗	-	-	-	-	-	↗
	Middle	↗↗↗	-	-	-	-	-	↗
	Bottom	↗↗↗	↗	↗↗	↗	-	-	↗↗↗
	Average	↗↗↗	↗	↗↗	↗	-	-	↗↗↗
Nitrate nitrogen (mg/L)	Surface	↗↗↗	-	↗↗	-	↗↗	-	↗↗↗
	Middle	↗↗↗	↗	↗↗	↗	↗↗	↗	↗↗↗
	Bottom	↗↗↗	↗	↗↗	↗	↗↗	↗	↗↗↗
	Average	↗↗↗	-	↗↗	↗	↗↗	↗	↗↗↗
Total inorganic nitrogen (mg/L)	Surface	-	-	-	-	-	-	↗↗↗
	Middle	-	-	-	-	-	-	↗↗↗
	Bottom	-	-	↗	-	↗	↗↗	↗↗↗
	Average	-	-	-	-	-	↗↗	↗↗↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-	-	↗↗↗
	Middle	-	-	-	-	-	-	↗↗↗
	Bottom	-	-	-	-	-	-	↗↗↗
	Average	-	-	-	-	-	-	↗↗↗
Total nitrogen (mg/L)	Surface	-	-	-	-	-	↘	↗↗↗
	Middle	-	-	-	-	-	-	↗↗↗
	Bottom	-	-	-	-	-	-	↗↗↗
	Average	-	-	-	-	-	-	↗↗↗
Orthophosphate phosphorus (mg/L)	Surface	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	-
	Middle	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	-
	Bottom	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	-
	Average	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	↘↘↘	-
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	↘	-	-	-	-	-
	Average	↘	↘	-	-	-	↘	-
Silica (mg/L)	Surface	-	-	-	-	-	↘↘↘	↗↗↗
	Middle	-	-	-	-	-	↘↘↘	↗↗↗
	Bottom	-	-	-	-	-	↘↘↘	↗↗↗
	Average	-	-	-	-	-	↘↘↘	↗↗↗
Chlorophyll- <i>a</i> (µg/L)	Surface	↗	-	↘↘	↘↘	-	↗↗↗	-
	Middle	-	-	↘↘	↘↘	-	↗↗↗	↘
	Bottom	-	-	↘↘	↘↘	-	↗↗↗	-
	Average	-	-	-	-	-	↗↗↗	-
<i>E. coli</i> (cfu/100mL)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Port Shelter WCZ, 1986 – 2016

Monitoring Station		PM1	PM2	PM3	PM4	PM6	PM7	PM8	PM9	PM11
Monitoring Period		1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1993 2016
Parameter	Water Depth									
Temperature (°C)	Surface	-	-	-	-	↗	-	↗	-	-
	Middle	↗	-	↗	↗	↗	↗	↗	↗	-
	Bottom	↗	-	↗	↗	↗	↗	↗	↗	-
	Average	↗	-	↗	↗	↗	↗	↗	↗	-
Salinity	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	↘	↘	↘	↘	-	↘	-
	Middle	-	-	-	↘	-	-	-	-	-
	Bottom	↘	-	-	↘	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Dissolved Oxygen (% saturation)	Surface	↗	↗	↗	↗	↗	↗	-	-	-
	Middle	↗	↗	↗	↗	-	-	-	-	-
	Bottom	↗	↗	↗	↗	-	-	-	-	-
	Average	↗	↗	↗	↗	-	-	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↘	↘	-	-	↘	-	-	-	-
Turbidity (NTU)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Suspended Solids (mg/L)	Surface	↘	-	↘	↘	↘	↘	↘	-	-
	Middle	↘	↘	↘	↘	↘	-	↘	-	-
	Bottom	↘	↘	↘	↘	↘	-	↘	-	-
	Average	↘	↘	↘	↘	↘	-	↘	-	-
Total volatile solids (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	-	↘
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	↘	-	-	↘	↘	-	-	↘
	Middle	-	↘	-	-	↘	↘	↘	↘	↘
	Bottom	-	↘	-	-	↘	↘	↘	↘	↘
	Average	-	↘	-	-	↘	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	↘	↘	↘	↘
	Average	-	-	-	-	-	↘	↘	↘	↘
Nitrate nitrogen (mg/L)	Surface	-	-	-	-	-	↘	↘	↘	↘
	Middle	-	-	-	-	-	↘	↘	↘	↘
	Bottom	-	-	-	-	-	↘	↘	↘	↘
	Average	-	-	-	-	-	↘	↘	↘	↘
Total inorganic nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	↘	↘	↘	↘
	Average	-	-	-	-	-	↘	↘	↘	↘
Total Kjeldahl nitrogen (mg/L)	Surface	-	↘	↘	↘	↘	-	↘	↘	-
	Middle	-	↘	↘	↘	↘	-	↘	↘	-
	Bottom	-	↘	↘	↘	↘	-	↘	↘	-
	Average	-	↘	↘	↘	↘	-	↘	↘	-
Total nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	↘	↘	↘	↘	-	-	-	-
	Middle	↘	↘	↘	↘	↘	-	-	-	-
	Bottom	↘	↘	↘	↘	↘	-	-	-	-
	Average	↘	↘	↘	↘	↘	-	-	-	-
Silica (mg/L)	Surface	↘	↘	↘	-	-	↘	↘	↘	-
	Middle	↘	↘	↘	-	-	↘	↘	↘	-
	Bottom	↘	↘	↘	-	-	↘	↘	↘	-
	Average	↘	↘	↘	-	-	↘	↘	↘	-
Chlorophyll- <i>a</i> (µg/L)	Surface	↘	↘	↘	↘	-	↘	↘	↘	-
	Middle	↘	↘	↘	↘	-	↘	↘	↘	-
	Bottom	↘	↘	↘	↘	-	↘	↘	↘	-
	Average	↘	↘	↘	↘	-	↘	↘	↘	-
<i>E. coli</i> (cfu/100mL)	Surface	-	↘	-	-	↘	-	-	-	-
	Middle	-	↘	-	-	-	-	-	-	-
	Bottom	-	↘	-	-	-	-	-	-	-
	Average	-	↘	-	-	↘	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	-	-	-	↘	-	-	-	-
	Middle	-	↘	-	-	↘	-	-	-	-
	Bottom	-	↘	-	-	↘	-	-	-	-
	Average	-	↘	-	-	↘	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Tolo Harbour and Channel WCZ, 1986 – 2016

Monitoring Station		TM2	TM3	TM4	TM5	TM6	TM7	TM8
Monitoring Period		1986 2016	1986 2016	1986 2016	1988 2016	1986 2016	1988 2016	1986 2016
Parameter	Water Depth							
Temperature (°C)	Surface	↗	↗	↗	↗	↗	↗	↗
	Middle	NA	↗	↗	NA	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↗	↘	↘	-	↘	↘	↘
	Average	↗	↘	↘	-	↘	↘	↘
Dissolved Oxygen (% saturation)	Surface	-	↘	↘	↘	↘	↘	↘
	Middle	NA	↗	↘	NA	↗	↘	↗
	Bottom	↗	↘	↘	↗	↗	↘	↗
	Average	↗	↘	↘	↗	↗	↘	↗
pH	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	-
Turbidity (NTU)	Surface	-	-	-	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Suspended Solids (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	-	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total volatile solids (mg/L)	Surface	↘	↘	↘	-	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	-	↘	↘	↘	↘	↘
	Average	↘	-	↘	-	↘	↘	↘
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	-	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↘	↘	↘	-	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	-	↘	↘	↘
	Average	↘	↘	↘	-	↘	↘	↘
Nitrate nitrogen (mg/L)	Surface	↘	↘	↘	-	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total inorganic nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	NA	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	-	↘	↘	-	-	-	-
	Middle	NA	-	-	NA	-	-	-
	Bottom	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	↘	↘	↘	-	↘	-	↘
	Middle	NA	↘	↘	NA	-	↘	↘
	Bottom	-	-	-	-	-	↘	↘
	Average	↘	-	↘	-	-	↘	↘
<i>E. coli</i> (cfu/100mL)	Surface	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	NA	↘	-	-
	Bottom	↘	↘	↘	↘	↘	-	-
	Average	↘	↘	↘	↘	↘	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	NA	-	-	-
	Bottom	↘	↘	↘	-	-	-	-
	Average	↘	↘	↘	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. NA (Not Applicable) indicates the measurement was not made due to shallow water

4. ↗ significant increase

5. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Southern WCZ, 1986 – 2016

Monitoring Station		SM1	SM2	SM3	SM4	SM5	SM6	SM7	SM9
Monitoring Period		1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1986 2016	1988 2016
Parameter	Water Depth								
Temperature (°C)	Surface	-	-	↗	↗	↗	↗	↗	↗
	Middle	-	-	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	↗
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	↘	-	-	-	-	-
Dissolved Oxygen (% saturation)	Surface	-	-	-	-	-	-	-	↗
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		-	↗	↗	↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-	-	-	-	↘
	Middle	-	-	-	-	-	-	-	↘
	Bottom	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	↘
Suspended Solids (mg/L)	Surface	-	-	↘	↘	-	-	-	↘
	Middle	-	-	-	↘	-	-	-	↘
	Bottom	-	-	-	↘	-	-	-	↘
	Average	-	-	-	↘	-	-	-	↘
Total volatile solids (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	-	-	↗	↗	-	-	-	-
	Middle	-	-	-	↗	-	-	-	-
	Bottom	-	-	-	-	-	-	-	↗
	Average	-	-	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	-	-	↘	-	↘
	Middle	↘	↘	↘	-	-	↘	-	↘
	Bottom	↘	↘	↘	-	-	↘	-	↘
	Average	↘	↘	↘	-	-	↘	-	↘
Nitrite nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	-	↗	↗
	Middle	↗	↗	↗	↗	↗	-	↗	↗
	Bottom	↗	↗	↗	↗	↗	-	↗	↗
	Average	↗	↗	↗	↗	↗	-	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	↘	-	-	-	-	↘	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Total nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	-	-	↘	-	-	↘	↘
	Middle	-	-	-	↘	-	-	↘	↘
	Bottom	↘	↘	↘	↘	↘	-	↘	↘
	Average	↘	↘	↘	↘	↘	-	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	↗	↗	-	-	↗	↗	↗
	Middle	-	-	-	-	-	-	↗	↗
	Bottom	-	-	-	-	-	-	↗	↗
	Average	-	-	-	-	-	-	↗	↗
<i>E. coli</i> (cfu/100mL)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	-	-	-	-	-	-	-
	Middle	-	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	↘	-	-	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Southern WCZ, 1986 – 2016 (continued)

Monitoring Station		SM10	SM11	SM12	SM13	SM17	SM18	SM19	SM20
Monitoring Period		1986 2016	1986 2016	1986 2016	1986 2016	1989 2016	1989 2016	1989 2016	1999 2016
Parameter	Water Depth								
Temperature (°C)	Surface	↗	↗	↗	↗	↗	↗	-	-
	Middle	NA	↗	↗	↗	↗	↗	-	-
	Bottom	↗	↗	↗	↗	↗	↗	-	-
	Average	↗	↗	↗	↗	↗	↗	-	-
Salinity	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	↘	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Dissolved Oxygen (% saturation)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	↗	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
pH	Surface	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	-	↗
Turbidity (NTU)	Surface	-	-	-	-	-	-	-	↘
	Middle	NA	-	-	-	-	-	-	↘
	Bottom	-	-	-	-	-	-	-	↘
	Average	-	-	-	-	-	-	-	↘
Suspended Solids (mg/L)	Surface	↘	↘	-	-	-	↘	-	↘
	Middle	NA	↘	-	-	↘	↘	-	↘
	Bottom	↘	↘	-	-	-	↘	-	↘
	Average	↘	↘	-	-	↘	↘	-	↘
Total volatile solids (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	↘	↘	-
	Bottom	-	-	-	-	-	↘	↘	-
	Average	-	-	-	-	-	↘	↘	-
Ammonia nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	↗	-	-	-	-	-	-
	Bottom	-	↗	-	-	-	-	-	-
	Average	-	↗	-	-	-	-	-	-
Unionised Ammonia (mg/L)	Surface	-	-	-	↘	↘	-	↘	-
	Middle	NA	-	-	-	↘	-	↘	-
	Bottom	-	-	-	-	↘	-	↘	-
	Average	-	-	-	-	↘	-	↘	-
Nitrite nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	↗	↗	↗
	Middle	NA	↗	↗	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗	↗	↗	↗
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	↗	↗	-
	Middle	NA	↗	↗	↗	↗	↗	↗	-
	Bottom	↗	↗	↗	↗	↗	↗	↗	-
	Average	↗	↗	↗	↗	↗	↗	↗	-
Total inorganic nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗	-	-	↗
	Middle	NA	↗	↗	↗	↗	-	-	↗
	Bottom	↗	↗	↗	↗	↗	-	-	↗
	Average	↗	↗	↗	↗	↗	-	-	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	↗
	Middle	NA	-	-	-	-	-	-	↗
	Bottom	-	-	-	-	-	-	-	↗
	Average	-	-	-	-	-	-	-	↗
Total nitrogen (mg/L)	Surface	-	-	-	-	-	-	-	↗
	Middle	NA	-	-	-	-	-	-	↗
	Bottom	-	-	-	-	-	-	-	↗
	Average	-	-	-	-	-	-	-	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	-
	Middle	NA	↘	↘	↘	↘	↘	↘	-
	Bottom	↘	↘	↘	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	↘	↘	↘	-
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	-	-
	Middle	NA	↘	↘	↘	-	-	-	-
	Bottom	↘	↘	↘	↘	↘	↘	-	-
	Average	↘	↘	↘	↘	↘	↘	-	-
Silica (mg/L)	Surface	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	↗	↗	↗	↗	↗	-	-	-
	Middle	NA	↗	↗	↗	↗	-	-	-
	Bottom	↗	↗	↗	↗	↗	-	-	-
	Average	↗	↗	↗	↗	↗	-	-	-
<i>E. coli</i> (cfu/100mL)	Surface	↘	↘	-	-	-	-	-	-
	Middle	NA	↘	↗	-	-	-	-	-
	Bottom	↘	↘	↗	-	-	-	-	-
	Average	↘	↘	↗	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	↘	↘	↗	-	-	-	-	-
	Middle	NA	↘	↗	-	-	-	-	-
	Bottom	↘	↘	↗	-	-	-	-	-
	Average	↘	↘	↗	-	-	-	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. NA (Not Applicable) indicates the measurement was not made due to shallow water

4. ↗ significant increase

5. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Victoria Harbour WCZ, 1986 – 2016

Monitoring Station		VM1	VM2	VM4	VM5	VM6
Monitoring Period		1988 2016	1988 2016	1988 2016	1986 2016	1988 2016
Parameter	Water Depth					
Temperature (°C)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Salinity	Surface	-	-	-	-	-
	Middle	↗	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Dissolved Oxygen (% saturation)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
pH	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Suspended Solids (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total volatile solids (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	↘	↘	-	-
	Middle	↘	↘	↘	-	-
	Bottom	↘	↘	↘	-	-
	Average	↘	↘	↘	-	-
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-	-	↘	↘
	Middle	↘	-	-	↘	-
	Bottom	-	-	-	↘	-
	Average	-	-	-	↘	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↘	↘	↘	-	-
	Middle	↘	↘	↘	-	-
	Bottom	↘	↘	↘	-	↘
	Average	↘	↘	↘	-	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-	↗	-	-
	Middle	-	-	↗	-	-
	Bottom	-	↗	-	-	-
	Average	-	-	-	-	-
<i>E. coli</i> (cfu/100mL)	Surface	↘	↘	↘	-	↘
	Middle	↘	↘	↘	-	↘
	Bottom	↘	↘	↘	-	↘
	Average	↘	↘	↘	-	↘
Faecal coliforms (cfu/100mL)	Surface	↘	↘	-	↗	-
	Middle	↘	↘	-	-	-
	Bottom	↘	↘	-	-	-
	Average	↘	↘	-	↗	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Victoria Harbour WCZ, 1986 – 2016 (continued)

Monitoring Station		VM7	VM8	VM1 2	VM1 4	VM1 5
Monitoring Period		1986 2016	1986 2016	1986 2016	1986 2016	1993 2016
Parameter	Water Depth					
Temperature (°C)	Surface	↗	↗	↗	↗	-
	Middle	↗	↗	↗	↗	-
	Bottom	↗	↗	↗	↗	-
	Average	↗	↗	↗	↗	-
Salinity	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	↗	-	↗
	Middle	↗	-	↗	↗	↗
	Bottom	↗	-	↗	↗	↗
	Average	↗	-	↗	↗	↗
Dissolved Oxygen (% saturation)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
pH	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Secchi disc depth (m)		↘	↘	↘	↘	↘
Turbidity (NTU)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	↘
	Average	-	-	-	-	↘
Suspended Solids (mg/L)	Surface	↘	↘	-	↘	↘
	Middle	↘	↘	-	↘	↘
	Bottom	↘	↘	-	↘	↘
	Average	↘	↘	-	↘	↘
Total volatile solids (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	↘	↘	-
	Middle	-	-	↘	↘	-
	Bottom	-	-	↘	↘	-
	Average	-	-	↘	↘	-
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	-
	Bottom	↗	↗	↗	↗	-
	Average	↗	↗	↗	↗	-
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	↗	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗	↗
	Average	↗	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	↗	-	↗	↘
	Middle	-	↗	-	↗	↘
	Bottom	-	↗	-	↗	↘
	Average	-	↗	-	↗	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘
	Middle	↘	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-	-	-	-
	Middle	-	-	-	-	-
	Bottom	-	-	-	-	-
	Average	-	-	-	-	-
<i>E. coli</i> (cfu/100mL)	Surface	-	-	↘	↘	↘
	Middle	-	-	-	↘	↘
	Bottom	-	↗	-	↘	↘
	Average	-	↗	-	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↗	↗	-	↘	↘
	Middle	↗	↗	-	↘	↘
	Bottom	↗	↗	-	↘	↘
	Average	↗	↗	-	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Eastern Buffer WCZ, 1986 – 2016

Monitoring Station		EM1	EM2	EM3
Monitoring Period		1986 2016	1986 2016	1988 2016
Parameter	Water Depth			
Temperature (°C)	Surface	↗	↗	↗
	Middle	↗	↗	↗
	Bottom	↗	↗	↗
	Average	↗	↗	↗
Salinity	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	-
	Middle	↗	↗	-
	Bottom	↗	-	-
	Average	↗	-	-
Dissolved Oxygen (% saturation)	Surface	↗	↗	-
	Middle	↗	↗	↗
	Bottom	↗	↗	-
	Average	↗	↗	-
pH	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Secchi disc depth (m)		↗	↗	-
Turbidity (NTU)	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
Suspended Solids (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	↘	-
	Average	↘	↘	-
Total volatile solids (mg/L)	Surface	-	-	-
	Middle	↘	-	-
	Bottom	↘	-	-
	Average	↘	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↘	-	-
	Middle	↘	-	-
	Bottom	↘	-	-
	Average	↘	-	-
Ammonia nitrogen (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	↘	-
	Middle	↘	↘	-
	Bottom	↘	↘	-
	Average	↘	↘	-
Nitrate nitrogen (mg/L)	Surface	↗	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
Total inorganic nitrogen (mg/L)	Surface	↘	↘	-
	Middle	↘	↘	-
	Bottom	↘	↘	-
	Average	↘	↘	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Silica (mg/L)	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-	-
	Middle	-	-	-
	Bottom	-	-	-
	Average	-	-	-
<i>E. coli</i> (cfu/100mL)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	↘	↘
	Middle	↘	↘	↘
	Bottom	↘	↘	↘
	Average	↘	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Western Buffer WCZ, 1986 – 2016

Monitoring Station		WM1	WM2	WM3	WM4
Monitoring Period		1988 2016	1988 2016	1986 2016	1986 2016
Parameter	Water Depth				
Temperature (°C)	Surface	↗	↗	↗	↗
	Middle	↗	↗	↗	↗
	Bottom	↗	↗	↗	↗
	Average	-	↗	↗	↗
Salinity	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	-	↘
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Dissolved Oxygen (% saturation)	Surface	-	-	↗	↘
	Middle	-	↗	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
pH	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Secchi disc depth (m)		↗	↗	↗	↗
Turbidity (NTU)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Suspended Solids (mg/L)	Surface	↘	↘	↘	↘
	Middle	-	↘	↘	↘
	Bottom	↗	↘	↘	↘
	Average	-	↘	↘	↘
Total volatile solids (mg/L)	Surface	-	-	↘	↘
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	↘	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	-	↗	-	↗
	Middle	-	↗	-	↗
	Bottom	↘	-	-	↗
	Average	-	↗	-	↗
Unionised Ammonia (mg/L)	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↗	↗	↗	↗
	Middle	-	↗	↗	↗
	Bottom	-	↗	↗	↗
	Average	↗	↗	↗	↗
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	↗
	Middle	-	↗	↗	↗
	Bottom	-	↗	↗	↗
	Average	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↗	↗	↗	↗
	Middle	-	↗	↗	↗
	Bottom	-	↗	↗	↗
	Average	-	↗	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	-	-	-	-
	Middle	↘	-	-	-
	Bottom	↘	-	↘	-
	Average	↘	-	-	-
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘	↘	↘
	Middle	↘	↘	↘	↘
	Bottom	↘	↘	↘	↘
	Average	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	-	-	-	-
	Average	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-	-	-
	Middle	-	-	-	-
	Bottom	↗	-	-	-
	Average	-	-	-	-
<i>E. coli</i> (cfu/100mL)	Surface	-	-	-	-
	Middle	↘	↗	-	-
	Bottom	↘	-	-	-
	Average	↘	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	↗	↗	-
	Middle	↘	↗	↗	↗
	Bottom	↘	↗	↗	↗
	Average	-	↗	↗	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Junk Bay WCZ, 1986 – 2016

Monitoring Station		JM3	JM4
Monitoring Period		1986 I 2016	1986 I 2016
Parameter	Water Depth		
Temperature (°C)	Surface	↗	↗
	Middle	↗	↗
	Bottom	↗	↗
	Average	↗	↗
Salinity	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Dissolved Oxygen (mg/L)	Surface	-	-
	Middle	↗	↗
	Bottom	↗	↗
	Average	↗	↗
Dissolved Oxygen (% saturation)	Surface	↗	↗
	Middle	↗	↗
	Bottom	↗	↗
	Average	↗	↗
pH	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Secchi disc depth (m)		↗	↘
Turbidity (NTU)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Suspended Solids (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total volatile solids (mg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Ammonia nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Unionised Ammonia (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Nitrate nitrogen (mg/L)	Surface	-	↘
	Middle	↗	↘
	Bottom	↗	↘
	Average	↗	↘
Total inorganic nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total Kjeldahl nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total nitrogen (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Total phosphorus (mg/L)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Silica (mg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-
	Middle	-	-
	Bottom	-	-
	Average	-	-
<i>E. coli</i> (cfu/100mL)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	↘
	Middle	↘	↘
	Bottom	↘	↘
	Average	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the Deep Bay WCZ, 1986 – 2016

Monitoring Station		DM1	DM2	DM3	DM4	DM5
Monitoring Period		1986 I 2016	1986 I 2016	1986 I 2016	1986 I 2016	1991 I 2016
Parameter	Water Depth					
Temperature (°C)	Surface	↗	↗	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	↗	↗	↗	↗	↗
Salinity	Surface	↘	↘	-	-	-
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↘	↘	-	-	↘
Dissolved Oxygen (mg/L)	Surface	-	↘	↘	↘	↘
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	NA	-
	Average	-	↘	↘	↘	-
Dissolved Oxygen (% saturation)	Surface	-	↘	↘	↘	↘
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	NA	-
	Average	-	↘	↘	↘	-
pH	Surface	↘	↘	↘	↘	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	↘	↘
	Average	↗	↘	↘	↘	↘
Secchi disc depth (m)		↗	-	-	-	-
Turbidity (NTU)	Surface	-	-	-	-	-
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	-	↘
	Average	-	-	-	-	↘
Suspended Solids (mg/L)	Surface	-	-	-	-	↘
	Middle	NA	NA	NA	NA	↘
	Bottom	NA	NA	NA	-	↘
	Average	-	-	-	↘	↘
Total volatile solids (mg/L)	Surface	-	↗	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	-	↗	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	↗	↗	-	NA	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	↗	↗	-	-	-
Ammonia nitrogen (mg/L)	Surface	-	-	-	↗	↗
	Middle	NA	NA	NA	↗	↗
	Bottom	NA	NA	NA	↗	↗
	Average	-	-	-	↗	↗
Unionised Ammonia (mg/L)	Surface	↘	-	-	NA	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	↘	-	-	-	-
Nitrite nitrogen (mg/L)	Surface	↗	↗	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	↗	↗	↗	↗	↗
Nitrate nitrogen (mg/L)	Surface	↗	-	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	↗	-	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	-	-	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	-	-	↗	↗	↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	-	-	-	-	-
Total nitrogen (mg/L)	Surface	-	-	↗	↗	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	↗	↗
	Average	-	-	↗	↗	↗
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	-	↗	-
	Middle	NA	NA	NA	NA	-
	Bottom	↘	↘	NA	-	-
	Average	↘	↘	-	-	-
Total phosphorus (mg/L)	Surface	↘	↘	↘	NA	↘
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	↘	↘	↘	-	↘
Silica (mg/L)	Surface	↗	↗	-	-	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	↗
	Average	↗	↗	-	-	-
Chlorophyll-a (µg/L)	Surface	↗	↗	-	-	↗
	Middle	NA	NA	NA	NA	↗
	Bottom	NA	NA	NA	-	↗
	Average	↗	↗	-	-	↗
<i>E. coli</i> (cfu/100mL)	Surface	-	-	-	NA	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	-	-
	Average	-	-	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	↗	-	↗	-
	Middle	NA	NA	NA	NA	-
	Bottom	NA	NA	NA	↗	-
	Average	-	↗	-	↗	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. NA (Not Applicable) indicates the measurement was not made due to shallow water

4. ↗ significant increase

5. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in the North Western WCZ, 1986 – 2016

Monitoring Station		NM1	NM2	NM3	NM5	NM6	NM8
Monitoring Period		1988 2016	1986 2016	1986 2016	1988 2016	1991 2016	1999 2016
Parameter	Water Depth						
Temperature (°C)	Surface	-	↗	↗↗	↗↗	↗↗	-
	Middle	-	-	-	-	-	-
	Bottom	-	-	↗↗	↗	↗↗	-
	Average	-	-	↗↗	↗	↗↗	-
Salinity	Surface	-	-	↘	-	-	-
	Middle	-	-	-	-	-	-
	Bottom	-	-	-	-	-	-
	Average	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	-	-	-	↘	-	-
	Middle	-	↘	↘	↘↘	-	-
	Bottom	-	↘	↘	↘↘	-	-
	Average	-	-	↘	↘↘	-	-
Dissolved Oxygen (% saturation)	Surface	-	-	-	↘	-	-
	Middle	-	↘	-	-	-	-
	Bottom	-	↘	-	-	-	-
	Average	-	-	↘	↘	-	-
pH	Surface	↘↘	↘↘	↘↘	↘↘	↘↘	↘↘
	Middle	↘↘	↘↘	↘↘	↘↘	↘↘	↘↘
	Bottom	↘↘	↘↘	↘↘	↘↘	↘↘	↘↘
	Average	↘↘	↘↘	↘↘	↘↘	↘↘	↘↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗
Turbidity (NTU)	Surface	-	↘↘	-	-	-	↘↘
	Middle	-	↘↘	-	-	-	↘↘
	Bottom	-	↘	↘	↘	-	↘↘
	Average	-	-	-	↘	-	↘↘
Suspended Solids (mg/L)	Surface	↘	↘↘	↘	↘	↘	-
	Middle	-	↘↘	-	-	↘	-
	Bottom	-	↘↘	↘↘	-	↘↘	-
	Average	-	↘↘	↘↘	-	↘↘	-
Total volatile solids (mg/L)	Surface	-	-	-	-	-	-
	Middle	-	↘	-	-	-	-
	Bottom	-	↘	-	-	-	-
	Average	-	↘	-	-	-	-
5-day Biochemical Oxygen Demand (mg/L)	Surface	-	-	-	-	-	-
	Middle	-	↘	-	-	-	-
	Bottom	-	↘	-	-	-	-
	Average	-	↘	-	-	-	-
Ammonia nitrogen (mg/L)	Surface	↗↗	↗	↗↗	↗	-	-
	Middle	↗↗	↗	↗↗	↗	-	-
	Bottom	↗↗	↗	↗↗	↗	-	-
	Average	↗↗	↗	↗↗	↗	-	-
Unionised Ammonia (mg/L)	Surface	↘↘	↘	-	-	-	-
	Middle	↘↘	↘	-	-	-	-
	Bottom	↘↘	↘	-	-	-	-
	Average	↘↘	↘	-	-	-	-
Nitrite nitrogen (mg/L)	Surface	↗↗	↗	↗↗	↗↗	↗↗	↗
	Middle	↗↗	↗↗	↗↗	↗↗	↗↗	↗
	Bottom	↗↗	↗↗	↗↗	↗↗	↗↗	↗
	Average	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
Nitrate nitrogen (mg/L)	Surface	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
	Middle	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
	Bottom	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
	Average	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
Total inorganic nitrogen (mg/L)	Surface	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
	Middle	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
	Bottom	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
	Average	↗↗	↗↗	↗↗	↗↗	↗↗	↗↗
Total Kjeldahl nitrogen (mg/L)	Surface	-	-	-	-	-	↗↗
	Middle	-	-	-	-	-	↗↗
	Bottom	-	-	-	-	-	↗↗
	Average	-	-	-	-	-	↗↗
Total nitrogen (mg/L)	Surface	-	↗	↗	↗↗	↗	↗↗
	Middle	-	↗	-	↗↗	-	↗↗
	Bottom	-	-	-	↗	-	↗↗
	Average	-	-	-	↗	-	↗↗
Orthophosphate phosphorus (mg/L)	Surface	↘	-	-	-	-	-
	Middle	↘	↘	-	-	↘	-
	Bottom	↘	↘	-	-	↘	-
	Average	↘	↘	-	-	↘	-
Total phosphorus (mg/L)	Surface	-	-	-	-	-	-
	Middle	↘	↘	↘	↘	↘	-
	Bottom	↘	↘	↘	↘	↘	-
	Average	↘	↘	↘	↘	↘	-
Silica (mg/L)	Surface	-	↗	-	-	-	↗↗
	Middle	-	↗	-	-	-	↗↗
	Bottom	-	↗	-	-	-	↗↗
	Average	-	↗	-	-	-	↗↗
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-	↗	-	-	-
	Middle	-	-	-	-	↗↗	-
	Bottom	-	-	-	-	↗↗	-
	Average	-	-	-	-	↗↗	-
<i>E. coli</i> (cfu/100mL)	Surface	-	↘↘	↘↘	-	-	-
	Middle	-	↘↘	↘↘	-	-	-
	Bottom	-	↘↘	↘↘	-	-	-
	Average	-	↘↘	↘↘	-	-	-
Faecal coliforms (cfu/100mL)	Surface	-	↘	-	↗↗	-	-
	Middle	-	↘	-	-	-	-
	Bottom	-	-	-	↗↗	-	-
	Average	-	-	-	↗↗	-	-

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. ↗ significant increase

4. ↘ significant decrease

Summary statistics for bottom sediment quality in the Tolo Harbour and Channel and Southern WCZs, 2012 – 2016

Parameter	Tolo Harbour and Channel				Hong Kong Island		West Lamma Channel	
	Harbour Subzone		Buffer Subzone	Channel Subzone	(South)			
	TS2	TS3	TS4	TS5	SS1	SS2	SS3	SS4
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	72 (57 - 83)	87 (71 - 95)	74 (54 - 92)	92 (84 - 97)	71 (40 - 95)	79 (62 - 88)	78 (66 - 93)	85 (62 - 95)
Electrochemical Potential (mV)	-302 (-408 - -179)	-318 (-418 - -165)	-303 (-419 - -128)	-331 (-444 - -239)	-239 (-433 - -85)	-253 (-443 - -91)	-221 (-419 - -77)	-245 (-424 - -112)
Total Solids (%w/w)	38 (31 - 47)	32 (28 - 43)	34 (23 - 45)	30 (28 - 34)	58 (53 - 60)	50 (45 - 54)	50 (42 - 54)	45 (42 - 51)
Total Volatile Solids (%TS)	9.6 (6.8 - 11.0)	9.9 (6.2 - 11.0)	10.0 (7.6 - 13.0)	10.8 (10.0 - 12.0)	5.5 (3.9 - 6.5)	6.8 (5.9 - 7.2)	6.6 (5.6 - 7.5)	7.1 (6.4 - 9.5)
Chemical Oxygen Demand (mg/kg)	21300 (18000 - 24000)	19000 (16000 - 21000)	16600 (15000 - 18000)	17000 (13000 - 21000)	9560 (8300 - 11000)	11610 (9100 - 15000)	12440 (9400 - 15000)	11960 (9600 - 14000)
Total Carbon (%w/w)	0.9 (0.6 - 1.2)	0.7 (0.6 - 0.8)	0.9 (0.5 - 1.2)	0.8 (0.8 - 0.9)	0.9 (0.8 - 1.0)	0.7 (0.6 - 0.9)	0.8 (0.6 - 1.0)	0.7 (0.6 - 1.1)
Ammonical Nitrogen (mg/kg)	5.96 (1.70 - 12.00)	5.02 (<0.05 - 8.20)	9.61 (1.30 - 27.00)	13.02 (8.50 - 25.00)	6.45 (0.26 - 16.00)	8.40 (1.50 - 43.00)	4.23 (<0.05 - 10.00)	4.26 (1.50 - 13.00)
Total Kjeldahl Nitrogen (mg/kg)	600 (440 - 780)	570 (440 - 690)	610 (400 - 740)	790 (690 - 940)	420 (340 - 500)	530 (410 - 600)	480 (420 - 580)	510 (440 - 610)
Total Phosphorus (mg/kg)	180 (150 - 210)	170 (140 - 190)	180 (140 - 220)	210 (190 - 250)	240 (170 - 290)	240 (190 - 270)	240 (220 - 260)	230 (190 - 250)
Total Sulphide (mg/kg)	129.2 (0.8 - 330.0)	96.2 (48.0 - 140.0)	154.8 (21.0 - 310.0)	93.3 (42.0 - 190.0)	30.0 (2.5 - 65.0)	33.5 (6.2 - 65.0)	23.7 (11.0 - 42.0)	23.5 (0.4 - 53.0)
Total Cyanide (mg/kg)	0.2 (<0.1 - 0.2)	0.2 (0.1 - 0.3)	0.2 (<0.1 - 0.3)	0.2 (0.1 - 0.3)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)
Arsenic (mg/kg)	9.6 (7.4 - 12.0)	10.2 (6.6 - 15.0)	8.5 (6.3 - 11.0)	6.1 (5.3 - 7.8)	6.5 (5.9 - 7.0)	8.2 (6.6 - 10.0)	6.6 (4.7 - 8.4)	7.8 (6.2 - 9.9)
Cadmium (mg/kg)	0.6 (0.4 - 0.9)	0.7 (0.5 - 0.8)	0.4 (0.2 - 0.6)	0.3 (0.2 - 0.4)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)
Chromium (mg/kg)	22 (15 - 26)	24 (14 - 27)	24 (19 - 32)	31 (26 - 35)	21 (18 - 23)	29 (27 - 32)	27 (21 - 34)	32 (25 - 38)
Copper (mg/kg)	33 (20 - 41)	39 (21 - 59)	23 (14 - 35)	21 (18 - 29)	9 (8 - 11)	18 (15 - 21)	15 (12 - 17)	27 (18 - 34)
Lead (mg/kg)	79 (46 - 96)	88 (50 - 110)	63 (36 - 91)	47 (35 - 54)	25 (22 - 28)	32 (20 - 38)	31 (20 - 37)	38 (25 - 46)
Mercury (mg/kg)	0.10 (<0.05 - 0.38)	0.09 (<0.05 - 0.30)	0.09 (<0.05 - 0.30)	0.08 (<0.05 - 0.30)	0.06 (<0.05 - 0.07)	0.09 (0.05 - 0.14)	0.10 (0.06 - 0.17)	0.13 (0.09 - 0.21)
Nickel (mg/kg)	13 (9 - 15)	14 (9 - 18)	16 (12 - 21)	23 (20 - 26)	15 (13 - 17)	19 (17 - 22)	18 (15 - 21)	20 (17 - 24)
Silver (mg/kg)	0.4 (0.3 - 0.6)	0.5 (0.3 - 0.8)	0.2 (<0.2 - 0.3)	0.2 (<0.2 - 0.3)	<0.2 (<0.2 - 0.2)	0.2 (<0.2 - 0.3)	<0.2 (<0.2 - 0.2)	0.3 (<0.2 - 0.4)
Zinc (mg/kg)	230 (150 - 330)	290 (180 - 360)	150 (120 - 170)	130 (100 - 150)	63 (51 - 70)	94 (84 - 130)	83 (67 - 110)	110 (92 - 120)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁴⁾	140 (90 - 430)	150 (90 - 410)	160 (90 - 670)	120 (93 - 220)	110 (90 - 180)	110 (90 - 190)	120 (94 - 180)	120 (95 - 270)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(5) (6)}	60 (16 - 100)	54 (28 - 93)	62 (17 - 160)	71 (41 - 110)	34 (20 - 50)	110 (27 - 280)	63 (26 - 110)	110 (46 - 210)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.

3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polycyclic aromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

5 High molecular weight polycyclic aromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(g,h,i)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary statistics for bottom sediment quality in the Southern, Junk Bay and Deep Bay WCZs, 2012 – 2016

	Lantau Island		Junk Bay	Inner Deep Bay		Outer Deep Bay	
Parameter	(East) SS5	(South) SS6	JS2	DS1	DS2	DS3	DS4
Number of samples	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (% w/w)	94 (82 - 99)	64 (55 - 71)	91 (69 - 99)	64 (36 - 98)	67 (33 - 93)	83 (54 - 95)	58 (34 - 81)
Electrochemical Potential (mV)	-262 (-408 - -88)	-227 (-379 - -73)	-249 (-395 - -75)	-321 (-401 - -92)	-320 (-420 - -52)	-313 (-387 - -172)	-328 (-398 - -170)
Total Solids (% w/w)	39 (33 - 43)	64 (59 - 68)	43 (40 - 50)	47 (36 - 56)	50 (46 - 55)	48 (40 - 57)	57 (48 - 63)
Total Volatile Solids (%TS)	7.7 (5.9 - 8.5)	3.7 (3.4 - 4.3)	7.5 (6.2 - 9.9)	6.3 (4.3 - 8.4)	6.9 (5.6 - 8.4)	6.9 (4.9 - 9.2)	5.7 (4.0 - 8.2)
Chemical Oxygen Demand (mg/kg)	12080 (9800 - 14000)	8060 (5900 - 10000)	13590 (8900 - 17000)	16500 (11000 - 21000)	13700 (10000 - 16000)	11430 (9300 - 14000)	11440 (7400 - 14000)
Total Carbon (%w/w)	0.6 (0.5 - 0.7)	0.5 (0.4 - 0.5)	0.7 (0.6 - 0.8)	0.6 (0.5 - 0.7)	0.6 (0.5 - 0.9)	0.6 (0.3 - 0.9)	0.6 (0.4 - 1.2)
Ammonical Nitrogen (mg/kg)	9.36 (0.79 - 13.00)	7.21 (<0.05 - 20.00)	4.89 (1.70 - 9.00)	26.69 (1.10 - 140.00)	3.91 (0.13 - 13.00)	4.22 (<0.05 - 23.00)	3.49 (0.11 - 9.00)
Total Kjeldahl Nitrogen (mg/kg)	580 (420 - 660)	360 (290 - 710)	570 (480 - 660)	570 (430 - 690)	440 (320 - 560)	450 (320 - 590)	370 (290 - 560)
Total Phosphorus (mg/kg)	210 (180 - 230)	200 (170 - 240)	220 (200 - 230)	320 (200 - 370)	270 (200 - 370)	250 (190 - 310)	200 (160 - 270)
Total Sulphide (mg/kg)	35.7 (6.4 - 88.0)	5.1 (0.8 - 17.0)	36.9 (15.0 - 81.0)	149.5 (14.0 - 450.0)	18.2 (2.5 - 44.0)	16.7 (0.5 - 30.0)	7.3 (1.0 - 30.0)
Total Cyanide (mg/kg)	0.2 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.3)	0.2 (<0.1 - 0.2)	0.2 (0.1 - 0.3)	0.1 (<0.1 - 0.2)
Arsenic (mg/kg)	7.3 (5.8 - 9.3)	5.6 (4.4 - 9.0)	7.6 (6.8 - 8.6)	9.7 (6.1 - 15.0)	12.7 (6.3 - 17.0)	11.6 (9.4 - 14.0)	9.7 (6.6 - 15.0)
Cadmium (mg/kg)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	0.1 (0.1 - 0.2)	0.3 (0.2 - 0.5)	0.2 (0.1 - 0.6)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)
Chromium (mg/kg)	37 (25 - 42)	19 (16 - 29)	41 (29 - 47)	42 (21 - 59)	37 (22 - 45)	38 (27 - 47)	28 (17 - 38)
Copper (mg/kg)	33 (26 - 44)	9 (7 - 14)	67 (55 - 78)	58 (22 - 83)	43 (24 - 73)	43 (25 - 59)	29 (15 - 53)
Lead (mg/kg)	47 (32 - 53)	23 (19 - 36)	47 (31 - 55)	46 (27 - 69)	55 (24 - 120)	43 (29 - 59)	34 (23 - 46)
Mercury (mg/kg)	0.15 (0.11 - 0.25)	0.06 (<0.05 - 0.10)	0.22 (0.18 - 0.30)	0.16 (0.07 - 0.29)	0.13 (0.06 - 0.19)	0.11 (0.10 - 0.14)	0.09 (<0.05 - 0.13)
Nickel (mg/kg)	22 (17 - 26)	12 (10 - 19)	20 (15 - 23)	23 (13 - 35)	21 (13 - 29)	23 (16 - 30)	17 (11 - 25)
Silver (mg/kg)	0.4 (0.2 - 0.5)	<0.2 (<0.2 - <0.2)	1.2 (0.9 - 1.6)	0.7 (0.3 - 1.1)	0.4 (0.2 - 1.1)	0.3 (<0.2 - 0.5)	0.2 (<0.2 - 0.3)
Zinc (mg/kg)	130 (85 - 170)	56 (46 - 99)	140 (97 - 160)	230 (100 - 350)	160 (97 - 270)	130 (87 - 160)	98 (64 - 120)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁴⁾⁽⁶⁾	130 (90 - 200)	100 (90 - 160)	160 (90 - 350)	130 (95 - 300)	120 (95 - 200)	100 (90 - 120)	110 (90 - 220)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁵⁾⁽⁶⁾	99 (51 - 190)	41 (19 - 120)	270 (130 - 390)	180 (76 - 450)	94 (54 - 130)	90 (37 - 140)	56 (37 - 93)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

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3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

5 High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(g,h,i)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary statistics for bottom sediment quality in the Port Shelter and Mirs Bay WCZs, 2012 – 2016

	Inner Port Shelter	Outer Port Shelter	Starling Inlet	Crooked Island	Port Island	Mirs Bay		
						(North)		
Parameter	PS3	PS5	PS6	MS1	MS2	MS7	MS17	MS3
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (% w/w)	86	65	77	93	95	93	93	76
	(12 - 97)	(39 - 81)	(56 - 90)	(82 - 99)	(88 - 99)	(86 - 99)	(85 - 98)	(13 - 94)
Electrochemical Potential (mV)	-294	-281	-276	-243	-289	-336	-246	-239
	(-399 - -96)	(-407 - -86)	(-400 - -86)	(-395 - -106)	(-401 - -105)	(-415 - -235)	(-388 - -85)	(-395 - -95)
Total Solids (%w/w)	35	49	47	38	33	31	33	44
	(32 - 39)	(41 - 61)	(38 - 53)	(36 - 42)	(30 - 36)	(26 - 36)	(29 - 38)	(36 - 51)
Total Volatile Soilds (%TS)	11.8	8.8	8.9	7.6	9.0	11.0	9.7	7.7
	(11.0 - 13.0)	(6.6 - 11.0)	(7.6 - 10.0)	(6.7 - 8.4)	(7.6 - 10.0)	(9.5 - 15.0)	(8.8 - 11.0)	(5.8 - 8.9)
Chemical Oxygen Demand (mg/kg)	14800	11450	11460	14000	14800	15600	13140	12330
	(10000 - 18000)	(7400 - 17000)	(9600 - 14000)	(11000 - 17000)	(11000 - 18000)	(13000 - 21000)	(8400 - 16000)	(8300 - 14000)
Total Carbon (%w/w)	1.1	1.5	1.2	0.6	0.7	0.8	0.8	0.8
	(1.0 - 1.2)	(1.1 - 2.4)	(0.9 - 1.4)	(<0.1 - 0.8)	(0.5 - 0.8)	(0.6 - 1.0)	(0.6 - 1.1)	(0.4 - 1.3)
Ammonical Nitrogen (mg/kg)	7.74	6.01	7.20	6.91	8.80	10.44	8.89	10.14
	(1.90 - 12.00)	(3.50 - 9.50)	(3.30 - 12.00)	(0.13 - 10.00)	(1.50 - 14.00)	(7.30 - 13.00)	(7.10 - 11.00)	(2.60 - 25.00)
Total Kjeldahl Nitrogen (mg/kg)	700	620	670	570	660	690	700	520
	(310 - 850)	(510 - 710)	(580 - 770)	(480 - 620)	(580 - 770)	(520 - 800)	(590 - 880)	(400 - 610)
Total Phosphorus (mg/kg)	210	220	240	190	190	200	210	200
	(180 - 240)	(190 - 280)	(210 - 280)	(180 - 200)	(180 - 210)	(170 - 240)	(190 - 280)	(160 - 220)
Total Sulphide (mg/kg)	36.7	21.0	34.2	32.1	57.8	70.1	37.3	51.7
	(22.0 - 69.0)	(2.1 - 35.0)	(3.8 - 54.0)	(2.7 - 88.0)	(3.4 - 150.0)	(1.3 - 180.0)	(5.7 - 88.0)	(1.3 - 170.0)
Total Cyanide (mg/kg)	0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.1
	(<0.1 - 0.2)	(<0.1 - 0.2)	(<0.1 - 0.2)	(<0.1 - 0.3)	(<0.1 - 0.2)	(<0.1 - 0.2)	(<0.1 - 0.3)	(<0.1 - 0.3)
Arsenic (mg/kg)	5.5	4.8	5.8	8.4	7.4	6.8	6.4	5.8
	(3.6 - 7.8)	(3.2 - 6.1)	(5.2 - 7.2)	(7.1 - 9.6)	(6.0 - 9.6)	(5.8 - 7.8)	(5.1 - 8.2)	(3.8 - 8.4)
Cadmium (mg/kg)	<0.1	<0.1	<0.1	0.2	0.3	0.4	<0.1	<0.1
	(<0.1 - 0.1)	(<0.1 - <0.1)	(<0.1 - <0.1)	(0.1 - 0.3)	(0.1 - 0.4)	(0.2 - 0.5)	(<0.1 - 0.1)	(<0.1 - 0.1)
Chromium (mg/kg)	25	21	25	30	33	32	33	26
	(16 - 29)	(14 - 31)	(21 - 28)	(19 - 36)	(27 - 39)	(27 - 34)	(26 - 37)	(21 - 33)
Copper (mg/kg)	21	11	11	22	22	21	16	11
	(10 - 26)	(6 - 16)	(9 - 14)	(17 - 27)	(19 - 27)	(13 - 26)	(12 - 17)	(7 - 15)
Lead (mg/kg)	36	27	32	46	45	40	42	26
	(21 - 44)	(18 - 38)	(24 - 40)	(34 - 54)	(34 - 51)	(27 - 45)	(34 - 46)	(20 - 34)
Mercury (mg/kg)	0.08	0.05	0.05	0.06	0.06	0.07	0.06	0.05
	(0.06 - 0.11)	(<0.05 - 0.06)	(<0.05 - 0.07)	(<0.05 - 0.07)	(<0.05 - 0.08)	(<0.05 - 0.10)	(<0.05 - 0.07)	(<0.05 - 0.06)
Nickel (mg/kg)	17	15	18	18	22	22	23	17
	(11 - 19)	(10 - 21)	(15 - 21)	(12 - 20)	(18 - 25)	(18 - 25)	(18 - 26)	(13 - 21)
Silver (mg/kg)	0.2	<0.2	<0.2	0.4	0.3	0.3	0.2	<0.2
	(<0.2 - 0.3)	(<0.2 - 0.2)	(<0.2 - <0.2)	(0.3 - 0.8)	(0.2 - 0.4)	(<0.2 - 0.3)	(<0.2 - 0.3)	(<0.2 - <0.2)
Zinc (mg/kg)	100	73	76	100	110	98	96	68
	(55 - 130)	(43 - 110)	(61 - 100)	(69 - 130)	(87 - 130)	(82 - 110)	(75 - 110)	(52 - 89)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18	18	18	18	18	18	18	18
	(18 - 18)	(18 - 18)	(18 - 18)	(18 - 18)	(18 - 18)	(18 - 18)	(18 - 18)	(18 - 18)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁴⁾	140	110	140	120	150	140	120	100
	(90 - 230)	(90 - 150)	(90 - 330)	(90 - 200)	(95 - 360)	(90 - 220)	(90 - 180)	(90 - 130)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(5) (6)}	74	38	49	52	59	67	60	40
	(32 - 160)	(18 - 78)	(18 - 91)	(29 - 90)	(32 - 88)	(31 - 110)	(31 - 140)	(18 - 66)

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6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary statistics for bottom sediment quality in the Mirs Bay WCZ, 2012 – 2016

Parameter	Mirs Bay (North)		Long Harbour	Waglan Island	Mirs Bay (South)		Mirs Bay (Central)	
	MS4	MS5	MS6	MS8	MS13	MS14	MS15	MS16
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	84 (18 - 96)	87 (77 - 97)	93 (83 - 98)	86 (52 - 95)	92 (89 - 96)	90 (75 - 99)	89 (83 - 98)	79 (68 - 95)
Electrochemical Potential (mV)	-232 (-392 - -68)	-228 (-383 - -95)	-271 (-394 - -115)	-215 (-366 - -79)	-238 (-360 - -123)	-233 (-389 - -128)	-234 (-384 - -120)	-211 (-380 - -110)
Total Solids (%w/w)	40 (34 - 57)	39 (35 - 44)	32 (28 - 36)	47 (38 - 51)	49 (44 - 53)	51 (49 - 53)	53 (50 - 54)	55 (52 - 57)
Total Volatile Solids (%TS)	7.9 (3.8 - 9.4)	8.1 (6.6 - 10.0)	10.9 (9.4 - 12.0)	6.7 (6.1 - 7.4)	6.5 (5.8 - 7.7)	6.2 (5.1 - 6.7)	6.2 (5.4 - 6.7)	6.2 (5.7 - 6.9)
Chemical Oxygen Demand (mg/kg)	12000 (10000 - 15000)	12150 (9500 - 14000)	14700 (11000 - 17000)	10140 (8600 - 12000)	8260 (6400 - 10000)	8050 (5200 - 9600)	8700 (5700 - 11000)	9200 (6300 - 12000)
Total Carbon (%w/w)	0.7 (0.6 - 0.9)	0.7 (0.6 - 0.8)	0.9 (0.8 - 1.3)	0.6 (0.5 - 0.7)	0.6 (0.5 - 0.7)	0.6 (0.5 - 0.6)	0.6 (0.5 - 0.7)	0.7 (0.6 - 0.8)
Ammonical Nitrogen (mg/kg)	6.71 (0.21 - 13.00)	6.22 (2.20 - 12.00)	9.53 (7.50 - 11.00)	3.58 (1.60 - 7.30)	3.23 (0.68 - 9.30)	3.10 (0.74 - 6.30)	3.81 (1.20 - 10.00)	3.91 (2.00 - 7.70)
Total Kjeldahl Nitrogen (mg/kg)	660 (590 - 730)	630 (560 - 670)	730 (680 - 810)	490 (400 - 550)	490 (430 - 550)	490 (400 - 550)	500 (440 - 560)	490 (400 - 540)
Total Phosphorus (mg/kg)	210 (190 - 240)	220 (200 - 240)	210 (190 - 230)	220 (190 - 240)	260 (230 - 360)	250 (230 - 270)	250 (210 - 270)	250 (200 - 280)
Total Sulphide (mg/kg)	20.0 (3.4 - 51.0)	21.3 (3.5 - 66.0)	41.3 (2.8 - 83.0)	12.7 (5.2 - 30.0)	11.0 (1.7 - 35.0)	9.1 (1.7 - 35.0)	8.3 (2.7 - 19.0)	8.4 (3.2 - 23.0)
Total Cyanide (mg/kg)	0.1 (<0.1 - 0.3)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.3)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)
Arsenic (mg/kg)	7.0 (6.1 - 8.5)	6.8 (6.0 - 8.0)	6.0 (5.4 - 6.9)	7.4 (6.6 - 8.7)	7.7 (6.3 - 8.7)	7.3 (5.6 - 8.7)	6.8 (4.9 - 8.1)	6.9 (5.2 - 9.9)
Cadmium (mg/kg)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium (mg/kg)	33 (29 - 38)	30 (24 - 37)	30 (28 - 35)	31 (28 - 33)	30 (24 - 34)	28 (19 - 33)	28 (19 - 35)	26 (18 - 40)
Copper (mg/kg)	15 (12 - 18)	14 (10 - 18)	17 (14 - 20)	14 (10 - 18)	12 (10 - 17)	11 (6 - 16)	11 (7 - 16)	9 (6 - 14)
Lead (mg/kg)	35 (29 - 41)	37 (26 - 45)	40 (32 - 45)	33 (30 - 37)	31 (25 - 35)	29 (21 - 32)	29 (22 - 33)	28 (22 - 35)
Mercury (mg/kg)	0.05 (<0.05 - 0.06)	0.05 (<0.05 - 0.06)	0.06 (<0.05 - 0.07)	0.05 (<0.05 - 0.06)	0.05 (<0.05 - 0.07)	<0.05 (<0.05 - <0.05)	0.05 (<0.05 - 0.06)	<0.05 (<0.05 - 0.05)
Nickel (mg/kg)	22 (18 - 24)	21 (17 - 24)	22 (19 - 25)	22 (20 - 24)	22 (19 - 24)	20 (14 - 23)	20 (13 - 24)	18 (12 - 28)
Silver (mg/kg)	<0.2 (<0.2 - <0.2)	<0.2 (<0.2 - 0.2)	<0.2 (<0.2 - <0.2)	<0.2 (<0.2 - <0.2)	<0.2 (<0.2 - <0.2)	<0.2 (<0.2 - <0.2)	<0.2 (<0.2 - <0.2)	<0.2 (<0.2 - <0.2)
Zinc (mg/kg)	86 (71 - 100)	86 (66 - 100)	100 (92 - 120)	79 (72 - 89)	78 (62 - 86)	72 (52 - 83)	69 (50 - 75)	63 (48 - 77)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁴⁾	130 (90 - 250)	120 (90 - 260)	110 (94 - 140)	130 (90 - 260)	150 (90 - 330)	120 (90 - 200)	120 (90 - 220)	110 (90 - 160)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(5) (6)}	64 (30 - 160)	54 (27 - 170)	70 (37 - 160)	77 (35 - 310)	49 (27 - 91)	37 (17 - 85)	37 (20 - 69)	34 (18 - 62)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.

3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

5 High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(g,h,i)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary statistics for bottom sediment quality in the North Western and Western Buffer WCZs, 2012 – 2016

	Pearl Island	Pillar Point	Urmston Road	Chek Lap Kok (North)	Tsing Yi (South)	Hong Kong Island (West)
Parameter	NS2	NS3	NS4	NS6	WS1	WS2
Number of samples	10	10	10	10	10	10
Particle Size Fractionation <63µm (% w/w)	57 (18 - 93)	57 (20 - 96)	53 (13 - 88)	74 (33 - 99)	77 (62 - 84)	85 (77 - 92)
Electrochemical Potential (mV)	-187 (-317 - -80)	-218 (-382 - -111)	-248 (-428 - -83)	-217 (-380 - -81)	-279 (-398 - -135)	-228 (-347 - -88)
Total Solids (%w/w)	50 (39 - 65)	51 (40 - 63)	56 (43 - 65)	57 (42 - 70)	46 (44 - 51)	46 (43 - 51)
Total Volatile Solids (%TS)	6.4 (4.8 - 9.2)	6.6 (4.9 - 9.6)	5.7 (4.2 - 7.5)	5.3 (3.0 - 7.2)	6.9 (6.5 - 7.5)	6.8 (6.1 - 7.6)
Chemical Oxygen Demand (mg/kg)	10780 (6900 - 15000)	12200 (11000 - 14000)	12610 (9100 - 17000)	10500 (8600 - 13000)	14900 (12000 - 21000)	12500 (10000 - 15000)
Total Carbon (%w/w)	0.8 (0.5 - 1.3)	0.7 (0.5 - 1.1)	0.6 (<0.1 - 0.8)	0.7 (0.4 - 1.2)	0.8 (0.5 - 1.1)	0.6 (0.5 - 0.9)
Ammonical Nitrogen (mg/kg)	4.69 (<0.05 - 13.00)	6.08 (0.07 - 31.00)	3.84 (0.17 - 8.90)	8.79 (0.07 - 76.00)	6.96 (1.50 - 14.00)	5.47 (0.69 - 12.00)
Total Kjeldahl Nitrogen (mg/kg)	400 (300 - 580)	450 (310 - 570)	410 (270 - 600)	400 (180 - 510)	520 (400 - 590)	520 (440 - 620)
Total Phosphorus (mg/kg)	210 (180 - 230)	220 (160 - 260)	210 (150 - 260)	200 (100 - 260)	220 (200 - 250)	220 (190 - 250)
Total Sulphide (mg/kg)	41.3 (0.9 - 190.0)	43.4 (1.6 - 130.0)	31.1 (4.5 - 76.0)	21.2 (0.2 - 51.0)	110.8 (75.0 - 160.0)	24.8 (2.0 - 72.0)
Total Cyanide (mg/kg)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.3)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)
Arsenic (mg/kg)	8.7 (7.0 - 11.0)	10.4 (7.7 - 16.0)	10.2 (7.8 - 13.0)	12.4 (7.1 - 17.0)	8.0 (6.8 - 9.4)	7.8 (7.1 - 8.9)
Cadmium (mg/kg)	<0.1 (<0.1 - 0.1)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.1)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium (mg/kg)	31 (20 - 43)	31 (19 - 45)	29 (23 - 36)	29 (17 - 40)	33 (25 - 37)	32 (27 - 35)
Copper (mg/kg)	30 (19 - 46)	30 (13 - 51)	26 (17 - 38)	20 (9 - 34)	46 (31 - 62)	24 (18 - 31)
Lead (mg/kg)	38 (23 - 49)	37 (26 - 53)	35 (29 - 40)	33 (20 - 43)	35 (29 - 40)	33 (24 - 39)
Mercury (mg/kg)	0.11 (0.06 - 0.16)	0.11 (0.07 - 0.14)	0.09 (0.06 - 0.12)	0.08 (<0.05 - 0.13)	0.22 (0.10 - 0.88)	0.12 (0.07 - 0.47)
Nickel (mg/kg)	18 (12 - 26)	19 (11 - 27)	17 (13 - 23)	18 (11 - 23)	18 (14 - 22)	21 (17 - 25)
Silver (mg/kg)	0.3 (<0.2 - 0.5)	0.3 (<0.2 - 0.4)	0.2 (<0.2 - 0.3)	<0.2 (<0.2 - 0.2)	0.6 (0.4 - 0.9)	0.2 (<0.2 - 0.3)
Zinc (mg/kg)	110 (83 - 140)	100 (68 - 160)	100 (83 - 120)	83 (42 - 110)	110 (83 - 130)	97 (79 - 110)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)	18 (18 - 18)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁴⁾	130 (90 - 350)	140 (90 - 500)	120 (90 - 240)	110 (90 - 150)	110 (90 - 170)	120 (90 - 210)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁵⁾⁽⁶⁾	69 (36 - 100)	150 (41 - 690)	98 (24 - 200)	52 (23 - 130)	250 (100 - 570)	180 (24 - 1100)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.

3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

5 High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(g,h,i)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary statistics for bottom sediment quality in the Eastern Buffer and Victoria Harbour WCZs, 2012 – 2016

Parameter	Eastern Buffer			Victoria Harbour			Rambler Channel	
	Chai Wan ES1	Tathong Channel ES2	ES4	(East) VS3	(Central) VS5	(West) VS6	VS9	VS10
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (%w/w)	66 (33 - 94)	66 (41 - 94)	60 (33 - 95)	58 (16 - 83)	85 (69 - 97)	69 (32 - 97)	71 (31 - 94)	75 (37 - 88)
Electrochemical Potential (mV)	-237 (-381 - -63)	-225 (-380 - -83)	-256 (-389 - -75)	-309 (-463 - -157)	-369 (-438 - -278)	-307 (-430 - -150)	-276 (-382 - -123)	-258 (-342 - -121)
Total Solids (%w/w)	56 (43 - 69)	53 (41 - 61)	57 (42 - 66)	49 (38 - 70)	41 (33 - 44)	45 (33 - 56)	45 (35 - 56)	43 (36 - 52)
Total Volatile Solids (%TS)	6.4 (4.2 - 9.0)	6.5 (4.2 - 8.5)	6.0 (5.0 - 8.1)	6.5 (3.7 - 8.6)	8.3 (6.7 - 8.9)	7.8 (6.0 - 9.2)	6.8 (5.3 - 10.0)	7.5 (6.2 - 8.5)
Chemical Oxygen Demand (mg/kg)	10090 (7200 - 15000)	9770 (7300 - 13000)	13030 (9300 - 24000)	16900 (12000 - 20000)	20600 (16000 - 23000)	17100 (13000 - 19000)	15400 (14000 - 18000)	16000 (12000 - 20000)
Total Carbon (%w/w)	1.2 (0.8 - 1.6)	0.8 (0.6 - 1.0)	1.1 (0.7 - 2.2)	0.8 (0.5 - 1.3)	0.9 (0.6 - 1.2)	0.9 (0.6 - 1.7)	0.7 (0.6 - 0.8)	0.8 (0.5 - 1.2)
Ammonical Nitrogen (mg/kg)	5.34 (0.22 - 14.00)	5.77 (1.10 - 19.00)	7.36 (0.80 - 12.00)	15.38 (2.70 - 58.00)	31.72 (9.20 - 70.00)	12.01 (0.11 - 41.00)	13.37 (2.90 - 34.00)	4.50 (1.40 - 17.00)
Total Kjeldahl Nitrogen (mg/kg)	430 (350 - 500)	480 (400 - 650)	490 (390 - 560)	530 (270 - 700)	750 (570 - 1200)	560 (400 - 700)	480 (380 - 550)	480 (370 - 600)
Total Phosphorus (mg/kg)	200 (130 - 250)	210 (180 - 240)	210 (180 - 240)	200 (140 - 250)	220 (190 - 250)	240 (200 - 280)	210 (180 - 260)	210 (180 - 250)
Total Sulphide (mg/kg)	37.5 (18.0 - 94.0)	36.3 (2.8 - 78.0)	111.3 (9.6 - 620.0)	222.2 (12.0 - 460.0)	310.0 (110.0 - 510.0)	246.0 (40.0 - 530.0)	132.4 (28.0 - 250.0)	42.7 (2.1 - 86.0)
Total Cyanide (mg/kg)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.2)	0.2 (<0.1 - 0.3)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.4)
Arsenic (mg/kg)	5.6 (3.6 - 8.7)	6.1 (4.5 - 9.5)	6.2 (4.2 - 8.6)	6.9 (2.8 - 8.8)	8.1 (7.2 - 8.9)	7.9 (6.5 - 9.1)	7.5 (4.4 - 10.0)	8.5 (6.8 - 11.0)
Cadmium (mg/kg)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	0.1 (<0.1 - 0.4)	0.3 (<0.1 - 0.5)	0.7 (0.4 - 0.9)	0.3 (<0.1 - 0.5)	0.2 (<0.1 - 0.2)	0.4 (0.2 - 0.9)
Chromium (mg/kg)	23 (14 - 36)	28 (19 - 49)	33 (24 - 42)	37 (16 - 51)	51 (38 - 61)	37 (24 - 47)	38 (28 - 48)	55 (42 - 67)
Copper (mg/kg)	26 (15 - 38)	18 (12 - 32)	57 (35 - 100)	94 (33 - 160)	150 (110 - 190)	76 (28 - 100)	59 (10 - 79)	130 (77 - 170)
Lead (mg/kg)	29 (15 - 43)	28 (18 - 42)	39 (25 - 66)	46 (29 - 59)	65 (55 - 72)	59 (29 - 100)	35 (25 - 40)	55 (37 - 78)
Mercury (mg/kg)	0.14 (0.08 - 0.31)	0.08 (<0.05 - 0.13)	0.21 (0.09 - 0.50)	0.32 (0.13 - 0.52)	0.49 (0.39 - 0.65)	0.57 (0.18 - 2.00)	0.22 (<0.05 - 0.86)	0.25 (0.09 - 0.75)
Nickel (mg/kg)	14 (9 - 21)	18 (12 - 31)	17 (13 - 23)	17 (7 - 24)	23 (18 - 28)	18 (13 - 23)	21 (14 - 26)	23 (17 - 28)
Silver (mg/kg)	0.4 (0.2 - 0.6)	0.2 (<0.2 - 0.4)	1.4 (0.6 - 4.2)	2.1 (0.7 - 4.1)	4.1 (2.4 - 7.9)	1.5 (0.3 - 2.2)	1.0 (<0.2 - 1.3)	2.6 (1.4 - 5.6)
Zinc (mg/kg)	74 (48 - 120)	75 (48 - 110)	120 (73 - 260)	150 (58 - 200)	270 (200 - 330)	160 (91 - 210)	130 (67 - 150)	180 (130 - 250)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18 (18 - 18)	18 (18 - 18)	18 (18 - 19)	19 (18 - 24)	21 (18 - 29)	20 (18 - 28)	18 (18 - 19)	18 (18 - 20)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ⁽⁴⁾	120 (90 - 220)	100 (90 - 150)	140 (98 - 250)	170 (110 - 430)	180 (110 - 300)	180 (110 - 340)	110 (93 - 170)	130 (94 - 250)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(5) (6)}	160 (48 - 280)	72 (38 - 190)	380 (110 - 1800)	970 (170 - 3300)	730 (500 - 1000)	1200 (320 - 3200)	160 (31 - 240)	400 (94 - 1100)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.

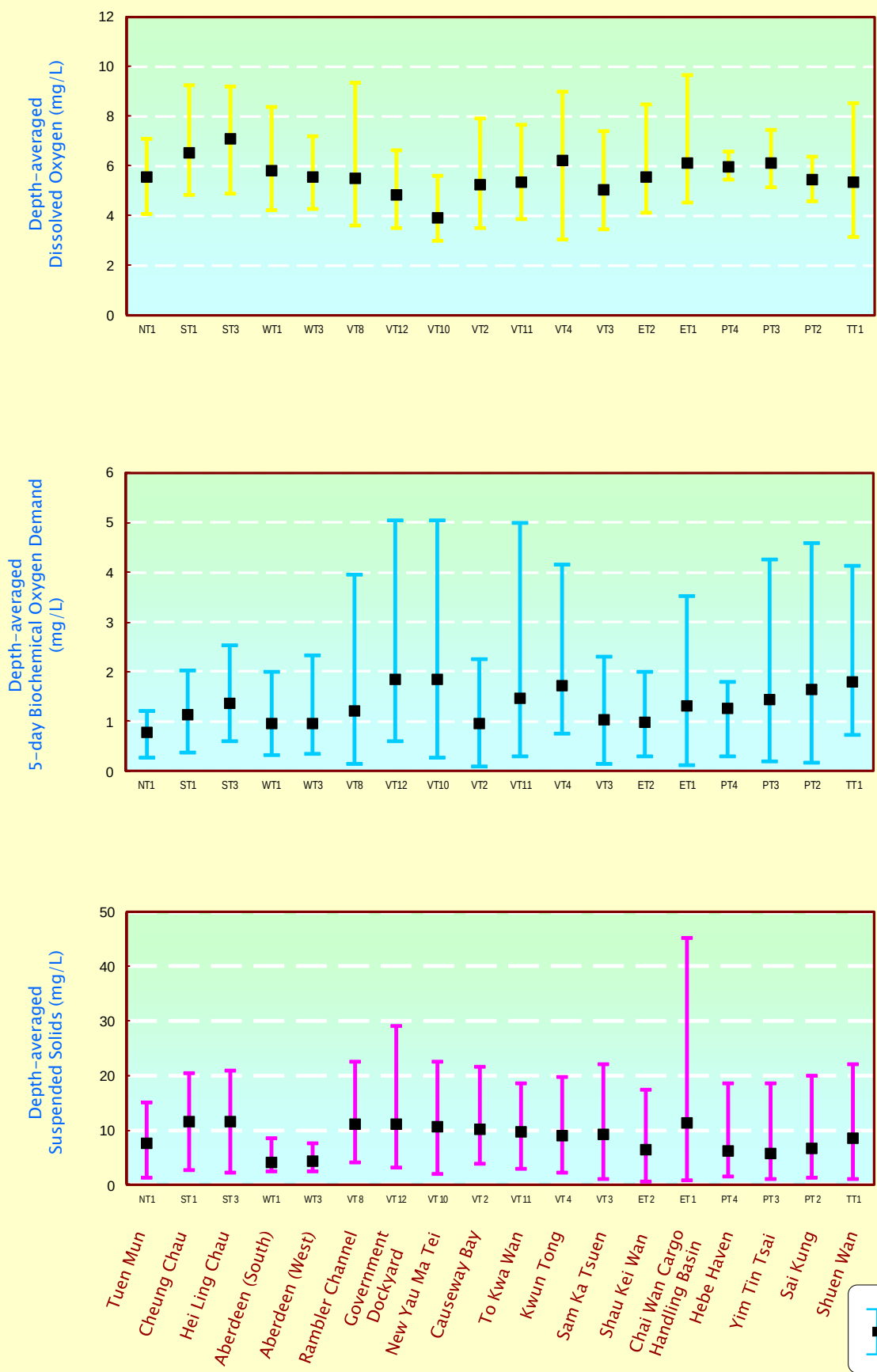
3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polyaromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

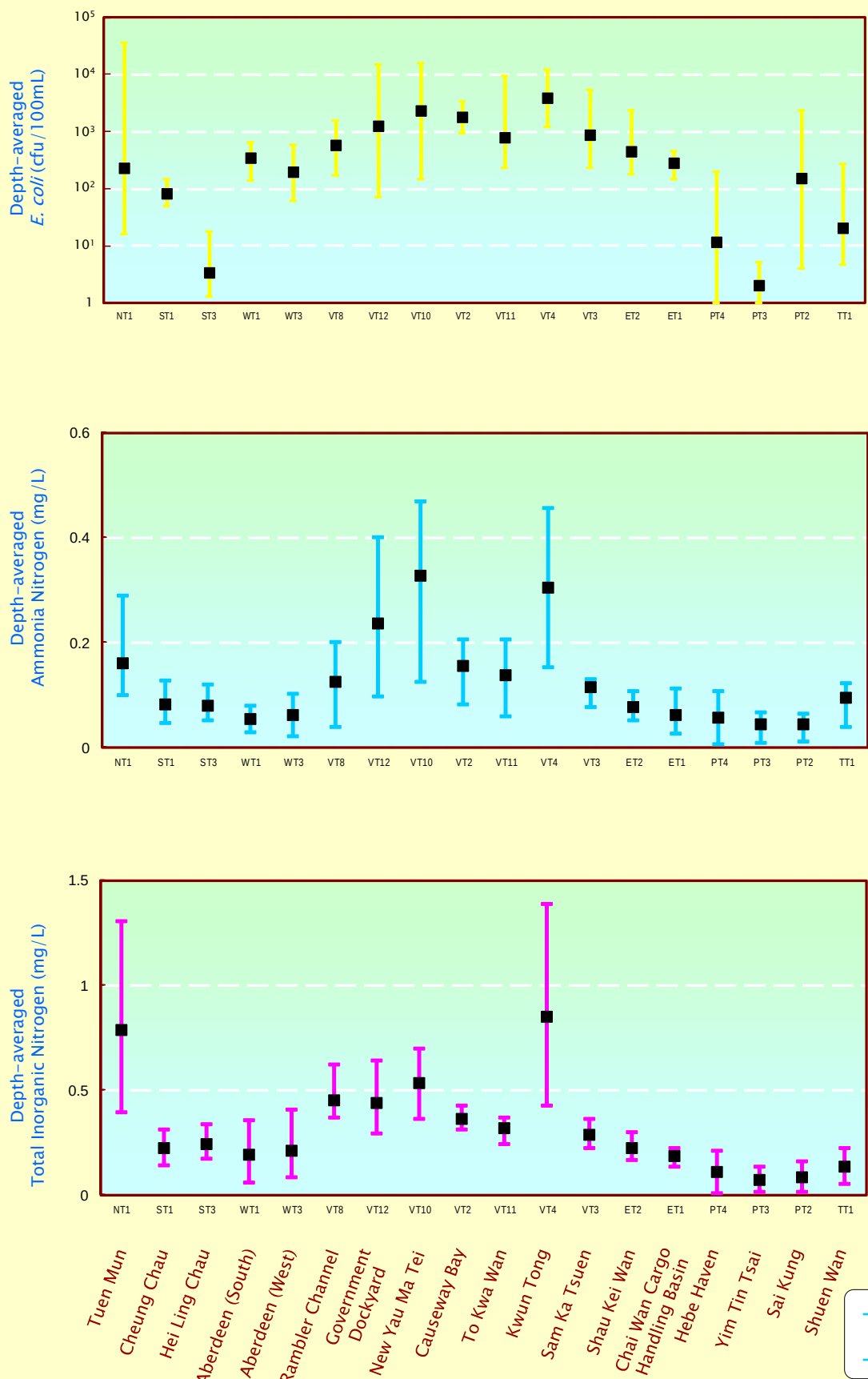
5 High molecular weight polyaromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(g,h,i)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Water quality of typhoon shelters, sheltered anchorages and Government Dockyard in Hong Kong in 2016



Water quality of typhoon shelters, sheltered anchorages and Government Dockyard in Hong Kong in 2016 (continued)



Results of the Seasonal Kendall Test for water quality trends in typhoon shelters, sheltered anchorages and Government Dockyard, 1986 – 2016

Monitoring Station		NT1	ST1	ST3	WT3	WT1	VT8	VT10	VT2	VT11
Monitoring Period		1986 I 2016	1986 I 2016	2000 I 2016	1986 I 2016	1986 I 2016	1986 I 2016	1993 I 2016	1986 I 2016	1994 I 2016
Parameter	Water Depth									
Temperature (°C)	Surface	↗	↗	-	-	-	↗	↗	↗	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	NA	↗	-	-	-	↗	↗	↗	-
	Average	↗	↗	-	-	-	↗	↗	↗	-
Salinity	Surface	-	-	-	-	-	-	-	-	-
	Middle	NA	-	↘	-	-	NA	NA	NA	-
	Bottom	NA	-	↘	-	-	-	-	-	-
	Average	-	-	↘	-	-	-	-	-	-
Dissolved Oxygen (mg/L)	Surface	↗	-	-	-	-	↗	-	↗	-
	Middle	NA	-	-	-	-	NA	NA	NA	↗
	Bottom	NA	-	-	-	↘	↗	↗	↗	↗
	Average	↗	-	-	-	-	↗	↗	↗	↗
Dissolved Oxygen (%)	Surface	↗	-	-	-	-	↗	-	↗	↗
	Middle	NA	-	-	-	↗	NA	NA	NA	↗
	Bottom	NA	-	-	↗	↗	↗	↗	↗	↗
	Average	↗	-	-	-	↗	↗	↗	↗	↗
pH	Surface	↘	↘	↘	↘	↘	↘	-	-	-
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	-
	Bottom	NA	↘	↘	↘	↘	-	-	-	-
	Average	↘	↘	↘	↘	↘	↘	-	-	-
Secchi disc depth (m)		↘	↘	↘	↘	↘	↘	↘	↘	-
Turbidity (NTU)	Surface	↘	↘	↘	↘	↘	-	↘	-	-
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	-
	Bottom	NA	↘	↘	↘	↘	-	-	-	-
	Average	↘	↘	↘	↘	↘	↘	↘	-	-
Suspended Solids (mg/L)	Surface	↘	-	-	-	-	↘	↘	-	-
	Middle	NA	↘	-	-	-	NA	NA	NA	-
	Bottom	NA	-	-	-	-	-	-	-	-
	Average	↘	-	-	-	-	↘	↘	-	-
Total volatile solids (mg/L)	Surface	↘	-	-	-	-	↘	↘	-	-
	Middle	NA	-	↗	-	-	NA	NA	NA	-
	Bottom	NA	-	-	↘	-	↘	↘	-	-
	Average	↘	-	-	↘	-	↘	↘	-	-
5-day Biochemical Oxygen Demand	Surface	↘	-	-	-	-	↘	↘	↘	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	NA	-	-	-	-	↘	↘	↘	↘
	Average	↘	-	-	-	-	↘	↘	↘	↘
Ammonia nitrogen (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	↘
	Middle	NA	-	-	-	-	NA	NA	NA	↘
	Bottom	NA	-	-	-	-	↘	↘	↘	↘
	Average	↘	-	-	-	-	↘	↘	↘	↘
Unionised Ammonia(mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	↘	↘	↘	NA	NA	NA	↘
	Bottom	NA	↘	↘	↘	↘	↘	↘	↘	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Nitrite nitrogen (mg/L)	Surface	↗	-	↗	-	-	↗	NA	-	↗
	Middle	NA	-	↗	-	-	NA	NA	NA	↗
	Bottom	NA	-	↗	-	-	↗	-	-	↗
	Average	↗	-	↗	-	-	↗	-	-	↗
Nitrate nitrogen (mg/L)	Surface	↗	↗	↗	-	-	↗	↗	↗	↗
	Middle	NA	-	↗	-	-	NA	NA	NA	↗
	Bottom	NA	-	↗	-	-	↗	↗	↗	↗
	Average	↗	↗	↗	-	-	↗	↗	↗	↗
Total inorganic nitrogen (mg/L)	Surface	↗	-	-	-	-	-	↗	↗	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	NA	-	-	-	-	-	↗	↗	-
	Average	↗	-	-	-	-	↗	↗	↗	-
Total Kjeldahl nitrogen (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	↘
	Middle	NA	-	-	-	-	NA	NA	NA	↘
	Bottom	NA	-	-	-	-	↘	↘	↘	↘
	Average	↘	-	-	-	-	↘	↘	↘	↘
Total nitrogen (mg/L)	Surface	-	-	↗	-	-	↗	↗	↗	-
	Middle	NA	-	↗	-	-	NA	NA	NA	-
	Bottom	NA	-	↗	-	-	↗	↗	↗	-
	Average	-	-	↗	-	-	↗	↗	↗	-
Orthophosphate phosphorus (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	↘
	Middle	NA	↘	-	-	-	NA	NA	NA	↘
	Bottom	NA	-	-	-	↘	↘	↘	↘	↘
	Average	↘	-	-	-	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	↘	-	-	-	-	↘	↘	↘	↘
	Middle	NA	-	-	-	-	NA	NA	NA	↘
	Bottom	NA	-	-	-	-	↘	↘	↘	↘
	Average	↘	-	-	-	-	↘	↘	↘	↘
Silica (mg/L)	Surface	-	-	-	-	-	-	-	-	-
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	NA	-	-	-	-	-	-	-	-
	Average	-	-	-	-	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	-	↗	-	-	↗	-	-	-
	Middle	NA	-	↗	-	-	NA	NA	NA	-
	Bottom	NA	-	-	-	-	-	↗	-	-
	Average	-	-	-	-	-	-	↗	-	-
E. coli (cfu/100mL)	Surface	↘	↘	-	↘	↘	↘	↘	↘	↘
	Middle	NA	↘	-	↘	↘	NA	NA	NA	↘
	Bottom	NA	↘	-	↘	↘	↘	↘	↘	↘
	Average	↘	↘	-	↘	↘	↘	↘	↘	↘
Faecal coliforms (cfu/100mL)	Surface	↘	-	↘	-	-	-	↘	↘	↘
	Middle	NA	-	-	-	-	NA	NA	NA	-
	Bottom	NA	-	-	-	-	-	↘	↘	-
	Average	↘	-	-	-	-	-	↘	↘	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. NA (Not Applicable) indicates the measurement was not made due to shallow water

4. ↗ significant increase

5. ↘ significant decrease

Results of the Seasonal Kendall Test for water quality trends in typhoon shelters, sheltered anchorages and Government Dockyard, 1986 – 2016 (continued)

Monitoring Station		VT12	VT4	VT3	ET2	ET1	PT4	PT3	PT2	TT1
Monitoring Period		2000 I 2016	1987 I 2016	1986 I 2016	1993 I 2016	1986 I 2016	1986 I 2016	1986 I 2016	1986 I 2016	1986 I 2016
Parameter	Water Depth									
Temperature (°C)	Surface	↗	↗	↗	-	-	-	-	-	↗
	Middle	NA	↗	NA	-	↗	NA	-	NA	-
	Bottom	-	↗	↗	-	↗	↗	↗	NA	-
	Average	-	↗	-	-	-	↗	-	-	↗
Salinity	Surface	-	-	-	-	-	-	-	-	-
	Middle	NA	-	NA	-	-	NA	-	NA	-
	Bottom	-	-	-	-	-	-	-	NA	-
	Average	-	-	-	-	-	-	-	NA	-
Dissolved Oxygen (mg/L)	Surface	-	↗	↗	↗	-	↘	-	-	↘
	Middle	NA	↗	NA	↗	↗	NA	-	NA	-
	Bottom	-	↗	↗	↗	↗	-	-	NA	-
	Average	-	↗	↗	↗	-	-	-	-	-
Dissolved Oxygen (%)	Surface	-	↗	↗	↗	-	-	-	-	↘
	Middle	NA	↗	NA	↗	↗	NA	-	NA	-
	Bottom	-	↗	↗	↗	↗	↗	-	NA	-
	Average	-	↗	↗	↗	↗	↗	-	NA	-
pH	Surface	-	↗	↗	↗	↗	↘	↘	↘	↘
	Middle	NA	-	NA	↗	↗	NA	↘	NA	↘
	Bottom	-	-	↗	↗	↗	↘	↘	NA	↘
	Average	-	↗	↗	↗	↗	↘	↘	NA	↘
Secchi disc depth (m)		↗	↗	↗	↗	↗	↗	-	↗	↗
Turbidity (NTU)	Surface	↘	↘	-	-	-	-	↘	↘	-
	Middle	NA	↘	NA	-	↘	NA	↘	NA	-
	Bottom	-	↘	-	-	-	-	↘	NA	-
	Average	↘	↘	-	-	-	-	↘	↘	-
Suspended Solids (mg/L)	Surface	-	↘	↘	↘	-	-	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	-	NA	↘
	Bottom	↘	↘	↘	↘	↘	-	-	NA	↘
	Average	-	↘	↘	↘	↘	↘	-	-	↘
Total volatile solids (mg/L)	Surface	-	↘	↘	-	-	-	-	-	↘
	Middle	NA	↘	NA	-	-	NA	-	NA	-
	Bottom	-	↘	↘	-	↘	↘	-	NA	-
	Average	-	↘	↘	-	↘	↘	-	-	↘
5-day Biochemical Oxygen Demand	Surface	-	↘	↘	↘	-	-	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	-	NA	↘
	Bottom	-	↘	↘	↘	-	↘	-	NA	↘
	Average	-	↘	↘	↘	-	↘	-	NA	↘
Ammonia nitrogen (mg/L)	Surface	-	↘	↘	↘	-	-	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	↗	NA	↘
	Bottom	-	↘	↘	↘	↘	-	-	NA	↘
	Average	↘	↘	↘	↘	↘	-	-	-	↘
Unionised Ammonia(mg/L)	Surface	-	↘	↘	↘	↘	↘	-	↘	↘
	Middle	NA	↘	NA	↘	↘	NA	-	NA	↘
	Bottom	↘	↘	↘	↘	↘	↘	-	NA	↘
	Average	↘	↘	↘	↘	↘	↘	-	↘	↘
Nitrite nitrogen (mg/L)	Surface	-	↗	-	-	-	-	-	-	-
	Middle	NA	↗	NA	-	-	NA	-	NA	-
	Bottom	-	↗	-	-	-	-	-	NA	-
	Average	-	↗	-	-	-	-	-	-	-
Nitrate nitrogen (mg/L)	Surface	-	↗	↗	-	-	-	-	-	-
	Middle	NA	↗	NA	-	-	NA	-	NA	-
	Bottom	-	↗	↗	-	↗	-	-	NA	-
	Average	-	↗	↗	-	↗	-	-	-	-
Total inorganic nitrogen (mg/L)	Surface	-	↘	↘	↘	-	-	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	↗	NA	↘
	Bottom	-	↘	↘	↘	-	-	-	NA	↘
	Average	-	↘	↘	↘	-	-	-	-	↘
Total Kjeldahl nitrogen (mg/L)	Surface	-	↘	↘	↘	-	↘	↗	↗	↘
	Middle	NA	↘	NA	↘	-	NA	↗	NA	↘
	Bottom	-	↘	↘	↘	-	↘	↗	NA	↘
	Average	-	↘	↘	↘	-	↘	↗	-	↘
Total nitrogen (mg/L)	Surface	-	↘	↘	↘	-	-	↗	↗	↘
	Middle	NA	↘	NA	↘	-	NA	↗	NA	↘
	Bottom	-	↘	↘	↘	-	-	↗	NA	↘
	Average	-	↘	↘	↘	-	↘	↗	↗	↘
Orthophosphate phosphorus (mg/L)	Surface	↘	↘	↘	↘	↘	↘	↘	-	↘
	Middle	NA	↘	NA	↘	↘	NA	-	NA	↘
	Bottom	↘	↘	↘	↘	↘	↘	↘	NA	↘
	Average	↘	↘	↘	↘	↘	↘	↘	↘	↘
Total phosphorus (mg/L)	Surface	-	↘	↘	↘	-	-	-	-	↘
	Middle	NA	↘	NA	↘	↘	NA	-	NA	↘
	Bottom	-	↘	↘	↘	↘	↘	-	NA	↘
	Average	-	↘	↘	↘	↘	↘	-	-	↘
Silica (mg/L)	Surface	-	-	-	↘	-	↗	-	-	-
	Middle	NA	↘	NA	↘	-	NA	-	NA	-
	Bottom	-	-	-	↘	-	-	-	NA	-
	Average	-	-	-	↘	-	-	-	-	-
Chlorophyll- <i>a</i> (µg/L)	Surface	-	↗	↗	-	-	-	-	-	↘
	Middle	NA	↗	NA	-	-	NA	-	NA	↘
	Bottom	↗	↗	↗	-	-	-	-	NA	↘
	Average	-	↗	↗	-	-	-	-	-	↘
E. coli (cfu/100mL)	Surface	-	↘	↘	↘	-	↘	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	-	NA	↘
	Bottom	-	↘	↘	↘	-	↘	-	NA	↘
	Average	-	↘	↘	↘	-	↘	-	-	↘
Faecal coliforms (cfu/100mL)	Surface	-	↘	↘	↘	-	↘	-	-	↘
	Middle	NA	↘	NA	↘	-	NA	-	NA	↘
	Bottom	-	↘	↘	↘	-	↘	-	NA	↘
	Average	-	↘	↘	↘	-	↘	-	-	↘

Note: 1. Results of the Seasonal Kendall Test statistically significant at $p < 0.05$

2. - indicates no significant trend

3. NA (Not Applicable) indicates the measurement was not made due to shallow water

4. ↗ significant increase

5. ↘ significant decrease

Summary of water quality statistics for typhoon shelters, sheltered anchorages and Government Dockyard in 2016

Parameter	Tuen Mun NT1	Cheung Chau ST1	Hei Ling Chau ST3	Aberdeen (South) WT1	Aberdeen (West) WT3	Rambler Channel VT8
Number of samples	6	6	6	6	6	6
Temperature (°C)	24.1 (16.8 - 28.3)	24.1 (17.3 - 28.2)	24.0 (17.0 - 28.4)	23.4 (17.3 - 27.2)	23.3 (17.3 - 27.2)	24.2 (17.3 - 28.3)
Salinity	23.9 (13.3 - 29.8)	30.4 (27.1 - 31.6)	30.1 (27.0 - 31.3)	30.1 (25.2 - 32.1)	30.5 (26.7 - 32.3)	28.6 (23.8 - 31.0)
Dissolved Oxygen (mg/L)	5.6 (4.1 - 7.1)	6.5 (4.8 - 9.3)	7.1 (4.9 - 9.2)	5.8 (4.2 - 8.4)	5.6 (4.3 - 7.2)	5.5 (3.6 - 9.3)
Bottom	NA	6.7 (5.0 - 9.7)	7.2 (5.2 - 10.4)	5.6 (3.4 - 7.8)	5.5 (3.7 - 7.0)	5.5 (3.7 - 9.4)
Dissolved Oxygen (% Saturation)	75 (58 - 92)	92 (73 - 138)	100 (75 - 137)	80 (63 - 105)	77 (63 - 91)	76 (54 - 116)
Bottom	NA	94 (73 - 142)	101 (78 - 153)	78 (50 - 98)	76 (55 - 90)	76 (55 - 116)
pH	7.8 (7.5 - 8.3)	7.9 (7.6 - 8.2)	7.9 (7.6 - 8.3)	7.8 (7.5 - 8.3)	7.7 (7.5 - 8.3)	7.8 (7.4 - 8.3)
Secchi Disc Depth (m)	2.0 (1.5 - 3.0)	2.4 (2.0 - 3.0)	2.3 (2.0 - 3.0)	3.0 (2.0 - 4.0)	3.0 (2.2 - 4.5)	2.2 (1.5 - 3.0)
Turbidity (NTU)	13.5 (1.1 - 51.0)	4.3 (2.3 - 7.5)	3.9 (1.8 - 7.4)	2.7 (0.8 - 4.7)	3.1 (1.0 - 5.4)	4.5 (1.8 - 10.5)
Suspended Solids (mg/L)	7.5 (1.3 - 15.0)	11.6 (2.7 - 20.3)	11.6 (2.2 - 21.0)	4.2 (2.4 - 8.4)	4.2 (2.4 - 7.6)	11.1 (4.1 - 22.5)
5-day Biochemical Oxygen Demand (mg/L)	0.8 (0.3 - 1.2)	1.1 (0.4 - 2.0)	1.4 (0.6 - 2.5)	1.0 (0.3 - 2.0)	1.0 (0.3 - 2.3)	1.2 (0.1 - 4.0)
Ammonia Nitrogen (mg/L)	0.160 (0.100 - 0.290)	0.083 (0.045 - 0.127)	0.079 (0.050 - 0.120)	0.054 (0.029 - 0.079)	0.061 (0.020 - 0.101)	0.124 (0.039 - 0.200)
Unionised Ammonia (mg/L)	0.005 (0.002 - 0.011)	0.003 (0.001 - 0.006)	0.003 (0.001 - 0.005)	0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.002)	0.005 (<0.001 - 0.011)
Nitrite Nitrogen (mg/L)	0.056 (0.021 - 0.099)	0.024 (0.003 - 0.059)	0.029 (0.004 - 0.070)	0.025 (<0.002 - 0.068)	0.031 (0.003 - 0.077)	0.045 (0.016 - 0.088)
Nitrate Nitrogen (mg/L)	0.570 (0.240 - 1.100)	0.117 (0.058 - 0.190)	0.137 (0.078 - 0.210)	0.112 (0.019 - 0.283)	0.116 (0.023 - 0.287)	0.284 (0.140 - 0.475)
Total Inorganic Nitrogen (mg/L)	0.79 (0.39 - 1.31)	0.22 (0.14 - 0.31)	0.24 (0.17 - 0.34)	0.19 (0.06 - 0.36)	0.21 (0.08 - 0.41)	0.45 (0.37 - 0.62)
Total Kjeldahl Nitrogen (mg/L)	0.34 (0.16 - 0.54)	0.39 (0.24 - 0.65)	0.40 (0.24 - 0.53)	0.40 (0.20 - 0.56)	0.38 (0.15 - 0.51)	0.37 (0.14 - 0.63)
Total Nitrogen (mg/L)	0.97 (0.64 - 1.55)	0.54 (0.40 - 0.75)	0.56 (0.42 - 0.77)	0.54 (0.37 - 0.84)	0.53 (0.32 - 0.81)	0.70 (0.53 - 0.89)
Orthophosphate Phosphorus (mg/L)	0.034 (0.004 - 0.056)	0.015 (0.009 - 0.023)	0.015 (0.006 - 0.025)	0.009 (0.004 - 0.019)	0.008 (0.004 - 0.016)	0.022 (0.008 - 0.032)
Total Phosphorus (mg/L)	0.09 (0.05 - 0.15)	0.04 (0.03 - 0.08)	0.05 (0.03 - 0.09)	0.05 (0.03 - 0.15)	0.05 (0.02 - 0.10)	0.08 (0.04 - 0.11)
Silica (as SiO ₂) (mg/L)	3.85 (1.20 - 8.40)	1.07 (0.27 - 1.68)	1.14 (0.65 - 2.20)	1.06 (0.47 - 2.07)	1.12 (0.29 - 2.23)	1.94 (1.20 - 3.40)
Chlorophyll- <i>a</i> (µg/L)	2.9 (1.0 - 8.7)	7.6 (2.0 - 24.0)	8.0 (2.1 - 25.3)	4.3 (0.6 - 11.3)	4.5 (0.6 - 17.0)	5.1 (0.9 - 13.5)
<i>E. coli</i> (count/100mL)	220 (16 - 35000)	81 (48 - 150)	3 (1 - 18)	340 (140 - 630)	200 (61 - 570)	570 (170 - 1500)
Faecal Coliforms (count/100mL)	780 (35 - 140000)	210 (78 - 490)	6 (3 - 22)	1200 (500 - 3000)	490 (93 - 2300)	1700 (250 - 6000)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

4. NA (Not Applicable) indicates the measurement was not made due to shallow water.

Summary of water quality statistics for typhoon shelters, sheltered anchorages and Government Dockyard in 2016 (continued)

Parameter	Government Dockyard VT12	New Yau Ma Tei VT10	Causeway Bay VT2	To Kwa Wan VT11	Kwun Tong VT4	Sam Ka Tsuen VT3
Number of samples	6	6	6	6	6	6
Temperature (°C)	24.1 (17.6 - 28.0)	24.1 (17.3 - 28.0)	23.9 (17.1 - 28.0)	23.7 (17.2 - 28.1)	24.1 (17.2 - 28.4)	23.5 (17.0 - 27.9)
Salinity	30.0 (27.8 - 31.1)	30.0 (27.5 - 31.1)	30.4 (28.4 - 31.4)	31.0 (29.7 - 31.9)	28.0 (23.9 - 29.5)	31.2 (29.5 - 32.5)
Dissolved Oxygen (mg/L)	4.8 (3.5 - 6.6)	3.9 (3.0 - 5.6)	5.2 (3.5 - 7.9)	5.3 (3.9 - 7.6)	6.2 (3.0 - 9.0)	5.0 (3.5 - 7.4)
Bottom	5.0 (3.5 - 7.0)	4.1 (2.9 - 6.4)	5.2 (3.5 - 7.9)	5.5 (4.0 - 8.2)	5.9 (2.9 - 8.8)	5.3 (3.6 - 8.5)
Dissolved Oxygen (% Saturation)	68 (51 - 94)	55 (44 - 70)	73 (53 - 99)	74 (59 - 95)	86 (44 - 128)	70 (50 - 92)
Bottom	69 (52 - 91)	57 (43 - 80)	72 (54 - 99)	76 (58 - 102)	82 (42 - 123)	73 (51 - 106)
pH	7.8 (7.4 - 8.1)	7.7 (7.3 - 8.0)	7.8 (7.4 - 8.2)	7.8 (7.4 - 8.2)	7.7 (7.3 - 8.1)	7.8 (7.4 - 8.2)
Secchi Disc Depth (m)	2.2 (2.0 - 2.5)	2.3 (2.0 - 2.5)	2.2 (2.0 - 2.5)	2.4 (2.0 - 2.8)	2.1 (1.5 - 2.5)	2.7 (2.0 - 3.4)
Turbidity (NTU)	4.1 (1.5 - 7.5)	3.4 (1.2 - 7.7)	3.5 (1.5 - 5.9)	3.0 (1.3 - 5.0)	6.7 (1.0 - 29.0)	7.0 (1.0 - 30.0)
Suspended Solids (mg/L)	11.0 (3.1 - 29.0)	10.7 (2.0 - 22.5)	10.1 (3.9 - 21.5)	9.8 (2.9 - 18.7)	8.9 (2.2 - 19.7)	9.3 (1.1 - 22.0)
5-day Biochemical Oxygen Demand (mg/L)	1.9 (0.6 - 5.1)	1.8 (0.3 - 5.1)	1.0 (0.1 - 2.3)	1.5 (0.3 - 5.0)	1.7 (0.8 - 4.2)	1.0 (0.2 - 2.3)
Ammonia Nitrogen (mg/L)	0.237 (0.097 - 0.400)	0.328 (0.125 - 0.470)	0.155 (0.083 - 0.205)	0.138 (0.060 - 0.207)	0.306 (0.153 - 0.457)	0.116 (0.077 - 0.131)
Unionised Ammonia (mg/L)	0.007 (0.001 - 0.015)	0.008 (0.002 - 0.015)	0.005 (0.001 - 0.009)	0.004 ($<0.001 - 0.008$)	0.006 (0.004 - 0.010)	0.003 ($<0.001 - 0.006$)
Nitrite Nitrogen (mg/L)	0.031 (0.021 - 0.061)	0.036 (0.020 - 0.069)	0.030 (0.013 - 0.070)	0.030 (0.019 - 0.071)	0.135 (0.069 - 0.213)	0.029 (0.008 - 0.069)
Nitrate Nitrogen (mg/L)	0.173 (0.140 - 0.205)	0.169 (0.120 - 0.210)	0.180 (0.145 - 0.220)	0.152 (0.133 - 0.180)	0.408 (0.200 - 0.743)	0.143 (0.092 - 0.200)
Total Inorganic Nitrogen (mg/L)	0.44 (0.29 - 0.64)	0.53 (0.36 - 0.70)	0.37 (0.31 - 0.42)	0.32 (0.24 - 0.37)	0.85 (0.43 - 1.39)	0.29 (0.22 - 0.36)
Total Kjeldahl Nitrogen (mg/L)	0.64 (0.36 - 0.88)	0.80 (0.49 - 1.03)	0.55 (0.43 - 0.71)	0.58 (0.41 - 0.72)	0.76 (0.45 - 1.35)	0.45 (0.33 - 0.62)
Total Nitrogen (mg/L)	0.85 (0.55 - 1.11)	1.01 (0.66 - 1.22)	0.76 (0.63 - 0.94)	0.76 (0.57 - 0.90)	1.30 (0.72 - 2.30)	0.62 (0.47 - 0.85)
Orthophosphate Phosphorus (mg/L)	0.029 (0.003 - 0.057)	0.034 (0.011 - 0.062)	0.021 (0.007 - 0.039)	0.021 (0.007 - 0.036)	0.108 (0.071 - 0.168)	0.021 (0.011 - 0.033)
Total Phosphorus (mg/L)	0.08 (0.07 - 0.09)	0.09 (0.07 - 0.13)	0.06 (0.04 - 0.10)	0.06 (0.05 - 0.08)	0.19 (0.14 - 0.31)	0.05 (0.03 - 0.08)
Silica (as SiO ₂) (mg/L)	1.23 (0.53 - 1.70)	1.28 (1.20 - 1.40)	1.31 (1.20 - 1.50)	1.09 (0.88 - 1.27)	1.97 (1.20 - 3.56)	1.11 (1.01 - 1.25)
Chlorophyll-a (µg/L)	7.6 (1.5 - 17.5)	6.6 (1.7 - 20.0)	4.2 (1.2 - 12.5)	4.9 (1.2 - 12.0)	9.1 (1.7 - 18.4)	3.2 (1.1 - 8.9)
<i>E. coli</i> (count/100mL)	1300 (69 - 15000)	2300 (140 - 15000)	1700 (930 - 3200)	800 (220 - 9400)	3800 (1200 - 12000)	870 (220 - 5300)
Faecal Coliforms (count/100mL)	3100 (220 - 25000)	5600 (870 - 22000)	5400 (2700 - 7600)	1600 (440 - 13000)	7300 (2100 - 30000)	2100 (520 - 18000)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

Summary of water quality statistics for typhoon shelters, sheltered anchorages and Government Dockyard 2016 (continued)

Parameter	Shau Kei Wan ET2	Chai Wan ET1	Hebe Haven PT4	Yim Tin Tsai PT3	Sai Kung PT2	Shuen Wan TT1
Number of samples	6	6	6	6	6	6
Temperature (°C)	22.9 (16.6 - 27.2)	23.0 (16.9 - 28.1)	24.2 (16.4 - 29.6)	23.6 (16.0 - 28.9)	24.1 (16.2 - 29.0)	23.9 (16.4 - 28.4)
Salinity	30.8 (27.5 - 32.0)	30.9 (29.3 - 31.6)	30.2 (27.9 - 32.7)	31.2 (29.5 - 33.7)	30.8 (29.4 - 32.7)	30.3 (27.3 - 31.8)
Dissolved Oxygen (mg/L)	5.6 (4.1 - 8.5)	6.1 (4.5 - 9.7)	6.0 (5.4 - 6.6)	6.1 (5.1 - 7.5)	5.5 (4.6 - 6.3)	5.3 (3.1 - 8.5)
Bottom	5.2 (3.1 - 8.0)	6.1 (4.1 - 9.5)	6.0 (5.1 - 6.8)	6.2 (4.7 - 7.4)	NA	4.0 (0.9 - 6.6)
Dissolved Oxygen (% Saturation)	76 (62 - 105)	84 (66 - 120)	85 (74 - 98)	85 (79 - 95)	78 (58 - 96)	74 (48 - 111)
Bottom	70 (47 - 99)	84 (60 - 118)	85 (75 - 94)	86 (72 - 100)	NA	55 (13 - 88)
pH	7.7 (7.4 - 8.0)	7.8 (7.6 - 8.2)	7.9 (7.5 - 8.1)	7.9 (7.6 - 8.2)	7.8 (7.6 - 8.0)	7.8 (7.5 - 8.1)
Secchi Disc Depth (m)	3.2 (3.0 - 4.0)	2.9 (2.4 - 4.0)	2.7 (2.0 - 3.5)	3.2 (2.5 - 4.0)	2.5 (2.0 - 3.0)	2.7 (1.5 - 4.0)
Turbidity (NTU)	2.6 (1.1 - 3.8)	4.1 (1.0 - 14.2)	2.9 (1.2 - 6.0)	2.2 (1.2 - 3.1)	2.4 (1.0 - 4.8)	4.3 (2.1 - 9.5)
Suspended Solids (mg/L)	6.4 (0.6 - 17.3)	11.3 (0.7 - 45.3)	6.2 (1.6 - 18.5)	5.6 (1.0 - 18.7)	6.5 (1.2 - 20.0)	8.5 (0.9 - 22.1)
5-day Biochemical Oxygen Demand (mg/L)	1.0 (0.3 - 2.0)	1.3 (0.1 - 3.5)	1.3 (0.3 - 1.8)	1.4 (0.2 - 4.3)	1.6 (0.2 - 4.6)	1.8 (0.7 - 4.1)
Ammonia Nitrogen (mg/L)	0.076 (0.050 - 0.107)	0.062 (0.026 - 0.112)	0.057 (0.005 - 0.107)	0.043 (0.009 - 0.065)	0.044 (0.010 - 0.064)	0.094 (0.038 - 0.123)
Unionised Ammonia (mg/L)	0.002 (<0.001 - 0.004)	0.002 (<0.001 - 0.004)	0.002 (<0.001 - 0.003)	0.001 (<0.001 - 0.002)	0.001 (<0.001 - 0.002)	0.002 (0.001 - 0.003)
Nitrite Nitrogen (mg/L)	0.023 (0.004 - 0.063)	0.022 (0.003 - 0.071)	0.007 (0.002 - 0.012)	0.007 (<0.002 - 0.018)	0.006 (<0.002 - 0.014)	0.010 (0.003 - 0.029)
Nitrate Nitrogen (mg/L)	0.122 (0.061 - 0.233)	0.099 (0.046 - 0.183)	0.045 (<0.002 - 0.100)	0.022 (<0.002 - 0.069)	0.034 (<0.002 - 0.093)	0.032 (0.003 - 0.097)
Total Inorganic Nitrogen (mg/L)	0.22 (0.17 - 0.30)	0.18 (0.13 - 0.23)	0.11 (<0.01 - 0.21)	0.07 (0.01 - 0.13)	0.08 (0.01 - 0.16)	0.14 (0.05 - 0.22)
Total Kjeldahl Nitrogen (mg/L)	0.48 (0.12 - 0.67)	0.39 (0.09 - 0.70)	0.34 (0.22 - 0.48)	0.37 (0.23 - 0.69)	0.43 (0.23 - 0.76)	0.53 (0.27 - 0.95)
Total Nitrogen (mg/L)	0.63 (0.37 - 0.85)	0.51 (0.29 - 0.86)	0.39 (0.27 - 0.48)	0.40 (0.30 - 0.70)	0.47 (0.24 - 0.79)	0.57 (0.28 - 0.96)
Orthophosphate Phosphorus (mg/L)	0.019 (0.007 - 0.028)	0.012 (0.005 - 0.022)	0.005 (<0.002 - 0.016)	0.004 (<0.002 - 0.009)	0.004 (<0.002 - 0.009)	0.012 (0.005 - 0.033)
Total Phosphorus (mg/L)	0.05 (0.03 - 0.08)	0.05 (0.03 - 0.08)	0.04 (<0.02 - 0.08)	0.04 (<0.02 - 0.07)	0.03 (<0.02 - 0.06)	0.06 (0.03 - 0.09)
Silica (as SiO ₂) (mg/L)	1.09 (0.78 - 1.40)	0.96 (0.73 - 1.17)	1.10 (0.68 - 1.75)	0.86 (0.34 - 1.27)	0.98 (0.75 - 1.40)	1.54 (1.17 - 2.20)
Chlorophyll-a (µg/L)	3.1 (0.6 - 6.4)	6.0 (0.6 - 12.1)	3.6 (1.3 - 7.6)	5.6 (1.6 - 20.7)	7.5 (1.5 - 31.0)	5.9 (0.7 - 10.6)
<i>E. coli</i> (count/100mL)	450 (180 - 2300)	270 (140 - 460)	12 (<1 - 190)	2 (<1 - 5)	150 (4 - 2300)	21 (5 - 270)
Faecal Coliforms (count/100mL)	970 (340 - 7800)	620 (250 - 1600)	30 (7 - 650)	4 (1 - 12)	1100 (160 - 5200)	60 (18 - 980)

Note : 1. Unless otherwise specified, data presented are depth-averaged (A) values calculated by taking the means of three depths: Surface (S), Mid-depth (M), Bottom (B).

2. Data presented are annual arithmetic means of the depth-averaged results except for *E. coli* and faecal coliforms which are annual geometric means.

3. Data in brackets indicate the ranges.

4. NA (Not Applicable) indicates the measurement was not made due to shallow water.

Summary statistics of bottom sediment quality of typhoon shelters, sheltered anchorages and Government Dockyard, 2012– 2016

Parameter	Tuen Mun NS5	Cheung Chau SS7	Hei Ling Chau SS8	Rambler Channel VS17	Government Dockyard VS21	New Yau Ma Tei VS19	Causeway Bay VS12
Number of samples	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (% w/w)	76 (53 - 95)	88 (65 - 98)	97 (92 - 99)	70 (23 - 99)	80 (3 - 99)	86 (78 - 92)	73 (26 - 99)
Electrochemical Potential (mV)	-262 (-386 - -92)	-291 (-465 - -74)	-270 (-403 - -79)	-296 (-395 - -169)	-337 (-402 - -230)	-374 (-412 - -320)	-317 (-405 - -193)
Total Solids (% w/w)	44 (36 - 53)	38 (28 - 48)	36 (34 - 40)	46 (37 - 70)	37 (30 - 46)	49 (41 - 54)	50 (35 - 68)
Total Volatile Solids (% w/w)	7.4 (5.5 - 9.2)	7.6 (6.2 - 11.0)	7.8 (6.4 - 8.7)	7.2 (4.1 - 9.0)	7.7 (6.4 - 9.2)	6.9 (6.0 - 8.2)	6.8 (3.8 - 10.0)
Chemical Oxygen Demand (mg/kg)	16500 (13000 - 27000)	14700 (13000 - 18000)	10850 (8900 - 14000)	19100 (15000 - 24000)	15600 (14000 - 20000)	15400 (12000 - 22000)	17900 (10000 - 33000)
Total Carbon (% w/w)	0.7 (0.5 - 0.9)	0.6 (0.5 - 0.9)	0.5 (0.4 - 0.6)	1.0 (0.7 - 1.9)	0.5 (0.4 - 0.6)	0.9 (0.8 - 1.1)	1.0 (0.6 - 2.0)
Ammonical Nitrogen (mg/kg)	8.93 (2.20 - 27.00)	7.78 (<0.05 - 23.00)	7.68 (<0.05 - 18.00)	3.75 (0.23 - 6.60)	10.29 (2.70 - 22.00)	21.71 (6.10 - 32.00)	16.43 (1.30 - 65.00)
Total Kjeldahl Nitrogen (mg/kg)	480 (270 - 580)	500 (410 - 680)	540 (450 - 650)	500 (200 - 710)	440 (370 - 520)	510 (380 - 590)	480 (300 - 750)
Total Phosphorus (mg/kg)	220 (150 - 290)	230 (170 - 420)	190 (160 - 230)	200 (120 - 260)	180 (160 - 190)	250 (170 - 360)	200 (130 - 310)
Total Sulphide (mg/kg)	89 (13 - 340)	74 (0 - 330)	31 (1 - 66)	170 (23 - 400)	96 (7 - 350)	110 (3 - 240)	140 (11 - 410)
Total Cyanide (mg/kg)	0.2 (0.1 - 0.3)	0.2 (0.1 - 0.3)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.5)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)
Arsenic (mg/kg)	8.9 (7.2 - 14.0)	7.2 (5.9 - 10.0)	6.8 (5.7 - 8.6)	8.5 (6.4 - 10.0)	7.6 (4.3 - 9.6)	5.6 (4.2 - 8.2)	6.7 (3.8 - 11.0)
Cadmium (mg/kg)	0.3 (0.2 - 0.6)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	0.7 (0.2 - 1.8)	0.4 (0.3 - 0.5)	0.4 (0.2 - 0.6)	0.4 (0.1 - 0.8)
Chromium (mg/kg)	33 (28 - 43)	44 (31 - 60)	37 (32 - 42)	110 (28 - 330)	42 (28 - 50)	35 (30 - 42)	36 (17 - 54)
Copper (mg/kg)	47 (31 - 67)	88 (56 - 130)	34 (28 - 39)	210 (33 - 730)	150 (89 - 190)	73 (36 - 100)	89 (37 - 170)
Lead (mg/kg)	46 (27 - 69)	65 (33 - 230)	45 (39 - 51)	69 (54 - 87)	47 (33 - 57)	37 (24 - 48)	54 (27 - 90)
Mercury (mg/kg)	0.10 (0.07 - 0.17)	0.17 (0.10 - 0.30)	0.13 (0.11 - 0.23)	0.21 (0.07 - 0.33)	0.20 (0.16 - 0.27)	0.16 (<0.05 - 0.22)	0.48 (0.12 - 0.69)
Nickel (mg/kg)	18 (14 - 24)	20 (15 - 24)	23 (20 - 27)	42 (12 - 160)	22 (19 - 25)	20 (17 - 22)	17 (7 - 28)
Silver (mg/kg)	1.2 (0.4 - 3.2)	0.3 (<0.2 - 0.4)	0.4 (0.2 - 0.4)	3.7 (0.4 - 12.0)	1.4 (1.0 - 1.9)	1.1 (0.5 - 1.8)	1.7 (0.4 - 4.4)
Zinc (mg/kg)	160 (110 - 260)	150 (110 - 180)	140 (120 - 220)	240 (110 - 400)	200 (110 - 290)	180 (110 - 300)	170 (71 - 300)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	18 (18 - 18)	18 (18 - 21)	18 (18 - 18)	32 (18 - 76)	21 (18 - 29)	18 (18 - 20)	67 (18 - 400)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(4) (6)}	170 (90 - 340)	110 (90 - 190)	110 (90 - 190)	210 (96 - 510)	130 (95 - 280)	120 (90 - 170)	180 (94 - 290)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(5) (6)}	250 (76 - 970)	220 (100 - 650)	91 (59 - 170)	1700 (260 - 5400)	240 (110 - 490)	200 (37 - 340)	680 (140 - 2100)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.

3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polycyclic aromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

5 High molecular weight polycyclic aromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(g,h,i)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

Summary statistics of bottom sediment quality of typhoon shelters, sheltered anchorages and Government Dockyard, 2012– 2016 (continued)

Parameter	To Kwa Wan VS20	Kwun Tong VS14	Sam Ka Tsuen VS13	Shau Kei Wan ES5	Chai Wan ES3	Hebe Haven PS4	Yim Tin Tsai PS2	Shuen Wan TS7
Number of samples	10	10	10	10	10	10	10	10
Particle Size Fractionation <63µm (% w/w)	79 (58 - 99)	79 (52 - 91)	84 (79 - 96)	93 (81 - 99)	89 (83 - 96)	91 (75 - 97)	79 (33 - 91)	76 (45 - 100)
Electrochemical Potential (mV)	-357 (-402 - -318)	-320 (-405 - -147)	-321 (-457 - -164)	-356 (-400 - -289)	-320 (-389 - -252)	-302 (-396 - -151)	-282 (-398 - -86)	-313 (-451 - -201)
Total Solids (% w/w)	42 (38 - 48)	36 (29 - 41)	38 (35 - 41)	29 (25 - 37)	38 (35 - 44)	42 (36 - 46)	48 (38 - 72)	38 (28 - 50)
Total Volatile Solids (% w/w)	9.8 (7.5 - 16.0)	10.1 (8.5 - 12.0)	9.6 (8.6 - 10.0)	9.3 (8.1 - 10.0)	8.3 (7.2 - 9.0)	9.5 (8.2 - 11.0)	9.8 (5.8 - 12.0)	8.5 (6.0 - 11.0)
Chemical Oxygen Demand (mg/kg)	27300 (21000 - 41000)	22600 (18000 - 32000)	22000 (17000 - 27000)	16900 (12000 - 21000)	20600 (16000 - 26000)	22300 (20000 - 24000)	15640 (9400 - 19000)	23000 (20000 - 29000)
Total Carbon (% w/w)	1.4 (0.6 - 2.8)	1.0 (0.7 - 1.8)	1.0 (0.7 - 1.2)	0.7 (0.5 - 0.9)	0.9 (0.8 - 1.0)	1.0 (0.7 - 1.3)	2.0 (0.9 - 6.0)	0.8 (0.6 - 1.0)
Ammonical Nitrogen (mg/kg)	10.87 (2.00 - 25.00)	17.27 (5.80 - 36.00)	15.60 (1.40 - 74.00)	30.90 (23.00 - 37.00)	4.17 (0.28 - 10.00)	5.40 (3.40 - 9.80)	10.36 (4.10 - 23.00)	7.88 (2.80 - 15.00)
Total Kjeldahl Nitrogen (mg/kg)	630 (500 - 760)	700 (500 - 1200)	630 (380 - 750)	560 (470 - 620)	570 (460 - 660)	760 (610 - 1000)	650 (170 - 950)	620 (430 - 780)
Total Phosphorus (mg/kg)	250 (210 - 340)	250 (190 - 420)	280 (170 - 310)	180 (150 - 220)	220 (190 - 240)	210 (170 - 250)	180 (79 - 260)	210 (160 - 320)
Total Sulphide (mg/kg)	190 (43 - 350)	280 (0 - 860)	290 (120 - 430)	510 (200 - 730)	220 (63 - 350)	52 (5 - 140)	32 (8 - 83)	130 (30 - 340)
Total Cyanide (mg/kg)	0.2 (<0.1 - 0.3)	0.3 (<0.1 - 0.5)	0.2 (<0.1 - 0.3)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.2)	0.2 (0.1 - 0.3)	0.1 (<0.1 - 0.2)	0.2 (<0.1 - 0.4)
Arsenic (mg/kg)	9.0 (7.2 - 14.0)	7.3 (6.2 - 8.8)	7.8 (6.7 - 11.0)	7.5 (6.7 - 8.5)	9.3 (7.5 - 11.0)	8.3 (6.5 - 10.0)	4.6 (3.5 - 5.9)	9.2 (7.1 - 11.0)
Cadmium (mg/kg)	0.9 (0.6 - 1.4)	2.6 (1.2 - 4.9)	0.8 (0.4 - 1.0)	0.4 (0.4 - 0.5)	0.5 (0.3 - 0.8)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	0.4 (0.2 - 0.6)
Chromium (mg/kg)	93 (67 - 140)	260 (130 - 510)	56 (46 - 69)	53 (46 - 68)	79 (58 - 100)	24 (17 - 27)	16 (13 - 21)	22 (15 - 33)
Copper (mg/kg)	600 (350 - 1100)	1700 (900 - 4300)	160 (120 - 200)	200 (150 - 300)	230 (140 - 280)	51 (24 - 61)	14 (11 - 17)	99 (41 - 140)
Lead (mg/kg)	140 (75 - 400)	130 (84 - 170)	86 (61 - 100)	66 (55 - 80)	84 (61 - 100)	39 (23 - 45)	31 (20 - 41)	97 (55 - 110)
Mercury (mg/kg)	1.31 (0.89 - 2.70)	0.89 (0.57 - 1.40)	1.01 (0.41 - 1.60)	0.26 (0.17 - 0.29)	0.41 (0.28 - 0.53)	0.13 (0.09 - 0.19)	0.06 (<0.05 - 0.08)	0.26 (0.06 - 1.50)
Nickel (mg/kg)	31 (20 - 51)	71 (37 - 160)	19 (16 - 22)	24 (20 - 31)	25 (17 - 30)	8 (5 - 10)	13 (9 - 21)	10 (7 - 13)
Silver (mg/kg)	4.2 (3.2 - 5.3)	8.0 (4.5 - 15.0)	2.2 (1.8 - 2.5)	4.6 (1.8 - 9.2)	12.0 (6.0 - 23.0)	0.3 (0.2 - 0.3)	0.2 (<0.2 - 0.3)	0.4 (0.2 - 0.5)
Zinc (mg/kg)	340 (190 - 980)	420 (310 - 640)	330 (280 - 390)	270 (230 - 370)	340 (240 - 490)	160 (81 - 250)	90 (53 - 250)	280 (190 - 450)
Total Polychlorinated Biphenyls (PCBs) (µg/kg) ⁽³⁾	88 (56 - 130)	210 (120 - 380)	40 (29 - 66)	20 (18 - 31)	34 (23 - 51)	18 (18 - 19)	18 (18 - 18)	19 (18 - 25)
Low Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(4) (6)}	43000 (1000 - 120000)	260 (96 - 550)	220 (110 - 430)	190 (100 - 350)	200 (100 - 280)	170 (93 - 390)	150 (90 - 400)	140 (90 - 290)
High Molecular Weight Polycyclic Aromatic Hydrocarbons (PAHs) (µg/kg) ^{(5) (6)}	440000 (9300 - 1200000)	1200 (790 - 2100)	830 (540 - 1200)	630 (390 - 1200)	600 (310 - 970)	110 (71 - 150)	59 (21 - 95)	130 (53 - 210)

Note: 1 Data presented are arithmetic means ; data in brackets indicate ranges.

2 All data are based on the analyses of bulk (unsieved) sediment and are reported on a dry weight basis unless stated otherwise.

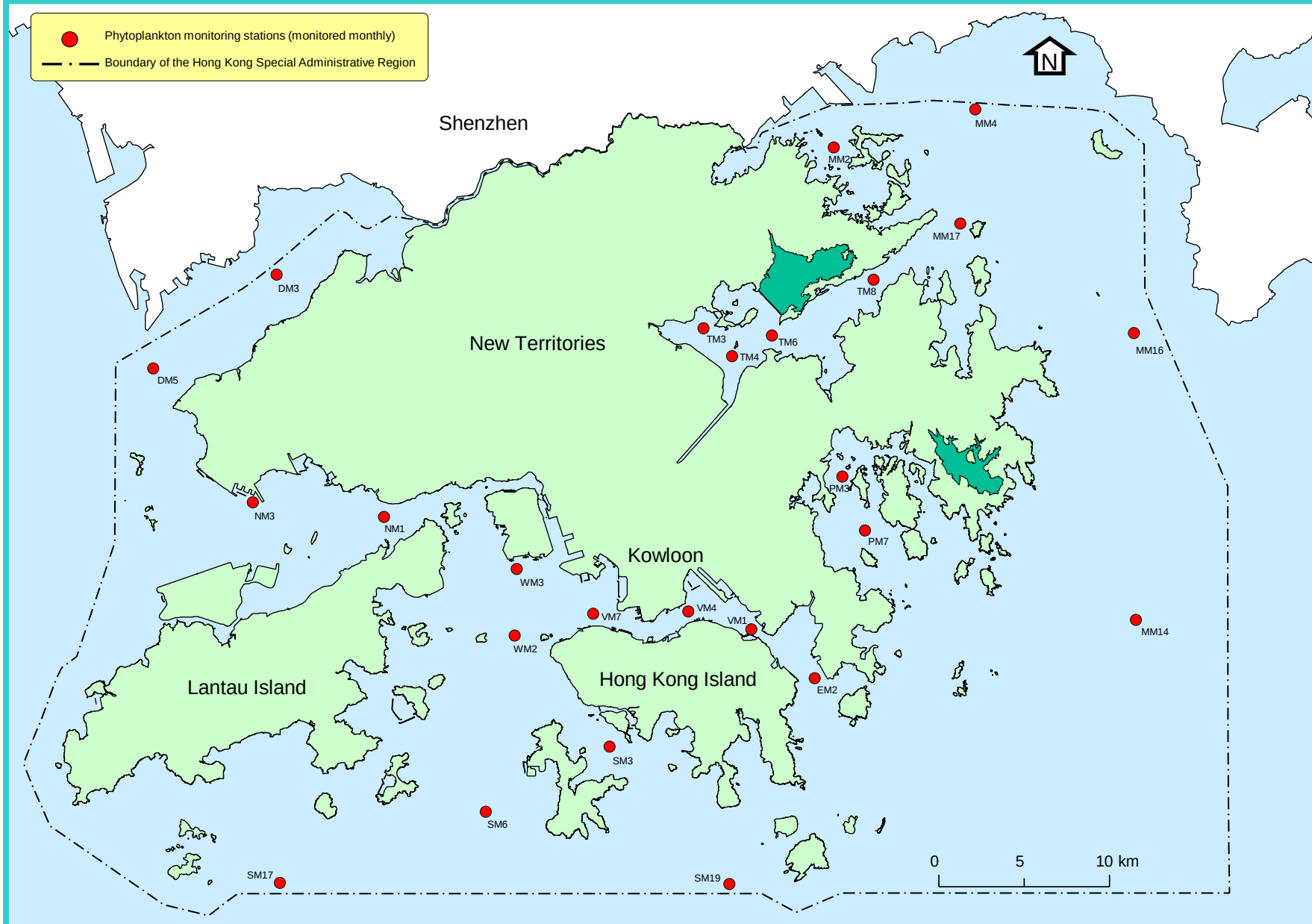
3 Total PCBs results are derived from the summation of 18 congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

4 Low molecular weight polycyclic aromatic hydrocarbons (PAHs) include 6 congeners of molecular weight below 200, namely : Acenaphthene, Acenaphthylene, Anthracene, Fluorene, Naphthalene and Phenanthrene.

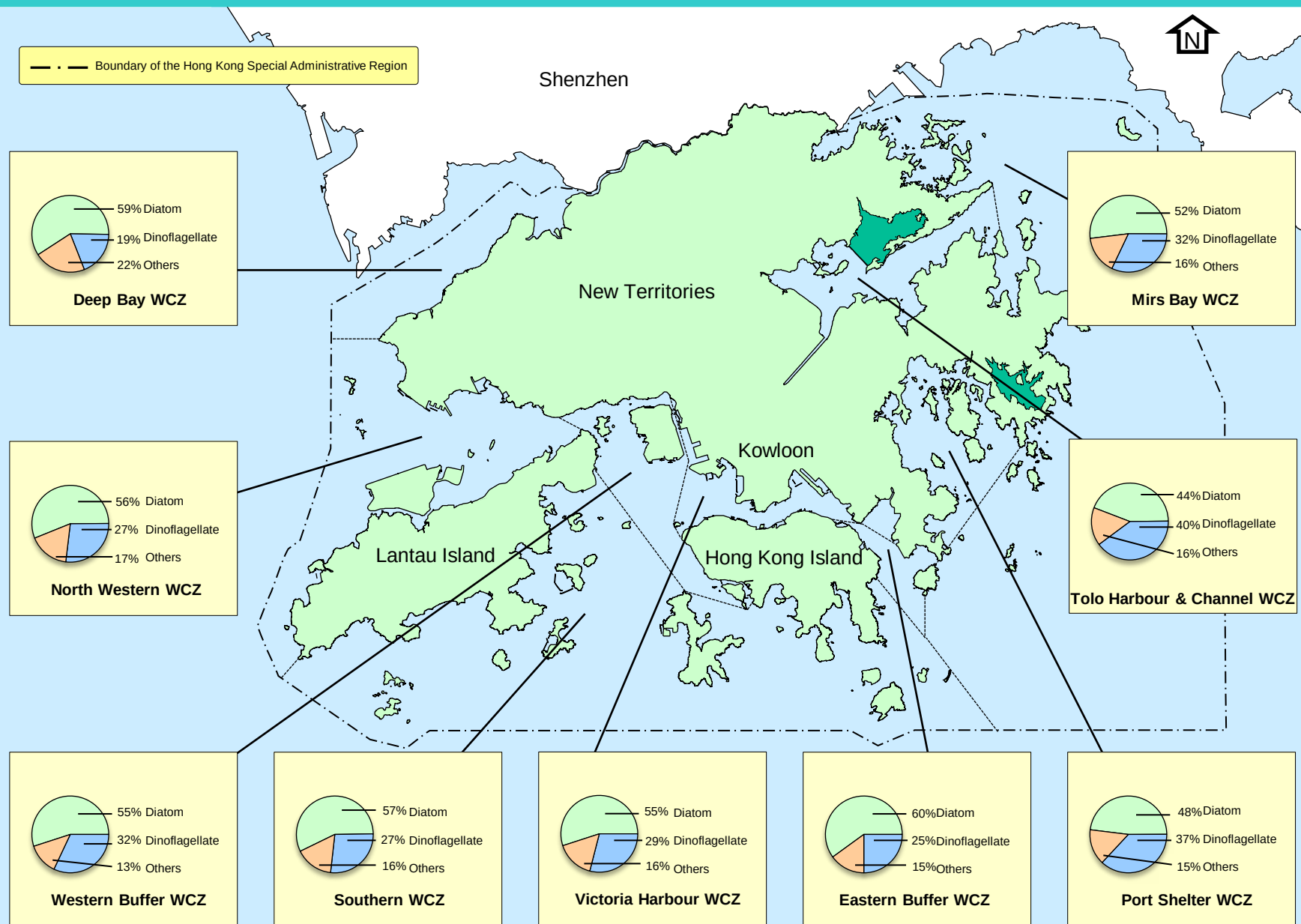
5 High molecular weight polycyclic aromatic hydrocarbons (PAHs) include 10 congeners of molecular weight above 200, namely : Fluoranthene, Pyrene, Benzo(a)anthracene, Chrysene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Benzo(a)pyrene, Dibenzo(a,h)anthracene, Benzo(q,h)perylene and Indeno(1,2,3-cd)pyrene.

6 Low and high molecular weight PAHs results are derived from the summation of the corresponding congeners. If the concentration of a congener is below report limit (RL), the result will be taken as 0.5xRL in the calculation.

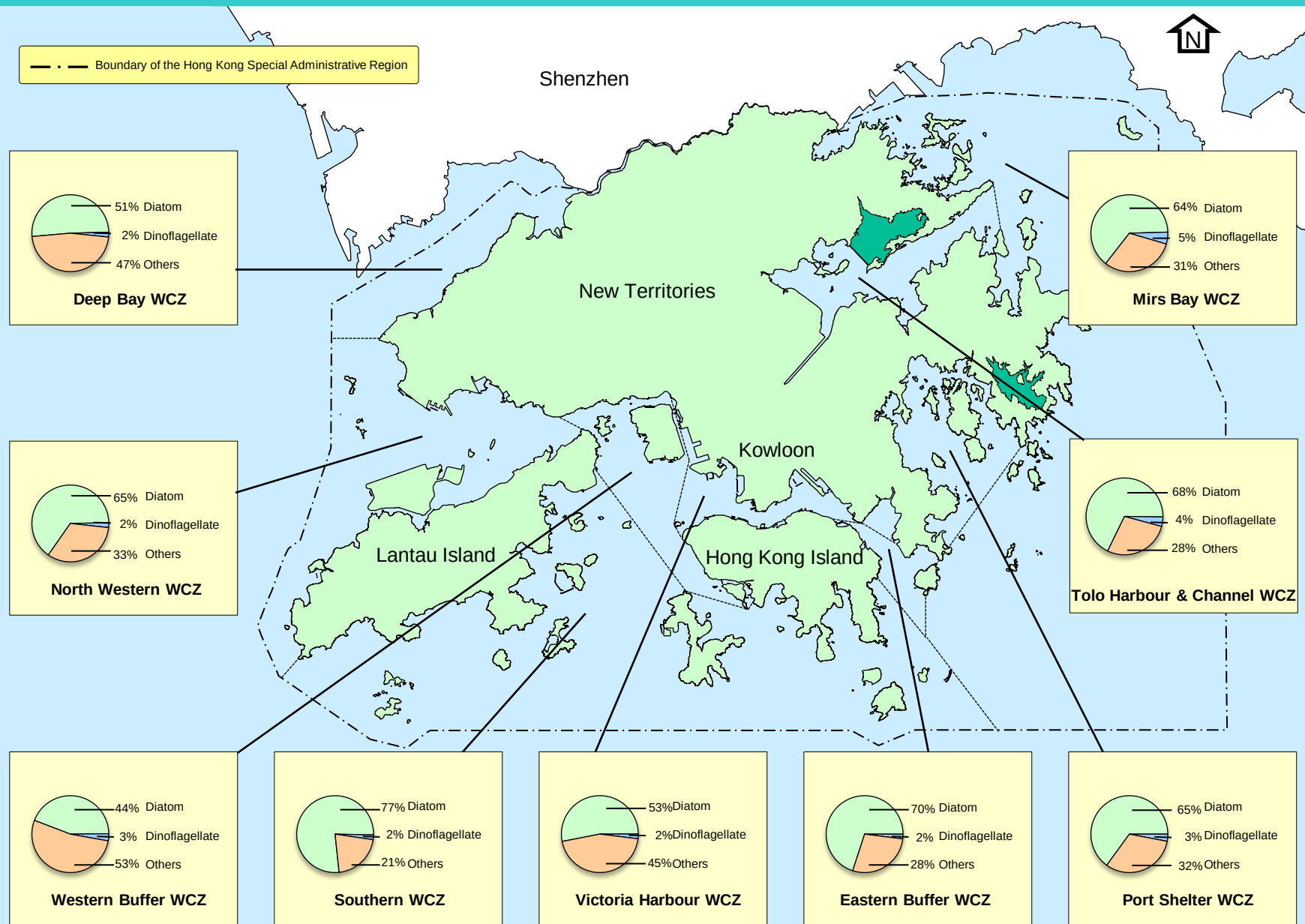
The 25 phytoplankton monitoring stations in Hong Kong waters in 2016



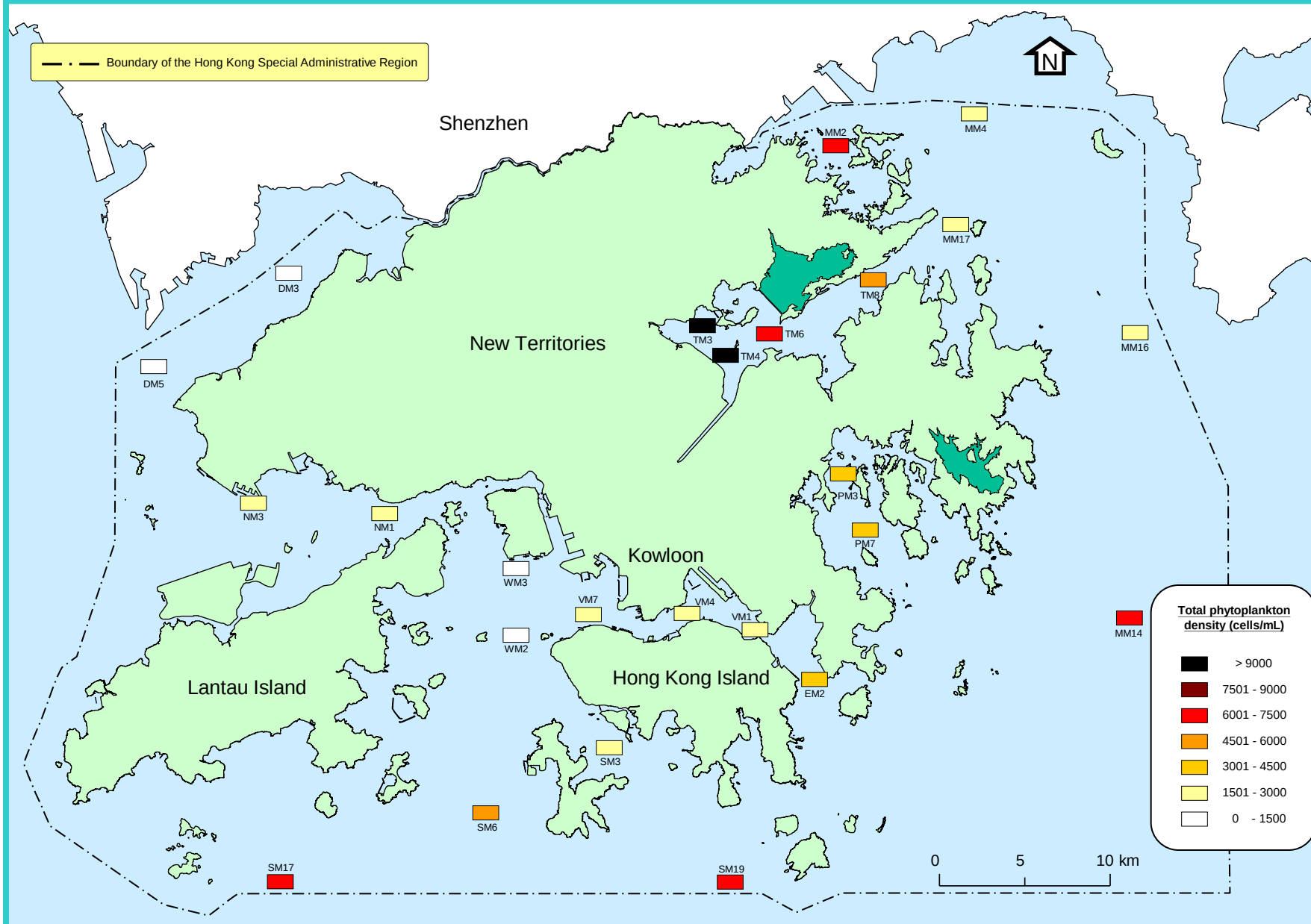
Percentage contribution of phytoplankton groups to the total number of species in the 9 Water Control Zones (2016)



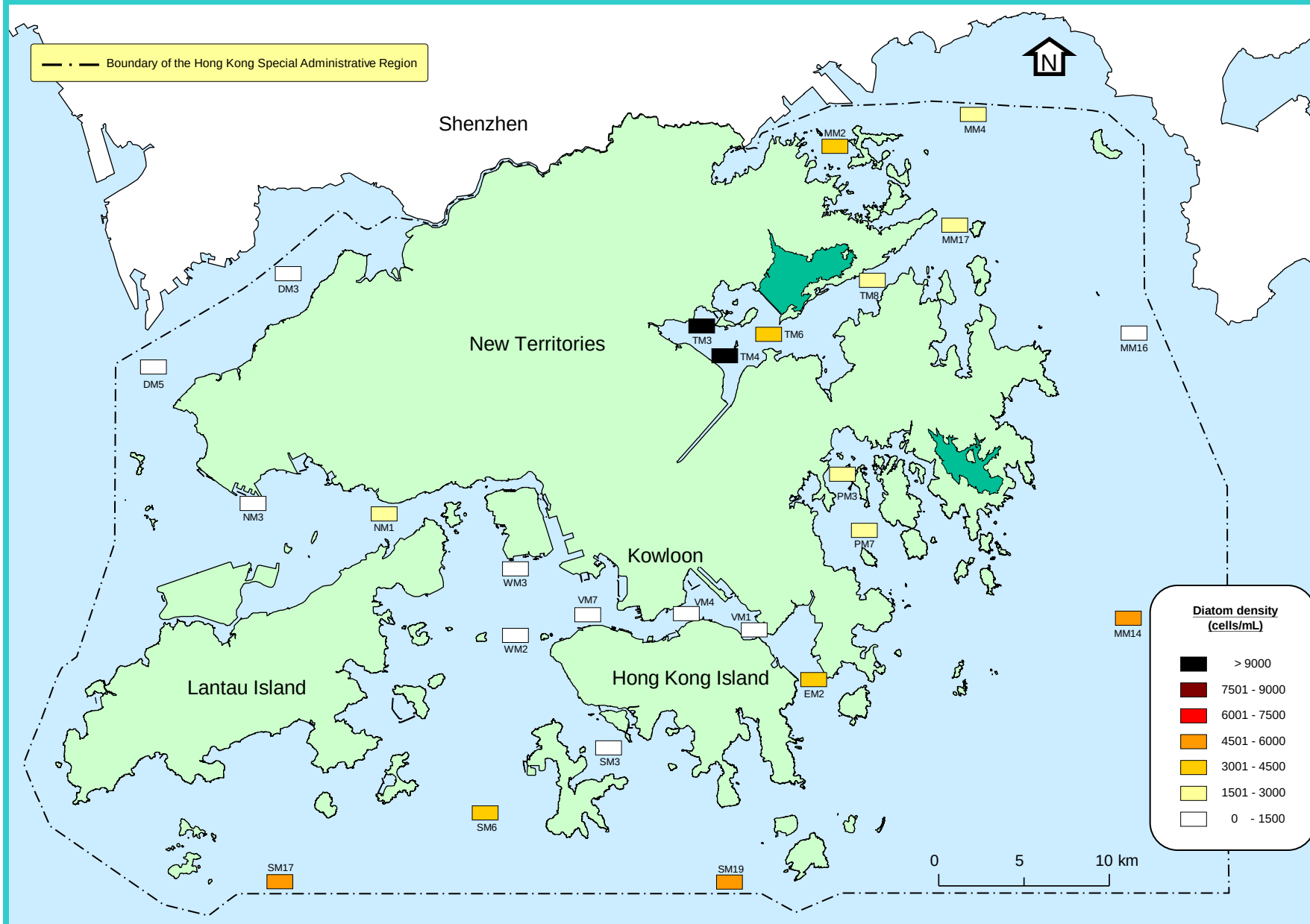
Percentage contribution of phytoplankton groups to the total density in the 9 Water Control Zones (2016)



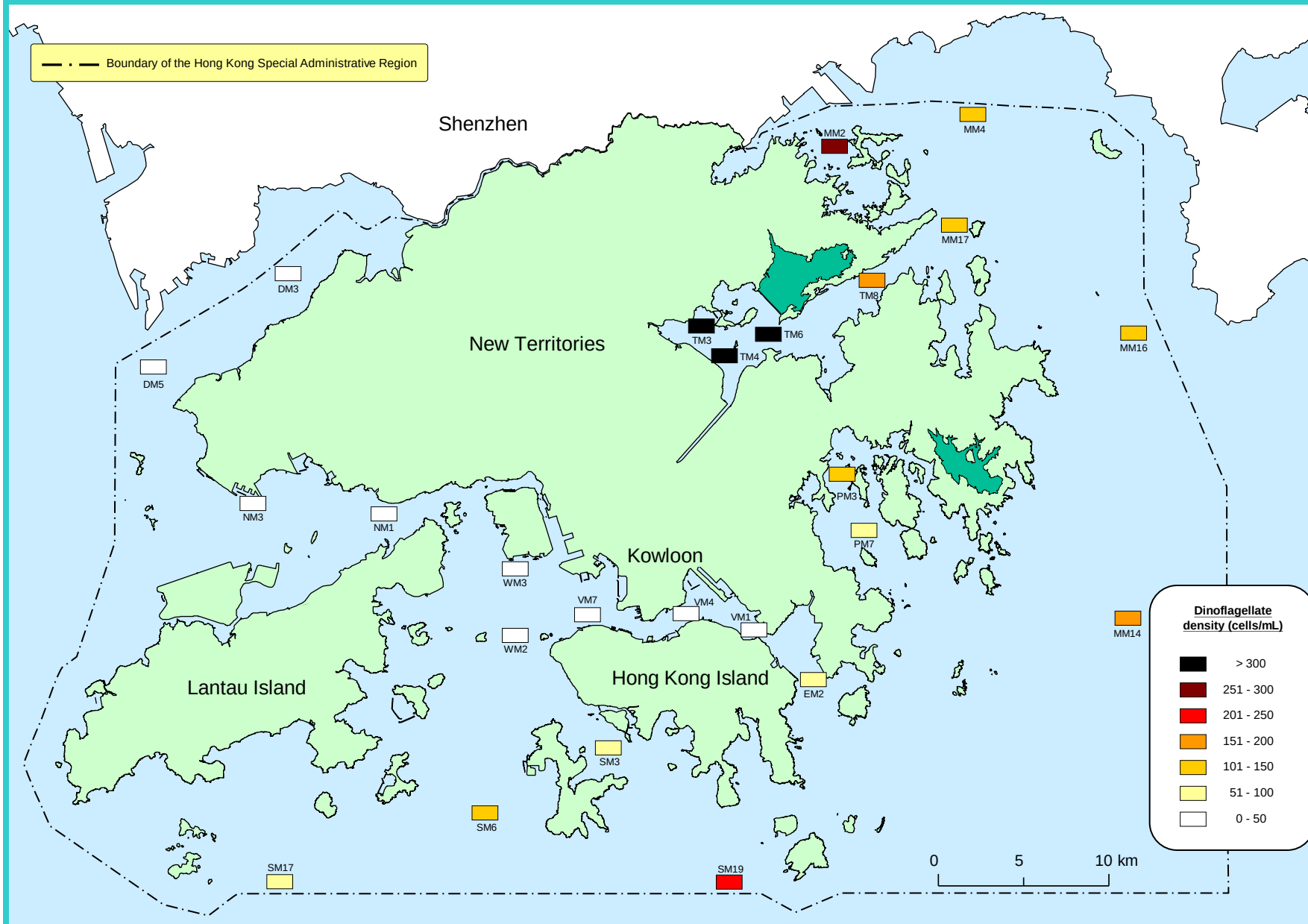
Annual mean densities (cells/mL) of total phytoplankton at 25 phytoplankton monitoring stations in Hong Kong waters in 2016



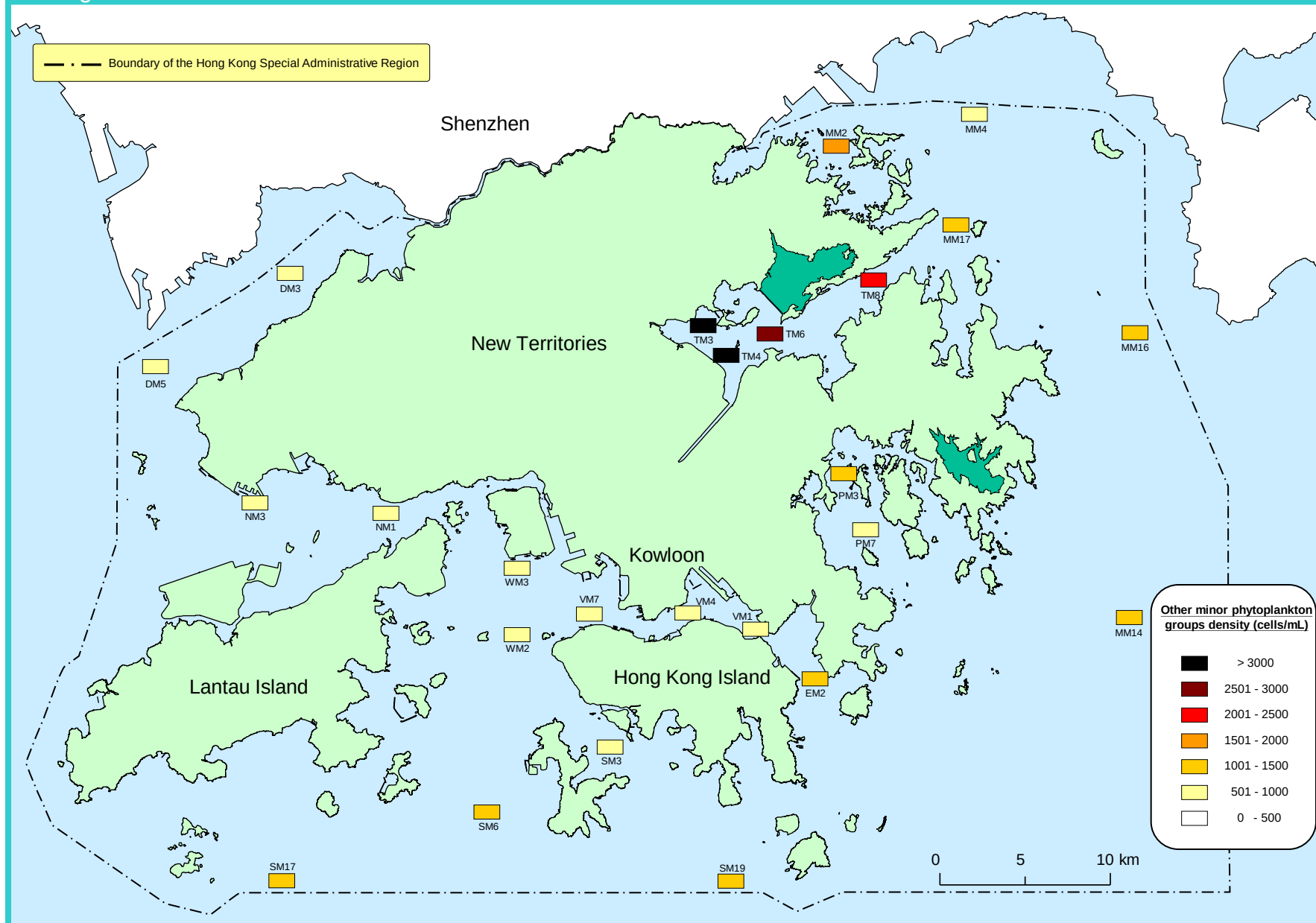
Annual mean densities (cells/mL) of diatoms at 25 phytoplankton monitoring stations in Hong Kong waters in 2016



Annual mean densities (cells/mL) of dinoflagellates at 25 phytoplankton monitoring stations in Hong Kong waters in 2016

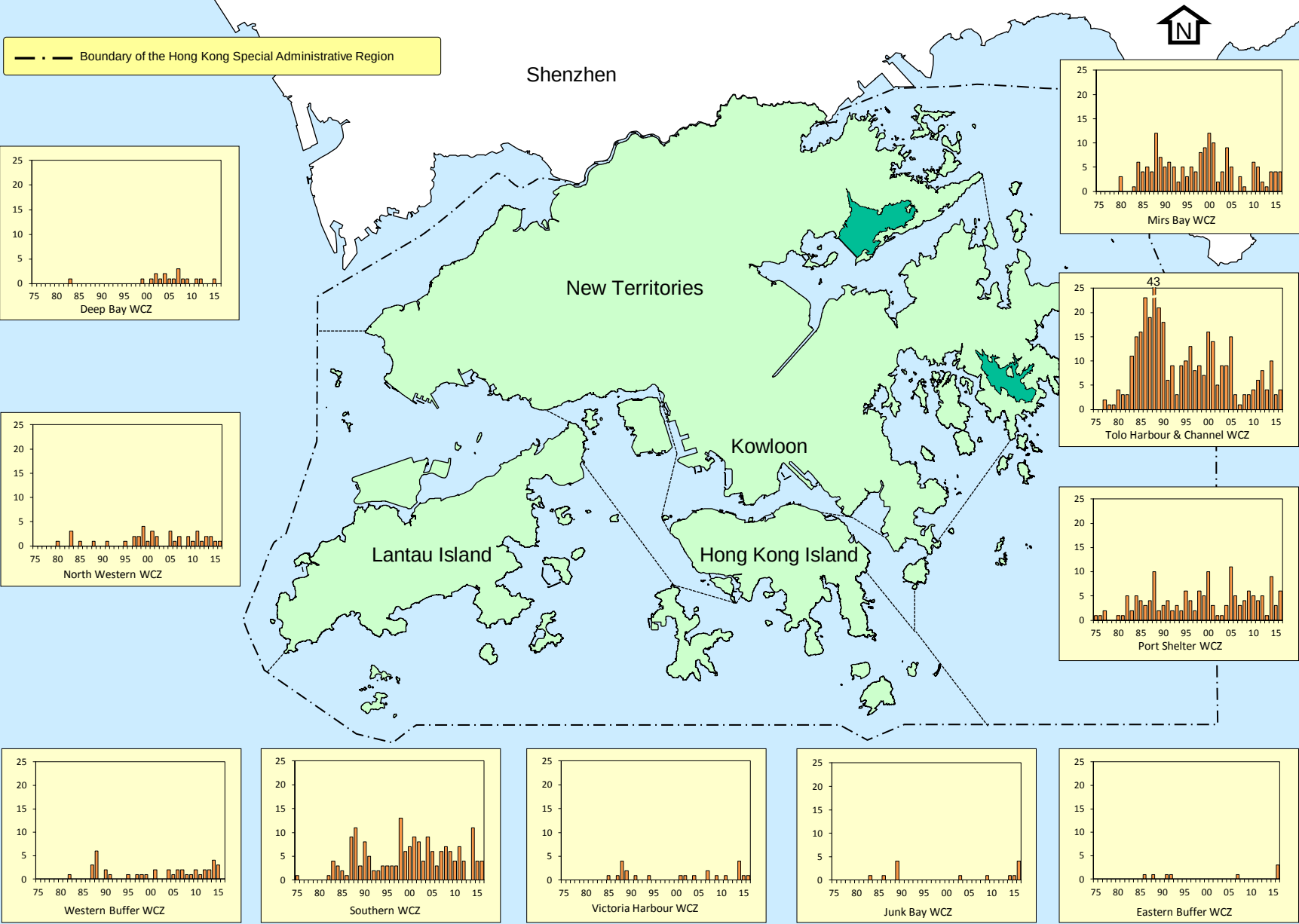


Annual mean densities (cells/mL) of other minor phytoplankton groups at 25 phytoplankton monitoring stations in Hong Kong waters in 2016



Occurrence of red tides in 10 Water Control Zones in Hong Kong waters, 1975 - 2016

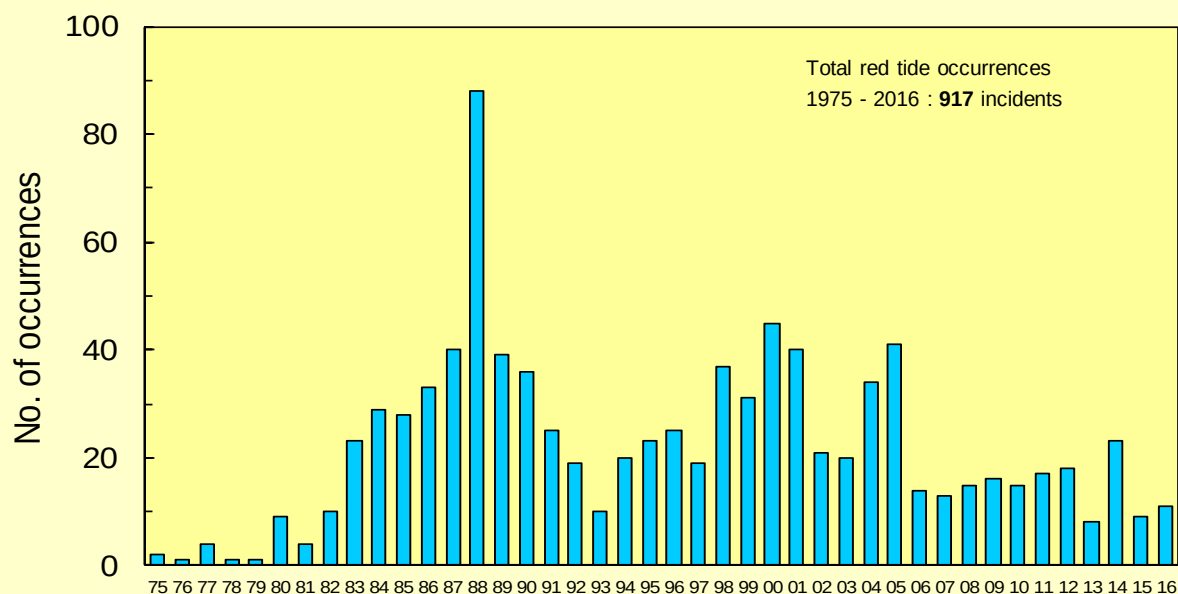
(Sources: Agriculture, Fisheries and Conservation Department and Environmental Protection Department)



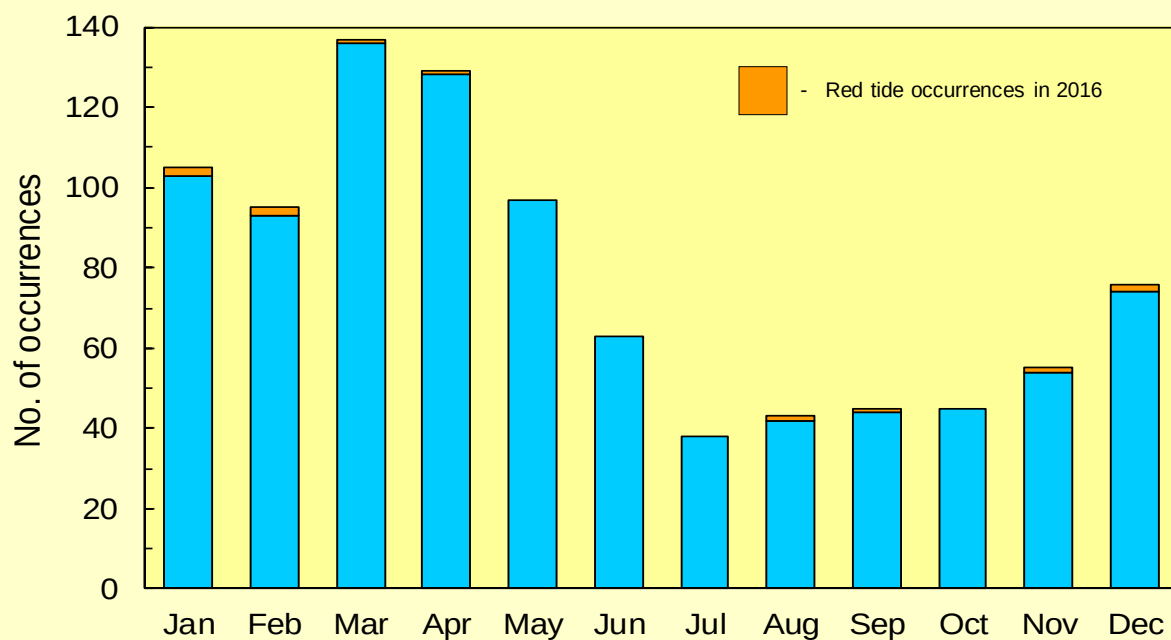
Occurrence of red tides in Hong Kong waters, 1975-2016

(Sources: Agriculture, Fisheries and Conservation Department and Environmental Protection Department)

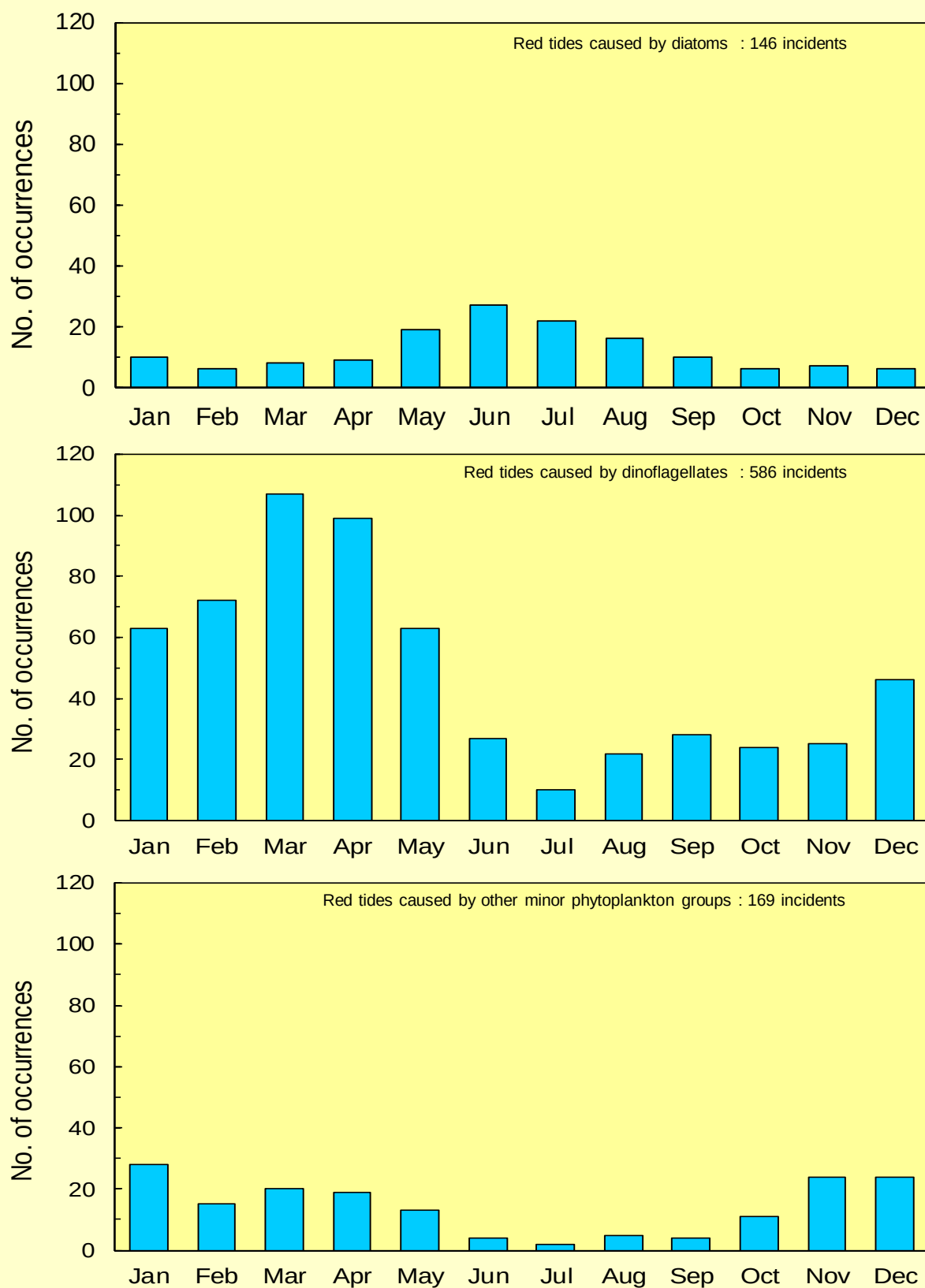
Yearly Distribution



Seasonal Distribution



Seasonal Distribution of red tides caused by different phytoplankton groups in Hong Kong, 1975-2016 (Sources: Agriculture, Fisheries and Conservation Department and Environmental Protection Department)

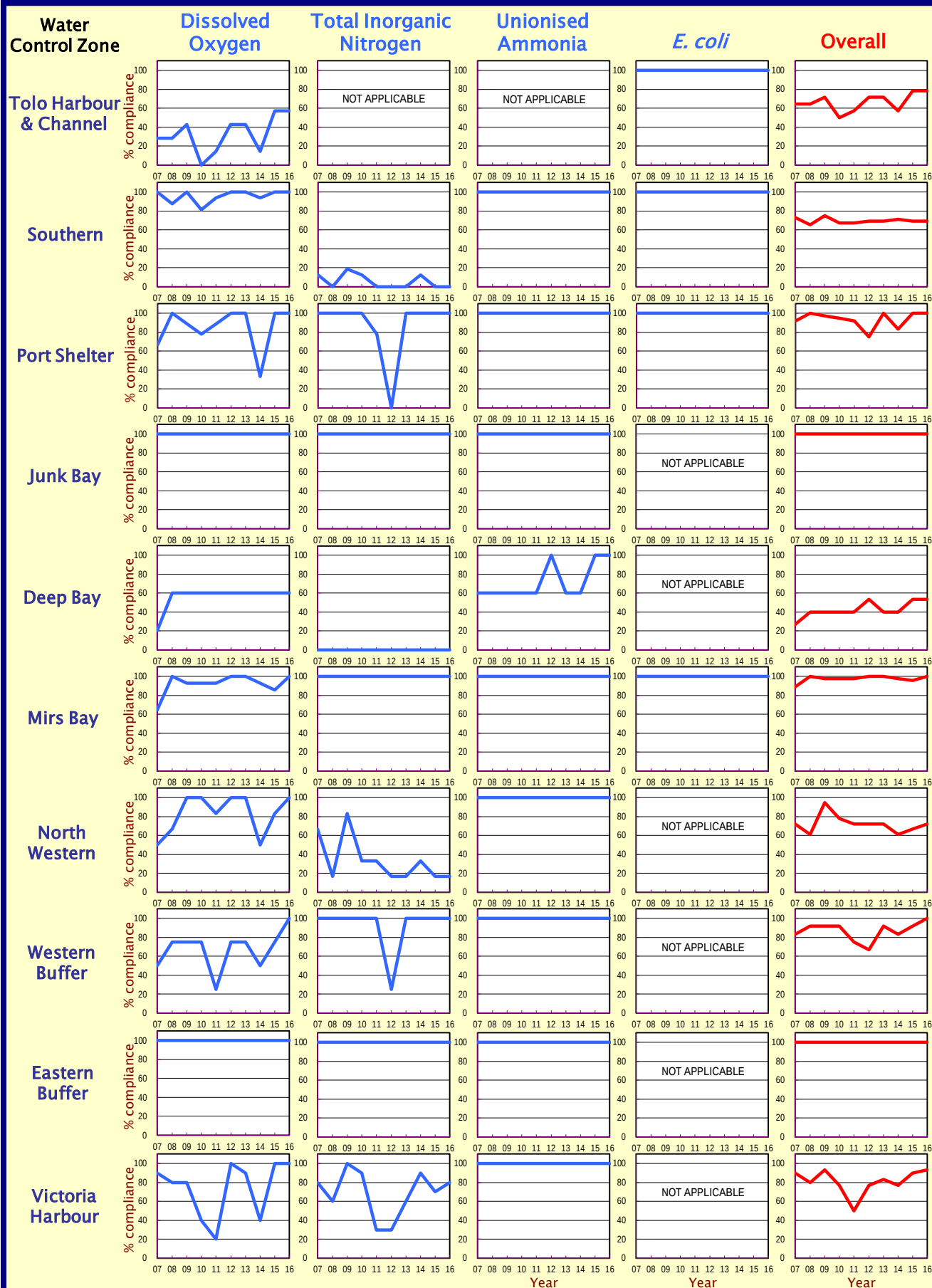


Abundance and frequency of the dominant phytoplankton species in different Water Control Zones in 2016

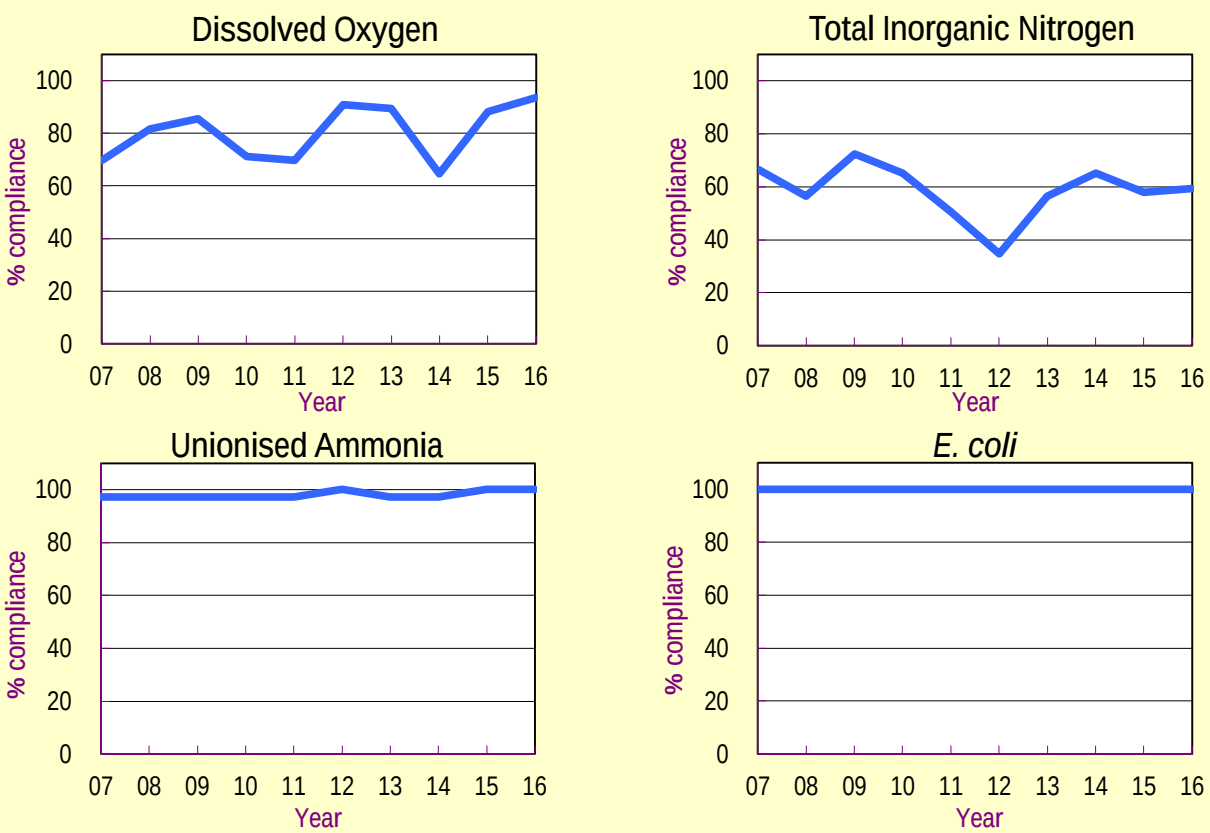
Species	% Abundance ¹	Frequency ²	Species	% Abundance ¹	Frequency ²	Species	% Abundance ¹	Frequency ²
Tolo Harbour & Channel			Port Shelter			North Western		
Diatoms			Diatoms			Diatoms		
<i>Chaetoceros</i> spp.	27.13	11	<i>Skeletonema costatum</i>	23.97	8	<i>Chaetoceros</i> spp.	45.37	11
<i>Thalassiosira</i> spp.	18.27	12	<i>Chaetoceros</i> spp.	22.83	12	<i>Thalassiosira</i> spp.	21.72	12
<i>Ceratoneis closterium</i>	14.82	8	<i>Thalassiosira</i> spp.	17.31	12	<i>Pseudo-nitzschia</i> spp.	13.38	8
Dinoflagellates			Dinoflagellates			Dinoflagellates		
<i>Gymnodinium</i> spp.	25.35	12	<i>Akashiwo sanguinea</i>	35.16	5	<i>Gymnodinium</i> spp.	75.32	12
<i>Karenia mikimotoi</i>	18.22	6	<i>Gymnodinium</i> spp.	24.23	12	<i>Amphidinium</i> spp.	7.36	6
<i>Karenia papilionacea</i>	8.89	2	<i>Amphidinium</i> spp.	14.48	10	<i>Protoperidinium</i> spp.	4.55	3
Others			Others			Others		
small flagellates	75.89	12	small flagellates	80.62	12	small flagellates	66.20	12
<i>Plagioselmis prolunga</i>	19.19	12	<i>Plagioselmis prolunga</i>	15.83	12	<i>Plagioselmis prolunga</i>	24.55	12
<i>Teleaulax acuta</i>	3.99	12	<i>Teleaulax acuta</i>	2.38	10	<i>Teleaulax acuta</i>	7.65	12
Mirs Bay			Victoria Harbour			Western Buffer		
Diatoms			Diatoms			Diatoms		
<i>Chaetoceros</i> spp.	23.40	12	<i>Thalassiosira</i> spp.	32.32	12	<i>Thalassiosira</i> spp.	41.06	12
<i>Thalassiosira</i> spp.	17.72	12	<i>Chaetoceros</i> spp.	23.88	11	<i>Skeletonema costatum</i>	13.07	8
<i>Pseudo-nitzschia</i> spp.	16.45	10	<i>Pseudo-nitzschia</i> spp.	13.69	10	<i>Chaetoceros</i> spp.	13.01	11
Dinoflagellates			Dinoflagellates			Dinoflagellates		
<i>Akashiwo sanguinea</i>	40.36	4	<i>Gymnodinium</i> spp.	44.71	12	<i>Gymnodinium</i> spp.	49.36	11
<i>Gymnodinium</i> spp.	32.87	12	<i>Akashiwo sanguinea</i>	20.04	3	<i>Amphidinium</i> spp.	16.49	9
<i>Amphidinium</i> spp.	14.98	11	<i>Amphidinium</i> spp.	16.37	10	<i>Akashiwo sanguinea</i>	15.30	3
Others			Others			Others		
small flagellates	77.06	12	small flagellates	81.14	12	small flagellates	76.91	12
<i>Plagioselmis prolunga</i>	18.80	12	<i>Plagioselmis prolunga</i>	14.91	12	<i>Plagioselmis prolunga</i>	15.63	12
<i>Teleaulax acuta</i>	2.40	12	<i>Teleaulax acuta</i>	3.38	11	<i>Teleaulax acuta</i>	6.91	11
Eastern Buffer			Southern			Deep Bay		
Diatoms			Diatoms			Diatoms		
<i>Pseudo-nitzschia</i> spp.	30.65	7	<i>Skeletonema costatum</i>	27.81	10	<i>Thalassiosira</i> spp.	59.05	12
<i>Thalassiosira</i> spp.	24.67	12	<i>Chaetoceros</i> spp.	24.98	12	<i>Skeletonema costatum</i>	12.56	10
<i>Chaetoceros</i> spp.	16.10	9	<i>Thalassiosira</i> spp.	20.12	12	<i>Navicula</i> spp.	10.62	11
Dinoflagellates			Dinoflagellates			Dinoflagellates		
<i>Gymnodinium</i> spp.	59.16	12	<i>Gymnodinium</i> spp.	39.85	12	<i>Gymnodinium</i> spp.	77.95	12
<i>Akashiwo sanguinea</i>	15.86	3	<i>Akashiwo sanguinea</i>	25.25	4	<i>Amphidinium</i> spp.	11.03	5
<i>Amphidinium</i> spp.	14.94	7	<i>Amphidinium</i> spp.	22.09	11	<i>Akashiwo sanguinea</i>	6.34	2
Others			Others			Others		
small flagellates	81.86	12	small flagellates	81.72	12	small flagellates	70.43	12
<i>Plagioselmis prolunga</i>	13.19	12	<i>Plagioselmis prolunga</i>	13.16	12	<i>Plagioselmis prolunga</i>	15.99	12
<i>Teleaulax acuta</i>	3.28	10	<i>Teleaulax acuta</i>	3.51	12	<i>Teleaulax acuta</i>	10.28	12

¹ % of species/group in diatoms, dinoflagellates and other minor phytoplankton groups² Number of occurrences out of 12 sampling occasions.

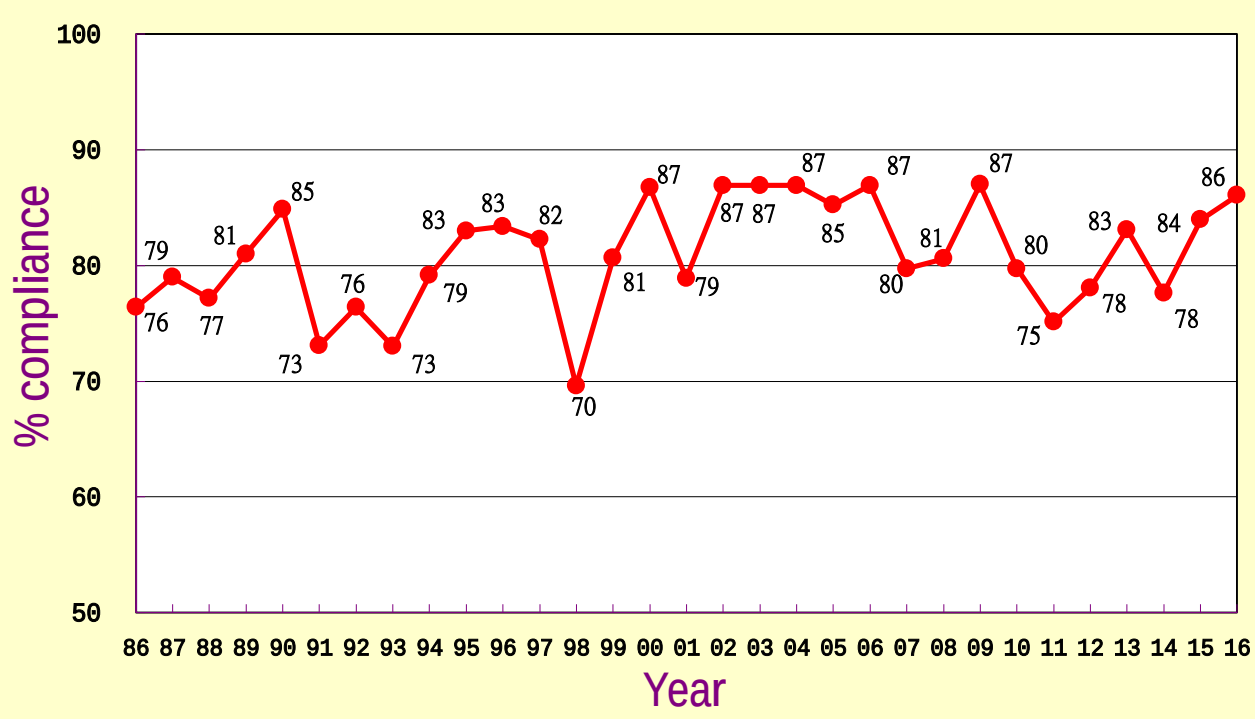
Level of compliance with key marine Water Quality Objectives for 10 Water Control Zones in Hong Kong, 2007 – 2016

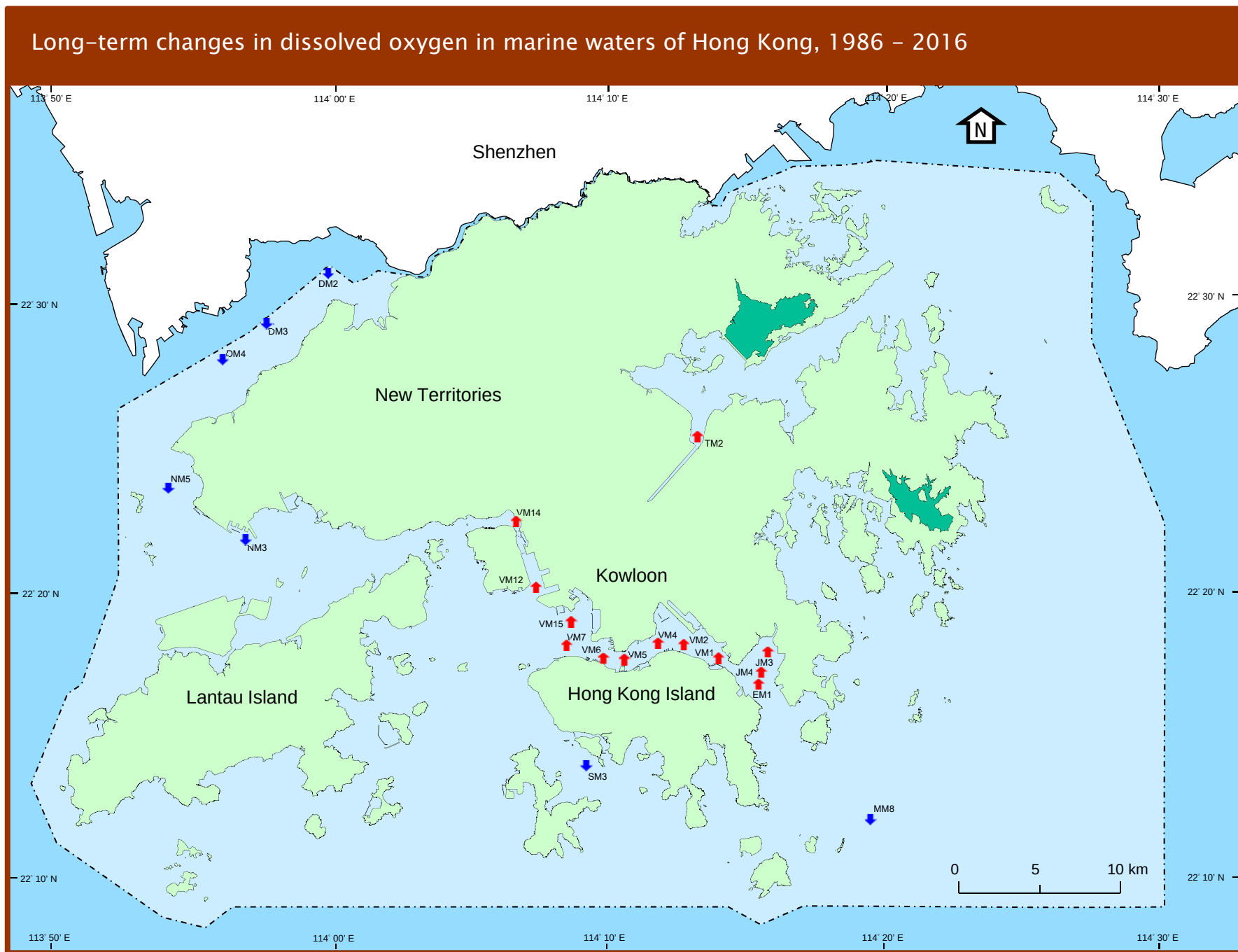


Level of compliance with key marine Water Quality Objectives in Hong Kong, 2007 – 2016



Overall level of compliance with key marine Water Quality Objectives in Hong Kong, 1986 – 2016





Long-term changes in 5-day biochemical oxygen demand in marine waters of Hong Kong, 1986 – 2016

